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MIL-HDBK-17-1D
NOTICE 1
29 SEPTEMBER 1995

MILITARY HANDBOOK
POLYMER MATRIX COMPOSITES
Volume 1. Guidelines

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1. Instructions for making Change Notice 1 revision to MIL-HDBK-17-1D are listed below.

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3. Vertical lines are used in Change Notice 1 to identify changes with respect to previous issue.

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20025

1. This Military Handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. Every effort has been made to reflect the latest information on polymeric composites. The Handbook is continually reviewed and revised to insure its completeness and currentness. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Department of the Army, U.S. Army Research Laboratory, Materials Directorate, Attn: AMSRL-MA-TS, MIL-HDBK-17 Coordinator, Aberdeen Proving Ground, MD 21005-5066, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. Documentation for the secretariat should be directed to: Materials Sciences Corporation, MIL-HDBK-17 Secretariat, 500 Office Center Drive, Suite 250, Fort Washington, PA 19034.

3. Military Handbook 17 (Part 1) provides guidelines and material properties for polymer (organic) matrix composite materials. This handbook currently focuses on, but is not limited to, polymeric composites intended for aircraft and aerospace vehicles. Metal matrix composites (MMC), ceramic matrix composites (CMC), and carbon/carbon composites (C/C) will be covered in separate military handbooks as developments occur.

4. This standardization handbook has been developed and is being maintained as a joint effort of the Department of Defense and the Federal Aviation Administration.

5. The information contained in this handbook was obtained from materials producers, industry, reports on Government sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity.

6. All information and data contained in this handbook have been coordinated with industry and the US Army, Navy, Air Force, NASA, and Federal Aviation Administration prior to publication.

7. Copies of this document and revisions thereto may be obtained from the Standardization Document Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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1.1 **INTRODUCTION.** The standardization of a statistically-based mechanical property data base, procedures, and overall material guidelines for characterization of composite material systems is recognized as being beneficial to both manufacturers and governmental agencies. It is also recognized that a complete characterization of the capabilities of any engineering material system is primarily dependent on the inherent material physical and chemical composition which precede, and are independent of, specific applications. Therefore, at the material system characterization level, the data and guidelines contained in this handbook are applicable to military and commercial products and provide the technical basis for establishing statistically valid design values acceptable to certifying or procuring agencies.

This handbook specifically provides statistically-based mechanical property data on current and emerging polymer matrix composite materials, provides guidelines for the analysis and presentation of data, and provides fabrication and characterization documentation to ensure repeatability of results or reliable detection of differences. The primary focus of MIL-HDBK-17 in the overall characterization/ design procedure as commonly applied to composites is shown in Figure 1.1.

The data and guidelines contained herein have been developed from experience gained in aerospace and aircraft vehicle design. Expansion into a broader class of applications is ongoing.

The data contained herein, or appearing in approved items in the minutes of MIL-HDBK-17 coordination meetings are acceptable to the Army, the Navy, the Air Force, and the Federal Aviation Administration. Approval by the certifying or procuring agency must be obtained for the use of data or guidelines not contained herein.

This standardization handbook has been developed and is maintained as a joint effort of the Department of Defense and the Federal Aviation Administration. It is oriented toward the standardization of methods used to develop and analyze mechanical property data on current and emerging composite materials.

MIL-HDBK-17 is divided into three volumes. The first volume is oriented toward guidelines for data development and analyses. Chapter 2 provides guidelines for the generation of material properties, including the qualification of alternate materials and pooling of data from different sources. Chapters 3 through 5 define acceptable procedures for the evaluation of composite constituents including reinforcement fibers, resins, and prepreg materials. Chapter 6 addresses acceptable procedures for the evaluation of lamina and laminate materials. Procedures for analyzing composite structural details, and in particular, bolted joints are presented in Chapter 7. Chapter 8 of Volume 1 addresses the analysis and presentation of composite material property data. Important references are cited at the end of each chapter.

The second volume of MIL-HDBK-17 provides a compilation of statistically-based material properties for current and emerging composite materials. The first two chapters of this volume address matrix material and reinforcement fiber constituent properties. The remaining chapters address composite system properties and they are defined according to the primary reinforcement fiber types. Properties for composite materials reinforced with glass fibers, aramid fibers, carbon fibers, and so on are covered in separate chapters.

Volume 3 provides guidelines for the application of the data which are presented in Volume 2. Chapter 2 addresses typical composite materials and processes. The matrix materials and reinforcement fibers of interest in MIL-HDBK-17 are reviewed, as are the typical product forms. Fabrication methods are also covered for information purposes. Chapter 3 reviews important issues related to quality control in the production of composite materials. Recommended manufacturing inspection procedures are reviewed, along with techniques for material property verification and

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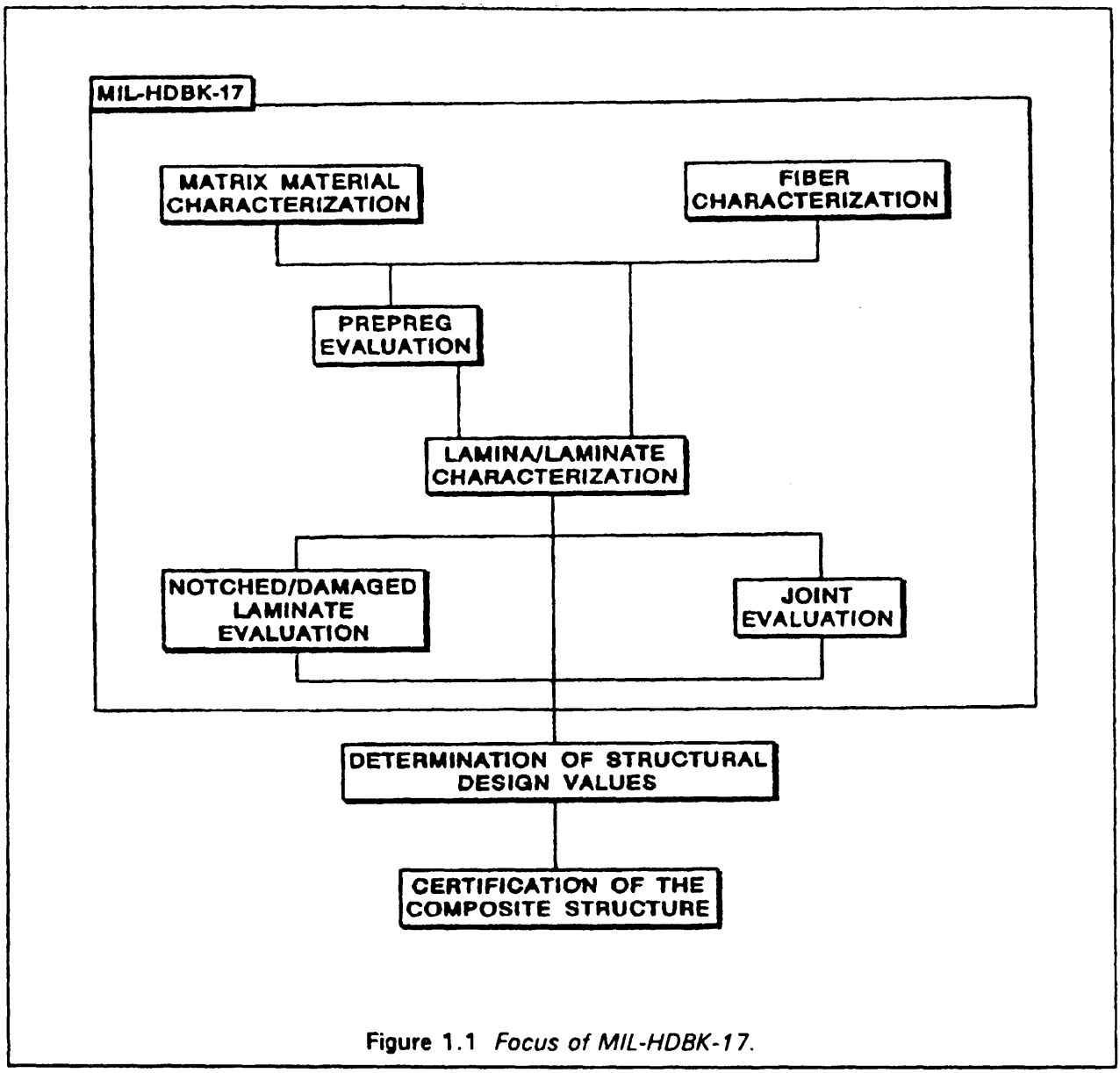


Figure 1.1 *Focus of MIL-HDBK-17.*

statistical quality control. Chapter 4 addresses the design and analysis of composite material systems. It provides an overview of the current techniques and describes how the various constituent properties reported in MIL-HDBK-17 are used in the design and analysis of a composite system.

The remainder of Chapter 1 of this volume provides additional information of importance to any user of MIL-HDBK-17. The purpose and scope of the document are defined and some comments are provided on its use and limitations. The accepted system of symbols, abbreviations, and units are also reviewed, and a large collection of terms pertinent to this handbook are defined.

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1.2 **PURPOSE.** The purpose of this Handbook is to provide a standard source of statistically-based mechanical property data for current and emerging composite materials, and to provide standard methods for general use in developing and analyzing mechanical property data. In order to accomplish this, documentation requirements on the fabrication and characterization of these composites are also defined.

Twice yearly MIL-HDBK-17 coordination meetings are held for the specific purpose of reviewing and approving new guidelines and data proposals. Participants represent DoD, NASA, FAA, industry, and academia, both national and international. Materials data approved and included in this handbook or the minutes of the MIL-HDBK-17 coordination group meetings are not mandatory but are considered acceptable to the Army, Navy, Air Force, and Federal Aviation Administration in showing compliance with respective agency requirements and are considered effective on the date approved by MIL-HDBK-17 coordination committee. The use of data or guidelines that are not approved and therefore not included in this document must be approved by the appropriate certification or procurement agency."

1.3 **SCOPE.** MIL-HDBK-17 is published in three volumes, and serves as a source for the following:

Volume 1 - Provides guidelines for the characterization of composite material systems to be used in vehicles and structures. Composite material systems must normally be evaluated in accordance with these, or equivalent guidelines, in order to be considered acceptable by government certification and procuring agencies.

Volume 2 - Provides a compilation of statistically-based mechanical property data for current and emerging composite material systems used in the aerospace industry. B-basis strength and strain-to-failure values will be presented along with related data.

Volume 3 - Provides information regarding materials and fabrication procedures, quality control, and design and analysis.

Specifically, Volume 1 provides guidelines for the physical, chemical, and mechanical characterization of composite materials. These guidelines address characterization of the fiber, matrix, and prepreg materials, which are the primary constituents in a typical composite system. The guidelines also address the characterization of the composite system, with particular emphasis on lamina properties, rather than laminate properties. Recommendations on the evaluation of bolted joints in composite materials are provided. Volume 1 also provides guidelines for the statistical analysis and presentation of data.

Volume 2 provides statistically-based mechanical property data for composite material systems used in the aerospace industry. Strength and strain-to-failure properties are reported either in terms of B-values or S-values (see Section 1.7). Stiffness properties are generally reported as typical values. The specific statistical significance of each of these quantities is defined in the first chapter of Volume 2. Physical, chemical, and mechanical properties of the composite constituents - the fibers, matrix material and prepreg - are reported where applicable. Later chapters include data summaries for the various composite systems. Individual chapters focus on particular reinforcement fibers.

Volume 3 provides guidelines for the design and analysis of composite materials. The chapters in Volume 3 address related topics of 1) Materials and Processes, 2) Quality Control of Production

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Composites, and 3) Design and Analysis. This information is included primarily for background information and is not offered for regulatory purposes.

Statistically-based strength properties are defined for each composite material system over a range of potential usage conditions. The intent is to provide data at the upper and lower limits of the potential environmental conditions for a particular material, so that applications issues do not govern the mechanical property characterizations. If data are also available at intermediate environmental conditions, they are used to more exactly define the relationship between the mechanical properties and the effect of the environment on those properties. The statistically-based strength data which are available are tabulated in Volume 2. These data are useful as a starting point for establishing structural design allowables when stress and strength analysis capabilities permit lamina level margin of safety checks. Depending on the application, some structural design allowables will have to be determined empirically at the laminate and composite level, since MIL-HDBK-17 does not provide these data.

Additional information and properties are added as they become available and are demonstrated to meet the guideline criteria. Typical property values, as well as S-values (see definitions) are included if they meet the approval of the MIL-HDBK-17 Coordination Group.

When the guidelines or data requirements of MIL-HDBK-17 cannot be followed, the certifying or procuring government agency should be contacted to determine data requirements and other documentation which may be necessary to justify data values proposed or used by the manufacturer.

1.4 **USE OF THE DOCUMENT AND LIMITATIONS.** The information contained in MIL-HDBK-17 is obtained from materials producers and fabricators, the aerospace industry, reports on government-sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity. All of the information and data contained in this document have been coordinated with representatives from industry and the Army, Navy, Air Force, and Federal Aviation Administration prior to publication. Every effort has been made to reflect the latest information on composite materials, vehicles, and structures. The handbook is continually reviewed and revised to insure its completeness and to keep it as current as possible.

All data included herein are based on test specimens only. Unless otherwise noted, test specimens dimensions conform with those specified for the particular test method which is used. Standard test methods are recommended where possible (ASTM standards are the primary source). The designer and all other users must be responsible for any translation of the data contained herein to other coupon dimensions, temperature, humidity, and other environmental conditions not covered in this document. Problems such as scale up effects and the influence of the test method selected on properties are also not addressed in this document. The manner in which S-basis values are used is also up to the discretion of the designer. In general, decisions concerning which properties to use for a specific application or design are the responsibility of the designer and are outside the scope of this handbook.

The data which are tabulated in this handbook are intended as an aid in assigning property values to a material. In specific cases where it is necessary or preferable to develop strength properties superior to those in this handbook acceptance of such values must be obtained from the appropriate procurement or certification agency. The applicability and interpretation of specific provisions of this handbook must also be defined by the appropriate procurement, regulatory, or certification agency.

Reference information which is cited in this handbook may not comply in every respect with the guidelines or other criteria specified in this document. References are provided at the end of each chapter primarily as a source of additional information in a given subject area.

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The use of tradenames and proprietary product names does not constitute an endorsement of those products by the Government.

1.4.1 Toxicity, Health Hazards, and Safety. Certain processing and test methods discussed in MIL-HDBK-17 may involve hazardous materials, operations or equipment. These methods may not address safety problems associated with their use. It is the responsibility of the user of these methods to establish appropriate safety and health practices and to determine the applicability of regulatory limitations prior to use. The user is referred to the Advanced Composite Materials U. S. Army Interim Health and Safety Guidance for a discussion of the health and safety issues involved in the processing and use of composite materials. This document is generated by the U. S. Army Environmental Hygiene Agency, Aberdeen Proving Ground, MD. Various composites user groups may also provide guidance on health and safety issues pertinent to composite materials. Restrictions on the use of ozone depleting chemicals are detailed in the Clean Air Act of 1991.

1.5 APPROVAL PROCEDURES. The MIL-HDBK-17 Coordination Group is a joint government-industry activity that meets twice yearly. At each meeting, this group acts upon proposed changes or additions to the document that are submitted by any of the working groups. The agenda is usually mailed to attendees four weeks prior to the meeting date, and the minutes four weeks following the meeting. Requests for consideration of material for inclusion in the handbook should be submitted to the appropriate working group and the Secretariat well in advance of the mailing date.

Attachments containing proposed changes or additions to the document shall include specific notation of the changes or additions to be made and adequate documentation of supporting data and analytical procedures. Reproducible copies of drawings or photographs intended for inclusion in the document shall be furnished to the Secretariat.

Requests for inclusion of data in MIL-HDBK-17 should be submitted to the Secretariat with the documentation specified in Section 8.1.2. Following analysis and review of the data by the Secretariat, the data will be presented at the next meeting of the Coordination Group.

The choice of new materials to be included herein is governed by the MIL-HDBK-17 Coordination Group. Practical considerations preclude inclusion of all advanced composite materials that may be of interest. Reasonable attempts will be made to add new materials of interest in a timely manner.

1.6 SYMBOLS, ABBREVIATIONS, AND SYSTEMS OF UNITS. This section defines the symbols and abbreviations which are used within MIL-HDBK-17 and describes the system of units which is maintained. Common usage is maintained where possible. References 1.6(a), 1.6(b), and 1.6(c) served as primary sources for this information.

1.6.1 Symbols and abbreviations. The symbols and abbreviations used in this document are defined in this section with the exception of statistical symbols. These latter symbols are defined in Chapter 8. The lamina/laminate coordinate axes used for all properties and a summary of the mechanical property notation are shown in Figure 1.6.1.

- The symbols f and m, when used as either subscripts or superscripts, always denote fiber and matrix, respectively.
- The type of stress (e.g., cy - compression yield) is always used in the superscript position.
- Direction indicators (e.g., x, y, z, 1, 2, 3, etc.) are always used in the subscript position.

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- Ordinal indicators of laminae sequence (e.g., 1, 2, 3, etc.) are used in the superscript position and must be parenthesized to distinguish them from mathematical exponents.

- Other indicators may be used in either subscript or superscript position, as appropriate for clarity.

- Compound symbols (i.e., basic symbol plus indicators) which deviate from these rules are shown in their specific form in the following list.

The following general symbols and abbreviations are considered standard for use in MIL-HDBK-17. Where exceptions are made, they are noted in the text and tables.

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.1 certification. The contractor shall furnish a certificate of 4.4.1

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open seams.

GPa - gigapascal(s)
g - (1) gram(s)
- (2) acceleration due to gravity (m/s^2,ft/s^2)
H/C - honeycomb (sandwich)
h - height dimension (mm,in) e.g. the height of a beam cross-section
hr - hour(s)
I - area moment of inertia (mm^4,in^4)
i - slope (due to bending) of neutral plane in a beam, in radians
in. - inch(es)
J - (1) torsion constant (= I_p for round tubes) (m^4,in^4)
- (2) Joule
K - (1) Kelvin
- (2) stress intensity factor (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
- (3) coefficient of thermal conductivity (W/m °C, BTU/ft^2/hr/in/°F)
- (4) correction factor
- (5) dielectric constant
K_{app} - apparent plane strain fracture toughness or residual strength (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_c - critical plane strain fracture toughness, a measure of fracture toughness at point of crack growth instability (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_{Ic} - plane strain fracture toughness (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_N - empirically calculated fatigue notch factor
K_s - plate or cylinder shear buckling coefficient
K_t - (1) theoretical elastic stress concentration factor
- (2) t_w/c ratio in H/C sandwich
Kv - dielectric strength (KV/mm, V/mil)
K_x,K_y - plate or cylinder compression buckling coefficient
k - strain at unit stress (m/m,in/in)
L - cylinder, beam, or column length (mm,in)
L' - effective column length (mm,in)
lb - pound
M - applied moment or couple (N-m,in-lbf)
Mg - megagram(s)
MPa - megapascal(s)
MS - military standard
M.S. - margin of safety
MW - molecular weight
MWD - molecular weight distribution
m - (1) mass (kg,lb)
- (2) number of half wave lengths
- (3) metre
- (4) slope
N - (1) number of fatigue cycles to failure
- (2) number of laminae in a laminate
- (3) distributed in-plane forces on a panel (lbf/in)
- (4) Newton
- (5) normalized
NA - neutral axis
NAS - National Aerospace Standard
n - (1) number of times in a set
- (2) number of half or total wavelengths
- (3) number of fatigue cycles endured

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P - (1) applied load (N,lbf)
- (2) exposure parameter
- (3) probability
- (4) specific resistance (Ω)

P^u - test ultimate load, (N,lb per fastener)

P^y - test yield load, (N,lb per fastener)

p - normal pressure (Pa,psi)

psi - pounds per square inch

Q - area static moment of a cross-section (mm^3,in^3)

q - shear flow (N/m,lbf/in)

R - (1) algebraic ratio of minimum load to maximum load in cyclic loading
- (2) reduced ratio

RA - reduction of area

R.H. - relative humidity

RMS - root-mean-square

RT - room temperature

r - (1) radius (mm,in)
- (2) root radius (mm,in)
- (3) reduced ratio (regression analysis)

S - (1) shear force (N,lbf)
- (2) nominal stress in fatigue (MPa,ksi)
- (3) S-basis for mechanical property values

S_a - stress amplitude in fatigue (MPa,ksi)

S_e - fatigue limit (MPa,ksi)

S_m - mean stress in fatigue (MPa,ksi)

S_{max} - highest algebraic value of stress in the stress cycle (MPa,ksi)

S_{min} - lowest algebraic value of stress in the stress cycle (MPa,ksi)

S_R - algebraic difference between the minimum and maximum stresses in one cycle (MPa,ksi)

S.F. - safety factor

s - (1) arc length (mm,in)
- (2) H/C sandwich cell size (mm,in)

T - (1) temperature (°C,°F)
- (2) applied torsional moment (N-m,in-lbf)

T_d - thermal decomposition temperature (°C,°F)

T_F - exposure temperature (°C,°F)

T_g - glass transition temperature (°C,°F)

T_m - melting temperature (°C,°F)

t - (1) thickness (mm,in)
- (2) exposure time (s)
- (3) elapsed time (s)

V - (1) volume (mm^3,in^3)
- (2) shear force (N,lbf)

W - (1) weight (N,lbf)
- (2) width (mm,in)
- (3) Watt

x - distance along a coordinate axis

Y - nondimensional factor relating component geometry and flaw size

y - (1) deflection (due to bending) of elastic curve of a beam (mm,in)
- (2) distance from neutral axis to given point
- (3) distance along a coordinate axis

Z - section modulus, I/y (mm^3,in^3)

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α - coefficient of thermal expansion (m/m/°C,in/in/°F)
γ - sheer strain (m/m,in/in)
Δ - difference (used as prefix to quantitative symbols)
δ - elongation or deflection (mm,in)
ϵ - strain (m/m,in/in)
ϵ - elastic strain (m/m,in/in)
ϵ - plastic strain (m/m,in/in)
μ - permeability
η - plasticity reduction factor
$[\eta]$ - intrinsic viscosity
η^* - dynamic complex viscosity
ν - Poisson's ratio
ρ - (1) density (kg/m3,lb/in3)
- (2) radius of gyration (mm,in)
ρ_c - H/C sandwich core density (kg/m^3,lb/in^3)
Σ - total, summation
σ - standard deviation
σ_{ij}, τ_{ij} - stress in j direction on surface whose outer normal is in i direction (i, j = 1, 2, 3 or x, y, z) (MPa,ksi)
T - applied shear stress (MPa,ksi)
ω - angular velocity (radians/s)
∞ - infinity

1.6.1.1 *Constituent properties.* The following symbols apply specifically to the constituent properties of a typical composite material.

E^f - Young's modulus of filament material (MPa,ksi)
E^m - Young's modulus of matrix material (MPa,ksi)
E_x^g - Young's modulus of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (MPa,ksi)
E_y^g - Young's modulus of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (MPa,ksi)
G^f - shear modulus of filament material (MPa,ksi)
G^m - shear modulus of matrix (MPa,ksi)
G_{xy}^g - shear modulus of impregnated glass scrim cloth (MPa,ksi)
G_{cx}' - shear modulus of sandwich core along X-axis (MPa,ksi)
G_{cy}' - shear modulus of sandwich core along Y axis (MPa,ksi)
ℓ - filament length (mm,in)
α^f - coefficient of thermal expansion for filament material (m/m/°C,in/in/°F)
α^m - coefficient of thermal expansion for matrix material (m/m/°C,in/in/°F)
α_x^g - coefficient of thermal expansion of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (m/m/°C,in/in/°F)
α_y^g - coefficient of thermal expansion of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (m/m/°C,in/in/°F)
ν^f - Poisson's ratio of filament material
ν^m - Poisson's ratio of matrix material

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ν_{xy}^{g} - glass scrim cloth Poisson's ratio relating to contraction in the transverse (or fill) direction as a result of extension in the longitudinal (or warp) direction

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ν^{g}_{yx} - glass scrim cloth Poisson's ratio relating to contraction in the longitudinal (or warp) direction as a result of extension in the transverse (or fill) direction
σ - applied axial stress at a point, as used in micromechanics analysis (MPa,ksi)
τ - applied shear stress at a point, as used in micromechanics analysis (MPa,ksi)

1.6.1.2 *Laminae and laminates.* The following symbols, abbreviations, and notations apply to composite laminae and laminates. At the present time the focus in MIL-HDBK-17 is on laminae properties. However, commonly used nomenclature for both laminae and laminates are included here to avoid potential confusion.

A_{ij} (i,j = 1,2,6) - extensional rigidities (N/m,lbf/in)
B_{ij} (i,j = 1,2,6) - coupling matrix (N,lbf)
C_{ij} (i,j = 1,2,6) - elements of stiffness matrix (Pa,psi)
D_x, D_y - flexural rigidities (N-m,lbf-in)
D_{xy} - twisting rigidity (N-m,lbf-in)
D_{ij} (i,j = 1,2,6) - flexural rigidities (N-m,lbf-in)
E_1 - Young's modulus of lamina parallel to filament or warp direction (GPa,Msi)
E_2 - Young's modulus of lamina transverse to filament or warp direction (GPa,Msi)
E_x - Young's modulus of laminate along x reference axis (GPa,Msi)
E_y - Young's modulus of laminate along y reference axis (GPa,Msi)
G_{12} - shear modulus of lamina in 12 plane (GPa,Msi)
G_{xy} - shear modulus of laminate in xy reference plane (GPa,Msi)
h_i - thickness of ith ply or lamina (mm,in)
M_x, M_y, M_{xy} - bending and twisting moment components (N-m/m, in-lbf/in in plate and shell analysis)
n_f - number of filaments per unit length per lamina
Q_x, Q_y - shear force parallel to z axis of sections of a plate perpendicular to x and y axes, respectively (N/m,lbf/in)
Q_{ij} (i,j = 1,2,6) - reduced stiffness matrix (Pa,psi)
u_x, u_y, u_z - components of the displacement vector (mm,in)
u°_x, u°_y, u°_z - components of the displacement vector at the laminate's midsurface (mm,in)
V_v - void content (% by volume)
V_f - filament content or fiber volume (% by volume)
V_g - glass scrim cloth content (% by volume)
V_m - matrix content (% by volume)
V_x, V_y - edge or support shear force (N/m,lbf/in)
W_f - filament content (% by weight)
W_g - glass scrim cloth content (% by weight)
W_m - matrix content (% by weight)
W_s - weight of laminate per unit surface area (N/m^2,lbf/in^2)
α_1 - lamina coefficient of thermal expansion along 1 axis (m/m/°C,in/in/°F)
α_2 - lamina coefficient of thermal expansion along 2 axis (m/m/°C,in/in/°F)
α_x - laminate coefficient of thermal expansion along general reference x axis (m/m/°C,in/in/°F)
α_y - laminate coefficient of thermal expansion along general reference y axis (m/m/°C,in/in/°F)
α_{xy} - laminate shear distortion coefficient of thermal expansion (m/m/°C,in/in/°F)
θ - angular orientation of a lamina in a laminate, i.e., angle between 1 and x axes (°)
λ_{xy} - product of ν_{xy} and ν_{yx}

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ν_{12} - Poisson's ratio relating contraction in the 2 direction as a result of extension in the 1 direction[1]

ν_{21} - Poisson's ratio relating contraction in the 1 direction as a result of extension in the 2 direction[1]

ν_{xy} - Poisson's ratio relating contraction in the y direction as a result of extension in the x direction[1]

ν_{yx} - Poisson's ratio relating contraction in the x direction as a result of extension in the y direction[1]

ρ_c - density of a single lamina (kg/m^3,lb/in^3)

$\bar{\rho}_c$ - density of a laminate (kg/m^3,lb/in^3)

ϕ - (1) general angular coordinate, (°)
- (2) angle between x and load axes in off-axis loading (°)

1.6.1.3 *Subscripts.* The following subscript notations are considered standard in MIL-HDBK-17.

1, 2, 3 - laminae natural orthogonal coordinates (1 is filament or warp direction)
A - axial
a - (1) adhesive
- (2) alternating
app - apparent
byp - bypass
c - composite system, specific filament/matrix composition. Composite as a whole, contrasted to individual constituents. Also, sandwich core when used in conjunction with prime (')
- (4) critical
cf - centrifugal force
e - fatigue or endurance
eff - effective
eq - equivalent
f - filament
g - glass scrim cloth
H - hoop
i - i^{th} position in a sequence
L - lateral
m - (1) matrix
- (2) mean
max - maximum
min - minimum
n - (1) n^{th} (last) position in a sequence
- (2) normal
p - polar
s - symmetric
st - stiffener
T - transverse
t - value of parameter at time t
x, y, z - general coordinate system

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[1] The convention for Poisson's ratio should be checked before comparing different sources as different conventions are used.

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Σ	- total, or summation
o	- initial or reference datum
( )	- format for indicating specific, temperature associated with term in parentheses. RT - room temperature (21°C,70°F); all other temperatures in °F unless specified.

1.6.1.4 *Superscripts.* The following superscript notations are considered standard in MIL-HDBK-17.

b	- bending
br	- bearing
c	- (1) compression
	- (2) creep
cc	- compression crippling
cr	- compression buckling
e	- elastic
f	- filament
g	- glass scrim cloth
is	- interlaminar shear
(i)	- i^{th} ply or lamina
lim	- limit, used to indicate limit loading
m	- matrix
ohc	- open hole compression
oht	- open hole tension
p	- plastic
pl	- proportional limit
rup	- rupture
s	- shear
scr	- shear buckling
sec	- secant (modulus)
T	- temperature or thermal
t	- tension
tan	- tangent (modulus)
u	- ultimate
y	- yield
'	- secondary (modulus), or denotes properties of H/C core when used with subscript c
CAI	- compression after impact

1.6.2 System of units. To comply with Department of Defense Instructive 5000.2, Part 6, Section M, "Use of the Metric System," dated February 23, 1991, the data in MIL-HDBK-17 are generally presented in both the International System of Units (SI units) and the U. S. Customary (English) system of units. ASTM E-380, Standard for Metric Practice, provides guidance for the application for SI units which are intended as a basis for worldwide standardization of measurement units (Reference 1.6.2(a)). Further guidelines on the use of the SI system of units and conversion factors are contained in the following publications (References 1.6.2(b) - (e)):

(1) DARCOM P 706-470, *Engineering Design Handbook: Metric Conversion Guide*, July 1976.

(2) NBS Special Publication 330, "The International System of Units (SI)," National Bureau of Standards, 1986 edition.

(3) NBS Letter Circular LC 1035, "Units and Systems of Weights and Measures, Their Origin, Development, and Present Status," National Bureau of Standards, November 1985.

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(4) NASA Special Publication 7012, "The International System of Units Physical Constants and Conversion Factors", 1964.

English to SI conversion factors pertinent to MIL-HDBK-17 data are contained in Table 1.6.2.

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Aramid -- A manufactured fiber in which the fiber-forming substance consisting of a long-chain synthetic aromatic polyamide in which at least 85% of the amide (-CONH-) linkages are attached directly to two aromatic rings.

Areal Weight of Fiber -- The weight of fiber per unit area of prepreg. This is often expressed as grams per square meter. See Table 1.6.2 for conversion factors.

Artificial Weathering -- Exposure to laboratory conditions which may be cyclic, involving changes in temperature, relative humidity, radiant energy and any other elements found in the atmosphere in various geographical areas.

Aspect Ratio -- In an essentially two-dimensional rectangular structure (e.g., a panel), the ratio of the long dimension to the short dimension. However, in compression loading, it is sometimes considered to be the ratio of the load direction dimension to the transverse dimension. Also, in fiber micro-mechanics, it is referred to as the ratio of length to diameter.

Autoclave -- A closed vessel for producing an environment of fluid pressure, with or without heat, to an enclosed object which is undergoing a chemical reaction or other operation.

Autoclave Molding -- A process similar to the pressure bag technique. The lay-up is covered by a pressure bag, and the entire assembly is placed in an autoclave capable of providing heat and pressure for curing the part. The pressure bag is normally vented to the outside.

Axis of Braiding: The direction in which the braided form progresses.

B-Basis (or B-Value) -- The mechanical property value above which at least 90 percent of the population of values is expected to fall, with a confidence of 95 percent, specifically the same as A-Basis except on the tenth percentile. (See Volume 1, Section 8.5.1.1)

B-Stage -- An intermediate stage in the reaction of a thermosetting resin in which the material softens when heated and swells when in contact with certain liquids but does not entirely fuse or dissolve. Materials are usually precured to this stage to facilitate handling and processing prior to final cure. (Sometimes referred to as resitol.)

Bag Molding -- A method of molding or laminating which involves the application of fluid pressure to a flexible material which transmits the pressure to the material being molded or bonded. Fluid pressure usually is applied by means of air, steam, water or vacuum.

Balanced Laminate -- A composite laminate in which all laminae at angles other than 0 degrees and 90 degrees occur only in ± pairs (not necessarily adjacent).

Batch (or Lot) -- For fibers and resins, a quantity of material formed during the same process and having identical characteristics throughout. For prepregs, laminae, and laminates, material made from one batch of fiber and one batch of resin.

Bearing Area -- The product of the pin diameter and the specimen thickness.

Bearing Load -- A compressive load on an interface.

Bearing Yield Strength -- The bearing stress at which a material exhibits a specified limiting deviation from the proportionality of bearing stress to bearing strain.

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Bend Test -- A test of ductility by bending or folding, usually with steadily applied forces. In some instances the test may involve blows to a specimen having a cross section that is essentially uniform over a length several times as great as the largest dimension of the cross section.

Binder -- A bonding resin used to hold strands together in a mat or preform during manufacture of a molded object.

Binomial Random Variable -- The number of successes in independent trials where the probability of success is the same for each trial.

Birefringence -- The difference between the two principal refractive indices (of a fiber) or the ratio between the retardation and thickness of a material at a given point.

Bleeder Cloth -- A nonstructural layer of material used in the manufacture of composite parts to allow the escape of excess gas and resin during cure. The bleeder cloth is removed after the curing process and is not part of the final composite.

Bobbin: A cylinder or slightly tapered barrel, with or without flanges, for holding tows, rovings, or yarns.

Bond -- The adhesion of one surface to another, with or without the use of an adhesive as a bonding agent.

Braid: A system of three or more yarns which are interwoven in such a way that no two yarns are twisted around each other.

Braid Angle: The acute angle measured from the axis of braiding.

Braid, Biaxial: Braided fabric with two-yarn systems, one running in the $+\theta$ direction, the other in the $-\theta$ direction as measured from the axis of braiding.

Braid Count: The number of braiding yarn crossings per inch measured along the axis of a braided fabric.

Braid, Diamond: Braided fabric with an over one, under one weave pattern, (1 x 1).

Braid, Flat: A narrow bias woven tape wherein each yarn is continuous and is intertwined with every other yarn in the system without being intertwined with itself.

Braid, Hercules: A braided fabric with an over three, under three weave pattern, (3 x 3).

Braid, Jacquard: A braided design made with the aid of a jacquard machine, which is a shedding mechanism by means of which a large number of ends may be controlled independently and complicated patterns produced.

Braid, Regular: A braided fabric with an over two, under two weave pattern (2 x 2).

Braid, Square: A braided pattern in which the yarns are formed into a square pattern.

Braid, Two-Dimensional: Braided fabric with no braiding yarns in the through thickness direction.

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Braid, Three-Dimensional: Braided fabric with one or more braiding yarns in the through thickness direction.

Braid, Triaxial: A biaxial braided fabric with laid in yarns running in the axis of braiding.

Braiding -- A textile process where two or more strands, yarns or tapes are intertwined in the bias direction to form an integrated structure.

Broadgoods -- A term loosely applied to prepreg material greater than about 12 inches in width, usually furnished by suppliers in continuous rolls. The term is currently used to designate both collimated uniaxial tape and woven fabric prepregs.

Buckling (Composite) -- A mode of structural response characterized by an out-of-plane material deflection due to compressive action on the structural element involved. In advanced composites, buckling may take the form not only of conventional general instability and local instability but also a micro-instability of individual fibers.

Bundle -- A general term for a collection of essentially parallel filaments or fibers.

C-Stage -- The final stage of the curing reaction of a thermosetting resin in which the material has become practically infusable and insoluble. (Normally considered fully cured and sometimes referred to as resite.)

Capstan: A friction type take-up device which moves braided fabric away from the fell. The speed of which determines the braid angle.

Carbon Fibers -- Fibers produced by the pyrolysis of organic precursor fibers such as rayon, polyacrylonitrile (PAN), and pitch in an inert atmosphere. The term is often used interchangeably with "graphite"; however, carbon fibers and graphite fibers differ in the temperature at which the fibers are made and heat-treated, and the amount of carbon produced. Carbon fibers typically are carbonized at about 2400°F (1300°C) and assay at 93 to 95% carbon, while graphite fibers are graphitized at 3450 to 5450°F (1900 to 3000°C) and assay at more than 99% elemental carbon.

Carrier: A mechanism for carrying a package of yarn through the braid weaving motion. A typical carrier consists of a bobbin spindle, a track follower, and a tensioning device.

Caul Plates -- Smooth metal plates, free of surface defects, the same size and shape as a composite lay-up, used immediately in contact with the lay-up during the curing process to transmit normal pressure and to provide a smooth surface on the finished laminate.

Censoring -- Data is right (left) censored at M, if, whenever an observation is less than or equal to M (greater than or equal to M), the actual value of the observation is recorded. If the observation exceeds (is less than) M, the observation is recorded as M.

Chain-Growth Polymerization -- One of the two principal polymerization mechanisms. In chain-growth polymerization, the reactive groups are continuously regenerated during the growth process. Once started, the polymer molecule grows rapidly by a chain of reactions emanating from a particular reactive initiator which may be a free radical, cation or anion.

Chromatogram -- A plot of detector response against peak volume of solution (eluate) emerging from the system for each of the constituents which have been separated.

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Circuit -- One complete traverse of the fiber feed mechanism of a winding machine; one complete traverse of a winding band from one arbitrary point along the winding path to another point on a plane through the starting point and perpendicular to the axis.

Cocuring -- The act of curing a composite laminate and simultaneously bonding it to some other prepared surface during the same cure cycle (see **Secondary Bonding**).

Coefficient of Linear Thermal Expansion -- The change in length per unit length resulting from a one-degree rise in temperature.

Coefficient of Variation -- The ratio of the population (or sample) standard deviation to the population (or sample) mean.

Collimated -- Rendered parallel.

Compatible -- The ability of different resin systems to be processed in contact with each other without degradation of end product properties. (See **Compatible**, Volume 1, Section 8.5.1.1)

Composite Class -- As used in the handbook, a major subdivision of composite construction in which the class is defined by the fiber system and the matrix class, e.g., organic-matrix filamentary laminate.

Composite Material -- Composites are considered to be combinations of materials differing in composition or form on a macroscale. The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other although they act in concert. Normally, the components can be physically identified and exhibit an interface between one another.

Compound -- An intimate mixture of polymer or polymers with all the materials necessary for the finished product.

Condensation Polymerization -- This is a special type of step-growth polymerization characterized by the formation of water or other simple molecules during the stepwise addition of reactive groups.

Confidence Coefficient -- See **Confidence Interval**.

Confidence Interval -- A confidence interval is defined by a statement of one of the following forms:

$$(1)\ P\{a<\theta\} \leq 1-\alpha$$
$$(2)\ P\{\theta<b\} \leq 1-\alpha$$
$$(3)\ P\{a<\theta<b\} \leq 1-\alpha$$

where $1-\alpha$ is called the confidence coefficient. A statement of type (1) or (2) is called a one-sided confidence interval and a statement of type (3) is called a two-sided confidence interval. In (1) a is a lower confidence limit and in (2) b is an upper confidence limit. With probability at least $1-\alpha$, the confidence interval will contain the parameter θ.

Constituent -- In general, an element of a larger grouping. In advanced composites, the principal constituents are the fibers and the matrix.

Continuous Filament -- A yarn or strand in which the individual filaments are substantially the same length as the strand.

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Coupling Agent -- Any chemical substance designed to react with both the reinforcement and matrix phases of a composite material to form or promote a stronger bond at the interface. Coupling agents are applied to the reinforcement phase from an aqueous or organic solution or from a gas phase, or added to the matrix as an integral blend.

Coverage: The measure of the fraction of surface area covered by the braid.

Crazing -- Apparent fine cracks at or under the surface of an organic matrix.

Creel: A framework arranged to hold tows, rovings, or yarns so that many ends can be withdrawn smoothly and evenly without tangling.

Creep -- The time dependent part of strain resulting from an applied stress.

Creep, Rate Of -- The slope of the creep-time curve at a given time.

Crimp: The undulations induced into a braided fabric via the braiding process.

Crimp Angle: The maximum acute angle of a single braided yarn's direction measured from the average axis of tow.

Crimp Exchange: The process by which a system of braided yarns reaches equilibrium when put under tension or compression.

Critical Value(s) -- When testing a one-sided statistical hypothesis, a critical value is the value such that, if the test statistic is greater than (less than) the critical value, the hypothesis is rejected. When testing a two-sided statistical hypothesis, two critical values are determined. If the test statistic is either less than the smaller critical value or greater than the larger critical value, then the hypothesis is rejected. In both cases, the critical value chosen depends on the desired risk (often 0.05) of rejecting the hypothesis when it is true.

Crossply -- Any filamentary laminate which is not uniaxial. Same as Angleply. In some references, the term crossply is used to designate only those laminates in which the laminae are at right angles to one another, while the term angleply is used for all others. In the handbook, the two terms are used synonymously. The reservation of a separate terminology for only one of several basic orientations is unwarranted because a laminate orientation code is used.

Cumulative Distribution Function -- See Volume 1, Section 8.5.1.1.

Cure -- To change the properties of a thermosetting resin irreversibly by chemical reaction, i.e., condensation, ring closure, or addition. Cure may be accomplished by addition of curing (cross-linking) agents, with or without catalyst, and with or without heat. Cure may occur also by addition, such as occurs with anhydride cures for epoxy resin systems.

Cure Cycle -- The schedule of time periods at specified conditions to which a reacting thermosetting material is subjected in order to reach a specified property level.

Cure Stress -- A residual internal stress produced during the curing cycle of composite structures. Normally, these stresses originate when different components of a lay-up have different thermal coefficients of expansion.

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Debond -- A deliberate separation of a bonded joint or interface, usually for repair or rework purposes. (See **Disbond, Unbond**).

Deformation -- The change in shape of a specimen caused by the application of a load or force.

Degradation -- A deleterious change in chemical structure, physical properties or appearance.

Delamination -- The separation of the layers of material in a laminate. This may be local or may cover a large area of the laminate. It may occur at any time in the cure or subsequent life of the laminate and may arise from a wide variety of causes.

Denier -- A direct numbering system for expressing linear density, equal to the mass in grams per 9000 meters of yarn, filament, fiber, or other textile strand.

Density -- The mass per unit volume.

Desorption -- A process in which an absorbed or adsorbed material is released from another material. Desorption is the reverse of absorption, adsorption, or both.

Deviation -- Variation from a specified dimension or requirement, usually defining the upper and lower limits.

Dielectric Constant -- The ratio of the capacity of a condenser having a dielectric constant between the plates to that of the same condenser when the dielectric is replaced by a vacuum; a measure of the electrical charge stored per unit volume at unit potential.

Dielectric Strength -- The average potential per unit thickness at which failure of the dielectric material occurs.

Disbond -- An area within a bonded interface between two adherends in which an adhesion failure or separation has occurred. It may occur at any time during the life of the structure and may arise from a wide variety of causes. Also, colloquially, an area of separation between two laminae in the finished laminate (in this case the term "delamination" is normally preferred.) (See **Debond, Unbond, Delamination.**)

Distribution -- A formula which gives the probability that a value will fall within prescribed limits. (See **Normal, Weibull,** and **Lognormal Distributions**, also Volume 1, Section 8.5.1.1).

Dry Fiber Area -- Area of fiber not totally encapsulated by resin.

Ductility -- The ability of a material to deform plastically before fracturing.

Elasticity -- The property of a material which allows it to recover its original size and shape immediately after removal of the force causing deformation.

Elongation -- The increase in gage length or extension of a specimen during a tension test, usually expressed as a percentage of the original gage length.

Eluate -- The liquid emerging from a column (in liquid chromatography).

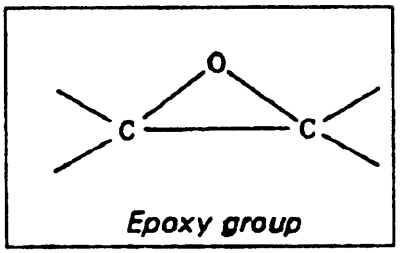
Eluent -- The mobile phase used to sweep or elute the sample (solute) components into, through, and out of the column.

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End -- A single fiber, strand, roving or yarn being or already incorporated into a product. An end may be an individual warp yarn or cord in a woven fabric. In referring to aramid and glass fibers, an end is usually an untwisted bundle of continuous filaments.

Epoxy Equivalent Weight -- The number of grams of resin which contain one chemical equivalent of the epoxy group.

Epoxy Resin -- Resins which may be of widely different structures but are characterized by the presence of the epoxy group. (The epoxy or epoxide group is usually present as a glycidyl ether, glycidyl amine, or as part of an aliphatic ring system. The aromatic type epoxy resins are normally used in composites.)



Epoxy group

Extensometer -- A device for measuring linear strain.

F-Distribution -- See Volume 1, Section 8.5.1.1.

Fabric, Nonwoven -- A textile structure produced by bonding or interlocking of fibers, or both, accomplished by mechanical, chemical, thermal, or solvent means, and combinations thereof.

Fabric, Woven -- A generic material construction consisting of interlaced yarns or fibers, usually a planar structure. Specifically, as used in this handbook, a cloth woven in an established weave pattern from advanced fiber yarns and used as the fibrous constituent in an advanced composite lamina. In a fabric lamina, the warp direction is considered the longitudinal direction, analogous to the filament direction in a filamentary lamina.

Fell: The point of braid formation, which is defined as the point at which the yarns in a braid system cease movement relative to each other.

Fiber: A general term used to refer to filamentary materials. Often, fiber is used synonymously with filament. It is a general term for a filament of finite length. A unit of matter, either natural or manmade, which forms the basic element of fabrics and other textile structures.

Fiber Content -- The amount of fiber present in a composite. This is usually expressed as a percentage volume fraction or weight fraction of the composite.

Fiber Count -- The number of fibers per unit width of ply present in a specified section of a composite.

Fiber Direction -- The orientation or alignment of the longitudinal axis of the fiber with respect to a stated reference axis.

Fiber System -- The type and arrangement of fibrous material which comprises the fiber constituent of an advanced composite. Examples of fiber systems are collimated filaments or filament yarns, woven fabric, randomly oriented short-fiber ribbons, random fiber mats, whiskers, etc.

Filament -- The smallest unit of a fibrous material. The basic units formed during spinning and which are gathered into strands of fiber, (for use in composites). Filaments usually are of extreme length and of very small diameter. Filaments normally are not used individually. Some textile filaments can function as a yarn when they are of sufficient strength and flexibility.

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Filamentary Composites -- A major form of advanced composites in which the fiber constituent consists of continuous filaments. Specifically, a filamentary composite is a laminate comprised of a number of laminae, each of which consists of a nonwoven, parallel, uniaxial, planar array of filaments (or filament yarns) embedded in the selected matrix material. Individual laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Filament Winding -- A reinforced-plastics process that employs a series of continuous, resin-impregnated fibers applied to a mandrel in a predetermined geometrical relationship under controlled tension.

Filament Wound -- Pertaining to an object created by the filament winding method of fabrication.

Fill (Filling): In a woven fabric, the yarn running from selvage to selvage at right angles to the warp.

Filler -- A relatively inert substance added to a material to alter its physical, mechanical, thermal, electrical, and other properties or to lower cost. Sometimes the term is used specifically to mean particulate additives.

Finish (or Size System) -- A material, with which filaments are treated, which contains a coupling agent to improve the bond between the filament surface and the resin matrix in a composite material. In addition, finishes often contain ingredients which provide lubricity to the filament surface, preventing abrasive damage during handling, and a binder which promotes strand integrity and facilitates packing of the filaments.

Fixed Effect -- A common systematic (nonrandom) shift in a group of measurements which is due to a controlled level change of a treatment or condition. (See Volume 1, Section 8.5.1.1.)

Flash -- Excess material which forms at the parting line of a mold or die, or which is extruded from a closed mold.

Former Plate: A die attached to a braiding machine which helps to locate the fell.

Fracture Ductility -- The true plastic strain at fracture.

Gage Length -- the original length of that portion of the specimen over which strain or change of length is determined.

Gel -- The initial jelly-like solid phase that develops during formation of a resin from a liquid. Also, a semi-solid system consisting of a network of solid aggregates in which liquid is held.

Gel Coat -- A quick-setting resin used in molding processes to provide an improved surface for the composite; it is the first resin applied to the mold after the mold-release agent.

Gel Point -- The stage at which a liquid begins to exhibit pseudo-elastic properties. (This can be seen from the inflection point on a viscosity-time plot.)

Gel Time -- The period of time from a pre-determined starting point to the onset of gelation (gel point) as defined by a specific test method.

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Glass -- An inorganic product of fusion which has cooled to a rigid condition without crystallizing. In the handbook, all reference to glass will be to the fibrous form as used in filaments, woven fabric, yarns, mats, chopped fibers, etc.

Glass Cloth -- Conventionally-woven glass fiber material (see Scrim).

Glass Fibers -- A fiber spun from an inorganic product of fusion which has cooled to a rigid condition without crystallizing.

Glass Transition -- The reversible change in an amorphous polymer or in amorphous regions of a partially crystalline polymer from (or to) a viscous or rubbery condition to (or from) a hard and relatively brittle one.

Glass Transition Temperature -- The approximate midpoint of the temperature range over which the glass transition takes place.

Graphite Fibers -- See Carbon Fibers.

Greige -- Fabric that has received no finish.

Hand Lay-up -- A process in which components are applied either to a mold or a working surface, and the successive plies are built up and worked by hand.

Hardness -- Resistance to deformation; usually measured by indention. Types of standard tests include Brinell, Rockwell, Knoop, and Vickers.

Heat Cleaned -- Glass or other fibers which have been exposed to elevated temperatures to remove preliminary sizings or binders which are not compatible with the resin system to be applied.

Heterogeneous -- Descriptive term for a material consisting of dissimilar constituents separately identifiable; a medium consisting of regions of unlike properties separated by internal boundaries. (Note that all nonhomogeneous materials are not necessarily heterogeneous).

Homogeneous -- Descriptive term for a material of uniform composition throughout; a medium which has no internal physical boundaries; a material whose properties are constant at every point, i.e., constant with respect to spatial coordinates (but not necessarily with respect to directional coordinates).

Horizontal Shear -- Sometimes used to indicate interlaminar shear. This is not an approved term for use in this handbook.

Humidity, Relative -- The ratio of the pressure of water vapor present to the pressure of saturated water vapor at the same temperature.

Hybrid -- A composite laminate comprised of laminae of two or more composite material systems. Or, a combination of two or more different fibers such as carbon and glass or carbon and aramid into a structure (tapes, fabrics and other forms may be combined).

Hygroscopic -- Capable of absorbing and retaining atmospheric moisture.

Hysteresis -- The energy absorbed in a complete cycle of loading and unloading.

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Inclusion -- A physical and mechanical discontinuity occurring within a material or part, usually consisting of solid, encapsulated foreign material. Inclusions are often capable of transmitting some structural stresses and energy fields, but in a noticeably different manner from the parent material.

Integral Composite Structure -- Composite structure in which several structural elements, which would conventionally be assembled by bonding or with mechanical fasteners after separate fabrication, are instead laid up and cured as a single, complex, continuous structure; e.g., spars, ribs, and one stiffened cover of a wing box fabricated as a single integral part. The term is sometimes applied more loosely to any composite structure not assembled by mechanical fasteners.

Interface -- The boundary between the individual, physically distinguishable constituents of a composite.

Interlaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., shear stress) referenced as existing or occurring between two or more adjacent laminae.

Interlaminar Shear -- Shearing force tending to produce a relative displacement between two laminae in a laminate along the plane of their interface.

Intermediate Bearing Stress -- The bearing stress at the point on the bearing load-deformation curve where the tangent is equal to the bearing stress divided by a designated percentage (usually 4%) of the original hole diameter.

Intralaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., temperature gradient) existing entirely within a single lamina without reference to any adjacent laminae.

Isotropic -- Having uniform properties in all directions. The measured properties of an isotropic material are independent of the axis of testing.

Jammed State: The state of a braided fabric under tension or compression where the deformation of the fabric is dominated by the deformation properties of the yarn.

Knitting: A method of constructing fabric by interlocking series of loops of one or more yarns.

Knuckle Area -- The area of transition between sections of different geometry in a filament wound part.

k-Sample Data -- A collection of data consisting of values observed when sampling from k batches.

Laid-In Yarns: A system of longitudinal yarns in a triaxial braid which are inserted between the bias yarns.

Lamina -- A single ply or layer in a laminate made up of a series of layers.

Laminae -- Plural of lamina.

Laminate -- A product made by bonding together two or more layers or laminae of material or materials.

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Laminate Orientation -- The configuration of a crossplied composite laminate with regard to the angles of crossplying, the number of laminae at each angle, and the exact sequence of the lamina lay-up.

Lattice Pattern -- A pattern of filament winding with a fixed arrangement of open voids.

Lay-up -- A process of fabrication involving the assembly of successive layers of resin-impregnated material.

Lognormal Distribution -- A probability distribution for which the probability that an observation selected at random from this population falls between a and b ($0 < a < b < B$) is given by the area under the normal distribution between ℓog a and ℓog b. The common (base 10) or the natural (base e) logarithm may be used. (See Volume 1, Section 8.5.1.1.)

Lower Confidence Bound -- See **Confidence Interval**.

Macro -- In relation to composites, denotes the gross properties of a composite as a structural element but does not consider the individual properties or identity of the constituents.

Macrostrain -- The mean strain over any finite gage length of measurement which is large in comparison to the material's interatomic distance.

Mandrel: A form fixture or male mold used for the base in the production of a part by lay-up, filament winding or braiding.

Mat -- A fibrous material consisting of randomly oriented chopped or swirled filaments loosely held together with a binder.

Material Acceptance -- The testing of incoming material to ensure that it meets requirements.

Material Qualification -- The procedures used to accept a material by a company or organization for production use.

Material System -- A specific composite material made from specifically identified constituents in specific geometric proportions and arrangements and possessed of numerically defined properties.

Material System Class -- As used in this handbook, a group consisting of material systems categorized by the same generic constituent materials, but without defining the constituents uniquely; e.g., the carbon/epoxy class.

Material Variability -- A source of variability due to variations in the spatial, consistency, and processing aspects of the material. (See Volume 1, Section 8.5.1.1.)

Matrix -- The essentially homogeneous material in which the fiber system of a composite is embedded.

Mean -- See **Sample Mean** and **Population Mean**.

Mechanical Properties -- The properties of a material that are associated with elastic and inelastic reaction when force is applied, or the properties involving the relationship between stress and strain.

Median -- See **Sample Median** and **Population Median**.

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Micro -- In relation to composites, denotes the properties of the constituents, i.e., matrix and reinforcement and interface only, as well as their effects on the composite properties.

Microstrain -- The strain over a gage length comparable to the material's interatomic distance.

Modulus, Chord -- The slope of the chord drawn between any two specified points on the stress-strain curve.

Modulus, initial: The slope of the initial straight portion of a stress-strain curve.

Modulus, Secant -- The slope of the secant drawn from the origin to any specified point on the stress-strain curve.

Modulus, Tangent -- The ratio of change in stress to change in strain derived from the tangent to any point on a stress-strain curve.

Modulus, Young's -- The ratio of change in stress to change in strain below the elastic limit of a material. (Applicable to tension and compression).

Modulus of Rigidity (also Shear Modulus or Torsional Modulus) -- The ratio of stress to strain below the proportional limit for shear or torsional stress.

Modulus of Rupture, in Bending -- The maximum tensile or compressive stress (whichever causes failure) value in the extreme fiber of a beam loaded to failure in bending. The value is computed from the flexure equation:

$$F^b = \frac{Mc}{I} \qquad 1.4(a)$$

where M = maximum bending moment computed from the maximum load and the original moment arm,
c = initial distance from the neutral axis to the extreme fiber where failure occurs,
I = the initial moment of inertia of the cross section about its neutral axis.

Modulus of Rupture, in Torsion -- The maximum shear stress in the extreme fiber of a member of circular cross section loaded to failure in torsion calculated from the equation:

$$F^s = \frac{Tr}{J} \qquad 1.4(b)$$

where T = maximum twisting moment,
r = original outer radius,
J = polar moment of inertia of the original cross section.

Moisture Content -- The amount of moisture in a material determined under prescribed condition and expressed as a percentage of the mass of the moist specimen, i.e., the mass of the dry substance plus the moisture present.

Moisture Equilibrium -- The condition reached by a sample when it no longer takes up moisture from, or gives up moisture to, the surrounding environment.

Mold Release Agent -- A lubricant applied to mold surfaces to facilitate release of the molded article.

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Molded-Edge -- An edge which is not physically altered after molding for use in final form and particularly one which does not have fiber ends along its length.

Molding -- The forming of a polymer or composite into a solid mass of prescribed shape and size by the application of pressure and heat.

Monolayer -- The basic laminate unit from which crossplied or other laminates are constructed.

Monomer -- A compound consisting of molecules each of which can provide one or more constitutional units.

NDE -- Nondestructive evaluation. Broadly considered synonymous with NDI.

NDI -- Nondestructive Inspection. A process or procedure for determining the quality or characteristics of a material, part, or assembly without permanently altering the subject or its properties.

NDT -- Nondestructive Testing. Broadly considered synonymous with NDI.

Necking -- A localized reduction in cross-sectional area which may occur in a material under tensile stress.

Negatively Skewed -- A distribution is said to be negatively skewed if the distribution is not symmetric and the longest tail is on the left.

Nominal Specimen Thickness -- The nominal ply thickness multiplied by the number of plies.

Nominal Value -- A value assigned for the purpose of a convenient designation. A nominal value exists in name only.

Normal Distribution -- A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \qquad 1.7(c)$$

between a and b. (See Volume 1, Section 8.5.1.1.)

Normalized Stress -- Stress calculated by multiplying the raw stress value by the ratio of measured fiber volume to the nominal fiber volume. This ratio is often approximated by the ratio of the nominal specimen thickness to the measured specimen thickness. Stresses for fiber-dominated failure modes are often normalized.

Observed Significance Level (OSL) -- The probability of observing a more extreme value of the test statistic when the null hypotheses is true.

Oligomer -- A polymer consisting of only a few monomer units such as a dimer, trimer, etc. or their mixtures.

One-Sided Tolerance Limit Factor -- See Tolerance Limit Factor.

Orthotropic -- Having three mutually perpendicular planes of elastic symmetry.

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Oven Dry -- The condition of a material that has been heated under prescribed conditions of temperature and humidity until there is no further significant change in its mass.

PAN Fibers -- Reinforcement fiber derived from the controlled pyrolysis of poly(acrylonitrile) fiber.

Parallel Laminate -- A laminate of woven fabric in which the plies are aligned in the same position as originally aligned in the fabric roll.

Parallel Wound: A term used to describe yarn or other material wound into a flanged spool.

Peel Ply -- A layer of resin free material used to protect a laminate for later secondary bonding.

pH -- A measure of acidity or alkalinity of a solution, with neutrality represented by a value of 7, with increasing acidity corresponding to progressively smaller values, and increasing alkalinity corresponding to progressively higher values.

Pick Count: The number of filling yarns per inch or per centimeter of woven fabric.

Pitch Fibers -- Reinforcement fiber derived from petroleum or coal tar pitch.

Plastic -- A material that contains one or more organic polymers of large molecular weight, is solid in its finished state, and, at some state in its manufacture or processing into finished articles, can be shaped by flow.

Plasticizer -- A material of lower molecular weight added to a polymer to separate the molecular chains. This results in a depression of the glass transition temperature, reduced stiffness and brittleness, and improved processability. (Note, many polymeric materials do not need a plasticizer.)

Plied Yarn: A yarn formed by twisting together two or more single yarns in one operation.

Poisson's Ratio -- The absolute value of the ratio of transverse strain to the corresponding axial strain resulting from uniformly distributed axial stress below the proportional limit of the material.

Polymer -- An organic material composed of molecules characterized by the repetition of one or more types of monomeric units.

Polymerization -- A chemical reaction in which the molecules of monomers are linked together to form polymers via two principal reaction mechanisms. Addition polymerizations proceed by chain growth and most condensation polymerizations through step growth.

Population -- The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for carbon/epoxy system A, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population. (See Volume 1, Section 8.5.1.1.)

Population Mean -- The average of all potential measurements in a given population weighted by their relative frequencies in the population. (See Volume 1, Section 8.5.1.1.)

Population Median -- That value in the population such that the probability of exceeding it is 0.5 and the probability of being less than it is 0.5. (See Volume 1, Section 8.5.1.1.)

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Population Variance -- A measure of dispersion in the population.

Porosity -- A condition of trapped pockets of air, gas, or vacuum within a solid material, usually expressed as a percentage of the total nonsolid volume to the total volume (solid plus nonsolid) of a unit quantity of material.

Positively Skewed -- A distribution is said to be positively skewed if the distribution is not symmetric and the longest tail is on the right.

Postcure -- Additional elevated temperature cure, usually without pressure, to improve final properties or complete the cure or both.

Pot Life -- The period of time during which a reacting thermosetting composition remains suitable for its intended processing after mixing with a reaction initiating agent.

Precision -- The degree of agreement within a set of observations or test results obtained. Precision involves repeatability and reproducibility.

Precursor (for Carbon or Graphite Fiber) -- Either the PAN or pitch fibers from which carbon and graphite fibers are derived.

Preply -- A composite material lamina in the raw material stage ready to be fabricated into a finished laminate. The lamina is usually combined with other raw laminae prior to fabrication. A preply includes all of the fiber system placed in position relative to all or part of the required matrix material that together will comprise the finished lamina. An organic matrix preply is called a prepreg.

Prepreg -- Ready to mold or cure material in sheet form which may be fiber, cloth, or mat impregnated with resin and stored for use. The resin is partially cured to a B-stage and supplied to the fabricator for lay-up and cure.

Pressure -- The force or load per unit area.

Probability Density Function -- See Volume 1, Section 8.5.1.1.

Proportional Limit -- The maximum stress that a material is capable of sustaining without any deviation from the proportionality of stress to strain (also known as Hooke's law).

Quasi-Isotropic Laminate -- A laminate approximating isotropy by orientation of plies in several or more directions.

Random Effect -- A common shift in a group of measurements due to a random level change of a usually uncontrollable factor. (See Volume 1, Section 8.5.1.1.)

Random Error -- That part of data variation due to level changes in uncontrolled factors which affect each observation separately and independently. (See Volume 1, Section 8.5.1.1.)

Reduction of Area -- The difference between the original cross sectional area of a tension test specimen and the area of its smallest cross section, usually expressed as a percentage of the original area.

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Refractive Index - The ratio of the velocity of light (of specified wavelength) in air to its velocity in the substance under examination. Also defined as the sine of the angle of incidence divided by the sine of the angle of refraction as light passes from air into the substance.

Reinforced Plastic -- A plastic with relatively high stiffness or very high strength fibers embedded in the composition. This improves some mechanical properties over that of the base resin.

Release Agent -- See **Mold Release Agent**.

Resilience -- A property of a material which is able to do work against restraining forces during return from a deformed condition.

Resin -- A solid or pseudo-solid organic material usually of high molecular weight which exhibits a tendency to flow when subjected to stress, usually has a softening or melting range, and fractures conchoidally.

Resin Content -- The amount of matrix present in a composite either by percent weight or percent volume.

Resin Starved Area -- Area of composite part where the resin has a non-continuous smooth coverage of the fiber.

Resin System -- A mixture of resin, with ingredients such as catalyst, initiator, diluents, etc. required for the intended processing and final product.

Roving: A number of strands, tows, or ends collected into a parallel bundle with little or no twist. In spun yarn production, an intermediate state between sliver and yarn.

S-Basis (or S-Value) -- The mechanical property value which is usually the specified minimum value of the appropriate government specification or SAE Aerospace Material Specification for this material.

Sample -- A small portion of a material or product intended to be representative of the whole. Statistically, a sample is the collection of measurements taken from a specified population. (See Volume 1, Section 8.5.1.1.)

Sample Mean -- The arithmetic average of the measurements in a sample. The sample mean is an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Median -- Order the observation from smallest to largest. Then the sample median is the value of the middle observation if the sample size is odd; the average of the two central observations if n is even. If the population is symmetric about its mean, the sample median is also an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Standard Deviation -- The square root of the sample variance. (See Volume 1, Section 8.5.1.1.)

Sample Variance -- The sum of the squared deviations from the sample mean, divided by n-1. (See Volume 1, Section 8.5.1.1.)

Sandwich Construction -- A structural panel concept consisting in its simplest form of two relatively thin, parallel sheets of structural material bonded to, and separated by, a relatively thick, light-weight core.

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Saturation -- An equilibrium condition in which the net rate of absorption under prescribed conditions falls essentially to zero.

Scrim (also called Glass Cloth, Carrier) -- A low cost fabric woven into an open mesh construction, used in the processing of tape or other B-stage material to facilitate handling.

Secondary Bonding -- The joining together, by the process of adhesive bonding, of two or more already-cured composite parts, during which the only chemical or thermal reaction occurring is the curing of the adhesive itself.

Selvage or Selvedge: The woven edge portion of a fabric parallel to the warp.

Set -- The strain remaining after complete release of the force producing the deformation.

Shear Fracture (for crystalline type materials) -- A mode of fracture resulting from translation along slip planes which are preferentially oriented in the direction of the shearing stress.

Shelf Life -- The length of time a material, substance, product, or reagent can be stored under specified environmental conditions and continue to meet all applicable specification requirements and/or remain suitable for its intended function.

Significant -- Statistically, the value of a test statistic is significant if the probability of a value at least as extreme is less than or equal to a predetermined number called the significance level of the test.

Significant Digit -- Any digit that is necessary to define a value or quantity.

Size System -- See **Finish**.

Sizing: A generic term for compounds which are applied to yarns to bind the fiber together and stiffen the yarn to provide abrasion-resistance during weaving. Starch, gelatin, oil, wax, and man-made polymers such as polyvinyl alcohol, polystyrene, polyacrylic acid, and polyacetatates are employed.

Skewness -- See **Positively Skewed, Negatively Skewed**.

Sleeving: A common name for tubular braided fabric.

Slenderness Ratio -- The unsupported effective length of a uniform column divided by the least radius of gyration of the cross-sectional area.

Sliver -- A continuous strand of loosely assembled fiber that is approximately uniform in cross-sectional area and has no twist.

Solute -- The dissolved material.

Specific Gravity -- The ratio of the weight of any volume of a substance to the weight of an equal volume of another substance taken as standard at a constant or stated temperature. Solids and liquids are usually compared with water at 4°C (39°F).

Specific Heat -- The quantity of heat required to raise the temperature of a unit mass of a substance one degree under specified conditions.

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Specimen -- A piece or portion of a sample or other material taken to be tested. Specimens normally are prepared to conform with the applicable test method.

Spindle: A slender upright rotation rod on a spinning frame, roving frame, twister or similar machine.

Standard Deviation -- See **Sample Standard Deviation**.

Staple -- Either naturally occurring fibers or lengths cut from filaments.

Step-Growth Polymerization -- One of the two principal polymerization mechanisms. In sep-growth polymerization, the reaction grows by combination of monomer, oligomer, or polymer molecules through the consumption of reactive groups. Since average molecular weight increases with monomer consumption, high molecular weight polymers are formed only at high degrees of conversion.

Strain -- the per unit change, due to force, in the size or shape of a body referred to its original size or shape. Strain is a nondimensional quantity, but it is frequently expressed in inches per inch, meters per meter, or percent.

Strand -- Normally an untwisted bundle or assembly of continuous filaments used as a unit, including slivers, tow, ends, yarn, etc. Sometimes a single fiber or filament is called a strand.

Strength -- the maximum stress which a material is capable of sustaining.

Stress -- The intensity at a point in a body of the forces or components of forces that act on a given plane through the point. Stress is expressed in force per unit area (pounds-force per square inch, megapascals, etc.).

Stress Relaxation -- The time dependent decrease in stress in a solid under given constraint conditions.

Stress-Strain Curve (Diagram) -- A graphical representation showing the relationship between the change in dimension of the specimen in the direction of the externally applied stress and the magnitude of the applied stress. Values of stress usually are plotted as ordinates (vertically) and strain values as abscissa (horizontally).

Surfacing Mat -- A thin mat of fine fibers used primarily to produce a smooth surface on an organic matrix composite.

Symmetrical Laminate -- A composite laminate in which the sequence of plies below the laminate midplane is a mirror image of the stacking sequence above the midplane.

Tack -- Stickiness of the prepreg.

Tape -- Prepreg fabricated in widths up to 12 inches wide for carbon and 3 inches for boron. Cross stitched carbon tapes up to 60 inches wide are available commercially in some cases.

Tenacity -- The tensile stress expressed as force per unit linear density of the unstrained specimen i.e., grams-force per denier or grams-force per tex.

Tex -- A unit for expressing linear density equal to the mass or weight in grams of 1000 meters of filament, fiber, yarn or other textile strand.

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Thermal Conductivity -- Ability of a material to conduct heat. The physical constant for quantity of heat that passes through unit cube of a substance in unit time when the difference in temperature of two faces is one degree.

Thermoplastic -- A plastic that repeatedly can be softened by heating and hardened by cooling through a temperature range characteristic of the plastic, and when in the softened stage, can be shaped by flow into articles by molding or extrusion.

Thermoset -- A plastic that is substantially infusible and insoluble after having been cured by heat or other means.

Tolerance -- The total amount by which a quantity is allowed to vary.

Tolerance Limit -- A lower (upper) confidence limit on a specified percentile of a distribution. For example, the B-basis value is a 95% lower confidence limit on the tenth percentile of a distribution.

Tolerance Limit Factor -- The factor which is multiplied by the estimate of variability in computing the tolerance limit.

Toughness -- A measure of a material's ability to absorb work, or the actual work per unit volume or unit mass of material that is required to rupture it. Toughness is proportional to the area under the load-elongation curve from the origin to the breaking point.

Tow -- An untwisted bundle of continuous filaments. Commonly used in referring to man-made fibers, particularly carbon and graphite fibers, in the composites industry.

Transformation -- A transformation of data values is a change in the units of measurement accomplished by applying a mathematical function to all data values. For example, if the data is given by x, then $y = x + 1$, x, $1/x$, $\log x$, and $\cos x$ are transformations.

Transition, First Order -- A change of state associated with crystallization or melting in a polymer.

Transversely Isotropic -- Descriptive term for a material exhibiting a special case of orthotropy in which properties are identical in two orthotropic dimensions, but not the third; having identical properties in both transverse directions but not the longitudinal direction.

Traveller -- A small piece of the same product (panel, tube, etc.) as the test specimen, used for example to measure moisture content as a result of conditioning.

Twist: The number of turns about its axis per unit of length in a yarn or other textile strand. It may be expressed as turns per inch (tpi) or turns per centimeter (tpcm).

Twist, Direction of -- The direction of twist in yarns and other textile strands is indicated by the capital letters S and Z. Yarn has S twist if, when held in a vertical position, the visible spirals or helices around its central axis are in the direction of slope of the central portion of the letter S, and Z twist is in the other direction.

Twist multiplier: The ratio of turns per inch to the square root of the cotton count.

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specified

Typical Basis -- A typical property value is a sample mean. Note that the typical value is defined as the simple arithmetic mean which has a statistical connotation of 50% reliability with a 50% confidence.

Unbond -- An area within a bonded interface between two adherends in which the intended bonding action failed to take place. Also used to denote specific areas deliberately prevented from bonding in order to simulate a defective bond, such as in the generation of quality standards specimens. (See **Disbond, Debond**).

Unidirectional Laminate -- A laminate with nonwoven reinforcements and all layers laid up in the same direction.

Unit Cell: The term applied to the path of a yarn in a braided fabric representing a unit cell of a repeating geometric pattern. The smallest element representative of the braided structure.

Upper Confidence Limit -- See **Confidence Interval**.

Vacuum Bag Molding -- A process in which the lay-up is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible sheet placed over it and sealed at the edges.

Variance -- See **Sample Variance**.

Viscosity -- The property of resistance to flow exhibited within the body of a material.

Void -- A physical and mechanical discontinuity occurring within a material or part which may be two-dimensional (e.g., disbonds, delaminations) or three-dimensional (e.g., vacuum-, air-, or gas-filled pockets). Porosity is an aggregation of micro-voids. Voids are essentially incapable of transmitting structural stresses or nonradiative energy fields. (See **Inclusion**.)

Warp -- The longitudinally oriented yarn in a woven fabric (see **Fill**); a group of yarns in long lengths and approximately parallel.

Weibull Distribution (Two - Parameter) -- A probability distribution for which the probability that a randomly selected observation from this population lies between a and b ($0 < a < b < \infty$) is given by Equation 1.7(d) where α is called the scale parameter and β is called the shape parameter. (See Volume 1, Section 8.5.1.1.)

$$\exp\left[-\left(\frac{a}{\alpha}\right)^{\beta}\right] - \exp\left[-\left(\frac{b}{\alpha}\right)^{\beta}\right] \qquad 1.7(d)$$

Wet Lay-up -- A method of making a reinforced product by applying a liquid resin system while the reinforcement is put in place.

Wet Strength -- The strength of an organic matrix composite when the matrix resin is saturated with absorbed moisture. (See **Saturation**).

Wet Winding -- A method of filament winding in which the fiber reinforcement is coated with the resin system as a liquid just prior to wrapping on a mandrel.

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Whisker -- A short single crystal fiber or filament. Whisker diameters range from 1 to 25 microns, with aspect ratios between 100 and 15,000.

Work Life -- The period during which a compound, after mixing with a catalyst, solvent, or other compounding ingredient, remains suitable for its intended use.

Woven Fabric Composite -- A major form of advanced composites in which the fiber constituent consists of woven fabric. A woven fabric composite normally is a laminate comprised of a number of laminae, each of which consists of one layer of fabric embedded in the selected matrix material. Individual fabric laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Yarn -- A generic term for strands or bundles of continuous filaments or fibers, usually twisted and suitable for making textile fabric.

Yarn, Plied -- Yarns made by collecting two or more single yarns together. Normally, the yarns are twisted together though sometimes they are collected without twist.

Yield Strength -- The stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain. (The deviation is expressed in terms of strain such as 0.2 percent for the Offset Method or 0.5 percent for the Total Extension Under Load Method.)

X-Axis -- In composite laminates, an axis in the plane of the laminate which is used as the 0 degree reference for designating the angle of a lamina.

X-Y Plane -- In composite laminates, the reference plane parallel to the plane of the laminate.

Y-Axis -- In composite laminates, the axis in the plane of the laminate which is perpendicular to the x-axis.

Z-Axis -- In composite laminates, the reference axis normal to the plane of the laminate.

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2.1 INTRODUCTION. The purpose of Chapter 2 is to provide guidelines for the characterization of composite material properties. While current procedures emphasize the development of a lamina-level database, this does not preclude higher level testing. Further, it is possible that qualification can be achieved at the application level of testing.

This chapter also provides guidelines for the screening of new material systems. Mechanical properties testing is proposed which provides a preliminary, typical value, evaluation of key stiffness and strength properties. The intent of this screening test program is to provide an initial evaluation of new material systems under worst-case environmental and loading test conditions. The screening includes both lamina (ply) and laminate (application) level testing and should be sufficient to eliminate deficient material systems and to reveal promising new material systems for planning of in-depth evaluations.

2.2 RECOMMENDATIONS FOR THE GENERATION OF PHYSICAL AND MECHANICAL PROPERTIES.

2.2.1 General guidelines. Generation of design allowables for advanced composite materials requires physical and chemical property characterization of the prepreg material as well as physical and mechanical property tests on cured lamina and laminates. These guidelines contain recommendations for the number and type of tests sufficient to establish B-basis mechanical properties for advanced composite unidirectional tape and woven fabric materials. Recommendations are also provided for vendor acceptance tests by the manufacturer of the prepreg materials.

These guidelines present the methods recommended by MIL-HDBK-17 for generating statistically-based material properties for lamina coupons. Other methods involving statistical analysis of mechanical property tests on oriented laminates are equally acceptable when agreed to by the contractor and certifying agency.

2.2.2 Moisture effects. Organic matrix composites absorb small amounts of moisture. The absorbed water causes a dimensional change (swelling), lowers the glass transition temperature of the resin, and results in a reduction of the matrix and matrix/fiber interface dependent mechanical properties of the composite, particularly at elevated temperatures. Except for the aramid fiber composites, it is usually assumed that the moisture absorption is limited to the organic matrix.

The degradation of composite properties as a result of exposure to high temperatures is nonlinear. A drastic reduction in properties occurs at a characteristic temperature level, defined here as the material operational limit (MOL). For composites, this behavior is more complicated because the introduction of moisture will result in more than one critical temperature, as illustrated in Figure 2.2.2(a). The various moisture contents are represented schematically by saturation at different relative humidity levels. There are no standard criteria for the determination of the MOL. One method (References 2.2.2(a) - (c)), illustrated in Figure 2.2.2(b), utilizes the glass transition temperature (T_g) with some safety tolerance K as the basis for the MOL. For epoxy composites, 50 F° (28 C°) has been proposed for the value of the safety tolerance. If service usage at high temperatures is limited to occasional excursions,it can be argued that lower safety tolerances are acceptable.

The elevated temperature wet testing required by MIL-HDBK-17 should be at the MOL temperature, corresponding to the highest practical relative humidity. For aircraft, the worst-case relative humidity is 85%. As can be seen from Figure 2.2.2(a), the effect of moisture is small for matrix-dependent properties at temperatures below room temperature (RT). However, if Figure 2.2.2(a) were produced for the fiber-dependent properties, degradation in properties would be observed below room temperature. Therefore, MIL-HDBK-17 requires testing at cold temperatures. Since the changes in

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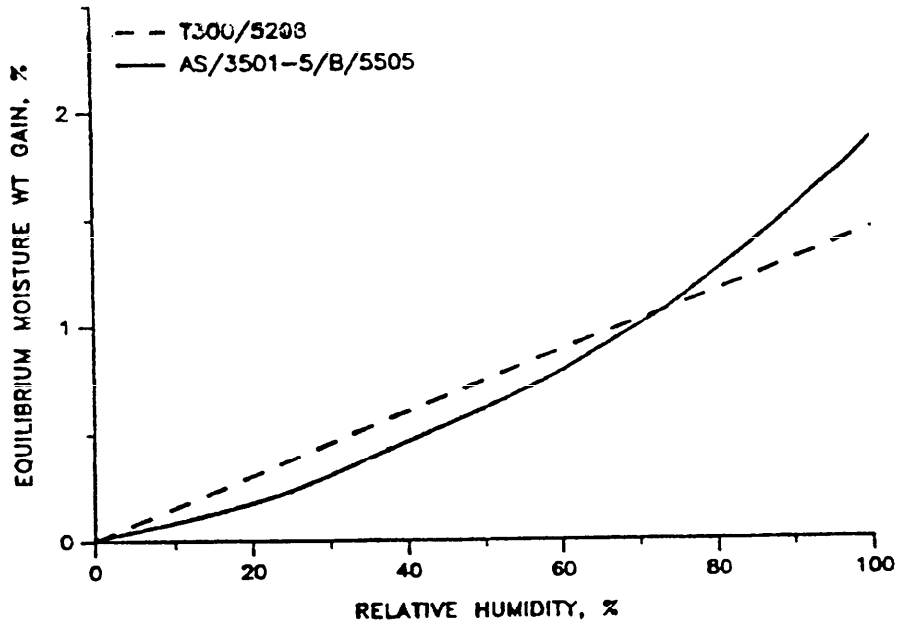


FIGURE 2.2.2(c) *Maximum moisture content as a function of relative humidity*

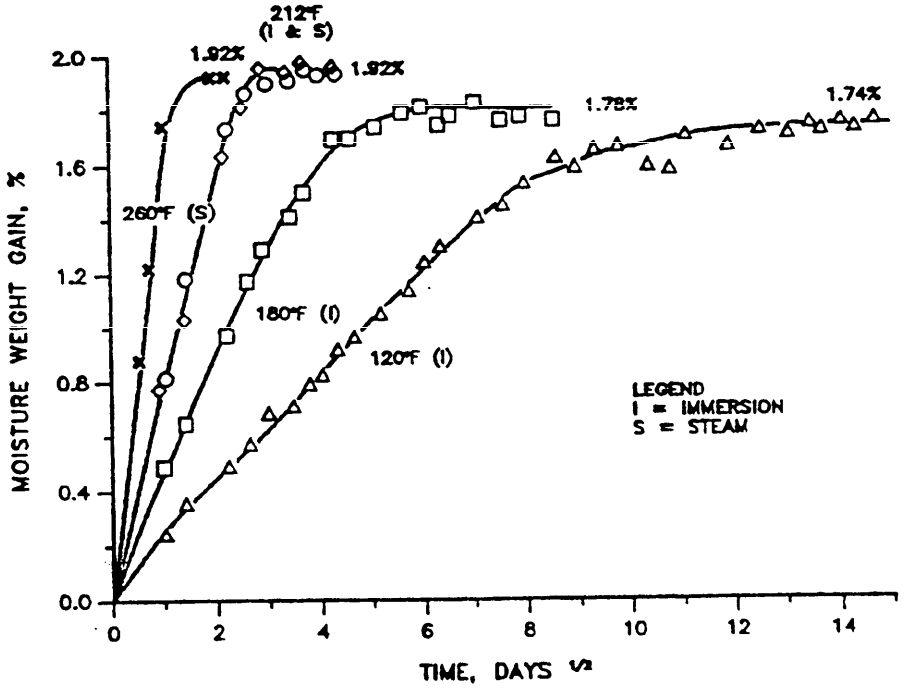


FIGURE 2.2.2(d) *Effect of temperature on moisture absorption rate in hybrid boron-graphite/epoxy (5505-AS/3501) laminate (3.0 x 0.5 x 12 in.; 76 x 13 x 3.0 mm) (Reference 2.2.2(b)).*

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$$G = 1 - \exp\left[-7.300\left(\frac{D_0 t}{h^2}\right)^{0.75}\right] \qquad 2.2.2(d)$$

where

h = slab thickness exposed on both sides to the same environment
t = exposure time

M can be obtained from Figure 2.2.2(c).

For time variable boundary conditions, the finite difference method (Reference 2.2.2(c)) can be used to numerically integrate Equation 2.2.2(b) with respect to time.

In the design of a composite structure, the effects of moisture are accounted for in the design allowables. The end-of-life moisture content (usually defined as the design service moisture content) is either calculated using Equation 2.2.2(b) or established by previous experimental or flight data. After the end-of-life moisture content is determined, mechanical property tests are performed with specimens preconditioned to that moisture level. The conditioning procedure to be performed is described in the next section.

Before end-of-life moisture contents can be calculated, environmental definition of component usage must be determined. An example is illustrated in Reference 2.2.2(d), where worldwide weather data and USAF aircraft basing data were gathered to define runway storage environmental spectra for each of the three classes of Air Force vehicles: fighters, bombers, and cargo/tankers. A ranking procedure was developed and applied to select baseline and worst-case locations with respect to the absorption of moisture by carbon/epoxy composite structures. A guide to environmental exposure parameters is provided in MIL-STD-810E (Reference 2.2.2(e)).

A moisture/temperature failure mode (no mechanical loads) that must be considered in establishing the maximum design temperature for a polymer matrix composite laminate is the steam pressure delamination failure (References 2.2.2(f)-(h)). Polymer matrix composites (thermosets and thermoplastics) contain some degree of porosity and absorb moisture. As the matrix absorbs moisture from the environment by the process of diffusion, the voided areas will partially fill with water. If the laminate is exposed to temperatures above the boiling point of water, the water converts to steam. When the temperature and associated steam pressure reaches the level where it exceeds the laminate wet interlaminar (i.e., flatwise) tensile strength of the material, delamination occurs.

The steam pressure delamination mode can occur over a range of temperatures depending on the amount of absorbed moisture as indicated in Figure 2.2.2(e). Failure can be predicted when the wet flatwise tensile strength curve (which is a function of the design relative humidity and moisture equilibrium level) intersects the steam pressure curve. To determine the maximum design temperature for a new material system for a range of design relative humidities, an experimental program similar to Figure 2.2.2(f) is recommended. Precondition panels to equilibrium at three relative humidity levels plus a dry condition. These panels are then exposed to the mission time-temperature profiles. One issue in conducting the panel thermal exposure test is that the time-temperature exposure should simulate the actual in-service heating conditions so that laminate moisture drying is representative of the design application. Panels that see a slower heating rate than the design condition may have more

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dryout and attain a fictitious higher temperature before delamination occurs. For high heating rates seen in missile applications, quartz lamps are recommended. For slower heating rates, a computer-controlled oven exposure may be acceptable. The allowable design temperature curve selected should include a safety tolerance (50°F in this example) below the temperature at which delaminations do occur.

2.2.2.1 *MOL criteria for high temperature composite systems.* The MOL for high temperature composite systems is dependent on other service environment conditions besides moisture. MOL is dependent on the mission life requirements of the actual applications. The life of the part is a function of time, temperature, pressure, and mechanical loading.

The wet Tg is one of the indicators of a high temperature composite material's MOL. Humidity does effect the elevated temperature properties and can induce thermal blistering in a thick laminate cross section. Thermal blister resistance is a function of the moisture content and thickness of the part, and the heat up rate the part will encounter.

The other indicators of a high temperature composite material's MOL is the transverse microcrack (TVM) resistance and thermal oxidative stability (TOS) properties. TVM occurs due to thermal cycling of the laminate over a temperature range. The TVM can develop because of the large difference in coefficient of expansion between the fiber and resin, and the relative low ductility of most high temperature resins. These thermal stresses can cause 90 degree ply failure, which occurs at the fiber matrix interface. This degradation primarily effects the resin/interface dominated properties like compression strength, in-plane shear strength, and interlaminar properties. The magnitude of TVM that

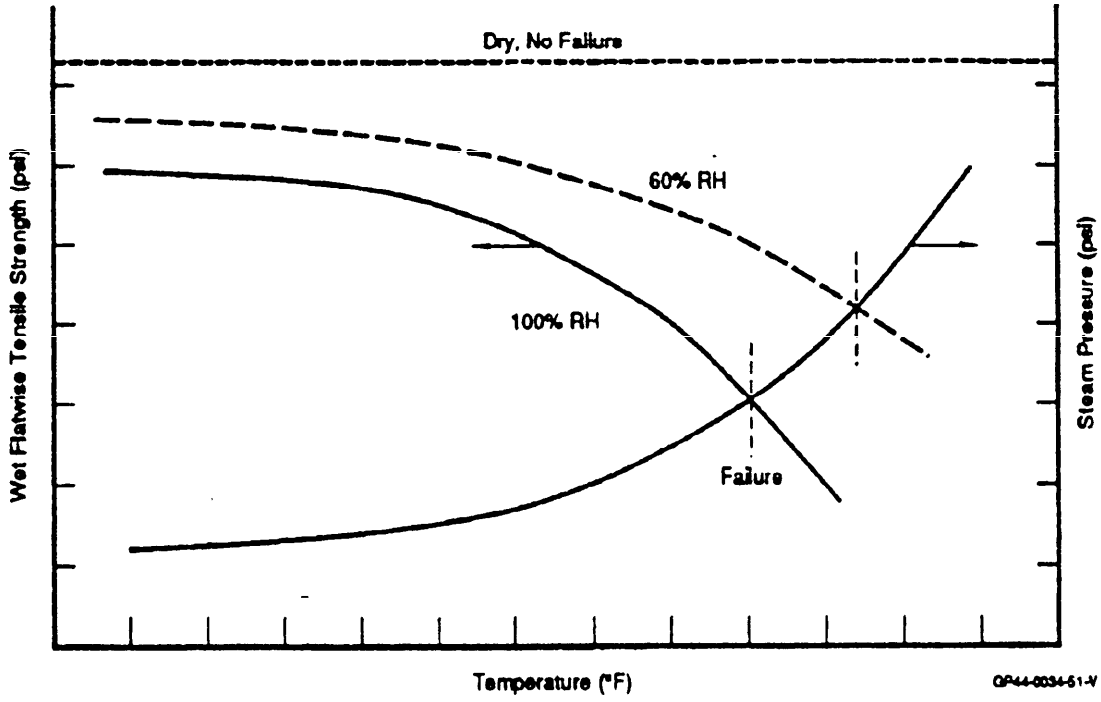


FIGURE 2.2.2(e) *Failure occurs when internal steam pressure exceeds flatwise tensile strength.*

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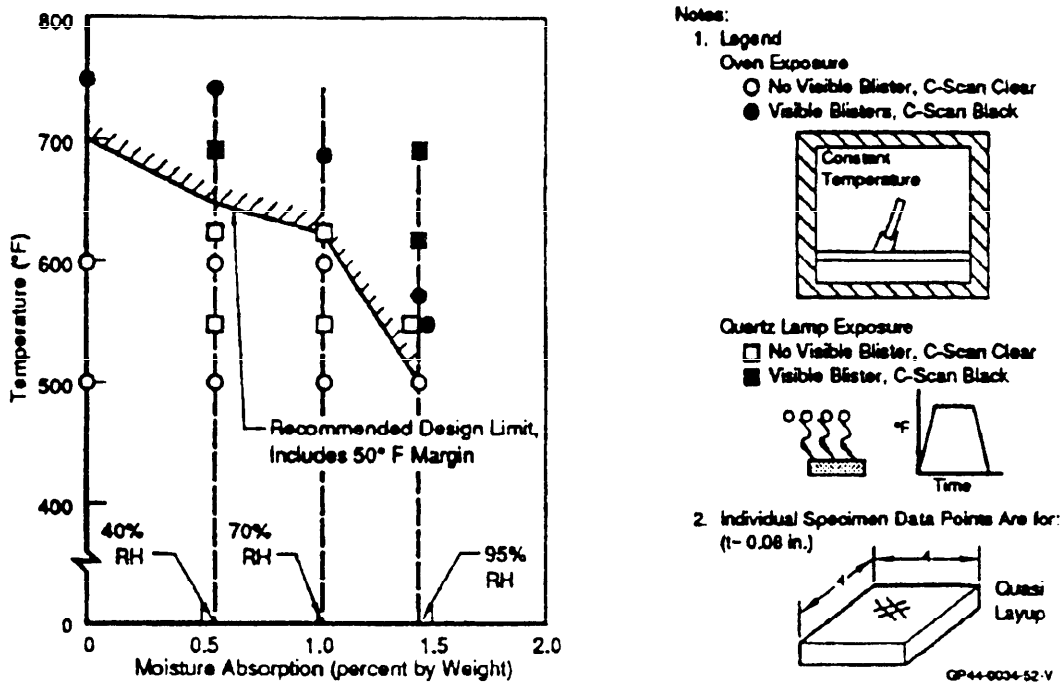


FIGURE 2.2.2(f) *Moisture content limits carbon/polyimide use temperature.*

will occur depends on the application temperature range, maximum operating temperature, and number of thermal cycles.

TOS is also an important property for high temperature composite systems. The thermal oxidation characteristics of a polymeric composite are a function of fiber, sizing, and resin. The constituents can be evaluated individually for thermal oxidative stability on a qualitative basis. The actual performance should be evaluated on the laminate level, since the fiber-matrix interface is the primary area that is degradated. All properties can be affected by TOS, although the interface dependent properties are most affected. The weight loss of a laminate is a good indication of the amount of TOS that has occurred for a particular system. The TOS performance of a material is a function of the time, temperature, and oxygen flow rate/pressure.

There can be synergistic effects among TVM, TOS, and hot/wet exposures for high temperature polymeric composites. In order to get an accurate assessment of a material's MOL, it is recommended these effects be combined in a realistic manner that reflects the actual application environment. For short term applications, the amount of degradation can be determined experimentally by exposing laminates to combined conditions of thermal cycling, aging at temperature, and humidity conditioning to the part's specific mission life. Specimens can be machined and tested from this environmentally exposed material, and the residual strength of the material can be assessed.

For long term applications, it may be difficult to perform this environmental exposure in real time. Durability modeling and accelerated testing may be required in order to predict end-of-life properties for these applications. Durability modeling can be used to predict the amount of damage that is generated as a function of mission exposure conditions, and the subsequent residual strength

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properties. Mission exposure testing can be accelerated by aging the material at higher temperature or pressures, in order to accelerate the oxidation of the material. It is important that the accelerated tests produce realistic damage mechanisms that will be evidenced in real time exposures. For this reason, it is recommended that some limited real time exposure testing be done in order to confirm the damage mechanisms and also be used to confirm the durability model's accuracy.

2.2.3 Conditioning of samples. To obtain test data for mechanical properties of wet material, specimens must be environmentally conditioned prior to mechanical testing. The goal for this procedure is a uniform moisture content throughout the thickness of the specimen. Uniformity in moisture content is achieved when the specimens reach an equilibrium moisture content. This moisture content is different at different humidity levels but is not, in general, a function of temperature, if the conditioning temperature is below the glass transition temperature of the material. The humidity level that is chosen depends on the application, the extreme being water immersion (100% RH). An 85% exposure would be more representative of worst usage conditions for aircraft based in the tropics. Because the moisture absorption rate is so strongly dependent on temperature, there is a tendency to increase conditioning temperatures to save time. However, long exposures to high temperatures combined with moisture are deleterious. It is essential that realistic exposure conditions be employed in conditioning samples such that moisture diffusion does not deviate significantly from Fickian behavior. Conditioning temperature and humidity specifications are material dependent. For epoxy matrices, the conditioning temperature should be kept below 170°F (77°C) for a 350°F (177°C) cure material and below 155°F (68°C) for a 250°F (121°C) cure material.

In addition to specifying conditioning temperature and RH, time of exposure has to be specified. The best way to specify conditioning is to require specimens to reach equilibrium, i.e. the maximum moisture content (MC) by weight that can be achieved at the given humidity and temperature level. A shortcut prevalent in the industry is the specification of a time period, e.g. thirty days. The drawbacks of this approach are illustrated in Figure 2.2.3 where, using the absorption rate and equilibrium level for IM6/3501-6 carbon/epoxy, the maximum moisture content has been calculated for various composite

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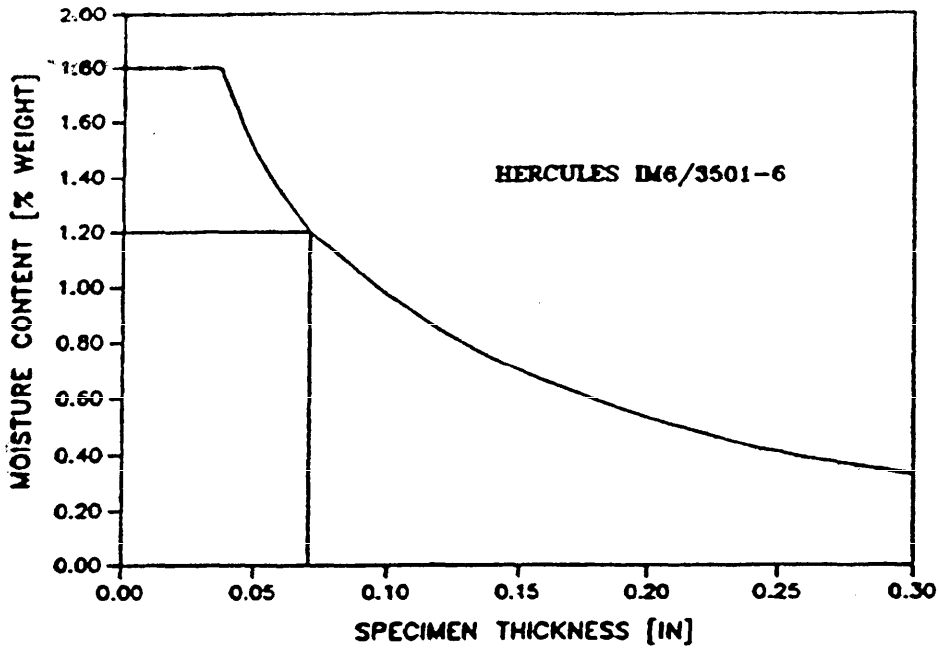


FIGURE 2.2.3 *Moisture absorption after 30 days at 95% RH and 140°F (60°C) for carbon/epoxy laminates with both sides exposed.*

laminate thicknesses exposed for 30 days at 95% relative humidity and 140°F (60°C). The required 1.2% MC (equilibrium at 78% RH) by weight can be obtained only if the laminate thickness is approximately 0.07 in. (1.8 mm). For thicknesses larger than 0.07 in. (1.8 mm), the MC will be lower than desired and, for thicknesses smaller than 0.07 in. (1.8 mm), the MC will be higher than required. Furthermore, the moisture distribution across the thickness will be non-uniform for all thicknesses greater than 0.035 in. (0.9 mm). Because a typical test matrix includes specimens with different thicknesses, different exposure times should be specified. Figure 2.2.3 was generated for a known material; such information may not be readily available for a new material system. Specifying final MC instead of time of exposure implies knowledge of the material properties at the resin and void content of the laminate.

Two-step conditioning schemes consisting of 95% RH exposure followed by a lower RH exposure can also be used. These save time and are perfectly acceptable if the last step is exposure at the required RH until equilibrium.

Moisture content measurements are taken by weighing either the actual specimens or travellers that have been cut from the same panel and conditioned at the same time as the specimens. To minimize ingress of moisture through the edges, the thickness of the travellers must be small in comparison to the length and width dimensions; a ratio of 1:10 is usually adequate. For greater thicknesses, the edges may have to be sealed. Because the weight gain is approximately 1%, measurements must be taken on a fine balance to an accuracy of 0.0001 g. If the specimens have fiberglass tabs, adhesive materials, etc., travellers must be used, unless the equilibrium moisture content at a given temperature and humidity is specified, and even in this case the specimen may be too large or heavy to be conveniently weighed.

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Testing, whether screening or qualification, is performed on specimens either in an unconditioned ("dry") or moisture-conditioned state. "Dry" is defined as the as-manufactured condition. Moisture conditioning is accomplished by 1) inserting the specimens in the chamber at the specified humidity level until equilibrium, or 2) in cases where absolute moisture content is required to be known, the drying procedure below should be used. The absolute moisture content can also be obtained using the first method by drying the specimens after moisture conditioning.

Both specimens and travellers should be dried in an oven prior to any additional conditioning. Care should be taken to avoid excessively high drying temperatures and high thermal excursions which may induce thermal cracking in specimens. For some but not all epoxy matrices, oven drying at temperatures up to 250°F (121°C) for 350°F (177°C) cure materials and 200°F (93°C) for 250°F (121°C) cure materials are often employed. Since the time from cure to completion of specimen fabrication may be long, a required time of drying cannot be specified. Periodic readings should be taken. In general, the time period between readings starts at twenty-four hours. Readings may be further apart towards the end of conditioning. For more information, see ASTM Standard Test Method for Moisture Absorption Properties and Equilibrium Conditioning of Polymer-Matrix Composite Materials (Reference 2.2.3(a)). When three consecutive readings are within 0.01% weight change, the travellers and specimens are considered to be dry. The dry weight should be obtained at this time. The specimens and travellers should then be stored in plastic bags with a desiccant. From this time on, the travellers must accompany the specimens throughout the conditioning history. For travellers and specimens that are conditioned until equilibrium, equilibrium is defined as the point when three consecutive weighings give weight changes that average less than 1% of the total increase in weight. ASTM Standard Test Method D 570-81 "Water Absorption of Plastics" and ASTM D 618-61 "Standard Methods of Conditioning Plastics and Electrical Insulating Materials for Testing" provide guidance for test procedures (References 2.2.3(b) and (c)).

The specimens or travellers should also be weighed just before and after mechanical testing, especially for elevated temperatures. At these temperatures, it is also important not to soak the specimens longer than needed so as not to drive-out moisture. A typical soak time after the thermocouples on the specimens reach the desired temperature is half a minute for carbon/epoxy for temperatures up to 350°F (177°C). Because thick laminates would require unacceptably long times to reach equilibrium moisture levels, thin laminates are usually used in moisture conditioning testing. The diffusion properties and equilibrium moisture content obtained from thin laminates are assumed to be applicable to thick laminates unless test data or calculations show otherwise.

2.2.4 Statistical development of mechanical properties. Data interpretation and statistical analysis procedures are defined in Chapter 8.

2.2.5 Data pooling requirements. The capability of pooling data may be desirable in order to obtain sufficient data to calculate B-basis values for material properties. Data to be pooled may be available for materials from different composite component manufacturers, different locations of one manufacturer, or slightly different processes from the same manufacturer. All properties should be examined for batch-to-batch variability (Section 8.5.3.2). A decision regarding the pooling of the data will be made in discussions with the certifying agency or, for MIL-HDBK-17 data, by the Data Review working group.

2.2.6 Test method acceptance criteria. While test methods used by contractors are agreed upon with customers and/or certifying agencies, specific criteria apply when submitting data to MIL-HDBK-17 for consideration for inclusion in Volume 2 of the Handbook. Ideally, a test method shall have undergone a rigorous review of its applicability, precision and bias by an independent voluntary consensus standards organization that may include representatives from material suppliers, end-users, academia, or government. This review, and the test method, should be available in a referenceable,

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open-literature publication, and include interlaboratory (round robin) testing. Many times test methods meeting the above criteria are not available, and methods which meet less rigorous criteria (2 or 3 below) must be selected for data submittal. The MIL-HDBK-17 Coordination Group has identified test methods to be used when submitting data for consideration for inclusion in Volume 2 of the Handbook. These methods will be designated or described in chapters 2 through 7, and will meet one or more of the following criteria:

1. Methods, applicable to advanced composites and in common use, which have completed the following:
   - Round robin testing under sponsorship of a recognized standards-making organization
   - Rigorous review of precision and bias
   - Publication in the open literature of a recognized standards-making organization

2. "Common practice" methods, which have not been standardized as in (1) above, but which are in common usage in the composite materials industry, are available in referenceable, open-literature publications, and have begun the process toward formal standardization.

3. Where no standards meeting the above criteria exist for specific structures or process/product forms, other test methods may have been selected by consensus of the MIL-HDBK-17 Coordination Group. Such methods may have been developed within the MIL-HDBK-17 Working Groups, or by other organizations, and will have begun the process toward formal standardization.

2.3 MATERIAL ACQUISITION AND PREPREG PHYSICAL PROPERTY CHARACTERIZATION. For material property characterization suitable for inclusion in MIL-HDBK-17, five prepreg batches of each material type shall be prepared by the material supplier using prepreg production facilities. The five prepreg batches shall be made with five different fiber lots impregnated with five different production resin batches. The contractor shall prepare and coordinate a preliminary material specification with the supplier prior to manufacture of the prepreg batches. The preliminary material specification shall define agreed-to laminate and specimen fabrication procedures, test procedures, and physical and mechanical property requirements.

The five-batch requirement only applies to material properties that are to be incorporated in MIL-HDBK-17. An alternate number of tests and batches may be employed upon approval of the certifying agency. However, mechanical property test data should be examined by the recommended statistical methods in this document (MIL-HDBK-17) to ascertain that they are statistically acceptable.

Recommended tests to be performed on the prepreg materials are shown in Table 2.3. When applicable, the tests should be standard ASTM procedures.

2.4 LAMINA PHYSICAL AND MECHANICAL PROPERTY TESTS. A minimum of two test panels for each prepreg batch shall be cured in two separate autoclave runs to be used in fabricating test coupons. The prepreg material shall be laid up, bagged, and cured in accordance with a process specification prepared by the contractor. Cured test panels shall be non-destructively evaluated (NDE) using ultrasonic inspection. Imperfections in the test panels shall be reported with questionable areas marked on the panels. Test coupons shall not be extracted from these areas.

Suggested physical and mechanical property tests are shown in Tables 2.4(a) and 2.4(b). A test procedures document should be prepared by the contractor prior to the start of testing. It is recommended that, whenever possible, the tests should be ASTM standard methods.

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TABLE 2.3 *Recommended physical and chemical property tests to be performed by material supplier and prime contractor.*

Test Property	Suggested Test Procedure[1]	Number of Tests per Batch[2]	Total Number of Tests
Resin Content, %	ASTM D 3529	3	15
Volatile Content, %	ASTM D 3530	3	15
Gel Time, min.	ASTM D 3532	3	15
Resin Flow, %	ASTM D 3531	3	15
Fiber Areal Wt., gm/M	†	3	15
Moisture Content, %	†	3	15
Tack	†	3	15
HPLC (High Performance Liquid Chromatography)	†	3	15
IR (Infrared Spectroscopy)	†	3	15
DMA (Dynamic Mechanical Analysis, neat resin only)	†	3	15
DSC (Differential Scanning Calorimetry)	†	3	15
RDS (Rheological Dynamic Spectroscopy)	†	3	15

[1]Test procedures shall be coordinated and agreed to prior to manufacture of prepreg material.
[2]Tests shall be performed on each of the five batches of prepreg material.
†Test procedures to be described at a later date.

The test matrix shown in Table 2.4(b) is based on thirty tests per data point (six tests per batch) to provide for parametric/nonparametric analysis in determining B-basis properties. Fewer tests per data point may be acceptable for non-critical properties as agreed to by the prime contractor and the certifying agency. For generating B-basis values, 0° and 90° tension, 0° and 90° compression, and in-plane shear tests are required. Short beam shear tests are required for determining process control test minimum values.

Extensometers or strain gages should be used to obtain strain measurements for each specimen. Two of each group of six compression specimens shall be strain gaged back-to-back with axial gages to assess possible premature buckling failure.

2.4.1 Mechanical property test matrix incorporating the use of regression analysis. The test matrix shown in Table 2.4.1 is a modification of Table 2.4(b) for the incorporation of regression analysis. The

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TABLE 2.4(a) *Cured lamina physical property tests.*

Physical Property	Suggested Test Procedure	Number of Tests Per Prepreg Batch[1]	Total Number of Tests
Fiber Volume, %	ASTM D 3171	3	15
Resin Volume, %	ASTM D 3171	3	15
Density, gm/cc	ASTM D 792	3	15
Measurements of per ply thickness	-	10	50
Glass Transition Temperature, °F (dry)[2]	-	3	15
Glass Transition Temperature, °F (wet)[2]	-	3	15

[1] Tests shall be performed on each of the five batches.

[2] Dry specimens are "as fabricated" specimens which have been maintained at ambient conditions in an environmentally-controlled test laboratory. Wet specimens are environmentally conditioned by exposing them in an elevated temperature humidity chamber until they attain an equilibrium moisture content agreed to by the contractor and customer, then packaged in a heat-sealed aluminized polyethylene bag until required for test. Tests shall be performed in a manner which maintains the moisture content in specimens at the levels agreed to by the contractor and certifying agency.

main benefit of this approach is the flexibility to calculate material allowables over a temperature range. This is useful when a material is used for multiple applications with varying maximum operating temperatures. This is also useful when an application temperature changes as the design process and substantiation data become available. Validity of regression analysis depends on a common failure mode exhibited between each test group.

The main differences between the point-specific test matrix in Table 2.4(b) and the regression test matrix in Table 2.4.1 are that two elevated test temperature conditions have been added and all testing is done in either the dry condition or wet condition. The new test conditions were added so that test points would be distributed over a temperature range. Examples of these test points are: for 350°F cure epoxy system: -65°F, 74°F, 180°F, 220°F, 250°F, for BMI system: -65°F, 74°F, 250°F, 350°F, 400°F, and for polyimide -65°F, 74°F, 350°F, 450°F, and 550°F. ET2 should represent the normal operating temperature of an application, e.g., 220°F test temperature for 350°F cure epoxy. ET3 should represent an upper end temperature of the material system but should be less than the dry T_g if the testing is done dry or should be less than the wet T_g if the testing is done wet.

For inclusion of data in the handbook, five batches of material should be tested similar to the matrix in Table 2.4(b). This test matrix is intended for dry testing, but wet testing can be performed by repeating this matrix in the wet condition or by performing wet tests only at the application specific

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TABLE 2.4(b) *Cured lamina mechanical property tests.*

Mechanical Property	Test Methods[1]	Test Condition[2] and Number of Tests Per Batch[3]			Number of Tests
	See Handbook Section	Min. Temp Dry	RT Dry	Max. Temp Wet	
0° Tension (warp)	6.6.2[4]	6	6	6	90
90° Tension (fill)	6.6.2[4]	6	6	6	90
0° Compression (warp)	6.6.3.4	6	6	6	90
90° Compression (fill)	6.6.3.4	6	6	6	90
In-plane Shear	6.6.5.4	6	6	6	90
0° Short Beam Shear	6.6.5.4	-	6	-	30
					480

[1] MIL-HDBK-17 is not currently in a position to make exclusive test method recommendations, but the referenced Handbook sections identify methods that are currently deemed acceptable for data submittals to MIL-HDBK-17.

[2] Minimum and maximum temperature tests shall be performed within ±5°F (±2.8°C) of the nominal test temperature. Nominal test temperatures will be as agreed to by contractor and certifying agency. Dry specimens are "as-fabricated" specimens which have been maintained at ambient conditions in an environmentally-controlled test laboratory. Wet specimens are environmentally-conditioned by exposing them in a humidity chamber until they attain an equilibrium moisture content agreed to by the contractor and certifying agency, and then packaging them in a heat-sealed aluminized polyethylene bag until required for test. Tests shall be performed in a manner which maintains the moisture content in specimens at the levels agreed to by the contractor and certifying agency.

[3] Tests shall be performed on each of the five batches.

[4] For 0° and 90° tension, ASTM D 3039 and SACMA Recommended Method (SRM) 4-88 are acceptable test methods for MIL-HDBK-17 data submittals.

[5] Short Beam Shear is for screening and quality control purposes only.

elevated test temperature. It is very important that specimen dry-out be measured and kept to a minimum during the elevated temperature wet tests.

This test matrix has been developed for generating A-Basis (95/99) allowables, using 90 data points from batches of material. This assumes the data generated has a coefficient of variation (C.V.) of less than 15%. If the C.V is larger, more data points will need to be added to the test matrix in order to

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TABLE 2.4.1 *Cured laminate mechanical property test matrix designed for regression analysis.*

A-Basis level matrix - 5 batches/90 data points per property

Mechanical Property	Test Methods[1]	Test Condition[2] and Number of Tests Per Batch[3]					Number of Tests
	See Handbook Section	Min	R.T.	ET1	ET2	ET3	
		Temp					
0° Tension (warp)	6.6.2[4]	3	4	3	4	4	90
90° Tension (fill)	6.6.2[4]	3	4	3	4	4	90
0° Compression (warp)	6.6.3.4	3	4	3	4	4	90
90° Compression (fill)	6.6.3.4	3	4	3	4	4	90
In-plane Shear	6.6.5.4	3	4	3	4	4	90
0° Short Beam Shear[5]	6.6.5.4	-	6	-	-	-	30 480

[1] MIL-HDBK-17 is not currently in a position to make exclusive test method recommendations, but the referenced Handbook sections identify methods that are currently deemed acceptable for data submittals to MIL-HDBK-17.

[2] Minimum and maximum temperature tests shall be performed within ±5°F (±2.8°C) of the nominal test temperature. Nominal test temperatures will be as agreed to by contractor and certifying agency. Dry specimens are "as-fabricated" specimens which have been maintained at ambient conditions in an environmentally-controlled test laboratory. Wet specimens are environmentally-conditioned by exposing them in a humidity chamber until they attain an equilibrium moisture content agreed to by the contractor and certifying agency, and then packaging them in a heat-sealed aluminized polyethylene bag until required for test. Tests shall be performed in a manner which maintains the moisture content in specimens at the levels agreed to by the contractor and certifying agency.

[3] Tests shall be performed on each of the five batches.

[4] For 0° and 90° tension, ASTM D 3039 and SACMA Recommended Method (SRM) 4-88 are acceptable test methods for MIL-HDBK-17 data submittals.

[5] Short Beam Shear is for screening and quality control purposes only.

Important Note: This matrix is intended for the generation of dry coupon data. Wet data can be generated by duplicating this test matrix in the wet condition or by generating hot/wet data at the application specific temperature.

Other important notes: Min Temperature is normally -65°F
ET2 is the maximum application temperature
ET3 should be less than Tg temperature of the material,
dry Tg if testing is done dry, wet Tg is the testing is done wet.

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calculate an A-Basis value. If B-Basis (95/90) values are being generated, only 30 data points are needed over a temperature range. The guidelines for this test matrix are that each batch should be distributed over the temperature range as uniformly as possible and that at least three batches are represented at any one test condition.

2.5 FILAMENT-WOUND MATERIALS PROPERTY TESTS. The test matrix shown in Table 2.5 contains the suggested mechanical property tests for filament wound structures.

TABLE 2.5 *Filament-wound materials property tests.*

Mechanical Property	Suggested Test Procedure[1]	Test Condition and Number of Tests Per Batch[2]			Number of Tests
Condition[3]		Min. Temp Dry	RT Dry	Max. Temp Wet	
0° Tension	ASTM D 3039	6	6	6	90
90° Tension	ASTM D 5450	6	6	6	90
0° Compression	ASTM D 3410 (Method B)	6	6	6	90
90° Compression	ASTM D 5449	6	6	6	90
In-plane Shear	ASTM D 5448	6	6	6	90
Transverse Shear	ASTM D 5379	6	6	6	90 540

1 Reader is referred to Section 6.7 Mechanical Property Tests for more information on these ASTM test methods

2 Tests shall be performed on each of the five batches.

3 Minimum and maximum temperature tests shall be performed within ±5°F (±2.8°C) of the nominal test temperature. Nominal test temperatures will be as agreed to by contractor and certifying agency. Dry specimens are "as-fabricated" specimens which have been maintained at ambient conditions in an environmentally-controlled test laboratory. Wet specimens are environmentally-conditioned by exposing them in a humidity chamber until they attain an equilibrium moisture content agreed to by the contractor and certifying agency, and then packaging them in a heat-sealed aluminized polyethylene bag until required for test. Tests shall be performed in a manner which maintains the moisture content in specimens at the levels agreed to by the contractor and certifying agency.

The JANNAF Composite Motorcase Subcommittee has recommended to filament wind the flat laminates used for the test articles for ASTM D 3039, ASTM D 3410 and the Iosipescu Shear test for the uniaxial material properties used in the design and analysis of filament wound structures. However, there are no universal standards describing the process. Consequently there are numerous methods used by industry and Government to manufacture the flat laminates, (References 2.5(a) and 2.5(b)).

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At these two meetings filament winders, both industry and government presented their techniques to prepare flat laminates for the purpose of testing for uniaxial mechanical material properties.

One main issue is whether to use a cylindrical or a rectangular winding mandrel. If a cylindrical mandrel is used, the diameter of the mandrel is a factor. The larger the diameter, the less the effects of shear when the laminates are removed from the winding mandrel and flattened for curing. If the mandrel is rectangular, the main concern is how tension of the fiber is maintained during winding.

The following issues have been identified as concerns for filament wound laminates:

- autoclave vs. non-autoclave cure
- cutting of fibers before or after cure
- whether to cure on the winding mandrel or to remove and cure on a separate fixture
- whether to use a caul plate
- whether to wind single layers, cut and stack versus winding entire thickness before cutting

Currently the winders appear to be using the technique that produces a panel that most closely simulates the process used on their finished part. The ASTM Task Group D30.04.05 is discussing and pursuing these same issues and developing a standard method to prepare filament wound laminates.

2.6 BRAIDED MATERIALS PROPERTY TESTS.

2.7 THICK-SECTION COMPOSITES PROPERTY TESTS.

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2.8 OTHER USEFUL TEST MATRICES.

2.8.1 Material system screening. The objective of this screening process is to reveal key mechanical property attributes and/or inadequacies in new material system candidates through use of a reduced test matrix. The screening process will identify the critical test and environmental conditions, and any special considerations for a particular composite material system. Proper test selection and environmental conditioning will enable comparisons with current productions systems.

Table 2.8.1 shows a typical mechanical property screening test matrix primarily intended for epoxy resin systems. The test matrix will vary depending on the purpose of the investigation. The 0° axial tensile tests examine fiber dominant properties, and 0° axial compression tests monitor matrix/fiber interactions. The 0° axial tensile tests may reveal matrix-fiber interaction in some materials at high strain rates. These tests will yield static strength and stiffness properties. To test matrix characteristics, ±45° tensile tests will be used. The ±45° tensile tests permit actual shear modulus and effective shear strength values to be obtained. Finally, damage tolerance will be evaluated using impact testing.

Testing will be conducted under three environmental conditions: cold temperature dry (CTD), room temperature ambient (RTA), and elevated temperature wet (ETW). These conditions are selected based on results for the epoxy resin systems currently in use which show that the CTD condition is a fiber critical environmental state and the ETW condition is the most severe test for the matrix. Moisture conditioning may be accomplished by placing the dried test specimens in an environmental chamber until an equilibrium moisture weight gain at the desired relative humidity is obtained.

TABLE 2.8.1 *Composite material static strength screening test matrix.*

Test	Number of Specimens			Evaluation Emphasis
	CTD	RTA	ETW	
Lamina:				
0° Tension	3	3	3	fiber
0° Compression		3		fiber/matrix
±45° Tension		3		fiber/matrix (0°/90° shear - lamina) (±45° - laminate)
Laminate:				
Open Hole Compression[1]		3	3	stress riser
Open Hole Tension[1]		3		stress riser
Bolt-Bearing[1]		3		bearing
Compression after Impact[2]		3		impact damage

[1]Fastener hole effects
[2]per NASA Reference Publication 1092.

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The screening process presented allows key mechanical properties and environmental conditions to be evaluated. Dependent upon application, sensitivity to exposures to operational fluids and other special issues, additional tests may also be included in the screening evaluation. However, a reduced test matrix for mean property evaluations should permit preliminary efficient and economical assessment of new composite material systems.

The approach to screening is the performance of key static tests to provide sufficient data for mean property evaluations of stiffness and strength. The recommended specimens provide data at the lamina (ply) and laminate (application) levels. The lamina level tests provide key generic material stiffness and strength properties commonly used in classical lamination plate point stress analysis including tension, compression, and shear loadings. The laminate level tests provide screening strength data on application issues relating to effects of stress concentrations, manufactured stress risers (fastener holes), bolt bearing, or impact damage. Additional laminate level tests provide screening stiffness data to verify the use of lamina data with classical lamination plate theory for laminate stiffness predictions. The damage tolerance screening is patterned after NASA Reference Publication 1092 (Reference 2.8.1).

All laminate level tests suggested are performed at the RTD test condition. For screening purposes, it is assumed the lamina (ply) level unnotched specimens will conservatively provide mean data to assess environmental effects on strength and stiffness. The laminate level tests will provide mean data on the effects of stress concentrations.

Table 2.8.1(b) shows the typical mechanical property test matrix intended for high temperature polymer matrix composites. The changes were made to Table 2.8.1(a) in order to properly assess the durability of high temperature polymer matrix composites during the screening stages of an evaluation. The test matrix will vary depending on the purpose of the investigation, but it is important that all exposure conditions be evaluated.

Prior to the mechanical test evaluation, it is necessary to evaluate prepreg physical and laminate properties. Test laminates should be carefully inspected for porosity content, dry T_g, and wet T_g. The recommended mechanical tests cover fiber-dominated, interface/resin-dominated, and damage tolerance properties. The elevated test temperature static test conditions should be set below the wet T_g of the system.

The wet exposure condition is 160°F/ 85% relative humidity to an equilibrium weight gain. It is very important that specimen dry-out be measured and kept to a minimum during the elevated temperature wet tests.

The thermal oxidative stability (TOS) test should be performed for a minimum of 1000 hours. Weight loss should be measured during testing at specified intervals of 100, 250, 500, 750, and 1000 hours. This test provides a measurement of the oxidation rate of the material.

Thermal cycling should be done for a minimum of 500 thermal cycles. The purpose of the test is to determine the rate of microcracking, not only if microcracking will occur. The minimum temperature should represent the minimum temperature of the potential application, e.g., -65°F.

The maximum exposure temperature for both the TOS and thermal cycling test should be between the wet T_g and dry T_g of the material system. If the exposure temperature is below the wet T_g, the test may not be discriminating enough and longer exposure times may be necessary. Exposures above the dry T_g of a material, normally provide an unrealistic damage mechanism that do not occur below the dry T_g of the material. Prior to the machining of specimens, laminates should be non-destructively inspected for porosity and delaminations. Microscopy should also be done in order to understand the

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damage mechanism associated with the specific exposure. This will include the measurement of microcrack density.

2.8.2 Qualification guidelines and requirements for alternate composite materials.

2.8.2.1 *Introduction.* These guidelines are intended to apply to the situation where one composite material system, from a single supplier, has been qualified and it is necessary or desirable to qualify an alternate system and/or supplier. The approach assumes the existence of a body of data and experience developed with the original material (none exists for the alternate system) from which the mechanical property allowables have been developed. It also assumes higher level tests have been performed to qualify a product and verify its performance.

A drastic change, such as switching fiber from E-Glass to aramid, is not covered by this guideline. The focus is on materials that will meet the original material specification. A fiber class change, or comparable substitution, is considered a major revision or redesign. Processing and tooling changes are also considered beyond the scope of this section.

2.8.2.2 *Goal.* The ultimate goal in qualifying an alternate material is to be able to exchange this material with the original system without compromising manufacturing or structural performance. To accomplish this goal it is necessary to define the key material parameters that govern performance during specific phases, such as processing, manufacturing, and service. The ideal is to perform this evaluation at the material constituent level, or composite lamina level; i.e., measure and compare the chemical composition, fiber strength, matrix strength, and composite strength. This may be possible in the future, but is not adequate with current technology.

Successful qualification of an alternate material will not, in itself, be sufficient to permit mixing of this system with the original material in a given part. Intermixing is prohibited unless appropriate evaluations are carried out to demonstrate compatibility.

The focus of MIL-HDBK-17 is "B" basis lamina properties. Adequate alternate material qualification may require going beyond this level of evaluation into more complex demonstrations involving analysis and tests. These may include laminate, coupon, element and subcomponent tests such as open hole, filled hole, bolt bearing, low velocity impact, fatigue, and panel buckling.

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TABLE 2.8.1(b) *Composite material test matrix for high temperature PMC's*

Mechanical Property	Dry Test Temperature			Wet[1]	TOS[2]	Thermal Cycle[3]
	Min Temp	75F	ET1	ET1	ET1	ET1
Tension	3	3	3	-	3	-
Compression or OHC	-	3	3	3	3	3
In-Plane Shear	-	3	3	3	-	3
Mode I Fracture Toughness or CAI	-	3	3	-	3	3

1 - ET1 Elevated test temperature should be less than the Wet Glass Transition Temperature of the material

2 - Laminates are to be thermally aged at a temperature greater than ET1 but less than the dry Tg for a minimum of 1000 hours or an accelerated test condition that represents a 1000 hour exposure. Weight loss should be recorded as a function of time, i.e., 100, 250 500, 750 and 1000 hours. Microscopy should be performed after exposure. Specimens are to be machined after exposure.

3 - Laminates are to be thermal cycled from Min Temp to a temperature greater than ET1 but less than the dry Tg. Laminates are to be cycled for a minimum of 500 thermal cycles. Microcrack density is to be measured after cycling. Specimens are to be machined from the laminate after exposure.

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TABLE 2.8.2.8(b) *Number of suggested laminate tests.*

Design Property	Loading		No. of Laminate Types		No. of Environments[1]	Replicates[2]	Total Number of Specimens	
	Tension	Compression	Tape	Fabric			Tape	Fabric
Static								
Unnotched laminate, strength and stiffness	X	X	1	1	2	3	12	12
Open Hole		X	1	1	2	3	6	6
Filled Hole	X		1	1	2	3	6	6
Impact Damage	X	X	1	1	1	3	6	6
Double Shear Bearing	X		1	1	1	3	3	3
Single Shear Bearing	X		1	1	1	3	3	3
							36	36
Fatigue[3]								
Open Hole	-	-	1	1	1	3	3	3
Impact Damage	-	-	1	1	1	3	3	3
							6	6
							42	42

[1] Where two environments are required, they should be RTD and worst case. Where one is required, it should be RTD.
[2] One batch of material is sufficient.
[3] Repeated load and residual strength: constant amplitude, R = -1, n = 1×10^6 cycles.

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intervals. Any of the mean differences which are statistically significant at the five percent level should be investigated.

To get a single number which measures the difference between the materials, let p_i be the OSL for the ith property as determined in Section 2.8.2.11, and let m be the number of properties compared. Calculate the following:

$$P = -2 \sum_{i=1}^{m} \ln (p_i) \qquad 2.8.2.10$$

The larger P is, the more evidence there is in the data for a difference between the materials. Compare P with the 95th percentile of a chi-square distribution with 2m degrees of freedom (Table 8.8.8) in order to determine if the differences in the means is significant at the five percent significance level.

This combined test is strictly valid only when the m sets of data are statistically independent of each other. However, since the tests are on the same material and the same batches, this independence will never exactly hold. In many situations, the tests will be approximately independent and the combined value P will provide a useful measure of the extent to which the two sets of tests differ. If it is apparent from examining the data that some batches are consistently high and others low, i.e. that independence does not hold, then the combined test should be interpreted with caution.

2.8.2.11 *Comparing two groups of batches.* This section considers the problem of testing whether a statistically significant difference exists between the means of two sets of measurements, where each set consists of several batches. The methods of this section might be applied, for example, to comparing the mean room temperature tensile strength of specimens made from three batches at one site to another set of measurements on the same mechanical property consisting of five batches manufactured at another site.

The two sets of data are represented by x_{ij} and y_{ij} where the first subscript indicates the batch and the second subscript denotes the values within each batch. We assume here that both the x and y sets of data are sampled from one-way, balanced, random effects models (see Section 8.6.2):

$$x_{ij} = \mu^{(1)} + b_i^{(1)} + e_{ij}^{(1)} \qquad 2.8.2.11(a)$$

where $i = 1,\ldots,k_1$ and $j = 1,\ldots,n_1$ and

$$y_{ij} = \mu^{(2)} + b_i^{(2)} + e_{ij}^{(2)} \qquad 2.8.2.11(b)$$

where $i = 1,\ldots,k_2$ and $j = 1,\ldots,n_2$. The number of batches and batch size are k_1 and n_1 for the x's and k_2 and n_2 for the y's.

The ANOVA model represents each observation as the sum of three components; $\mu^{(\ell)}$ is the overall mean, $\mu^{(\ell)} + b_i^{(\ell)}$ is the population average for the ith batch, and $e_{ij}^{(\ell)}$ represents the variation within each batch, where ℓ equals one for the x data and two for the y data. The error terms $e_{ij}^{(\ell)}$ are assumed to be independently distributed normal random variables with mean zero and variance σ_e^2 (the within batch variance).

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The batch means $b_i^{(1)}$ are assumed to be independent random variables following a normal distribution with zero and a variance of σ_b^2 (the between batch variance). The within-batch variance is assumed to be the same for all batches.

Denote the batch averages for the x's by $\bar{x}_i$, for $i = 1,\ldots,k_1$, and the batch averages for the y's by $\bar{y}_i$, for $i = 1,\ldots,k_2$. the test statistic uses the following four quantities:

$$\bar{x} = \frac{1}{k_1}\sum_{i=1}^{k_1}\bar{x}_i \qquad 2.8.2.11(c)$$

$$\bar{y} = \frac{1}{k_2}\sum_{i=1}^{k_2}\bar{y}_i \qquad 2.8.2.11(d))$$

$$S_x^2 = \frac{n_1}{k_1 - 1}\sum_{i=1}^{k_1}(\bar{x} - \bar{x}_i)^2 \qquad 2.8.2.11(e)$$

$$S_y^2 = \frac{n_2}{k_2 - 1}\sum_{i=1}^{k_2}(\bar{y} - \bar{y}_i)^2 \qquad 2.8.2.11(f)$$

If $k_1 = 1$, then let $S_x^2 = 0$; if $k_2 = 1$, let $S_y^2 = 0$; if $k_1 = k_2 = 1$, then the method of this subsection should not be used. In terms of the statistics in Equations 2.8.2.11(c) - (f), the test statistic is

$$T = \frac{|\bar{x} - \bar{y}|}{\left(\frac{S_x^2}{k_1 n_1} + \frac{S_y^2}{k_2 n_2}\right)^{0.5}} \qquad 2.8.2.11(g)$$

To test the hypothesis that $\mu^{(1)} = \mu^{(2)}$ at the α significance level, compare T with $t_{1-\alpha/2,\gamma}$, the $100(1 - \alpha/2)$ quantile of a central t random variable with $\gamma = k_1 + k_2 - 2$ degrees of freedom (Table 8.8.17). If T does not exceed this t quantile, then conclude that the data are consistent with the hypothesis that the population means are equal, otherwise conclude that there is a statistically significant difference in the population means (at the α level of significance).

A $100(1 - \alpha)$ confidence interval is

$$|\bar{x} - \bar{y}| \pm t_{1-\alpha/2,\gamma}\left(\frac{S_x^2}{k_1 n_1} + \frac{S_y^2}{k_2 n_2}\right)^{0.5} \qquad 2.8.2.11(h)$$

The observed significance level, or OSL, is the probability of observing a value of T as large or larger than the T actually observed if indeed the hypothesis of equal means is true. An OSL which is less than the significance level α indicates that the null hypothesis can be rejected at the α level of significance. The OSL is a function of T and $\gamma = k_1 + k_2 - 2$. For γ greater than ten, the following approximation is usually adequate. Calculate

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$$U = \frac{T\left(1 - \frac{1}{4\gamma}\right)}{\left(1 + \frac{T^2}{2\gamma}\right)} \qquad 2.8.2.11(i)$$

Determine the probability P that a standard normal random variable is less than U. this probability can be determined from a table of the normal distribution, such as Table 8.8.6. The OSL is equal to 2(1 - P). If γ is less than 10, then the above approximation is not sufficiently accurate and the OSL should be obtained from Table 8.8.15.

For example, consider the strength measurements in Table 2.8.2.11. The specimens tested to give these data were taken from a group of three consecutive batches and a group of five consecutive batches. The second group of batches was produced more than a year after the first group. Because of this time difference, one should not consider these data to be eight random batches from a single population without justification. A more prudent approach is to regard these test results as a random sample of three batches from one population and a random sample of five batches from a possible different population.

TABLE 2.8.2.11 *Strength measurements from two groups of consecutive batches.*

Set 1			Set 2		
Mean	Variance	n	Mean	Variance	n
402.2	138.7	5	408.4	40.8	5
387.8	1002.2	5	395.8	113.2	5
389.4	321.8	5	357.2	451.7	5
			376.2	119.7	5
			377.0	189.5	5

For the data in Table 2.8.2.11, Equations 2.8.2.11(c) - (g) give the following:

$$\bar{x} = 393 \qquad 2.8.2.11(j)$$

$$\bar{y} = 383 \qquad 2.8.2.11(k)$$

$$S_x^2 = 311 \qquad 2.8.2.11(l)$$

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$$S_y^2 = 1946$$

$$T = \frac{|393 - 383|}{\left(\frac{311}{(3)(5)} + \frac{1946}{(5)(5)}\right)^{0.5}} = 1.007 \qquad 2.8.2.11(n)$$

From Table 8.8.17, the 97.5 percentile of the t distribution with 5 + 3 - 2 = 6 degrees of freedom is $t_{0.975,6} = 2.45$. Since 1.007 is less than 2.45, one concludes that there is no statistically significant difference in the mean strength for the two sets of data at the five percent significance level. Because $k_1 + k_2 - 2 = 6$ is less than ten, use Table 8.8.5 to obtain an OSL of 0.42. A 95 percent confidence interval for the difference in the means is given by

$$|393 - 383| \pm (2.45)\left(\frac{311}{(3)(5)} + \frac{1946}{(5)(5)}\right)^{0.5} \qquad 2.8.2.11(o)$$

$$10 \pm 24.3$$

Note that the confidence interval must contain zero for this example since the difference in the means is not significant at the 95 percent level.

2.8.3 Design value leveraging requirements.

2.8.3.1 *The direct use of MIL-HDBK-17, or other sources of, allowables.* In order to use MIL-HDBK-17 material allowables (or allowables from other sources) directly in design, it is necessary to demonstrate the ability to consistently produce the same material as that manufactured during the material allowable generation. The tests identified in Table 2.4(b) should be conducted for this purpose. A total of six specimens per loading condition are required, either using two batches or two panels within a single batch, processed independently. The statistical procedures used to validate that the data are from the same population, as that for which the allowable was calculated, are shown in Section 2.8.2.10. The testing procedures and fabrication processes should be similar, i.e., cure cycle, tooling, etc. Any changes to the amount of substantiation required in the lamina level testing, e.g., a reduction in the number of loading conditions evaluated will depend upon agreements reached between the contractor and the certifying agency.

It is assumed that higher levels of testing (i.e., element, subcomponent, component tests) and assessments of the repeatability of the fabrication processes, leading to component validation will be planned and conducted.

2.9 GENERIC LAMINATE LEVEL CHARACTERIZATION.

2.9.1 Introduction. A simplified flow chart, Figure 2.9.1, overviews the building-block flow of a typical material/structural qualification process. A series of evaluations is required to assess the adequacy of a material system for production usage. These multi-purpose assessments, often performed in parallel, range from material performance to producibility and cost.

Depending on the sophistication of the application design concept (e.g., a flat monolithic panel versus an integrally cocured semi-monocoque fuselage section), an extensive, progressively more complex, building-block approach to certification testing may be required to evaluate material and structural design risks. It is recognized that the complete structural qualification of composite material systems for design allowables is often highly dependent on the application for which the material will be used. The historical lessons-learned dictate that composite production hardware design programs must evaluate and discover material, structural, and producibility design deficiencies early in the design

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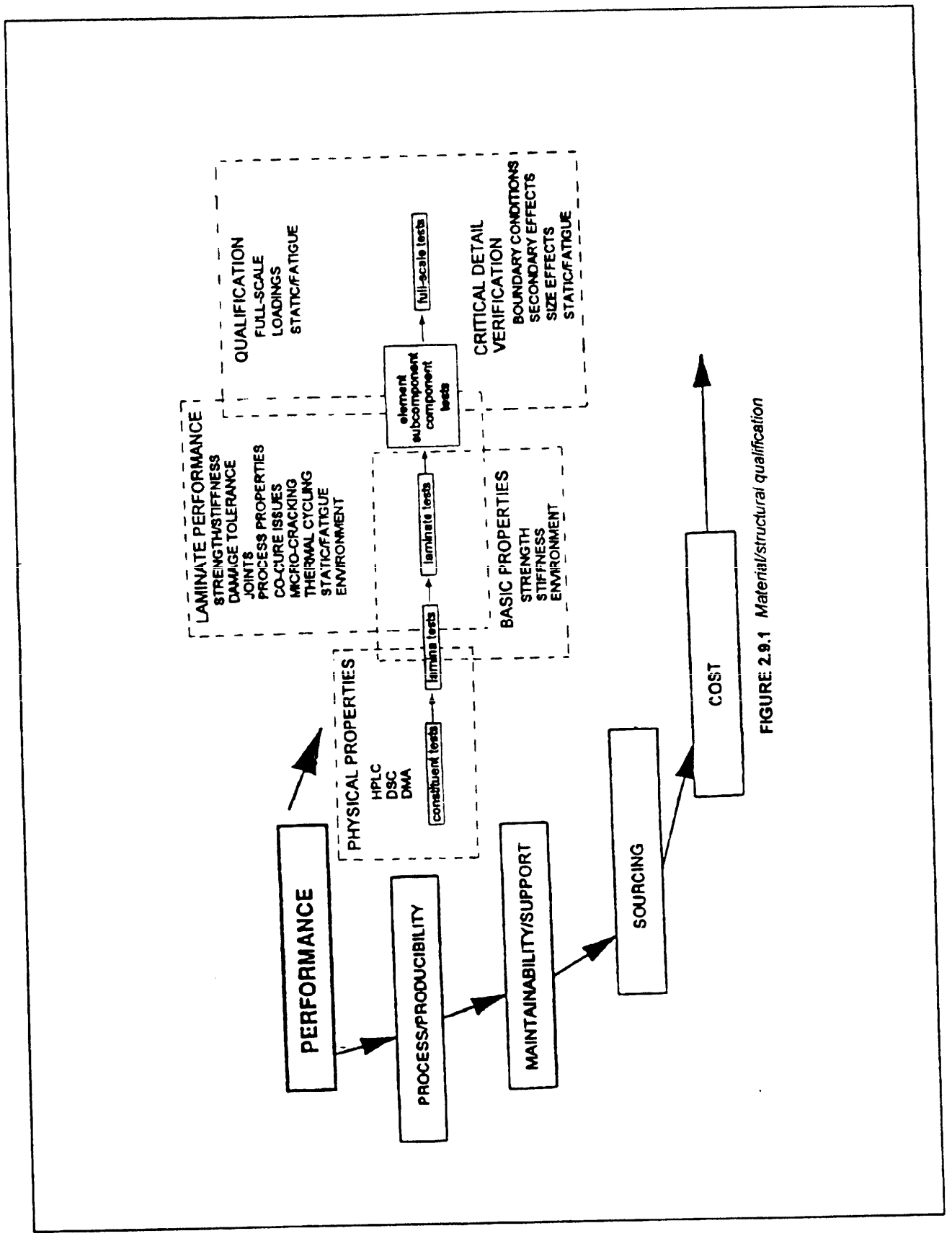


FIGURE 2.9.1 *Material/structural qualification*

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development program to meet cost, performance, and schedule goals. Toward this end, it is extremely desirable on any program to establish early, with high confidence, the material design allowables. If this is successfully accomplished, the design development program can then focus on detail design, higher level design development tests, and producibility issues. The most *adverse* situation for any program to experience is a material development or reselection effort *in parallel* with detail design development.

This section addresses that part of the Figure 2.9.1 process which assesses the mechanical property characterization at the laminate level. The intent is to define a series of laminate level test matrices which complements both the ply level mechanical property characterization test matrix and the lamina/laminate screening test matrix, previously defined in Volume 1, Sections 2.4 and 2.8.1.

The basis for the test matrices of this section is that a significant number of similar, laminate level, coupon tests are performed in almost all hardware design development programs prior to extended production. The additional laminate level test data are necessary for theory/test correlations to substantiate mathematical models used to predict design allowables using lamina (ply) stiffness and strength input data (Section 2.4), or are needed to establish empirical trending where mathematical models do not exist or are deemed deficient. In either case, some coupon laminate-level data have been historically required to substantiate or establish the design allowables essential to structural qualification. Historically, these costly and time-consuming tests are often repeated in each new application program. Because a significant number of these tests are performed at the coupon level, *the test data generated should, once generated, apply to a wide range of applications, and be acceptable to certifying agencies in other application programs.*

These standard tests, once performed (with test matrices in Sections 2.4 and 2.8.1), are intended to further reduce the cost and time of new material characterization efforts, and *establish a generic database for the tested material system applicable to other proposed applications.*

2.9.2 Overview. Two laminate level test matrices are defined: (1) laminate strength, and (2) bolt bearing and bearing/bypass strength. Together, these test matrices should provide a statistically significant laminate-level database. The test matrices are defined for selective 3-batch assessments of either tape or fabric prepreg materials. Dependent upon the availability of *validated* analytic models for strength prediction, and the degree to which they use only ply-level strength and stiffness input data, batch effects may be accounted for at the lamina level and not require multiple batch testing in the following test matrices. Thus, with certification agency approval, a single-batch test plan variation may be proposed with replicates of 5 specimens per test condition as implied in Section 7, Volume 1. However, the following test matrices are based on a multiple 3-batch (with 3 specimens/test condition), selective laminate, assessment of analytic models using lamina data for strength predictions. Additionally, it is noted that other load conditions, such as in-plane shear, may require additional testing at higher levels in the building-block assessment and are not covered by these test matrices.

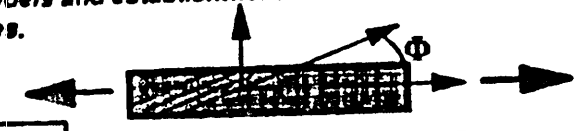
2.9.2.1 *Laminate strength test matrix.* As detailed in Table 2.9.2.1, a series of selected orientation laminate unnotched strength tests are recommended for both tensile and compressive loadings at selective cold temperature dry (CTD), room temperature dry (RTD), and elevated temperature wet (ETW) test conditions. For two laminate configurations, three replicate tests are repeated for each of three batches of the material system. Two additional laminates are selectively tested with 5 specimens per test condition using one batch of material. The matrix emphasizes fiber dominant laminate evaluations at the extremes of material environmental capabilities (CTD and ETW) and provides baseline data at room temperature dry (RTD) test conditions. The intent is to provide data to permit a selective validation of stiffness and strength analytic models over a representative range of application relevant laminates and test conditions.

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TABLE 2.9.2.1 *Laminate strength test matrix.*

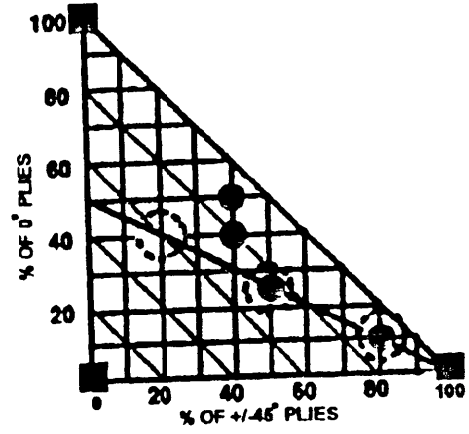
OBJECTIVE:

Create a generic, laminate-level, database to assess an application range of unnotched laminates for correlation of laminate stiffness and strength analytic models and establishment of a selective, but statistically significant, database for empirical approaches.



LAY-UP (%0,%±45,%90 degree plies)	THICK-NESS	LOAD ANGLE Φ	COMPR RTD	COMPR ETW	TENS CTD	TENS RTD
25/50/25	T1	0	9	9	9	9
	T2	0	9	9	9	9
	T1	22.5	-	5	5	-
50/40/10	T1	0	9	9	9	9
	T2	0	9	9	9	9
	T1	22.5	-	5	5	-
40/40/20	T1	0	-	5	5	-
10/80/10	T1	0	-	5	5	-
SUBTOTALS =			36	58	56	36
TOTALS =						184

◆ ORIENTATION OF 0 DEGREE PLIES TO LOAD DIRECTION



■ PLY (LAMINA) LEVEL TEST POINTS (VOL. I, SECT. 2.4)

● LAMINATE LEVEL TEST POINTS (THIS TEST MATRIX)
RTD - room temp., dry test condition
CTD - cold temp., dry test condition
ETD - elevated temp., dry test condition

◌ FABRIC TEST POINTS

NOTES:

Assumes 0, +45, -45, & 90 degree family of ply orientations; balanced & symmetric
9 implies 3 specimens/batch; 3 batches
5 implies 5 specimens/batch; 1 batch
T1 represents laminate thickness of approximately 2.0-6.6 mm (.08-.25 inch)
T2 represents second laminate thickness; *may be optional* depending on upper bound of application laminates
For fabric — 40/40/20 replaces 55/35/10 laminate and 40/40/20 is deleted

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The limited number of tests implies that data pooling using regression analysis across test conditions will be employed (Volume 1, Section 8.7). The exact specification of critical temperatures and moisture conditioning is determined either by the minimum/maximum material operational capabilities (MOL) established for lamina level tests (Sections 2.2 & 2.4) or application considerations jointly established by the manufacturer and the procuring agency.

Four general laminate configurations are specified for tape characterization testing; three laminates for fabric material forms. As illustrated in the carpet plot of Table 2.9.2.1, the selection of the four laminates is intended to span the usual application range of structural laminates with emphasis on the fiber dominant orthotropic and quasi-isotropic laminate constructions. Additionally, the solid circles indicate the solely 0-degree, 90-degree, or ±45-degree (lamina level) evaluations specified in Section 2.4. For fabric characterization testing, the bold line in the carpet plot of Table 2.9.2.1 represents the reduced range of possible laminate constructions and the 55/35/10 and 40/40/20 tape laminates are replaced by a 40/20/40 fabric laminate construction in the test matrix of Table 2.9.2.1. Stacking sequences as specified in Section 7.2 on mechanically fastened joints are also recommended for laminates in this section.

The test matrix also requires two of the laminates to be tested at 22.5 degrees off the principal material axis to assess off-axis material behavior at the critical environmental test conditions. Additionally, if the application range of thickness exceeds significantly (say by a factor greater than two) the basic "T1" thickness range of 0.08-0.24 inches (2-6 mm), a second three-batch series of "T2" laminate thickness tests is specified in Table 2.9.2.1 for all test conditions. However, if the application range of laminate thickness are contained within a 0.16 inch (4 mm) variation, it is believed only one test matrix thickness is required (perhaps different from the "T1" range suggested in Table 2.9.2.1). This would reduce the matrix to a total of 114 tests for tape laminates (104 for fabric). If the range of application thickness is significantly broader, the second series (T2) of laminate thickness tests should also be performed. This would result in a total of 184 tests for tape laminates (174 for fabric).

2.9.2.2 *Bolt bearing and bearing/bypass strength test matrix.* The test matrix, detailed in Table 2.9.2.2, is intended to provide strength data to assess the effects of fasteners under bearing and bearing/bypass strain concentrations on laminate strength. In addition to generating design data to establish allowables for composite bolted joint analysis, this test matrix provides baseline reference strength data for the effect of manufacturing anomalies and impact damage on laminate strength. The test recommendations in this section include all tests found in Volume 1, Section 7, Tables 7.2.5.2, 7.2.6.2, and 7.2.7.3. However, there are additional tests required by Table 2.9.2.2 because of the desire to empirically obtain laminate design allowables directly from these data. This necessitates selective testing of three batches of material at the worst environments.

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TABLE 2.9.2.2 *Bearing/bypass*

LAY-UP	THICK NESS	HOLE SIZE DIAMETER	FASTENER HEAD TYPE	COMPRESSION BEARING/BYPASS						
				100% BYPASS			75% BYPASS		50% BYPASS	
				RTD	ETW	ETW + ΔT	RTD	ETW	RTD	ETW
25/50/25	T1	D1	H1	5	9	9	5		5	5
25/50/25	T1	D2	H1	5	9					5
25/50/25	T1	D2,W/D = 8	H1	5						
25/50/25	T1	D3	H1	5	9					5
25/50/25	T1	D3,W/D = 4	H1	5						
25/50/25	T1	D1	H2	5	9					5
25/50/25	T2	D1	H1		9	9				5
25/50/25	T2	D2	H1							
25/50/25	T2	D3	H1		9					5
25/50/25	T3	D1	H1							
25/50/25	T3	D2	H1							
25/50/25	T3	D3	H1							
50/40/10	T1	D1	H1	5	9	9	5		5	5
50/40/10	T1	D2	H1	5	9					5
50/40/10	T1	D1	H2	5	9					5
50/40/10	T1	D3	H1	5	9					5
50/40/10	T2	D1	H1		9	9				5
50/40/10	T2	D2	H1							
50/40/10	T2	D3	H1		9					5
10/80/10	T1	D1	H1	5	5		5		5	
10/80/10	T1	D1	H2	5	5					
10/80/10	T2	D1	H1		5					
TOTALS				60	123	36	15	0	15	60

NOTES:
1. T1, D1, and H1 are the primary values of laminate thickness, fastener diameter, and fastener head type. T2, T3, D2, D3 and H2 *may be optional* depending on the range of laminate thicknesses and fastener geometries.

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laminate strength test matrix.

TENSION BEARING/BYPASS								NUMBER OF TESTS
100% BYPASS		75% BYPASS		50% BYPASS		0% BYPASS		
CTD	RTD	CTD	RTD	CTD	RTD	RTD	ETW	
9	5		5		5	5	9	76
9	5					5	9	47
	5							10
9	5					5	9	47
	5							10
9	5					5	9	47
9						5	9	46
						5	5	10
9							9	32
						5		5
						5		5
						5		5
9	5		5		5	5	9	76
9						5	9	42
9	5					5	9	47
9							9	37
9						5	9	46
						5	5	10
9							9	32
5	5		5		5	5	5	50
5	5					5	5	30
5							5	15
123	50	0	15	0	15	80	133	725

2. "9" represents 3 specimens/batch with 3 batches tested; "5" represents 5 specimens/batch with only 1 batch tested.

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Three of the laminate constructions previously tested under the unnotched laminate strength matrix of Section 2.9.2.1 are also specified for testing tape and fabric materials under tensile and compressive bolt bearing and bearing/bypass loadings. Single fastener joint evaluations cover a range of fastener load transfer test condition including filled hole (100% bypass), bearing/bypass interactions (75% and 50% bypass), and pure bearing (0% bypass). The test matrix is designed to emphasize critical material environmental conditions with most test data collected at either the hot-wet (ETW) condition for matrix-dominated failure modes or the cold-dry (CTD) condition for fiber-dominated failure modes. Sufficient data are also specified to establish a room temperature dry baseline condition. Tension and compression load conditions are specified. Pure bearing tests are only performed under tensile loadings.

The test philosophy is to first evaluate the effect of hole size on laminate strength. Three fastener diameters which span the range of application hardware are specified. Tension and compression strength data are gathered for filled hole tension/compression, bearing/bypass, and pure bearing test conditions. Specimens detailed in Figures 7.2.6.3, 7.2.7.2, and 7.2.5.2 of Section 7.2, Volume 1, respectively, should be employed. The fastener diameter which represents the majority of application fastener diameters should be selected as the baseline diameter D1 in Table 2.9.2.2. The remaining two fastener diameters (D2 and D3 of Table 2.9.2.2) should bound all other application fastener usage. Based on historical aerospace industry practice with carbon-epoxy, a baseline joint geometry of width/diameter (W/D) of 6 and edge-distance/diameter (e/D) of 3 is recommended for almost all test specimens (2 additional W/D ratios tested for the 25/50/25 lay-up; also as carbon-epoxy systems evolve or for other material systems these values may be different). The fastener head style, protruding or countersink, that represents the majority of application usage should also be selected as the baseline (H1 in Table 2.9.2.2) for all specimen configurations. Additionally, the same "T1" laminate thickness, as discussed in Section 2.7.2.1, will be used as the baseline for all specimens. If necessary to cover application design variations, a second series of laminate specimens, of thickness T2 in Table 2.9.2.2, may be required. Stacking sequences as specified in Section 7.2 on mechanically fastened joints are also recommended for laminates in this section.

The initial testing should be performed on the isotropic (25/50/25) laminate specimens for the 100% bypass and pure bearing (0% bypass) load conditions at RTD environment. *This should permit initial correlation (or calibration) of analytic or empirical models used for strength prediction of laminates under tension or compression loadings.* It is anticipated that these calibrated strength prediction models will then be used to predict, prior to test, the results to be obtained from the remaining bearing/bypass tests of the isotropic laminate and the full range of tests on the remaining two orthotropic laminates (tape - 50/40/10 and 10/80/10, and fabric - 40/20/40 and 10/80/10). These initial evaluations would be followed by the full range of environmental (CTD or ETW) tests and provide the statistical validation or basis for establishing critical design condition allowables. These same tests, compared to RTD test results, would verify consistency and types of laminate failure modes. Finally, if required, the second series of laminate thicknesses (T2 and T3) and fastener head style (H2) tests would be performed at the critical environmental test conditions to establish additional calibration of analytic models or empirical "knockdown" factors for design allowables.

Based on typical design considerations, pure bearing tests under tensile loading conditions provide conservative strength values and similar failure modes as compared to pure bearing tests under compressive loading, as long as the specimen a/D ratio is specified as 3 or greater. Accordingly, this test matrix requires pure bearing tests (0% bypass) only under tensile loading conditions. Similarly, design values based on the intermediate 50% level of bearing/bypass load interactions under compressive loading are generally conservative compared to design values based on the 75% level or intermediate levels of tensile loading. Thus, in some cases, only tests of 50% bearing/bypass load interaction under compressive loading may be sufficient and the compressive tests at 75% levels and the tests of 50% bearing/bypass conditions under tensile loading may be eliminated. For current

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composite material systems, this is felt to be realistically conservative for both the acknowledged nonlinear behavior under compression bearing/bypass load conditions and the relatively linear material behavior under tension bearing/bypass load conditions. Should material behavior or design weight goals require a less conservative approach, more experimental evaluation is necessary and other bearing/bypass ratio test conditions should be tested. The reader is referred to Section 7.2.7 for additional guidance.

As for the unnotched laminate test matrix of Section 2.9.2.1, the limited number of tests implies that data pooling using regression analysis across environmental test conditions will be employed (reference Section 8.7, Volume 1). The exact specification of critical temperatures and moisture conditioning is determined either by the minimum/maximum material operational capabilities (MOL) established for lamina level tests (Sections 2.2 & 2.4) or application considerations jointly established by the manufacturer and the procuring agency. to assure upper limits of temperature effects are understood for application laminates, an additional set of "ETW + ΔT" tests are performed selectively on the matrix sensitive open hole compression test specimens.

A total of 395 tests are specified if only one laminate thickness and one fastener head style are required to cover application design variables. As in Section 2.9.1.1, for two laminate configurations, three replicate tests are repeated for each of three batches of the material system at critical environmental test conditions. Two additional laminates are selectively tested with 5 specimens per test condition using one batch of material. An additional 124 tests are recommended to cover a second fastener head style, and a further set of 206 tests are recommended if a second (191 tests) and third (15 tests) thickness evaluation is required.

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Chapter 6. LAMINA AND LAMINATE CHARACTERIZATION

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6.1 INTRODUCTION. The use of composite materials continues to increase as new performance, reliability, and durability requirements drive hardware designs to higher levels of structural efficiency. Additionally, government requirements are becoming more stringent to ensure proper levels of structural integrity are maintained. These design drivers, among others, have resulted in a growing recognition that certification or qualification of aerospace structure requires an extensive combination of analysis, testing, and documentation.

Further, because of the large number of design variables inherent to composite structure, analytic models are even more necessary than for metallic structure to ensure completeness of the hardware qualification process. Inherent in all structural analysis models are material, physical, and mechanical property characterization data. Ideally, these analytic models would permit analysts to predict full-scale structural response (e.g. stability, deflections, strength, life) directly from a generic (lamina) material database. In truth, test data is required at design development (element, subcomponent, component) and full-scale article test levels as well as the generic (coupon) levels of evaluation.

The purpose of Chapter 6 is to provide guidelines on testing procedures for characterization of physical and mechanical lamina (ply) properties. While current procedures emphasize development of a lamina-level database, this does not preclude higher-level testing.

A laminate is a product made by bonding together two or more layers of material or materials, and a lamina is a single ply or layer in a laminate. The material forming each layer typically consists of a carbon, glass, or organic (polymeric) fiber reinforcement embedded in a thermoplastic or thermosetting resin matrix. While retaining their identities in the composite, the constituents combine to provide specific characteristics and properties.

Many techniques are used to characterize the chemical, physical, and mechanical properties of composite materials. The purpose of this chapter is to provide information on techniques that may be used to analyze and evaluate these properties. The test methods discussed in each section may not be appropriate for all types of composite materials. Currently, more studies are being conducted to investigate how variations in reinforcement and resin chemistry and morphology may affect the physical properties and long term performance of composites. Where possible, the limitations of existing test methods are discussed.

6.2 NON-DESTRUCTIVE EVALUATION. A variety of non-destructive testing (NDT) techniques are available for detecting both surface and interior flaws in composites. Visual inspection and liquid penetrant methods can be used for identifying surface defects, while more sophisticated techniques are required for detecting internal flaws (i.e. voids, delamination, foreign inclusions, disbonds). These techniques include ultrasonics, radiography, thermography, acoustic emission, X-ray and eddy-current testing. The basic principles and procedures for these methods are covered in the MIL-HDBK-728 series, while more specific information on the theory and interpretation of data can be found in the following:

MIL-HDBK-731 Thermography
MIL-HDBK-732 Acoustic Emission
MIL-HDBK-733 Radiography
MIL-HDBK-787 Ultrasonic

These documents do not discuss the recent advances in NDT techniques, which is currently an active field of research and development.

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6.3 PHYSICAL ANALYSIS TECHNIQUES. The physical analysis methods for laminae and laminates provide information on the integrity of the fabricated composite. Thermal analysis methods are used to determine the glass transition and crystalline melt temperatures, coefficient of thermal expansion, and residual heat of reaction. Additional analytical methods discussed in the following sections are used to determine fiber volume, void volume, density, dimensional stability, and moisture weight gain.

6.3.1 Thermal analysis. The thermal analytical techniques described in chapter 4, section 4.3.1 may also be used to evaluate composite materials. Information obtained from thermal analysis includes the glass transition temperature, crystalline melt temperature, expansion/contraction properties, thermal stability, and extent of cure for thermosets.

6.3.1.1 *Glass transition temperature* (T_g).

6.3.1.1.1 *Overview.* The glass transition of a polymer matrix composite is a temperature-induced change in the matrix material from the glassy to the rubbery state during heating, or from a rubber to a glass during cooling. A change in matrix stiffness of two to three orders of magnitude occurs during the glass transition, due to the onset or freezing out of long range molecular mobility of the polymer chains. The temperature at which the glass transition occurs is a function of the molecular architecture and crosslink density of the polymer chains, but it is also dependent on the heating or cooling rate used in the measurement, and on test frequency if a dynamic mechanical technique is employed. In addition to the change in stiffness, the glass transition is marked by a change in the heat capacity and the coefficient of thermal expansion of the material, and so has at least some characteristics of a second order thermodynamic transition (see Reference 6.3.1.1.1).

The glass transition is frequently characterized by a glass transition temperature (T_g), but since the transition often occurs over a broad temperature range, the use of a single temperature to characterize it may give rise to some confusion. The experimental technique used to obtain the T_g must be described in detail, especially temperature scanning rate and frequency used. The method by which T_g is calculated from the data must also be clearly stated. Reported T_g may reflect onset of the glass transition or midpoint temperature depending on the data reduction method.

Upon exposure to high humidity environments, polymer matrices will absorb environmental moisture and be plasticized by it. One effect of this plasticization is the depression of T_g, frequently by a significant amount. A highly crosslinked resin (one based for instance on a tetrafunctional epoxide such as TGMDA) may have a high initial T_g, but it may be depressed more strongly than that in a less highly crosslinked system. Measurement of the T_g in a composite material plasticized by absorbed moisture poses some difficult experimental challenges. Heating the test specimen as required by the measurement will drive off at least some of the absorbed moisture, thereby affecting the measured properties.

Due to the decrease in matrix stiffness that occurs at the glass transition and to the low strength of these polymer matrices in the rubbery state, the matrix can no longer function effectively to transfer load to the fibers or suppress fiber buckling above the glass transition. T_g is, therefore, frequently used to define the upper use temperature of a composite material, although the time-dependent properties of the material such as creep compliance may be more sensitive to temperature within the glass transition range than are the quasi-static mechanical properties. A safety margin of 50F° (28C°) between the T_g and the material operational limit (MOL) has been proposed for epoxy matrix composites (see Section 2.2.2). This approach is useful for initially estimating the MOL, or for verifying a previously chosen MOL. However, since glass transition frequently occurs over a temperature range, and the measured value of T_g is highly dependent on method, supplemental mechanical property tests should be considered, particularly for new material systems (see Section 2.2.2).

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6.3.1.1.2 *T_g Measurements.* Several different methods have been used to characterize the glass transition in polymeric materials, and most of these are also applicable to fiber reinforced materials.

6.3.1.1.2.1 *Differential scanning calorimetry (DSC).* Since the heat capacity of a composite material changes at the glass transition, differential scanning calorimetry (DSC) may be used to determine T_g. The glass transition is detected as a shift in the heat flow versus temperature curve (see Figure 6.3.1.1.2.1). Many calorimeters are supplied with software which may be used to calculated T_g. T_g of neat resin specimens is relatively easy to detect with DSC, but in composite specimens the resin content in the specimen is small, and the more highly crosslinked the resin, the smaller the change in heat capacity. It is, therefore, sometimes difficult to detect T_g in highly crosslinked cured composites (see Reference 6.3.1.1.2.1).

6.3.1.1.2.2 *Thermomechanical analysis (TMA).* Thermomechanical techniques such as expansion, flexure, or penetration thermomechanical analysis (TMA) may also be used to determine T_g. In expansion TMA, the coefficient of thermal expansion α is measured as a function of temperature. As noted above, α undergoes a change during the glass transition, and T_g is determined by the point of intersection of lines fit to the thermal expansion data above and below the glass transition range. Figure 6.3.1.1.2.2 illustrates the specimen geometries and data reduction methods used for various TMA techniques.

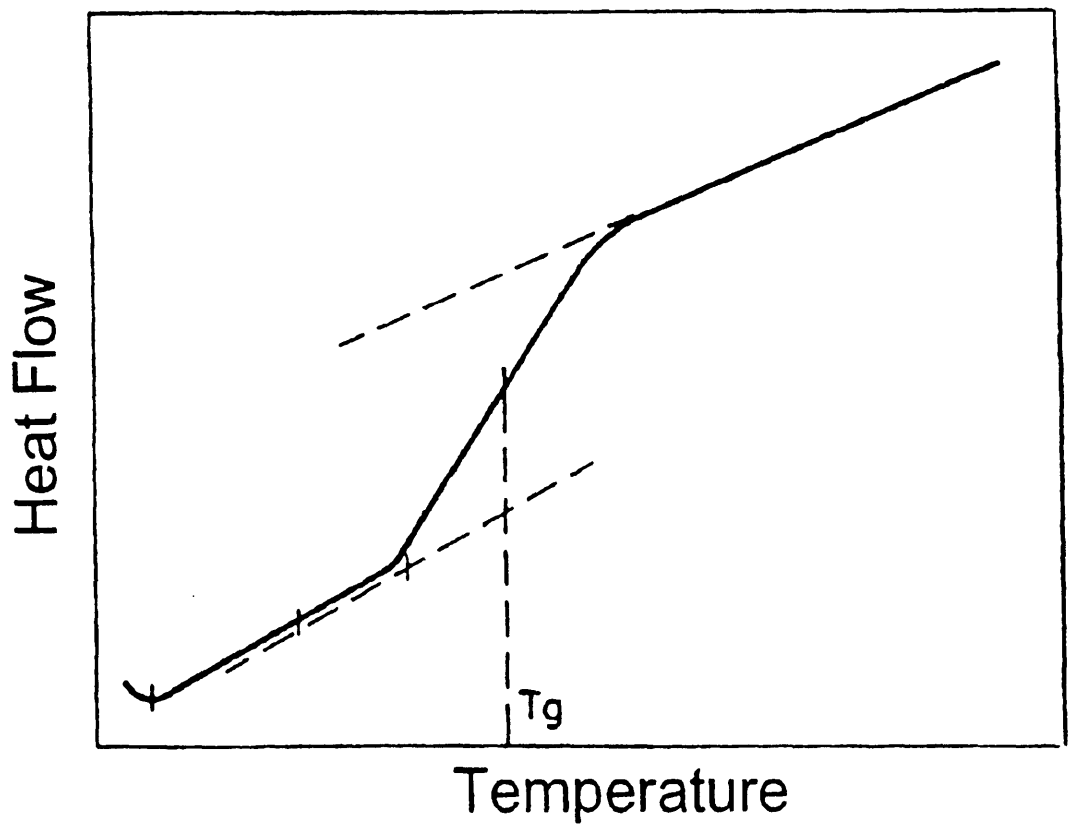


FIGURE 6.3.1.1.2.1 *Differential scanning calorimetry (DSC).*

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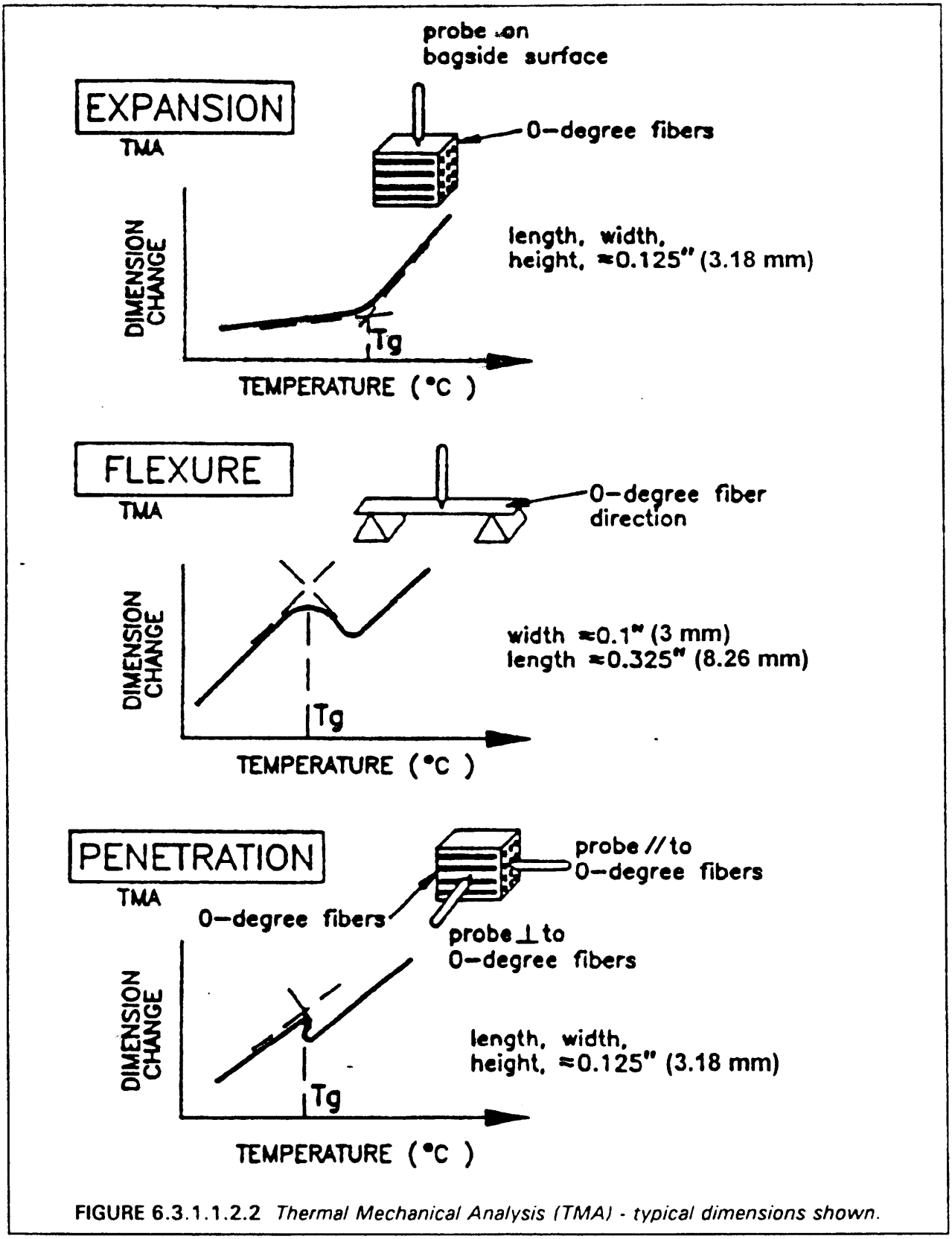


FIGURE 6.3.1.1.2.2 *Thermal Mechanical Analysis (TMA) - typical dimensions shown.*

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In flexural TMA, a rectangular specimen is loaded in bending and the dimensional change is measured as a function of temperature. A curve fitting technique as illustrated in Figure 6.3.1.1.2.2 is used to calculate T_g. Flexural TMA measurement of T_g is similar to heat distortion temperature (HDT) measurement, since in both cases the specimens are loaded in flexure. An HDT specimen may be a full-size flexural test coupon, and is loaded in three-point bending or as a cantilever beam. Displacement is measured as a function of temperature, and the HDT is the temperature at which the displacement reaches some predetermined value. Use of a full-size coupon minimizes moisture loss during the HDT test, but flexural TMA and HDT measurement share the disadvantage that values of T_g or HDT obtained will be sensitive to the modulus of the reinforcing fibers in the composite sample and they will give different results depending on the nature of the fiber.

As shown in Figure 6.3.1.1.2.2, penetration mode TMA measures the hardness of the material. One disadvantage of this technique is that if the probe is touching a reinforcing fiber, an accurate measurement of the T_g of the matrix will not be obtained.

6.3.1.1.2.3 *Dynamic mechanical analysis (DMA).* Dynamic mechanical analysis (DMA) is the most common and preferred method of characterizing the glass transition of organic matrix composites. There are several types of DMA which have been used with composites, including torsion pendulum analysis (TPA) and other resonant techniques, and forced oscillation measurements in tension, torsion, and shear. These forced measurements are made using a number of DMA instruments, manufactured by DuPont, Perkin Elmer, Polymer Laboratory, Rheometrics, TA Instruments, and others.

All these DMA techniques produce curves of dynamic storage and loss modulus and loss tangent (tan δ) or log decrement (Λ) as a function of temperature (see Figure 6.3.1.1.2.3(a)). Tan δ and Λ are proportional to the ratio of the loss modulus (E" or G") to the storage modulus (E' or G'). They reflect the amount of energy dissipated during each cycle of loading, and go through a peak value during the glass transition. T_g may be determined from DMA data in several different ways, and this may be a source of differences in reported values for T_g. As shown in Figure 6.3.1.1.2.3(a), T_g may be determined as the temperature at the onset or the midpoint of the transition based on the storage modulus curve, at the maximum in tan δ, or at the maximum in loss modulus. Clearly the method used for calculating T_g could produce markedly different values for the same set of DMA data. The temperature scanning rate and frequency employed will also affect the results, as discussed above.

An ASTM standard (D 4065-85) is available for DMA of plastics, covering both forced and resonant techniques (Reference 6.3.1.1.2.3(a)). The test techniques described in this standard practice are the same as those used for fiber reinforced plastics. In addition, a newly released SACMA method (SRM 18R-94) recommends the use of DMA for the measurement of T_g in oriented fiber-resin composites (Reference 6.3.1.1.2.3(b)). SACMA SRM 18R-94 specifies a forced oscillation measurement at 1 Hz, a heating rate of 5C° (9F°) per minute, and calculation of an onset T_g from the dynamic storage modulus curve. If a consistent material operational limit (MOL) is to be calculated from T_g, standards for these experimental variables should be specified along with a temperature safety margin. Otherwise the measured T_g may be shifted by increasing or decreasing heating rate or frequency.

As discussed above, measurement of T_g in a wet composite is made more difficult by the drying which occurs as the specimen is heated. Techniques which seek to prevent this drying by sealing the specimen in some way may be helpful in slowing the weight loss, but it cannot be prevented completely. If the specimen is sufficiently thick, the drying will occur primarily at the outside surface, resulting in a broadened or even bimodal glass transition (see Figure 6.3.1.1.2.3(b)). The lower temperature region will reflect the T_g of the interior of the specimen which is still wet, and the higher temperature region will reflect the T_g of the dried material. The loss tangent or log decrement curve will be broadened, or will exhibit two peaks or a peak and a shoulder, with the relative peak heights indicating the amounts of wet and dried material present in the specimen. In measuring T_g of a wet

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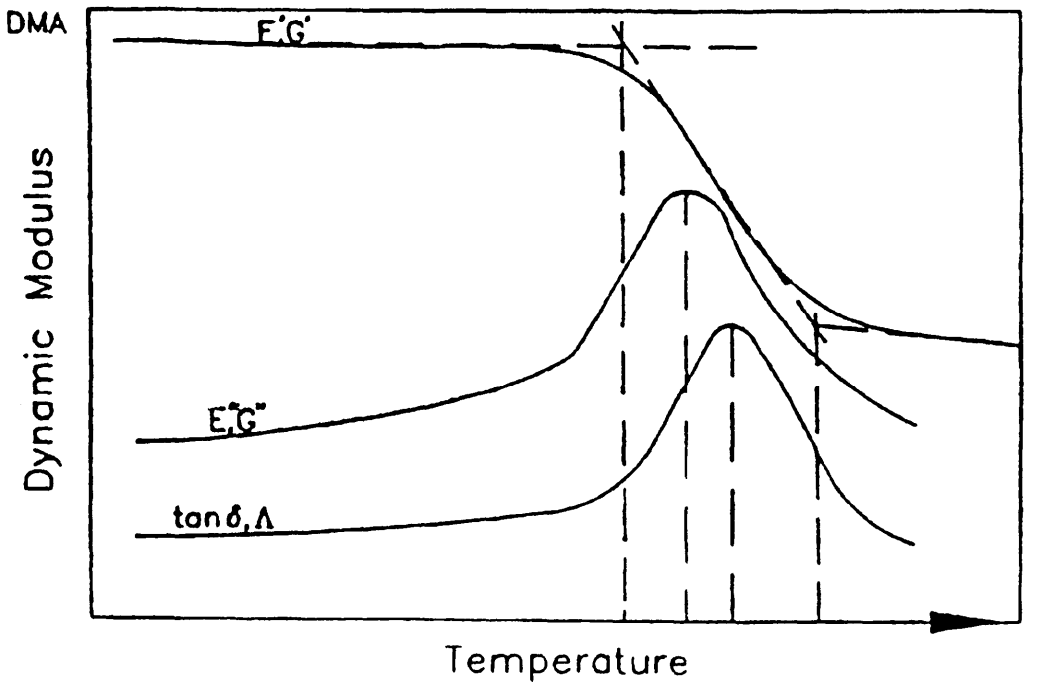


FIGURE 6.3.1.1.2.3(a) *Dynamic mechanical analysis (DMA).*

specimen, the lower temperature part of the transition may be the region of interest, suggesting that calculation of an onset T_g would be the appropriate and conservative approach.

6.3.1.1.3. *Glass transition test methods for MIL-HDBK-17 data submittal.* Data generated by DMA as described above are currently being accepted by MIL-HDBK-17 for consideration for inclusion in Volume 2. In addition to the specific apparatus used for the measurement, the heating rate and frequency must be included, and the method used to calculate T_g from the data must be specified. If a resonant method such as torsion pendulum is used, the frequency in the glassy region should be included with the data.

6.3.1.2 *Crystalline melt temperature.* The crystalline melt temperature (T_m) of semi-crystalline thermoplastic composites can be obtained from DSC or DTA experiments. In addition, an estimate of the degree of crystallinity can be made. This becomes an important parameter since the properties of semi-crystalline thermoplastic composites may be dependent upon the degree of crystallinity of the matrix resin. The heating required for processing prepregs into composite structures may have an affect on the degree of crystallinity as well as the crystal structure.

6.3.1.3 *Coefficient of thermal expansion.* The coefficient of thermal expansion (CTE) can be obtained from TMA. The CTE is obtained from the slope of the linear expansion versus temperature curve. Thermoset composites typically exhibit different linear expansion regions above and below T_g. ASTM Test Method E 831-86 "Linear Thermal Expansion of Solid Materials by Thermomechanical Analysis" describes the technique (Reference 6.3.1).

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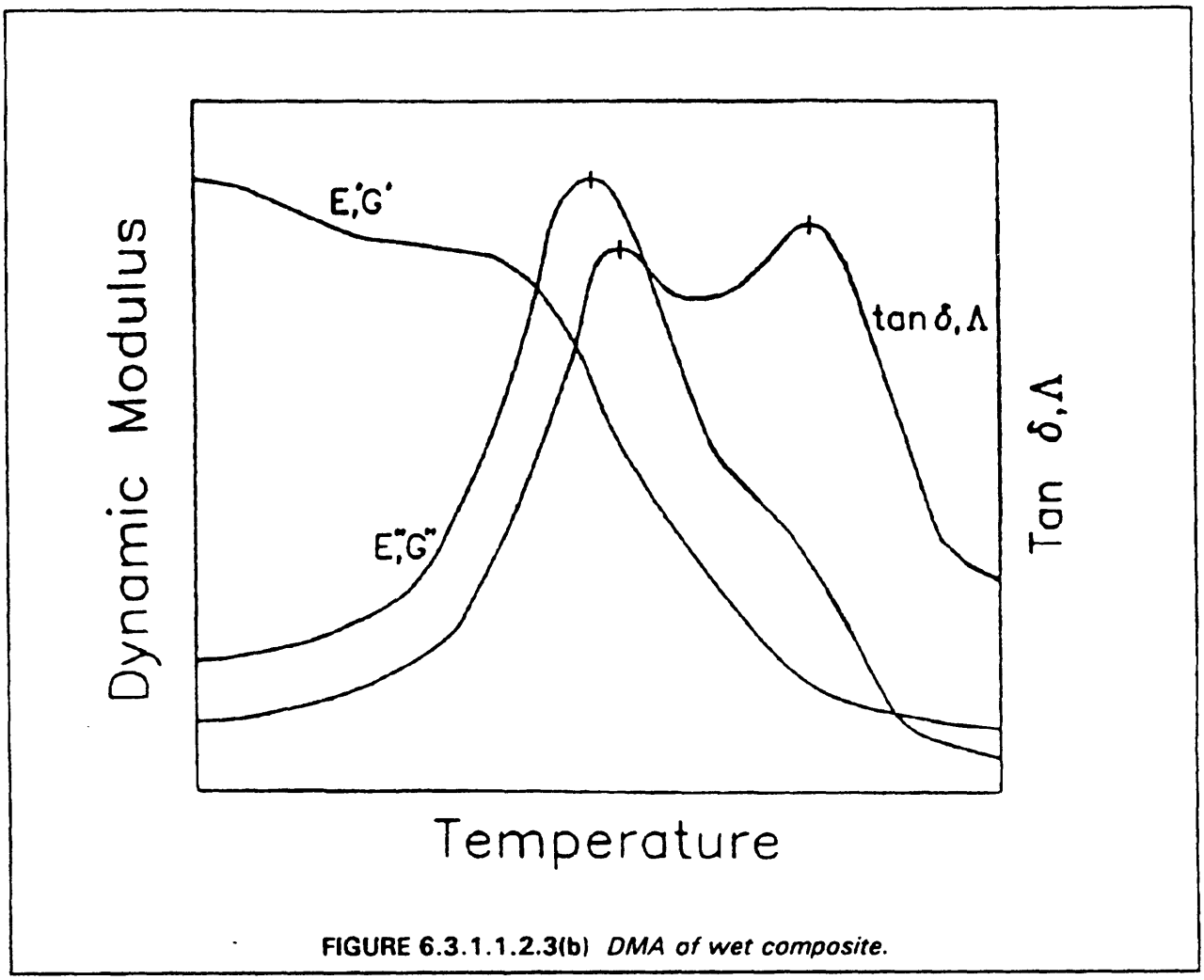


FIGURE 6.3.1.1.2.3(b) *DMA of wet composite.*

6.3.1.4 *Extent of cure for thermosetting composites.* The extent of cure for thermosetting composites can be evaluated by thermal analysis. Dynamic DSC or DTA can be used to detect residual heat of reaction, indicating incomplete cure. TMA and DMA may also be used to obtain relative information on the extent of cure for thermosets.

6.3.2 Fiber volume. The fiber volume of composites may be obtained from ASTM Test Method D 3171-76 "Fiber Content of Resin-Matrix Composites by Matrix Digestion" (Reference 6.3.3(a)). Values in terms of volume require determining the density of the composite sample, which may be obtained by a liquid displacement technique described in ASTM Test Method D 792-86 (Reference 6.3.3(b)). Another method based on known values of areal weight and fiber density has recently been published (Reference 6.3.3 (c)). Elastostatic measurements (Reference 6.3.3(d)) and ultrasonic methods (Reference 6.3.3(e)) have also been used for determining fiber volume.

6.3.3 Void volume. The void volume of composite materials is not easily determined. Ultrasonic NDT methods can estimate void size and give qualitative information, however quantitative analysis is not currently possible. Other methods have been tried with varying degrees of success. These include, sonic methods, radiography, electrical properties, microwave techniques, and infrared/thermal NDT methods (References 6.3.4(a) - 6.3.4(d)). Methods such as ASTM Test Method D 2734-70 "Void

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Content of Reinforced Plastics", require accurate determinations of the specific gravity of the resin and fibers in order to calculate the void content (Reference 6.3.4(e)).

6.3.4 Density. The density or specific gravity of composite samples may be determined from the following test methods (References 6.3.3(b) and 6.3.5):

ASTM D 792-86 "Specific Gravity and Density of Plastics by Displacement".

ASTM D 1505-85 "Density of Plastics by the Density-Gradient Technique".

6.3.5 Dimensional stability. Dimensional changes in composite materials are typically a function of temperature and/or moisture. Changes in length or volume of a sample can be detected by mechanical, optical or electrical transducer and recorded as a function of temperature or time. Several techniques for measuring linear expansion, such as dial gauges, micrometers, telescopes, linear variable differential transformers, interferometers, and X-ray diffraction patterns, have been used. As discussed in Section 6.3.1, TMA is a good technique for determining linear thermal expansion. Standard test methods for obtaining thermal expansion values are (References 6.3.1, 6.3.6(a) and (b)):

ASTM D 864-52 "Coefficient of Cubical Thermal Expansion of Plastics".

ASTM D 696-79 "Coefficient of Linear Thermal Expansion of Plastics".

ASTM E 831-86 "Linear Thermal Expansion of Solid Materials by Thermo-mechanical Analysis".

6.3.6 Moisture weight gain. The absorbance of moisture in polymer composite materials, when exposed to humid environments or immersed in water, may have a deleterious effect on physical and mechanical properties (References 6.4(a) - 6.4(g)). The diffusion controlled absorption process results in an equilibrium moisture level for the material which is expressed as an increase in weight. The absorption rate and equilibrium level will depend on the type of matrix and/or matrix/fiber material system. The importance of determining these values and recommendations for making measurements are presented in Sections 2.2.2 and 2.2.3.

6.4 ENVIRONMENTAL EXPOSURE TESTING. The evaluation of environmental effects on composite materials has been an area of investigation for a number of years (References 6.4 (a) - 6.4(g)). Both field exposure and in the laboratory accelerated exposure methods have been used. Typically, hygrothermal effects are being measured, and they are expressed as a change in the mechanical and/or physical properties of the material. However, the determination of changes in the chemical properties of the resin and/or fiber material should also be evaluated, as well as exposure to other environments (References 6.4(h) - 6.4(l)).

The type of environment used for exposure testing should be realistic with respect to the materials intended application. Although not intended for materials testing, MIL-STD-810E provides some information on environmental testing and possible effects of various exposure conditions on military equipment performance (Reference 6.4(m)). ASTM standards which may be applicable include (References 6.4(n) - (q)) :

ASTM D 543-87 "Resistance of Plastics to Chemical Reagents", provides guidance for determining changes in materials upon exposure to standard reagents.

ASTM D 756-78(83) "Determination of Weight and Shape Changes of Plastics Under Accelerated Service Conditions", seven procedures for exposing plastics to combinations of extreme humidity and temperature that will accelerate the material

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changes taking place in a sheltered space subjected to humidity and temperature variations.

ASTM D 4364-84 "Performing Accelerated Outdoor Weathering of Plastics Using Concentrated Natural Sunlight", provides a procedure for performing accelerated outdoor exposure testing of plastics using a Fresnel-reflector accelerated outdoor weathering test machine.

ASTM D 1435-85 "Outdoor Weathering of Plastics", defines test conditions for the exposure of plastic materials to weathering.

Physical effects of exposure may include changes in surface properties, electrical properties, dimensions and weight. Visual inspection of the material can give qualitative information on color and/or gloss changes, crazing/cracking and fiber blooming. Weight and dimensional changes will yield information on the type of interaction between the material and the environment (fluid absorption, corrosion, erosion, chemical reaction). Chemical property changes may result from thermo-oxidative, photo-oxidative, or chemical degradation modes. Mechanical property changes are usually observed as changes in tensile, compressive, shear and flexural strength, as well as decreased resistance to crack growth.

The mechanical test methods used for determining the effects of exposure must be sensitive to the type of deterioration process being investigated. Changes in the resin/fiber interface or matrix material would require tensile and compression testing in transverse direction, and in-plane shear. Section 6.6 on Lamina, Laminate, and Fabric Mechanical Test Methods provides information on the types of standard testing techniques currently in use. In addition, the application of a static and/or dynamic stress during exposure should be considered. This type of testing may require special equipment or test fixtures which will maintain the desired load during environmental exposure. Ideally, the validity of laboratory tests would need to be established by comparison with field tests and in service inspection data.

6.5 FLAMMABILITY. The flammability characteristics of organic polymer matrix and fiber materials is an important consideration when designing composite structures (Reference 6.5(a)). Along with a loss of mechanical properties, these materials may release toxic gases and dense smoke during pyrolysis. The flammability properties can be altered by the addition of flame retardant additives (Reference 6.5(b)). The additives may or may not be incorporated into the backbone of the polymer matrix, depending upon the resin formulation. Test methods used for evaluating plastic materials are listed below. These methods were not developed for testing composites (References 6.5(c) - (g)) .

ASTM D 568-77 "Rate of Burning and/or Extent and Time of Burning of Flexible Plastics in a Vertical Position".

ASTM D 635-81 "Rate of Burning and/or Extent and Time of Burning of Self-Supporting Plastics in a Horizontal Position".

ASTM D 2863-87 "Measuring the Minimum Oxygen Concentration to Support Candle-Like Combustion of Plastics (Oxygen Index)".

ASTM D 2843-88 "Density of Smoke from the Burning or Decomposition of Plastics".

ASTM F 814-84B "Specific Optical Density of Smoke Generated by Solid Materials for Aerospace Applications".

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6.6 LAMINA, LAMINATE, AND FABRIC MECHANICAL PROPERTY TESTS.

6.6.1 General. Section 6.6 discusses test methods for determining mechanical properties of laminated composites. The purpose of this section is to provide brief commentaries on the most commonly used methods, to alert the reader to the limitations of the various methods, and to encourage uniformity in the use of standard test methods with the ultimate goal of combinability of experimental data obtained from multiple sources. The reader is referred to Chapter 8 for statistical data analysis requirements for reporting of data to MIL-HDBK-17.

The section reflects the current dynamic state of test methods development for composite materials. Many of the methods were originally developed for testing of reinforced plastics, and modifications have been (or are being) made for applicability to advanced composites. In recent years there has been a tendency for users to unilaterally modify existing standards without a formal standardization process, leading to uncontrolled test results. In general, these modified standards are not discussed in Section 6.6 except where a specific modification is in common use, and where discussion of the technique is deemed constructive. The test methods included are representative of procedures used in the composite materials industry, and were selected after review of standards documents and user material specifications.

It is important to make a distinction between methods that are discussed in Section 6.6, and methods for data submittal to MIL-HDBK-17:

- Test methods used by contractors are agreed upon with customers and/or certifying agencies. Section 6.6 reviews many methods in order to provide the reader with awareness of the broad range of procedures in common use. Some of these have been formally standardized (ASTM and other standards) and some are "common practice" methods. Some have distinct limitations, and these are indicated as a matter of information. Mention or omission of a particular method does not, of itself, require or restrict usage. Specific methods are included to allow the user to perform tests consistent with industry practice; however, inclusion of these standards should not be considered an endorsement of any standard or organization by MIL-HDBK-17.

- When submitting data to MIL-HDBK-17 for consideration for inclusion in Volume 2 of the Handbook, specific methods must be used. Tables at the end of most subsections of 6.6 indicate which methods are acceptable for such submittals. These methods have been chosen in accordance with the criteria given in section 2.2.6. Readers are encouraged to also use these methods in contract and internal work to promote standardization.

When selecting and using a particular mechanical strength test method, the importance of obtaining the proper failure mode cannot be overemphasized. While universal definitions of "proper" and "valid" have not been established for most types of tests, further analysis must be employed when unexpected or questionable modes are observed or suspected. If the type of failure is different from what is expected from the test, the data may not represent the property being evaluated. Furthermore, if the failure mode varies within a group of specimens, statistical analysis of the resulting data will not be meaningful due to the introduction of an additional source of variability not related to the property being tested. Therefore, it is crucial that failure modes be reported, and that data be disqualified and discarded when analysis has indicated an unacceptable mode.

It should be noted that failure mode analysis is not necessarily limited to physical examination of the failed test specimens. Other evidence might be obtained from review of additional factors such as:

1. Bending curves from back-to-back strain gage data

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2. A check of test machine and/or test fixture alignment

3. A review of the exact procedure used to install and properly align the specimens in the test fixture

4. A check for possible damage to, or malfunction of, the test fixture

ASTM has begun to incorporate failure mode examples and codes into its standard test methods. For example, the 1993 revision of ASTM D 3039 (*Standard Test Method for Tensile Properties of Polymer Matrix Composite Materials*) depicts nine types of failures of the specimen, and defines a three-character coding system that describes various failures. The first letter of the code identifies the type of failure (angled, grip, delamination, etc.), the second indicates the area of the failure (gage, at grip, etc.), and the third denotes the failure location (top, bottom, middle, etc.). In the particular case of tensile testing, a failure of the tab or tab adhesive would be an unacceptable mode since the ultimate tensile strength of the laminate was not measured.

Rather than duplicate failure mode examples within the subsections of section 6.6, the reader is advised to be conscientious regarding the documentation of failure modes, and to refer to examples provided within specific test methods where such examples exist.

6.6.2 Tensile tests.

6.6.2.1 *General considerations.* Tensile testing of laminates will generally be accomplished in accordance with ASTM Test Method D 3039 (Reference 6.6.2.1). This test employs a straight-sided specimen with bonded end tabs for gripping, and is applicable to 0° and 90° unidirectional constructions as well as bidirectional and fabric laminates. Details for specific orientations are

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discussed in Sections 6.6.2.2, 6.6.2.3, and 6.6.2.4. The test yields tensile ultimate strength, tensile modulus, and the major Poisson's ratio.

Experience within the aerospace industry has indicated that the critical factors in obtaining consistent tensile test results are tab design, tab application, fiber alignment, and specimen machining quality. With respect to machining, it is important that methods be used that will not introduce cracks, as these could cause premature failure due to stress concentration. If rough cutting is performed, a secondary operation should be used that will remove any edge damage. Variables that can affect the degree of damage and effectiveness of damage removal are cutting tool design, cutting/grinding tool speed, feed rate, amount of material removed per pass, and use of liquid coolants. In general, liquid coolants are recommended to prevent thermal damage, but may not be necessary when removing small amounts of material per pass. Any coolant used shall not adversely affect the composite being machined. Tool design is particularly important when machining aramid reinforced composites (which are often difficult to cut cleanly). Although specific machining procedures depend on coupon configuration and material, it is generally advantageous to remove small amounts of material per pass and use a slow feed rate and relatively high tool speed, thereby keeping surface temperatures below the intended service temperature of the composite.

The purpose of tabbing is to introduce uniform loading into the specimen. The degree to which this is realized depends upon a number of factors including tab bevel angle, and the mechanical properties of the tab material relative to the laminate being tested. In general, the tab material should be less stiff than the test laminate.

Surface preparation prior to tab bonding may include light sanding or grit blasting to promote adhesion. Care must be taken not to abrade any part of the laminate gage section (any area not covered by the tab), as this damage could cause premature failure.

Thermal considerations are also important. If the coefficient of thermal expansion of the tab material is significantly different from that of the laminate, bond failure could occur either during bond cycle cool-down, or during elevated temperature testing. In addition, bond cycles that will advance the cure state of the laminate cannot be used, as this will invalidate the data. For this reason, many companies use 250°F (120°C) curing adhesives when tabbing 350°F (180°C) cured laminates. This is of less concern when postcured laminates are being tested.

6.6.2.2 *0° Unidirectional laminae.* When preparing 0° specimens, care must be taken to ensure that the fibers are aligned parallel to the coupon load axis. A misalignment of only 0.5° has been shown to affect ultimate strength by as much as 5%. To accommodate capacities of standard composite test machines, the width of 0° unidirectional specimens is usually specified as 0.500 inches (12.7 mm).

6.6.2.3 *90° Unidirectional laminae.* As specified in ASTM Test Method D 3039, the 90° tensile specimen is wider and shorter than the 0° coupon. Since this test is totally matrix dependent, the handling and precise alignment of the specimen are critical. Special alignment fixtures are often used in conjunction with hydraulic or pneumatic test grips to prevent introduction of out-of-plane forces.

As an alternate, a sandwich beam specimen in accordance with ASTM Test Method C 393 may be used for 90° testing (Reference 6.6.2.3). This specimen is less susceptible to handling damage, but panel warpage is a problem.

6.6.2.4 *Bidirectional and fabric laminates.* The specimen for these laminates is wider than the 0° unidirectional coupon. For fabrics it is mandatory to orient the warp and fill for each ply in the same direction, and to report the direction tested. A modified ASTM D 3039 dog-bone shaped specimen is

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frequently utilized for testing fabrics. As larger tow, fewer ends per inch fabrics have become more common, it has become practical to increase the gage width to 1.000 inch (25.4 mm) to reduce scatter of test results.

6.6.3 Compression tests.

6.6.3.1 *Overview.*

In-Plane 1-direction Compression Properties

E_1, ν_{12}^c, F_1^{cu}, ε_1^{cu}

In-Plane 2-direction Compression Properties

E_2, ν_{21}^c, F_2^{cu}, ε_2^{cu}

The compression response of composite materials has been the subject of research efforts and test programs since the early 1960's. Even with this long term study there exist numerous methods to test composites in compression and no consensus on a single most recommended method to use. Compression tests are conducted on composite materials to determine compressive modulus, Poisson's ratio, ultimate compressive strength or strain-at-failure utilizing appropriate instrumentation. These properties are determined through use of test fixturing that is typically designed: 1) to be as simple to use and fabricate as possible; 2) to minimize stress concentrations; 3) to minimize specimen volume, and 4) to introduce a uniform state of uniaxial stress in the specimen test section. Compression data is used for various purposes including research, quality control, and generation of design allowables.

While one measure of the acceptability of a compression test method is the magnitude of the compression strength it yields, this measure alone is not enough to determine acceptability. The coefficient of variation on strength and modulus data provided by a compression test method should also be considered when determining the acceptability of the method. When considering compression strength, some methods yield values considered "artificially high" due to fixture restraint that may suppress certain compression failure modes. On the other hand, fixtures are designed to intentionally inhibit some failure modes (such as end brooming and column buckling) to eliminate "artificially low" strengths and induce failure in the test section. This tradeoff between just enough restraint versus too much restraint and artificially high versus artificially low compression strength is the reason for the myriad of possible test methods and the lack of agreement on one acceptable method. There are differences in opinion on how to balance these trade-offs.

The compression strength for a single material system has been shown to differ when determined by different test methods. Such differences can be attributed to fabrication practices, control of fiber alignment, improper coupon machining and specimen preparation, improper placement of coupons in test fixtures or fixtures in testing machines, and improper use of test fixtures even when efforts have been made to minimize these effects.

A review of the numerous compression test methods available reveals they can be broadly classified into three groups; 1) those that introduce load into the specimen test section through shear, and 2) those that introduce load into the specimen test section through end-loading, and 3) those that introduce load into the specimen test section through a combination of compression and shear. The three compression test methods for high modulus ($>3\times10^6$ psi) fiber composites currently sanctioned by ASTM Committee D-30 in D 3410-87 introduce load into the test section of the specimen primarily by shear. Most of the other published compression test methods introduce load into the test section through endloading. Compression test methods can be further classified as having a supported or

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unsupported test section. An unsupported test section is defined as one inherently free from global buckling with nothing in contact with the faces of the specimen in the test section throughout the entire compression test. A supported test section is one with support on the specimen faces in the test section provided by the test fixture or ancillary equipment. All of the test methods discussed in this section require specimens with unsupported test sections, with the exception of the D 3410-87 Method C, the Sandwich Beam method. A more complete discussion of compression test methodology and a description of test methods not covered here can be found in References 6.6.3.1(a)-(e).

6.6.3.2 *In-plane compression tests.* The in-plane compression test methods described below can be used to generate the ultimate compression strength, strain-at-failure, longitudinal modulus, and Poisson's ratio of [0] and [90] coupons, over a typical thickness range of 0.050 to 0.125 in. With the exception of D 3410-87 Method C (sandwich beam method), all of the test methods discussed below will also accommodate $[0/90]_s$ style laminates; testing of these laminates has become a popular means for eliminating specimen and fixture related sensitivities associated with unidirectional specimens. If lamina compressive data is desired from $[0/90]_s$ laminates, data reduction procedures are required. A discussion on the use of [0/90] laminates for determining lamina properties and associated data reduction methods can be found in Volume 1, Section 8.3.2.4. Standard test methods for laminate configurations other than $[0/90]_s$ are not available, but laminate compression testing is common. Common practice is to extend the standardized test methods discussed below (or the other published test methods described in References 6.6.3.1(a)-6.6.3.1(e) to laminates. Compression testing of laminates can be less complex than testing unidirectional coupons due to the lower strength of laminates, but issues associated with laminate testing must be considered. The major issues to consider are the stability of an unsupported gage length, free-edge effects (Reference 6.6.3.2(a)), tab restraint and Poisson expansion effects (Reference 6.6.3.2(b)), and stress decay effects (Reference 6.6.3.2(c)). The standardization of laminate compression test methods is under development in ASTM Committee D-30. Test methods for specimen thicknesses greater than 0.125 inches exist, but have not been standardized or used as extensively as those discussed, and additional information on test methods for laminates thicker than 0.400 inches can be found in Volume 3, Chapter 7 of the handbook.

General Limitations of Compression Testing

Test Method Sensitivity--Compression strength for a single material system has been shown to differ when determined by different test methods. Such differences can be attributed to specimen alignment effects, specimen geometry effects and fixture effects even though efforts have been made to minimize these effects. Examples of the difference in test results between three procedures in ASTM D 3410-87 can be found in References 6.6.3.2(e)-(f).

Material and Specimen Preparation--Compression modulus, and especially compression strength, are sensitive to poor material fabrication practices, damage induced by improper coupon machining and lack of control of fiber alignment. Fiber alignment relative to the specimen coordinate axis should be maintained as carefully as possible, although no standard procedure to insure this alignment exists. Procedures found satisfactory include the following: fracturing a cured unidirectional laminate near one edge parallel to the fiber direction to establish the 0° direction, or laying in small filament count tows of contrasting color fiber (aramid in carbon laminates and carbon in aramid or glass laminates) parallel to the 0° direction either as part of the prepreg production or as part of panel fabrication.

6.6.3.2.1 *ASTM D 3410-87, Compressive Properties of Unidirectional or Crossply Fiber-Resin Composites.* Three compression test methods are published by ASTM in Test Method D 3410-87 and have historically been called the Celanese (D 3410-87 Method A), IITRI (Illinois Institute of Technology Research Institute, D 3410-87 Method B), and Sandwich Beam (D 3410-87 Method C) methods. The

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Celanese and IITRI methods, as with many other published methods, originally carried the names of the organizations under which the method was developed. The Celanese and IITRI methods address the use tabbed, rectangular coupons and transfer load into the specimen via wedge type grips. The sandwich beam method consists of two, thin face sheets, one of which is the compression specimen, bonded to a thick core material to make a sandwich beam. This beam is loaded in bending to apply compressive stress to the specimen face sheet, causing a compression failure on this sheet.

Limitations of the ASTM D 3410-87, Compression Test Method

Material Form--Limited to polymer matrix composites reinforced with oriented, continuous, high-modulus ($>3 \times 10^6$ psi) fibers and made primarily of prepreg or similar product forms. Limited to laminates that are unidirectional ([0] or [90]) or balanced crossply laminates ($[0/90]_{ns}$).

Test Fixture Characteristics--Although both Methods A and B in this test method transmit load to the specimen via tapered wedge grips, the wedges in Procedure A are conical and the wedges in Procedure B are rectangular. The conical wedges from Procedure A are known to be prone to cone-to-cone seating problems (Reference 6.6.3.2(e)). The rectangular wedge grip design used in Procedure B was employed to eliminate this wedge seating problem (Reference 6.6.3.2(e)). In addition to these differences, the fixture used for Procedure A is much smaller in size and weight than the fixture used for Procedure B. A fixture characteristic that can have a significant effect on test results is the surface finish of the mating surfaces of the wedge grip assembly. Since these surfaces undergo sliding contact they must be polished, lubricated, and nick free.

Strain Measuring Devices--While extensometers are not ruled out, practical considerations make use of strain gages essentially required. Back-to-back gages are highly recommended for Methods A and B.

ASTM D 3410-87, Method A

This test method requires a test fixture that consists of a pair of matched conical wedge grips that are seated in a cylindrical housing (Figure 6.6.3.2.1(a)). The test specimen used in this fixture is a tabbed coupon of rectangular cross-section. The specimen dimensions are nominally 5.5 inches long and 0.25 inches wide. The total thickness of specimen plus tabs for this method is 0.157 inches. After correctly placing the specimen in the test fixture, a compressive load is applied to the ends of the fixture in a standard testing machine. The load applied to the fixture is transferred from the wedge grips to the specimen tabs through shear, and from the tabs to the test specimen through shear. The complex stress state in the tabbed region of the specimen changes to uniaxial compression in the specimen test section. Compression strength is determined from load at failure while modulus and strain-at-failure are determined when strain gages or compressometers are employed.

The fixture design for the test method makes it susceptible to cone-to-cone seating problems on the conical wedge grips (Reference 6.6.3.2(e)). This difficulty has resulted in a decreasing use of this method since Method B was introduced to eliminate this problem. Other modifications of this test method continue to evolve to alleviate its inherent shortcomings (Reference 6.6.3.2.1).

Limitations of the ASTM D 3410-87, Method A

Specimen Dimensions--Due to the conical geometry of the wedge grips in this test method, the total specimen thickness must be held to a fixed dimension of 0.157 inches.

Material Form--The fixed gage length and width of this specimen limits its use for fabric based materials.

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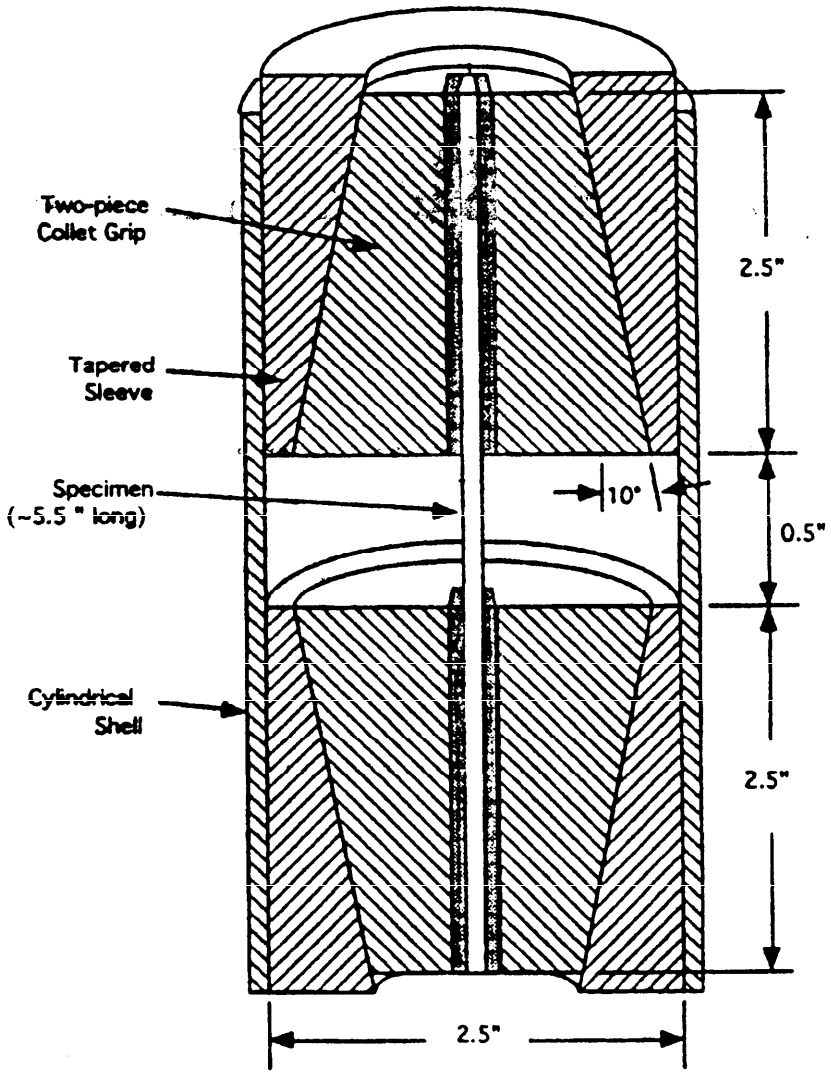


FIGURE 6.6.3.2.1(a) *Schematic of ASTM D 3410-87 method A test fixture and specimen.*

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Tabbing and Tolerances--The data resulting from this test method have been shown to be sensitive to the flatness and parallelism of the tabs, so care should be taken to assure that the specimen tolerance requirements are met. This usually requires precision grinding of the tab surfaces after bonding them to the specimen.

ASTM D 3410-87, Method B

The fixture design, specimen configuration, and loading principle for this test method are based on the same concepts as Method A described above. The fixture for this test method was designed principally to eliminate the seating problems associated with the conical wedge grips in Method A (Reference 6.6.3.2(e)). In place of conical wedge grips, the fixture for this test method consists of a pair of matching rectangular wedge grips seated in a rectangular housing (Figure 6.6.3.2.1(b)). The fixture for this method is much larger and heavier than for Method A. The test specimen used in this fixture is a tabbed coupon of rectangular cross-section nominally 5.5 inches long, 0.25-1.5 inches wide, 0.050 to 0.125 inches thick, and with a of 0.5 inch gage length. Thicker specimens and specimens with a longer gage length (and longer overall length) are allowed with this test method and are commonly used[1]. As with the Method A, the load that is applied to the fixture is transferred from the wedge grips to the specimen tabs through shear, and from the tabs to the test specimen through shear. The complex stress state in the tabbed region of the specimen changes to uniaxial compression in the specimen test section. Compression strength is determined from load at failure while modulus and strain-at-failure are determined when strain gages or compressometers are employed.

Limitations of the ASTM D 3410-87, Method B

Tabbing and Tolerances--The data resulting from this test method have been shown to be sensitive to the flatness and parallelism of the tabs, so care should be taken to assure that the specimen tolerance requirements are met. This usually requires precision grinding of the tab surfaces after bonding them to the specimen.

ASTM D 3410-87, Method C

The sandwich beam method consists of a honeycomb-core sandwich beam that is loaded in four-point bending placing the upper face sheet in compression (Figure 6.6.3.2.1(c)). The compressive face sheet (upper sheet) is a 6-ply unidirectional laminate and the lower face sheet should be the same material and twice as thick. The two face sheets are separated by and bonded to a deep aluminum honeycomb core. When the beam is subjected to four point bending, the upper face sheet is designed to fail in compression. The beam is loaded to failure in bending, resulting in the measurement of compression strength, compression modulus and strain-at-failure if strain gages or compressometers are employed.

Limitations of the ASTM D 3410-87, Method C

Material Form--This test procedure is limited to unidirectional material only.

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[1] Although the existing standard allows for specimens of any gage length with anti-buckling guides as needed, it is ambiguous on the use and effect of anti-buckling guides. Recommended specimen dimensions are under review in ASTM Committee D-30 and will be included in future revisions of D 3410. For practical purposes, this fixture has been commonly used for specimens up to 1 inch wide, and with gage lengths up to 1 inch long with acceptable results and without the need for anti-buckling guides.

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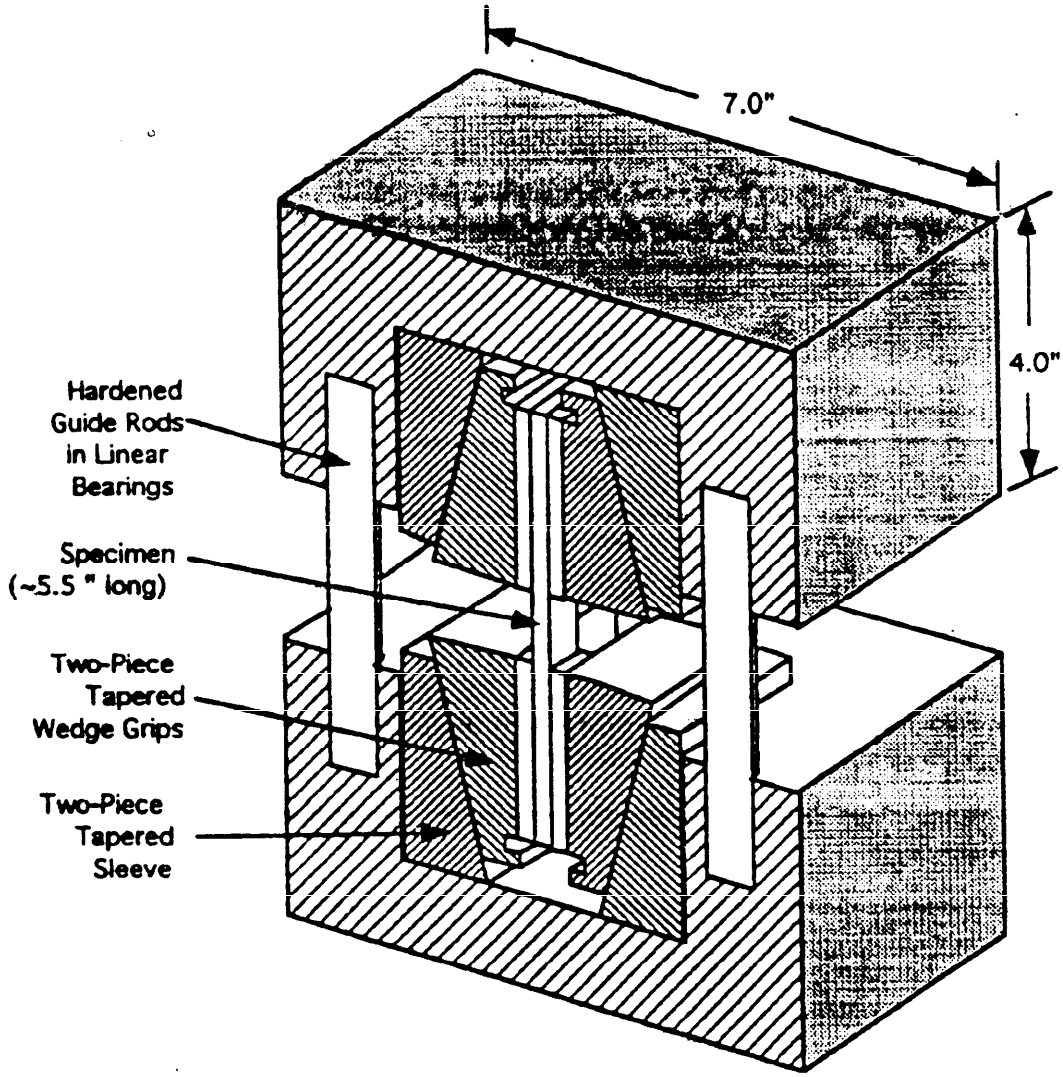


FIGURE 6.6.3.2.1(b) *Schematic of ASTM D 3410-87 method B test fixture and specimen.*

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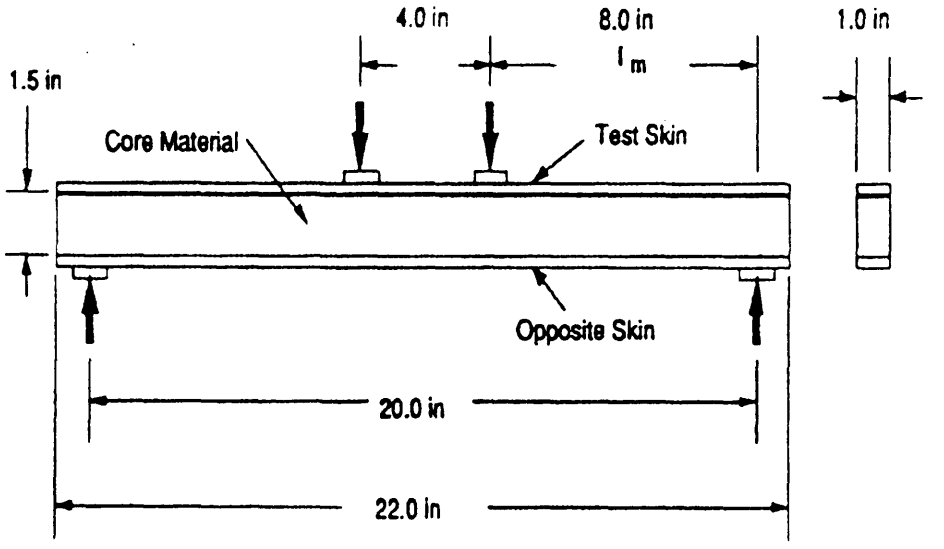


FIGURE 6.6.3.2.1(c) *Schematic of ASTM D 3410-87 Method C specimen.*

Specimen Complexity--Specimen is much larger, and specimen preparation is more complex and expensive than for Methods A or B.

Compression Strength--Compression strength for unidirectional materials are typically higher (10-15%) than for Methods A or B. This is believed to be attributable to the one-sided support provided by the core material in the test specimen.

Poisson's Ratio--The validity of Poisson's ratio from this method has been questioned due to anticlastic bending.

6.6.3.2.2 *ASTM D 695-89, Compressive Properties of Rigid Plastics.* This method was developed by ASTM Committee D-20 for compression testing of unreinforced and reinforced rigid plastics. Two types of specimens can be used for this method. The first is typically used for unreinforced plastics and is in the form of a right cylinder or prism whose length is twice its principal diameter or width. Preferred specimen sizes are 0.50 in. by 0.50 in. by 1 in. for a prism and 0.50 in. dia. by 1 in. for a cylinder. Smaller diameter rods or tubes may also be tested provided they are of sufficient length to allow a specimen slenderness ratio of 11:1 to 16:1. The specimen is tested by placing it between the hardened steel faces of a compression tool and loading it to failure.

The second test specimen in the standard is documented as being for "*Reinforced Plastics, Including High-Strength Composites and Highly Orthotropic Laminates*" < 0.125 in. thick. It uses a flat, untabbed specimen with a reduced width test section. Two I-shaped support plates with longitudinal grooves are clamped to the faces of the specimen, and are slightly shorter than the specimen as shown is Figure 6.6.3.2.2. After positioning the specimen between the support plates, a compressive load is applied to the end of the specimen until failure to determine ultimate compression strength.

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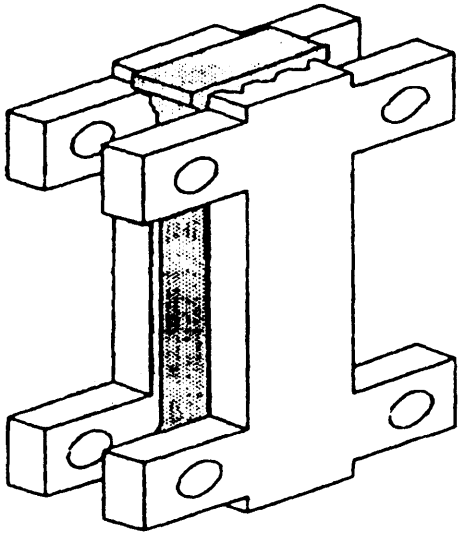


FIGURE 6.6.3.2.2 *Schematic of ASTM D 695-89 specimen and test fixture.*

The flat coupon, fixture supported method was evaluated in a D30 round robin for [0] AS/3501 and [0] E-Glass/1002 laminates. The conclusion from this study was that this test method is not adequate for determining the compression strength of high-modulus composite materials. In an attempt to modify this portion of the test method for use with high-modulus composites, a straight-sided, tabbed coupon has been developed. In addition, an L-shaped base to support the fixture-specimen assembly has also been added to the test method. A discussion of these modifications is included below in the section on SRM1-88.

Limitations of the ASTM D 695-89 Compression Test Method

Material Form--The published scope of this document states it is limited to unreinforced and reinforced rigid plastics, including high-modulus composites. Round-robin testing conducted by Committee D-30 found this method unacceptable for the measurement of strength of high modulus composites (Reference 6.6.3.2(f)).

6.6.3.2.3 *SACMA SRM 1-88, Compressive Properties of Oriented Fiber-Resin Composites.* A variation on the ASTM D 695-89 "Standard Test Method for Compressive Properties of Rigid Plastics" for continuous high-modulus fiber composites has been developed and documented by SACMA as SRM 1-88. While essentially retaining the simple fixturing of the D 695-89 method, the variation utilizes straight-sided tabbed specimens for compression strength and an L-shaped base for support of the fixture-specimen assembly. A separate, untabbed specimen must be used for the measurement of modulus. Both specimens are smaller than the D 3410-87 Methods A and B specimens and are 3.18 inches long, 0.5 inches wide and 0.040 to 0.120 inches thick. Although the test section in this method is unsupported, it is the shortest test section (0.188 inches) of any test method of this class, possibly contributing to the higher values of compression strength from this method than from D 3410-87 Methods A and B. A schematic of this test method with the compression strength specimen in place is shown in Figure 6.6.3.2.3.

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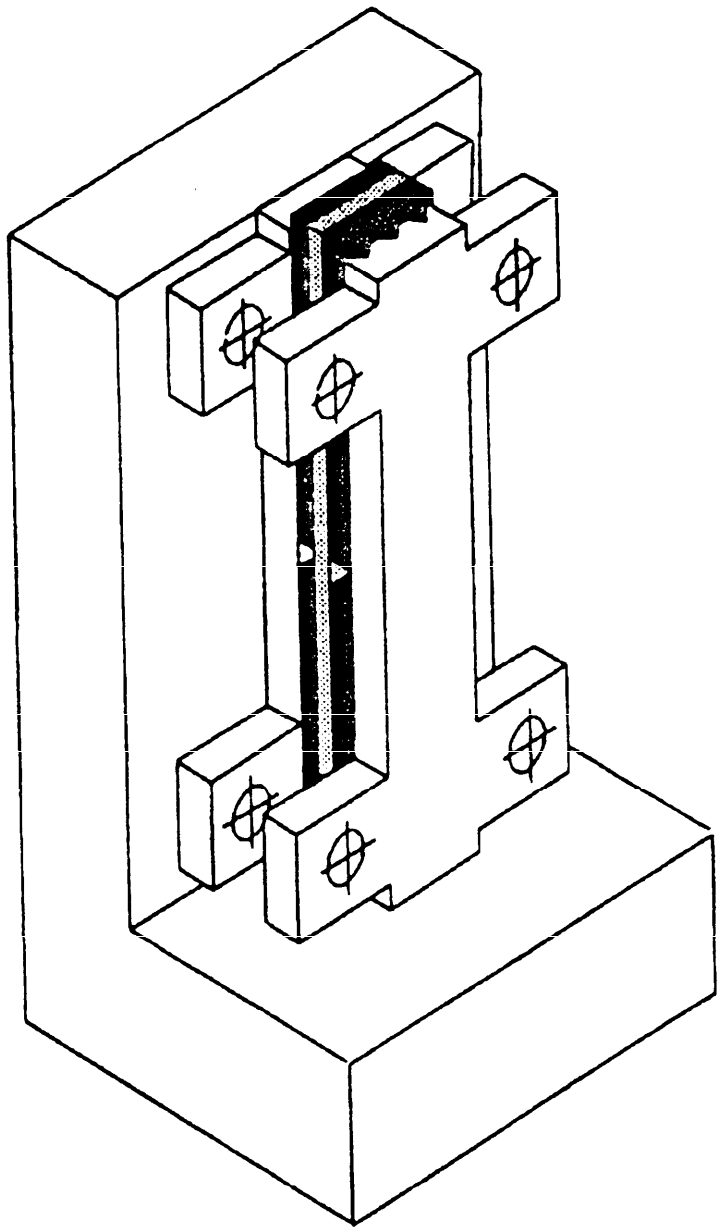


FIGURE 6.6.3.2.3 *Schematic of SACMA SRM 1-88 test fixture and specimen.*

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Limitations of the SRM 1-88 Compression Test Method

General--Separate strength and modulus specimens are required for this test method. The short specimen gage length and narrow width result in a small test section volume for the strength specimen.

Material Form--Limited to polymer matrix composites reinforced with oriented, continuous, high-modulus ($>3 \times 10^6$ psi) fibers, and made primarily of prepreg or similar product forms. The short gage length and narrow specimen width limits its use for fabric based materials.

Compression Strength--Compression strength for unidirectional materials are typically higher than for the methods in ASTM D 3410-87[2]. This is believed to be due to the short and narrow gage length of this test specimen, and the effect this geometry has in suppressing failure mechanisms that are present in longer, unsupported gage lengths such as in D 3410-87 Methods A and B.

Strain-at-Failure--This test method will not provide strain-at-failure since the gage region of the strength specimen is not large enough for a strain gage, and the modulus specimen geometry is not suitable for loading to failure. Consequently, stress-strain response can not be observed over most of the actual curve.

6.6.3.3 *Through-thickness compression tests.* Due to an historical lack of need for through-thickness compression data, there are no standardized or widely accepted test methods to determine through thickness (z-direction) compression strength, modulus or Poisson's ratio of composite laminates. These data have been reported to a limited extent in the literature (Reference 6.6.3.3(a)-(b)), and simple rectilinear specimens cut from thick-section laminates have been used to obtain these properties.

6.6.3.4 *Test methods for developing MIL-HDBK-17 data submittal.* Data provided by the following test methods are currently being accepted by MIL-HDBK-17 for consideration for inclusion in Volume 2.

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[2] Although this test method has been successfully used for strength measurement of unidirectional carbon reinforced composites, specimen preparation procedures, tabbing procedures, fixture/specimen alignment practices, and fixture loading practices that have been established through experience were followed. These procedures and practices have not been standardized or documented in SRM 1-88 and are factors that will effect the results from this test method.

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TABLE 6.6.3.4 *Compression test methods for MIL-HDBK-17 data.*

Property	Symbol	Fully Approved, Interim and Screening Data	Screening Data Only
In-plane Strength	F_1^{cu}, F_2^{cu}	D3410-87 A, B, C SRM1-88	
In-plane Strain-at-Failure	e_1^{cu}, e_2^{cu}	D3410-87 A, B, C	
In-plane Modulus	E_1^c, E_2^c	D3410-87 A, B, C SRM1-88	
In-plane Poisson's Ratio	ν_{12}^c, ν_{21}^c	D3410-87 A, B SRM1-88	

6.6.4 Flexure tests. There is not a recommended test method for determining the flexural properties of composite laminates. Even though there are approved flexure test methods, there is some debate as to the validity of the results.

Within the aerospace industry, flexure testing is primarily used for quality control. ASTM Test Method D 790, "Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials", was originally written for plastics but has since been modified and approved for composites (Reference 6.6.4). In some cases, ASTM Test Method C 393, "Flexure Test of Flat Sandwich Constructions", has been adapted for use with composite laminates (Reference 6.6.2.3).

6.6.5 Shear tests.

6.6.5.1 *Overview.*

In-Plane Shear Properties:

G_{12}, F_{12}^{su}, e_{12}^{su}

Out-of-Plane Shear Properties:

G_{13}, F_{13}^{su}, e_{13}^{su}

G_{23}, F_{23}^{su}, e_{23}^{su}

Shear testing of composite materials has proven to be one of the most difficult areas of mechanical property testing in which to define a rigorously correct test (especially in the out-of-plane direction). A number of test methods have been devised only some of which are described below. Many of these methods were originally developed for materials other than continuous fiber reinforced composites, such as metal, plastic, wood, or adhesive. Several of the methods are not yet fully standardized for composite materials and none of the methods is without deficiency, though some are clearly more deficient than others.

While there is general agreement regarding the accuracy of shear modulus measurements, the biggest difficulty with shear testing of composites is the determination of shear strength. Edge effects, material coupling effects, nonlinear behavior of the matrix or the fiber/matrix interface, and the presence of normal stresses combine to make shear strength determinations from existing shear test

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methods highly questionable. The use of such strengths in shear applications should be reviewed on a case-by-case basis.

A growing body of experience with composite shear testing, both published and unpublished, has led to a greater understanding of the strengths and weaknesses of each test method. At the Fall 1991 ASTM Committee D-30 meetings, discussions in the D30.04.03 Section on Shear Test Methods led to the first two conclusions. During the Spring 1993 meetings, this committee added the third determination. These philosophies are being included in existing and future ASTM standard shear test methods.

1. There are no standard (or non-standard) test methods that are capable of producing a perfectly pure shear stress condition to failure for every material, although some test methods can come acceptably close on specific material systems, as judged by the end-user for a given engineering purpose.

2. The strengths resulting from test methods that do not consistently produce reasonable approximation of pure shear, or that cause failure by a non-shear failure mode, should not be termed "shear strength".

3. Since ultimate strength values from existing shear tests are no longer believed able to provide an adequate criterion for comparison of material systems, the addition of a 0.2% offset strength is now suggested.

Because of the highly nonlinear shear stress-strain behavior of many filamentary composites, with high elongation material systems it is common to terminate a shear test prior to coupon failure due to.

Practical usage of structural laminates--Typical structural laminates are designed with fibers in all of the major load-carrying directions, ideally causing in-plane material coordinate system shear loads in a given ply to be carried by tension or compression of fibers in other plies that are oriented 45° from the ply of interest. The carrying of shear loads by fiber tension or compression puts a practical upper bound on the value of shear strain of twice the tensile or compressive fiber strain. As the most ductile structural carbon fibers currently fail at tensile or compressive strains of no more than 2.5%, a practical upper limit for shear strain in a shear test would be 5%. This saves time during testing with little loss in measured ultimate strength.

Limitations of common test methods--There are kinematic limitations in both the ±45° shear test and the V-notched beam shear test due to excessive scissoring of the fibers at high elongations, resulting in a practical limit of 5% shear strain, as supported by a detailed evaluation of the ±45° shear test (Reference 6.6.5.1), which concluded that the mechanics of this test become increasingly invalid in the high strain region. Also, common strain gages, if used for strain measurement in the shear test methods, are limited to about 3% extensional strain, which is equivalent to 6% shear strain.

6.6.5.2 *In-plane shear tests.*

6.6.5.2.1 *±45° tensile shear tests:*

1) ASTM D 3518/D 3518M-91, Practice for In-Plane Shear Stress-Strain Response of Unidirectional Polymer Matrix Composites.

2) SACMA SRM 7-88, In-Plane Shear Stress-Strain Properties of Oriented Fiber-Resin Composites.

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This test (References 6.6.5.2.1(a)-(b)) for strength and modulus consists of an ASTM Test Method D 3039 tensile coupon with a laminate ply orientation of +45 degrees and -45 degrees in a balanced and symmetric layup.[3] Away from the gripping region the in-plane shear stress is a simple function of the average applied tensile stress, thus allowing for straightforward calculation of the shear response of the material. This test has the advantages of a simple test coupon, no fixturing, and the flexibility to use either extensometers or strain gages. While the standard as it is currently written allows for its use only with unidirectional materials, it is generally agreed that the method is equally applicable to woven fabrics, and this is being addressed in a proposed revision to the standard.

Good modulus agreement with other shear test methods have been reported (References 6.6.5.2.1(c)-(e)), although the stress-strain response has been shown to be underestimated at strain levels above 1.3% (Reference 6.6.5.2.1(f)). There is a feeling by many in the composite structures community that while the stress state of this coupon may not be "pure", it responds in a manner that mimics the actual stress state and ply interaction within a structural laminate. The resulting response yields an "effective" shear modulus that may be of more practical use to the designer.

Strength determination is another matter. Though previous test method surveys had generally considered strengths from this test to be reliable, significant problems with strength data from this test have since been documented, as discussed in more detail below. Although ASTM Committee D-30 is taking action to modify Practice D 3518 accordingly, the 1991 version of the standard does not adequately address this issue.

It has been shown that the failure mode of the coupon is not due to shear and that strength results are sensitive to material toughness, ply stacking sequence, number of plies, edge effects, surface ply constraints.

Limitations of the $\pm 45^\circ$ tensile shear test:

Material Form--Limited to materials that are available in a fully balanced and symmetric $\pm 45^\circ$ tensile coupon. As discussed below, the stacking sequence and total number of plies has a direct effect on strength data.

Inhomogeneous Materials--The material is assumed homogeneous with respect to the size of the test section. Material forms which have relatively coarse features with respect to the test section dimensions, such as fabrics or braided materials with a coarse repeating pattern, require larger specimen widths.

Impurity of Stress State--The material in the gage section of this coupon is not in a state of pure in-plane shear, as an in-plane normal stress component is present throughout the gage section and a complex stress field is present close to the free edges of the specimen. Although this test method is believed to provide reliable initial material response and can establish shear stress-shear strain response well into the nonlinear region, the calculated shear stress values at failure do not represent true material strength values and should only be used with caution. Despite attempts to minimize these effects, the shear stress at failure obtained from this test method, even for otherwise identical materials that differ only in cured ply thickness or fiber areal weight, may have

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[3] A currently fully equivalent, but more restricted, subset of Practice D 3518 has been promulgated by the composite materials suppliers as SACMA SRM 7-88. However, it is expected that, barring a parallel revision to SRM 7-88, the two documents will diverge as a result of the on-going revision to Practice D 3518. A significant revision of D 3518 is currently being balloted.

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differing failure modes and may not be able to be statistically pooled. As an example, the results from four different laminates of the same material are shown below, taken from the paper by Kellas and Morton (Reference 6.6.5.1):

Laminate	Mean Initial Peak Shear Stress, MPa
$[\pm 45/\pm 45]_s$ (8 plies)	80
$[\pm 45/\pm 45]_{2s}$ (16 plies)	83
$[\pm 45/\pm 45]_{3s}$ (24 plies)	85
$[\pm 45/\pm 45]_{4s}$ (32 plies)	91

It is clear from these results that the initial peak shear stress resulting from this coupon is only a lower bound of the material shear strength. This indicates the difficulty with statistically combining strength data from this test method from coupons that have different ply counts or significantly differing ply thicknesses.

Effects of In-Plane Normal Stress Field--Of particular concern is the in-plane stress component normal to the fiber direction. This component of stress is present in all plies and throughout the gage section of the specimen. The effect of this stress on a given ply is minimized by the fiber reinforcement of the neighboring plies. Since the ply constraint is reduced with increasing ply thickness, the thickness of the individual plies is an important parameter that influences both the shear stress-shear strain response and the ultimate failure load of this specimen.[4] Moreover, the surface plies of a given specimen, being constrained by only one neighboring ply, as opposed to two for the interior plies within the same specimen, represent the weakest link in a $\pm 45°$ specimen. Since the first ply failures in these specimens loaded in tension consist primarily of normal (or mixed mode) failures rather than pure shear failures. The actual material shear strength cannot be obtained from this test. Except for the case of materials capable of sustaining large axial strains (greater than about 3.0%), the shear stress at failure is believed to underestimate the actual material shear strength.

Total Thickness Effects--As a result of the failure processes discussed above, the shear stress-shear strain response at higher strain levels depends upon the total number of plies. The deleterious effect of the two weak surface plies decreases in proportion to the total number of plies in the $\pm 45°$ specimen. After the surface plies of the laminate fail the load is redistributed to the remainder of the intact plies. The higher the total number of plies, the greater the chance that the remaining plies will be able to carry the load without immediate ultimate failure of the coupon. However, with each successive ply matrix failure the number of remaining intact plies diminishes, to the point where the applied load can no longer be carried. Due to this process higher ply count specimens tend to achieve higher failure loads.[5]

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[4] Repeating plies (adjacent plies at the same ply orientation) have an effect similar to thick plies, therefore, the proposed revision to Practice D 3518 prohibits constructions with repeating plies.

[5] In order to minimize these effects, the proposed revision to Practice D 3518 requires the use of a homogeneous stacking sequence and requires a fixed number of plies, for which the only repeating plies are the two required for symmetry on opposite sides of the laminate mid-plane.

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Effects of Large Deformation--Note that extreme fiber scissoring can occur in this specimen for the cases of ductile matrices, weak fiber/matrix interfaces, thick specimens with a large number of repeated plies, or a combination of the above. Kellas and Morton suggest that a general rule of thumb for this specimen is that a fiber rotation of 1° takes place for every 2% of axial strain (about 3.5% shear strain for commonly tested materials). The implication of fiber rotation is to invalidate the equation used to calculate the shear stress. If there is a shear stress peak for a given material in a $[\pm 45/\ \pm 45]_{2,3,\text{or } 4s}$ laminate, it will normally occur prior to this strain level.[6] Further details of the effects of stacking sequence, specimen geometry, and, in particular, specimen and ply thickness, are presented in (Reference 6.6.5.1).

Effects of Edge Stresses--Even though interlaminar stresses reach a maximum value near the free edges of this laminate, the effect of interlaminar stresses on the failure process of ±45° laminates is insignificant when compared to the effect of the normal stress component perpendicular to the fiber direction in the plane of the specimen. Therefore, the effect of specimen width is much less important than stacking sequence and specimen thickness effects.

Effect of Axial Stress Non-Uniformity--Both the shear stress and the shear modulus calculations depend upon the uniformity of the applied axial stress. Since the average applied load is used to calculate the shear stress, this will not necessarily correspond to the stress in the vicinity of the measured shear strain, unless the axial stress is uniform throughout the volume of the stress material. Therefore, the greater the degree of material inhomogeneity, such as with coarsely woven fabrics or materials with significant resin-rich regions, the greater the potential for inaccuracies in the measured response.

Determination of Failure--As described above, failure is not always obvious in certain materials or configurations. The stress-strain behavior for thick laminates of brittle matrices, following the initial peak and subsequent plastic strain, will stiffen and begin to take load again, finally failing at a tensile load value that is unrelated to shear stress at all, due to the significant amount of deformation occurring. For such materials it is the shear stress at the initial peak that should be reported as the estimate of shear strength.

6.6.5.2.2 *Iosipescu shear test:*
ASTM D 5379/D 5379M-93, Test Method for Shear Properties of Composite Materials by the V-Notched Beam Method.

The V-notched beam shear test (often called the Iosipescu test in the literature) has been standardized for composites by ASTM Committee D-30 in ASTM D 5379/D 5379M-93 (Reference 6.6.5.2.2(a)). The concept for the V-notched beam shear test for strength and modulus was originally identified in the late 1950's and early 1960's by Arcan (References 6.6.5.2.2(b)-(d)) and Iosipescu (References 6.6.5.2.2(e)-(g)) for use on metals. Subsequent usage was limited until detailed investigations were begun on an improved specimen and fixture at the University of Wyoming under NASA funding during the early 1980's (References 6.6.5.2.2(h)-(i)). The fixture was subsequently modified (References 6.6.5.2.2(j)-(k)), and this latter Wyoming configuration formed the basis for the ASTM standard. This method has been investigated extensively; see References 6.6.5.2.2(l)-(r) for additional investigations. Early historical perspectives are given in References 6.6.5.2.2(h)-(s). However, the remainder of the discussion focuses on the configuration that has been standardized.

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[6] As a result the proposed revision to Practice D 3518 recommends use of only these laminates, and termination of the test data at the 5% calculated shear strain level.

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In this method, a material coupon in the form of a rectangular flat strip with symmetrical centrally located V-notches, shown schematically in Figure 6.6.5.2.2(a), is loaded in a mechanical testing machine by a special fixture, shown schematically in Figure 6.6.5.2.2(b). Either in-plane or out-of-plane shear properties may be evaluated, depending upon the orientation of the material coordinate system relative to the loading axis.

The specimen is inserted into the fixture with the notch located along the line-of-action of loading via an alignment tool that centers the specimen in the fixture. The upper head of the fixture is attached to and driven downward by the cross-head of the testing machine, while monitoring load. The relative displacement between the two fixture halves loads the notched specimen. By placing two strain gage elements, oriented at $\pm 45^{\circ}$ to the loading axis, in the middle of the specimen (away from the notches) and along the loading axis, the shear response of the material can be measured.

The object of the VNB concept can be seen in the idealization of the applied loading as asymmetric flexure, shown in the shear and bending moment diagram of Figure 6.6.5.2.2(c). The coupon gage area is in the region of constant shear and zero moment. The coupon notches influence the shear strain along the loading direction, making the shear distribution more uniform than would be seen without the notches.[7] The degree of uniformity in the shear distribution is a function of material orthotropy; the best overall in-plane shear results have been obtained on $[0/90]_{ns}$-type laminates. However, while the point-loading idealization indicates constant shear loading and zero bending moment in the gage section of the specimen, in practice the fixture applies distributed loads to the specimen that contribute to an asymmetry in the shear strain distribution and to a component of normal stress which is particularly deleterious to $[90]_n$ specimens.

Limitations of the V-notched beam shear test:

Inhomogeneous Materials--The material is assumed homogeneous with respect to the size of the test section. Materials that have relatively coarse features with respect to the test section dimensions, such as fabrics using large filament count tows (such as tows of 12,000 filaments or more) or certain braided structures, should not be tested with this specimen size.

Uniformity of Strain Field--The calculations assume a uniform shear strain state between the notches. The actual degree of uniformity varies with the level of material orthotropy and the direction of loading. A new strain gage grid configuration has recently been developed specially for use with this test method. The active grid on this gage extends from notch-to-notch and provides an improved estimation of the average strain response. When using conventional strain gages the most accurate measurements of in-plane shear modulus for unidirectional materials have been shown to result from the $[0/90]_{ns}$ specimen.

Load Eccentricity--Twisting of the specimen during loading can occur, affecting strength results, and especially, elastic modulus measurement. It is recommended that at least one specimen of each sample be tested with back-to-back rosettes to evaluate the degree of twist.

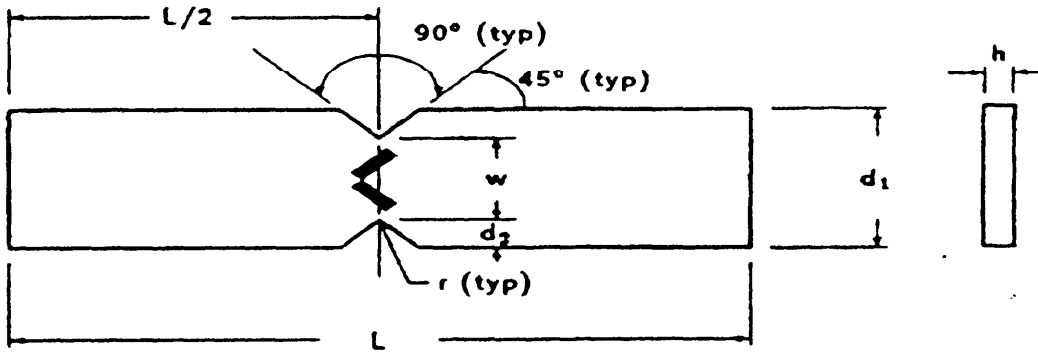
Determination of Failure--Failure is not always obvious in certain materials or configurations. See the standard test method (Reference 6.6.5.2.2(a)) for more information.

Instrumentation: Strain gages are required.

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[7] An isotropic beam in shear has a parabolic shear stress profile.

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Nominal Specimen Dimensions		
d_1	=	20.0 mm [0.75 in.]
d_2	=	4.0 mm [0.15 in.]
h	=	as required
L	=	76.0 mm [3.0 in.]
r	=	1.3 mm [0.05 in.]
w	=	12.0 mm [0.45 in.]

FIGURE 6.6.5.2.2(a) *V-notched beam test coupon schematic.*

6.6.5.2.3 *Rail shear tests:*
ASTM D 4255-83, Guide for Testing Inplane Shear Properties of Composite Laminates.

In 1983 ASTM Committee D-30 published the D 4255 Guide for Testing Inplane Shear Properties of Composite Laminates (Reference 6.6.5.2.3(a)), covering two types of rail shear testing. However, as part of the round robin evaluation testing of the test method prior to release as a standard, Committee D-30 found too much variability of results between laboratories with these tests to recommend this technique. However, Guide D 4255 was completed and published, with caveats, in order to provide a common ground for those who wish to use rail shear testing. Guide D 4255 is currently being re-written to correct errors and update the requirements.

The two standard methods are the two-rail shear test (Figure 6.6.5.2.3(a)) and the three-rail shear test (Figure 6.6.5.2.3(b)). In both cases the sides of a flat laminate are clamped or bonded between two (or three) stiff steel rails. A vertical load is applied to one of the rails and reacted by the other rail(s). In the center of the laminate, between rails, an approximate state of pure shear is obtained as the disturbances due to the edge boundary conditions and load application points damp out. This makes this test suitable for obtaining the shear modulus.

However, as the shear stress state is not uniform through the coupon, and as failures are often noted to begin outside the center of gage section (such as at the restrained corners of the plate) this test does not always produce reliable shear strength data (Reference 6.6.5.2.3(b)). The three-rail test has a purer state of stress (Reference 6.6.5.2.3(c)), although it requires a larger specimen size of approximately 150 mm by 150 mm (6 in. by 6 in.).

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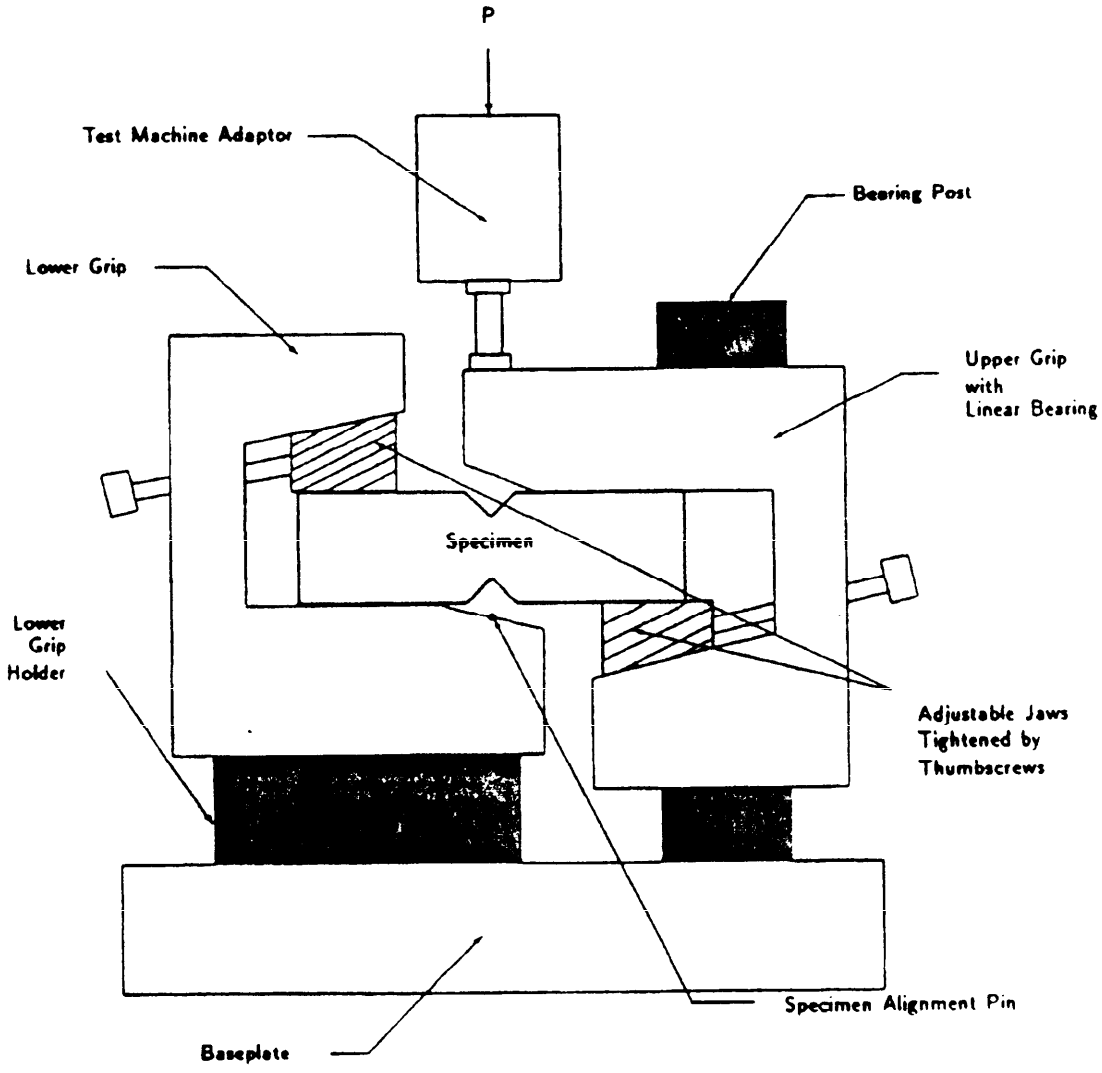
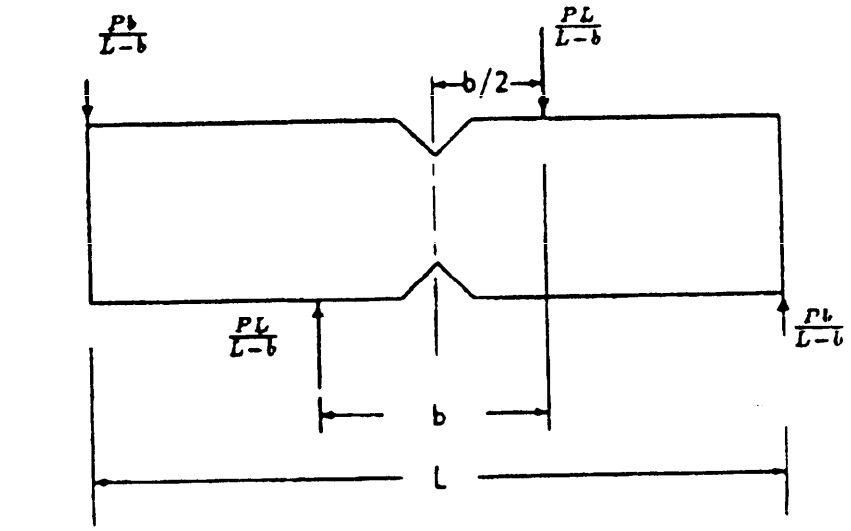
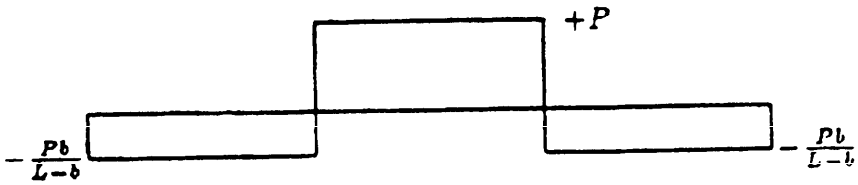


FIGURE 6.6.5.2.2(b) *V-notched beam test fixture schematic.*

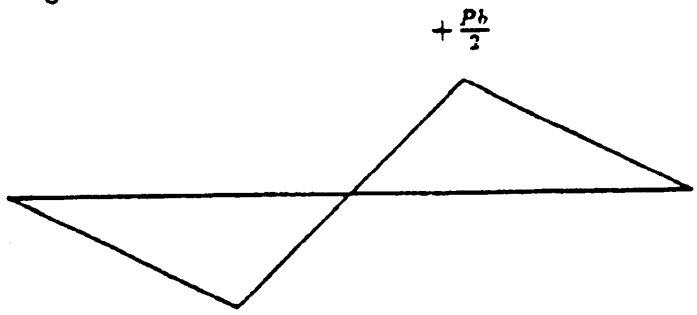
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Force Diagram



Shear Diagram



Moment Diagram $-\frac{Pb}{2}$

FIGURE 6.6.5.2.2(c) *Idealized force, shear, and moment diagram.*

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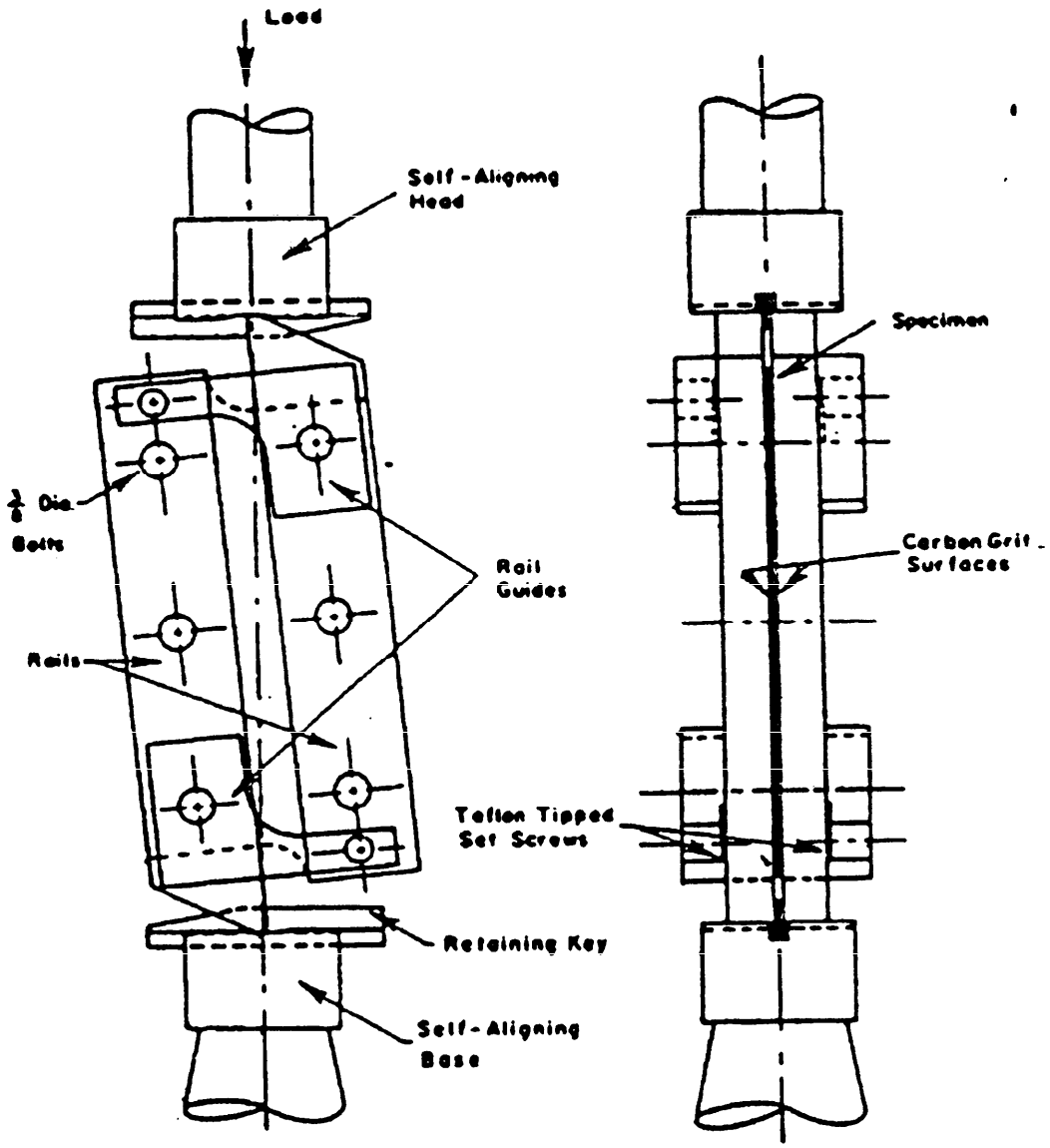


FIGURE 6.6.5.2.3(a) *ASTM D 4255-Method I (Two-Rail) Shear Test (Reference 6.6.5.2.3(a)).*

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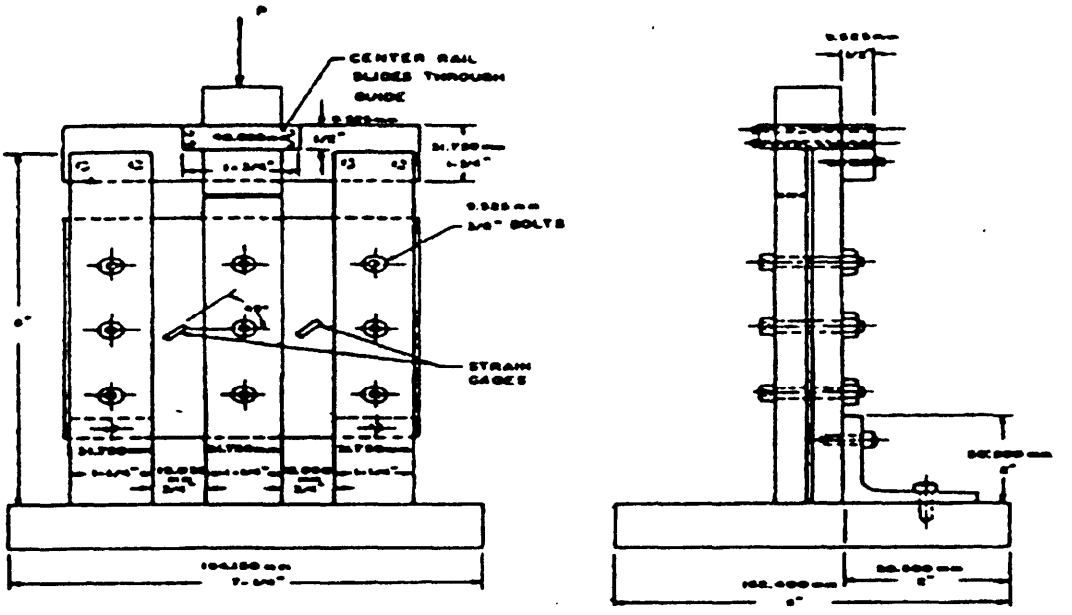


FIGURE 6.6.5.2.3(b) *ASTM D 4255-Method II (Three-Rail) Shear Test (Reference 6.6.5.2.3(a)).*

Limitations of the D 4255-83 rail shear tests:

Specimen Size: Both versions require larger specimens than other shear tests.

Instrumentation: Strain gages are required.

Stress State: The stress state is known to be non-uniform, and the failure mode is typically influenced by non-shear failures starting outside of the gage section.

Data Scatter: High data scatter from round-robin tests cast doubt upon the ability of these methods to produce repeatable data, at least in their current form.

6.6.5.2.4 *Ten-degree off-axis shear test.*

This method, first reported by Chamis and Sinclair (Reference 6.6.5.2.4) uses a straight-sided, rectangular unidirectional tensile specimen with the fiber oriented at ten degrees to the loading direction (Figure 6.6.5.2.4). Note that the material coupon is limited to unidirectional filamentary laminates. This specimen, like the ASTM Practice D 3518 specimen above, is also not under a state of pure shear and suffers from the effects of a combined stress state. This test produces results of generally higher modulus and significantly lower strengths than the other shear test methods such as ASTM Practice D 351 or ASTM Test Method D 5379.

Limitations of the 10° off-axis shear test:

Material Form: Limited to unidirectional laminates.

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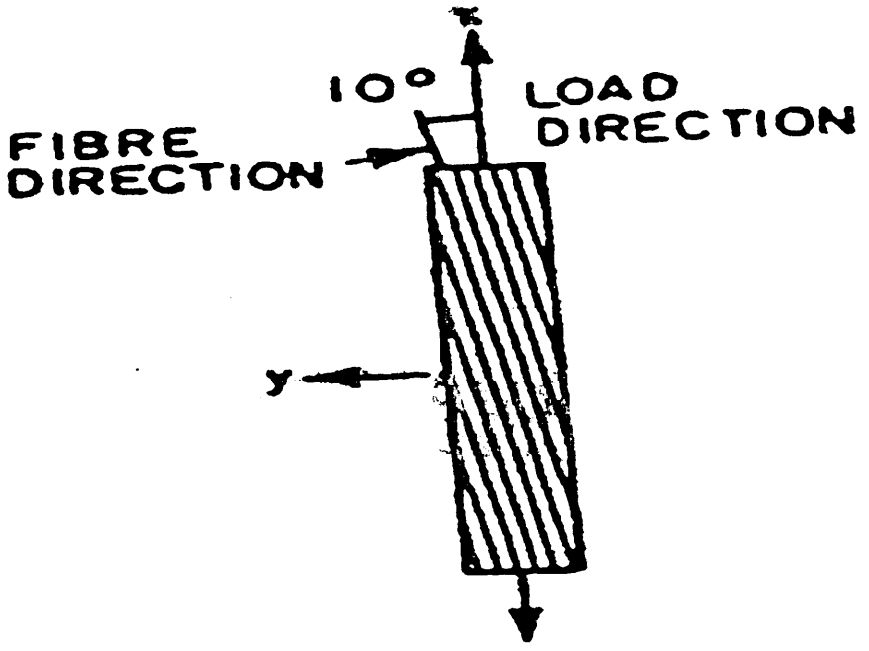


FIGURE 6.6.5.2.4 *Ten-degree off-axis tension shear test.*

Stress State: Known to have a significantly biased stress state producing an overly stiff initial response and premature failure.

Lack of Standardization: Has never been standardized.

6.6.5.2.5 *Tubular torsion tests:*

1) ASTM E 143-87, Test Method for Shear Modulus at Room Temperature
2) MIL-STD-375, Test Method for In-Plane Shear Properties of Hoop Wound Polymer Matrix Composite Cylinders
3) ASTM D 5448/D 5448M-93, Test Method for In-Plane Shear Properties of Hoop Wound Polymer Matrix Composite Cylinders.

Torsion testing of tubes has been standardized by ASTM since 1959 by Test Method E 143-87 (Reference 6.6.5.2.5(a)). While broad in scope, and technically not exclusive of composites, Test Method E 143 was primarily developed for metals. However the concept has also been applied to composites, where the challenge becomes in applying load to the specimen without producing a grip-induced failure; a typical gripping arrangement is shown in Figure 6.6.5.2.5. A torsion test specifically for wound composite tubes was developed and released as Military Standard, MIL-STD-375 (Reference 6.6.5.2.5(b)). MIL-STD-375 was submitted to ASTM for non-military standardization, and with minor changes was approved as ASTM D 5448/D 5448M-93 (Reference 6.6.5.2.5(c)). Test Method D 5448 consists of a 100 mm (4 in.) nominal diameter hoop-wound tube, which is gripped at each end and twisted via a fixture until failure. This test has been shown to produce good results and is the theoretical ideal for determining both in-plane shear strength and modulus.

Limitations of the torsional tube methods:

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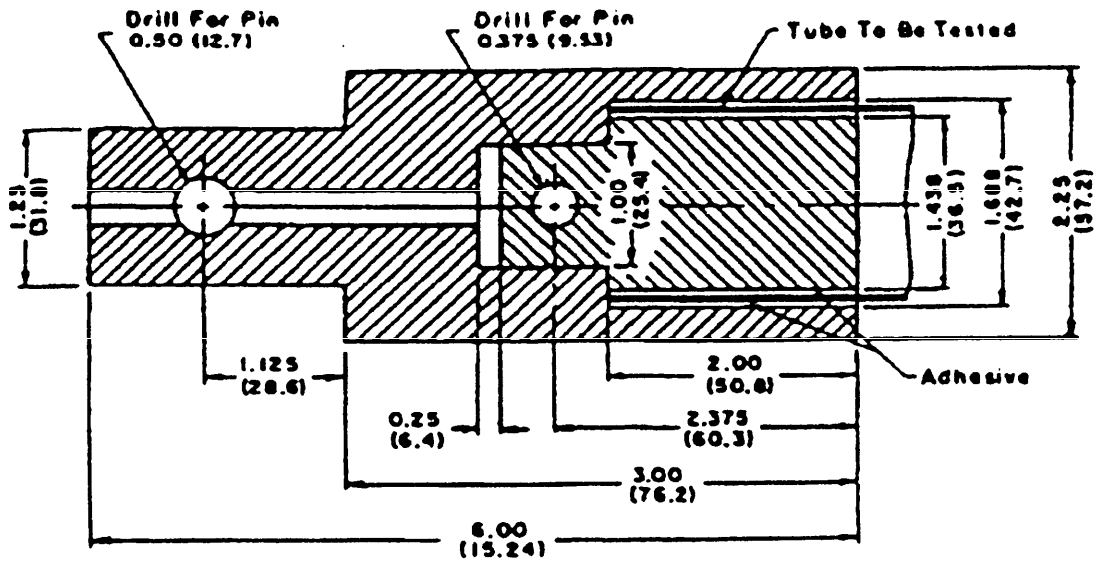


FIGURE 6.6.5.2.5 *Typical torsion tube shear fixture (Reference 6.6.5.2.5(d)).*

Material Form: If not using filament-wound materials the process required to create the tube may be significantly different than that used in the structure.

Cost of Specimen Fabrication: Fabrication of the specimen can be a significant undertaking requiring unusual expense.

Stress Concentration: A stress concentration exists at the end grips, as noted by Guess and Haizlip (Reference 6.6.5.2.5(e)), tending to result in failures in the gripping area, unless extreme precautions are taken.

Instrumentation: Strain gages are required.

6.6.5.3 *Out-of-plane shear tests.*

6.6.5.3.1 *Short-beam strength tests:*
1) ASTM D 2344-84, Test Method for Apparent Interlaminar Strength of Parallel Fiber Composites by Short-Beam Method
2) SACMA SRM 8-88, Apparent Interlaminar Shear Strength of Oriented Fiber-Resin Composites by the Short-Beam Method.

ASTM Test Method D 2344 (Reference 6.6.5.3.1(a)), commonly known as the short-beam strength (SBS) test, attempts to quantify the interlaminar (out-of-plane) shear strength of parallel fiber reinforced

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composites.[8] The specimen for this test is a short, relatively deep beam cut from a flat laminate. The specimen is mounted as a simply supported beam and loaded at the midpoint of the span of the specimen. The intent is to minimize bending stresses while maximizing out-of-plane shear stresses by using short, deep "beam".

However, the contact stresses induced at the load points greatly interfere with the strain distribution both through the depth of the beam and axially along the length of the beam. The resulting failure is rarely, if ever, a true pure shear failure but instead results from the complex stress state present in the specimen, as shown by Berg et al (Reference 6.6.5.3.1(b)), and others.

Unfortunately this test has commonly been used in the past (and is still used by some) to develop design allowables for structural design criteria. In the absence of any other choice this is understandable, though regrettable. However, the availability of the V-notched beam method, discussed in 6.6.5.3.2, makes the use of the short-beam strength test for property determination obsolete.

The short beam strength test should only be used for qualitative testing such as material process development and control. As a quality control test use of laminate configurations other than unidirectional are common, though currently non-standard.

Limitations of the short beam strength test include:

Stress State: The stress state is known to be significantly disruptive and three-dimensional. The resultant strengths are a poor estimation of the out-of-plane shear strength.

Failure Mode: The failure mode is most often multi-mode.

No Modulus/Material Response: Instrumentation of this specimen is not practical, therefore modulus and stress-strain data cannot be obtained.

6.6.5.3.2 *Iosipescu shear test:*
ASTM D 5379/D 5379M-93, Test Method for Shear Properties of Composite Materials by the V-Notched Beam Method.

This test method and the specimen geometry are described for in-plane shear testing in 6.6.5.2.2. When testing for out-of-plane shear properties the orientation of the fibers in the laminate is changed so as to cause a shearing action in the desired transverse plane. This test method is the only acceptable out-of-plane shear test available.

6.6.5.3.3 *ASTM D 3846-79, Test Method for In-Plane Shear Strength of Reinforced Plastics.*

ASTM Test Method D 3846 (Reference 6.6.5.3.3), despite the title, is not normally used as an in-plane shear strength test (using the most common definition of in-plane in the terminology of advanced composites) but is in fact an out-of-plane shear strength test and as such is covered in this section on out-of-plane shear tests.

---

[8] A currently fully equivalent, but more restricted, subset of Test Method D 2344 has been promulgated by the composite materials suppliers as SACMA 8-88 [30]. However, it is expected that, barring a parallel revision to SACMA 8-88, the two documents will diverge as a result of the on-going revision to Test Method D 2344.

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This test is primarily intended for use on randomly-dispersed fiber-reinforced thermosetting sheet plastics as a substitute to the short-beam strength test, Test Method D 2344 (Reference 6.6.5.3.1(a)) described in 6.6.5.3.1. The test consists of a doubly notched specimen loaded compressively in a supporting jig (the same fixture used in the Test Method D 695 compression test). Failure occurs in out-of-plane shear in the plane of the specimen between the two centrally located opposing square notches.

While this specimen can be (and has been) used for testing continuous-fiber laminated reinforced plastics, it is not recommended for use on advanced composite laminates. The notches which are machined into the specimen to force failure of the laminate in shear, were found by Herakovich et al (Reference 6.6.5.2.2(n)) to negatively influence the stress distribution in the coupon. As a result, a non-uniform, uniaxial stress state exists in the gage section, making a true strength calculation suspect at best.

Limitations of the D 3846 notched compression test:

Stress State: A highly three-dimensional, non-uniform stress state in the gage section cause strength values from this test to be unusually poor estimations of the true out-of-plane shear strength.

No Modulus/Material Response: Instrumentation of this specimen is not practical, therefore modulus and stress-strain data cannot be obtained.

6.6.5.4 Shear test methods for MIL-HDBK-17 data submittal.

Data produced by the test methods in Table 6.6.5.4 are currently being accepted by MIL-HDBK for consideration for inclusion in Volume 2.

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TABLE 6.6.5.4 *Shear test methods for MIL-HDBK-17 data.*

Property	Symbols	Fully Approved, Interim, and Screening Data	Screening Data Only
In-Plane Shear Strength	F_{12}^{su}, e_{12}^{su}	D 3518 SRM 7-88 D 5379 D 5448 MIL-STD-375	--
In-Plane Shear Modulus	G_{12}	D 3518 SRM 7-88 D 5379 D 4255 D 5448 MIL-STD-375	--
Out-of-Plane Shear Strength Out-of-Plane Shear Modulus	F_{13}^{su}/e_{13}^{su}, F_{23}^{su}/e_{23}^{su} G_{13}, G_{23}	D 5379 D 5379	-- --
Short Beam Strength	F^{SBS}	--	D 2344 SRM 8-88

6.6.6 Fatigue. Static testing of unidirectional composite coupons is useful for material characterization, comparison of materials, and for prediction of application laminate properties through the use of lamination plate theory. In the area of fatigue, however, no generalized methodology has yet been devised to predict laminate behavior from unidirectional coupon data. Hence, the development of fatigue design values becomes a unique problem for each application lay-up. Many studies have been undertaken, and much has been written concerning life prediction for specific laminates under cyclic loading spectra. Even at this level, empirical methods have been favored due to the inadequacy of results obtained from cumulative damage models, fracture mechanics analyses, and other theoretical approaches (References 6.6.6(a) and (b)).

ASTM Test Method D 3479, "Tension - Tension Fatigue of Oriented Fiber, Resin Matrix Composites", is a generalized coupon testing method (Reference 6.6.6(c)). However, because composite fatigue is so application dependent, it is important that the laminates represent the application and that the laminates testing account for the service load spectra and environmental conditions. Currently this is accomplished in composite hardware programs through a "building block" test approach involving coupon, element, and component specimens, all representative of full-scale structural details.

It is important to note that, for the majority of current aircraft composite structure, fatigue capability does not become a limiting factor if all static strength concerns have been thoroughly and successfully addressed. Exceptions to this are high-cycle components such as those found in helicopter dynamic systems.

6.6.7 Creep. Creep is defined as the change in a property over time when subjected to a constant forcing function. Here we are interested in the mechanical aspects of creep and monitor the change in measured strain with time when the specimen is subjected to a constant load. This change in strain

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must be indicative of an effective change in stiffness of the test specimen. This change may be (or may not be) recoverable upon unloading.

Creep should be considered if the end use involves high stress in the matrix-dominated direction, high temperature, or exposure to a harsh chemical environment - in other words, if there is a chance of matrix softening. In composites with a thermoplastic matrix, concern for creep is important, particularly if the service temperature is near or above the glass transition temperature. In a thermoset matrix, creep is expected to be small due to cross-linking. In general, creep testing does not provide primary design data. Designs should be checked for creep deformation if the working load involves major shearing action, e.g., short beam bending, etc. In composites, large shear stresses can be generated near a structural discontinuity; however, creep can be beneficial in some of these instances in relaxing the stress and avoiding catastrophic failure.

In fiber-reinforced plastics (composites), one can assume that creep will be more important when the composite is loaded in a matrix dominated manner than in a fiber dominated manner. For instance, the creep of a unidirectional specimen tensile loaded in the fiber direction is expected to be small and hence only of secondary importance. However, loading a specimen in a matrix dominated manner is not as straightforward as one would expect. Testing a unidirectional specimen in a transverse tensile manner one would think must essentially load the matrix, and this is not so. There are several explanations. One of the explanations is that loading a transverse specimen puts the matrix in a bidirectional state of stress (tensile) because the fibers prevent the matrix from laterally contracting (i.e., Poisson's effect) and thus the amount of creep response is restricted. Another argument for low creep response in transverse specimens is that the specimens are weak and the strains are small, so the change in strain would also be small. Another way to load the matrix is in shear and the creep response should be large. The most convenient way to load the matrix in shear is to load a $[\pm 45]$ specimen in tension. Although there is some argument that this test does not produce pure intralaminar shear, it at least produces some shearing and can be thought analogous to loading a unidirectional laminate in shear. Experience has shown that the resulting creep is significant. Other loadings that would be interesting to examine with respect to creep response would be compression of unidirectional specimens in the fiber direction and three-point bend loading of unidirectional specimens (in both of these methods shear plays a role).

Common experimental procedure is to apply a dead weight tensile load to a $[\pm 45]$ specimen at 35 MPa, 70 MPa, or 105 MPa (5, 10, or 15 ksi) and monitor the strain as a function of time. The strain reading at the first application of the full load is designated as the strain at zero time. Subsequent measurements are timed from that zero time reading. Readings are taken at 1, 2, 3, 10, 20, 30, 60, 100, and 200 minutes, and then as convenience dictates. Strain as a function of time is plotted on semi-log graph paper and the test continued for at least 30,000 minutes (or 3 weeks). Testing should be done at controlled (constant) humidity and temperature conditions (References 6.6.7(a) and (b)). Generally, specimens are 25 mm wide, 150 mm long and ~1 - 1.5 mm thick (1.0 in. wide, 6 in. long, and ~0.04 - 0.06 in. thick). These dimensions are open to question; there is some evidence that wider samples will creep less than narrow samples.

6.6.8 Fracture toughness tests.

6.6.8.1 *Overview.* Fracture in structural solids, such as wood, glass, metals, rock and concrete, is usually initiated by some crack or notch-like flaws, which cause high stresses in the neighborhood of such flaws. Inglis (Reference 6.6.8.1(a)) pointed out the significance of the localized concentration of stress near the tip of a sharp notch. A criterion of fracture based on the first law of thermodynamics was proposed by Griffith (Reference 6.6.8.1(b)), who postulated that the reduction in strain energy due to propagation of a crack is used to create new crack surfaces. Strain energy release rate, G, is defined as the reduction in strain energy (or increase in potential energy) due to an

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infinitesimal self-similar extension of the crack and catastrophic propagation of the crack will occur when this rate reaches a critical value, G_c. For a through crack of length 2a in a thin or a thick plate, subjected to a tensile stress σ, the energy release rate can be expressed in terms of σ, a and the properties of the material. Irwin (Reference 6.6.8.1(c)) pointed out that in isotropic materials, three independent kinematic movements are possible, by which the upper and lower crack surfaces can displace with respect to each other. These movements are schematically depicted in Figure 6.6.8.1. Only the first mode (Mode I or opening mode) was considered by Griffith. Irwin showed that the crack tip stresses can be expressed by a three parameter set of equations. These parameters, K_I, K_{II}, K_{III}, called the Mode I, Mode II and Mode III stress intensity factors, are functions of the crack dimensions and the applied loads and critical values of these parameters govern the phenomenon of unstable crack growth. The concept of failure based on the critical value of a stress intensity factor has been shown to be equivalent to that proposed by Griffith in terms of the critical strain energy release rate (G_{Ic}, G_{IIc}, or G_{IIIc}). Irwin also suggested the use of the critical value of the total energy release rate as the parameter governing failure, provided failure occurs by self-similar crack propagation. These concepts have also been extended to orthotropic materials (Reference 6.6.8.1(d)).

Use of fracture mechanics has gained wide acceptance in predicting failure in metal structures (References 6.6.8.1(e) - 6.6.8.1(g)) and various test methods have been developed for determining K_{Ic} (or G_{IC}) as well as crack growth resistance curves for cases when stable crack growth is possible. Further, over the years, there has emerged a fracture mechanics design procedure for fatigue of metal aeronautical structures, which is based on periodic inspection for monitoring visible cracks and predicting residual life using crack growth laws of the power law type (References 6.6.8.1(h) and 6.6.8.1(i)). Various attempts have been made to use fracture mechanics based methods for predicting failure of thin laminates with through cracks or notches (Reference 6.6.8.1(j)). However, it has been found that linear elastic fracture mechanics treating the thin laminates as orthotropic or isotropic plates is not useful because of considerable subcritical surface damage near the crack tips. Semi-empirical corrections are often employed for predicting strength of thin notched laminates (Reference 6.6.8.1(k)). On the other hand, fracture mechanics approaches appear to yield better results for thick laminates containing through cracks or deep surface flaws (References 6.6.8.1(l) - 6.6.8.1(n)) . It has also been shown that crack growth resistance or the R-curve concept, originally proposed for modeling stable crack growth (with increasing load) in metals, is useful for predicting fracture in notched chopped fiber composites (Reference 6.6.8.1(o)). Although use of fracture mechanics in the problems just described has been very limited, it is now being widely used in the industry for dealing with various problems involving delamination fracture. Delaminations (in resin rich regions between the plies in a laminate) can exist as manufacturing defects or can be created due to various reasons; namely, (i) coalescence of small voids at interfaces, (ii) foreign object impact and (iii) peculiar stress fields near discontinuities such as free edges, holes, ply drops, transverse ply cracks or bonded joints. The basic concepts of delamination fracture are the same as those discussed earlier. However, the strain energy release rate, which is the energy released due to infinitesimal extension (as described earlier for through cracks in a plate) of a delamination is commonly used for prediction of catastrophic fracture and various test methods have been proposed for determination of its critical value (often called the toughness) for each of the three modes of loading (I, II, and III as shown in Figure 6.6.8.1). Some tests have also been devised for determining the criteria of failure (mode interaction) under mixed mode conditions. The next section gives some general discussions on the test methods and use of the properties in practical applications. Subsequent sections deal with some of the test methods.

6.6.8.2 General discussion. It should be noted that although the subject is quite advanced at this point and various attempts have also been made to obtain (i) R curves for modeling stable delamination growth under increasing load and (ii) delamination growth law for predicting delamination extension under cyclic loading, only one test method has been standardized. The test specimens usually contain an implanted delamination in the form of a nonadhesive insert and have been used widely for

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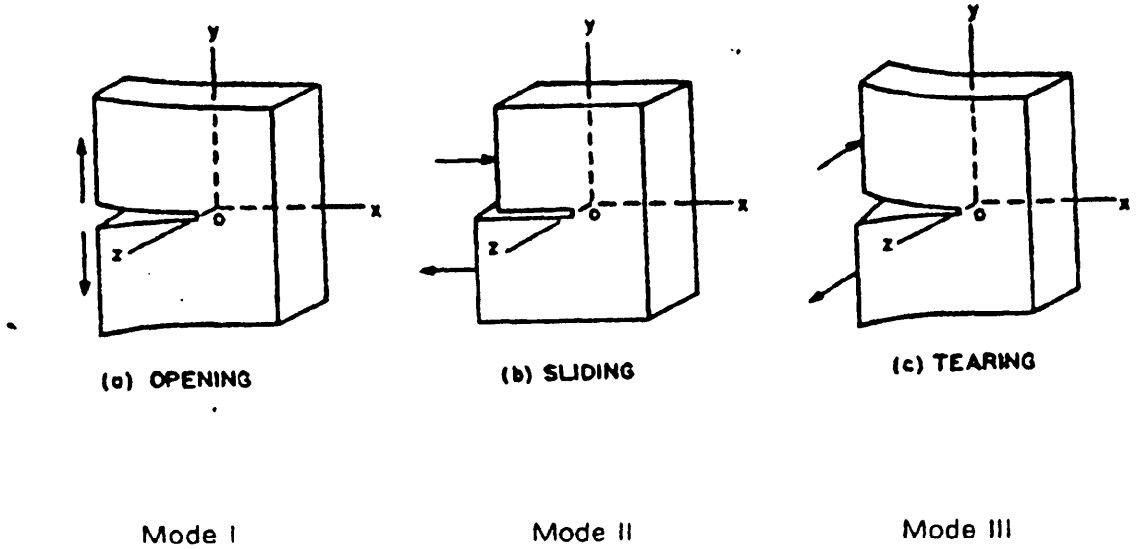


FIGURE 6.6.8.1 *Basic modes of crack extension.*

unidirectional glass or carbon fiber reinforced composites. The tests are designed such that delamination growth direction coincides with the fiber direction. Toughness values or other characteristics may sometimes vary depending on the tendency of the delamination to wander around various phases of multiple phase matrix materials. Also, brittle matrix composites with tough adhesive interleaves may yield different properties depending on the region where the delamination propagates, i.e., interleaves, brittle matrix or the interface. Use of non-unidirectional specimens with implanted delaminations between two off-axis plies of the same orientation or between two plies of different orientations may cause the delamination to shift its path through ply cracks and interpretation of data for such cases is difficult. It is also likely that the properties will differ if the delamination propagates in a direction other than the fiber direction, even though it remains coplanar. Woven fabric composites may show more scatter as compared to that for unidirectional laminates and increasing tendency of stable delamination growth (R-curve) because of the typical structural arrangement in such materials.

The main reason for observed resistance to delamination growth (for stable growth commonly observed in Mode I tests) in unidirectional brittle matrix composites is the phenomenon of fiber bridging across the delamination plane. Such bridging is caused by fiber nesting that is typically present in unidirectional composites and, hence, the toughness value for initiation should be identified separately from those at later stages of delamination growth. This value should be representative of the toughness for a natural delamination and it is often used to obtain a conservative design criterion. R-curves obtained from tests are not commonly used to obtain generic material property data. They are, however, often used to compare the degree of fiber matrix bonding between specimens, panels or batches of the same material or to compare composites with the same fiber, but different matrices. Poor bonding usually results in greater fiber bridging and, hence, a greater increase in G_c with delamination length.

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In materials not stroggly influenced by fiber bridging, a competing mechanism may dominate. When the delamination starts to grow from the insert, a pop-in type of behavior (yielding higher toughness values than that for subsequent growth) is observed. For Mode I tests, the first point is neglected in such cases and results for subsequent growth (made possible by displacement controlled tests) are utilized. For Mode II or mixed mode tests, failure is usually catastrophic and for this reason precracking (extension of the delamination beyond the insert) by Mode I loading, wedge insertion, or other methods is usually employed.

Most of the tests utilize beam type specimens with a single delamination tip. As the load is increased, load point deflections are measured and delamination growth is observed visually or using other aids or devices. In some cases, catastrophic delamination growth is observed (as in Mode II tests) and the maximum load reached is noted. Otherwise (in displacement controlled tests), load-displacement plots remain linear up to a point beyond which the delamination extends and a load drop occurs. In some materials, the onset of nonlinearity may be noticed before any delamination growth is noticeable. Such behavior may occur due to inelastic material response or subcritical damage growth ahead of the tip. The measured values of loads, deflections, and delamination lengths at the point of onset of nonlinearity or delamination growth are utilized to compute the critical energy release rates associated with fracture or subcritical damage growth. In some cases, approximate closed form expressions of energy release rates in terms of load, deflection, delamination length and/or material (or beam) stiffnesses are utilized for data reduction. When the deflection can not be measured and there is some uncertainty about the stiffness, an alternate approach is to first perform a compliance calibration on the specimen (or similar specimens) where the compliance

$$C = \frac{\delta}{P} \qquad 6.6.8.2(a)$$

δ being the deflection associated with the applied load P, for various values of delamination, length a is fitted to an exact or approximate relation (based on the principles of mechanics or a polynomial). The energy release rate is given by

$$G = -\frac{1}{b}\frac{dU}{da} \qquad 6.6.8.2(b)$$

or

$$G = \frac{1}{b}\frac{dV}{da} \qquad 6.6.8.2(c)$$

where b is the specimen width and a the delamination length. U and V are the strain and potential energies, respectively, both expressed in terms of displacements.

Now,

$$U = \frac{1}{2}P\delta = \frac{1}{2}\frac{\delta^2}{C} \qquad 6.6.8.2(d)$$

and

$$V = -\frac{1}{2}P\delta = -\frac{1}{2}\frac{\delta^2}{C} \qquad 6.6.8.2(e)$$

Therefore, it follows that for displacement controlled tests (δ prescribed)

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$$G = \frac{\delta^2}{2bc^2}\frac{dC}{da} = \frac{P^2}{2b}\frac{dC}{da} \qquad 6.6.8.2(f)$$

It can be shown that the same relation holds for load controlled tests. This approach is not suitable for mixed mode tests because the individual components of the energy release rates can not be calculated in this manner, unless their contribution to the total energy release rate is known.

As discussed in the previous section, catastrophic fracture in a single mode can be predicted using the critical strain energy release rate for the particular mode provided, of course, the energy release rate is constant along the delamination front. If these rates vary over the front, a conservative estimate can be obtained by equating the maximum value of this quantity at a point as determined from finite element or other stress analyses to the critical value (numerous investigations are reported in literature for such calculations, but they are beyond the scope of this section). Such estimates are often adequate for design purposes unless the delamination gets arrested because of structural or other constraints. A similar approach can be employed for estimating stable growth pattern and instability point using the R-curve concept. Growth of such delaminations under cyclic loading can be estimated using experimentally determined power law type growth laws, where the maximum value (or range) of the energy release rate is the controlling parameter. Uses of R-curves for predicting stable crack growth and power laws for estimating growth per cycle are, however, not yet accepted in the industry, since there exists some evidence that they do not always yield generic characterizations (Reference 6.6.8.2(a)).

The problem becomes more complicated when mixed mode conditions are present and the toughness values for the three modes differ significantly from one another. Various fracture criteria have been proposed for prediction of quasistatic fracture. A survey of such criteria for combined action of Modes I and II can be found in Reference 6.6.8.2(b). Power law type growth laws with the maximum value (or the range) of a scalar function f of the energy release rates ($f = G_I + G_{II} + G_{III}$ is a simple example) or the stress intensity factors have been suggested in different studies for modeling growth of delaminations under cyclic loading.

6.6.8.3 *Mode I tests.*

6.6.8.3.1 *Double cantilever beam (DCB) test, ASTM D 5528 (Reference 6.6.8.3.1(a)).* The test set up is schematically shown in Figure 6.6.8.3.1(a), which illustrates two types of loading attachments. The specimen is about 5 in. (125 mm) long, 0.8 - 1 in. (20-25 mm) wide and 0.12 - 0.20 in. (3-5 mm) thick. The applied load P to the two arms, the corresponding displacement δ, and typical load-displacement traces obtained are shown in Figure 6.6.8.3.1(b). The numbers on these traces indicate results for various delamination lengths and are obtained as the delamination progresses. NL indicates the onset of nonlinearity, which is usually caused by subcritical crack growth or material nonlinearities. The traces are often utilized to perform a compliance calibration. Various procedures for data reduction, other details, and restriction on specimen dimensions to avoid geometric nonlinearities are documented in References 6.6.8.3.1(a) and 6.6.8.3.1(b). As mentioned in the previous section, this test has been found to be adequate for unidirectional specimens. Some care should be taken when it is to be used for other lay-ups or material forms. Midplane symmetry is a requirement for pure Mode I deformation.

An asymptotic expression for the energy release rate for large a/h (a being the delamination length and h is the laminate thickness) is given by References 6.6.8.3.1(c) and 6.6.8.3.1(d).

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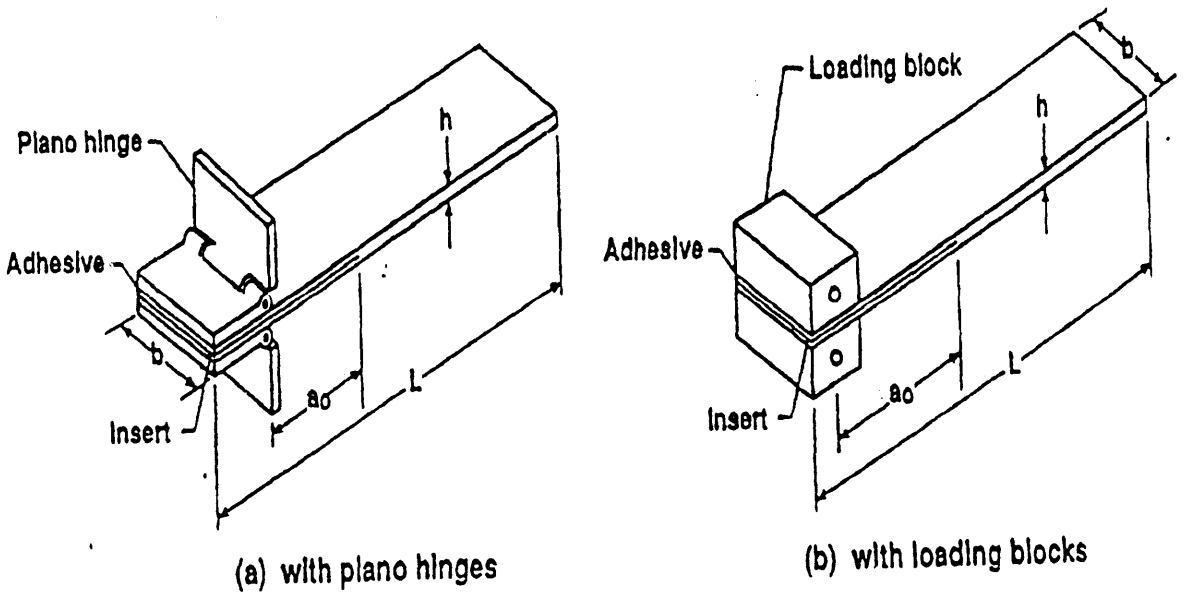


FIGURE 6.6.8.3.1(a) *Double cantilever beam specimen.*

$$G_I = \frac{96P^2(a+\alpha h)^2}{E_{11}b^2h^3} \qquad 6.6.8.3.1(a)$$

where b is the beam width, E_{11} is the axial Young's modulus of the unidirectional composite and α is a constant which depends on the ratio of the axial shear and Young's moduli. The relation in Equation 6.6.8.2(e) can also be used when each of the beam arms are balanced midplane symmetric laminates without any bending twisting coupling if E is replaced by the equivalent flexural modulus for the arms. Substituting Equation 6.6.8.2(e) in Equation 6.6.8.2(d) and integrating with respect to a, one obtains the following equation for the compliance C for large a/h.

$$C = C_o + \frac{64(a+\alpha h)^3}{E_{11}bh^3} \qquad 6.6.8.3.1(b)$$

where C_o is the integration constant. The constant α can be chosen as

$$\alpha = 1.45\sqrt{\frac{E_{11}}{G_{13}}} \qquad 6.6.8.3.1(c)$$

where G_{13} is the through the thickness shear modulus.

It has been suggested (References 6.6.8.3.1(b) and 6.6.8.3.1(d)) that for common glass and carbon fiber reinforced composites C_o can be chosen equal to zero. Also αh and the term in the denominator in the second term of Equation 6.6.8.3.1(b) or (a) should be determined by fitting measured

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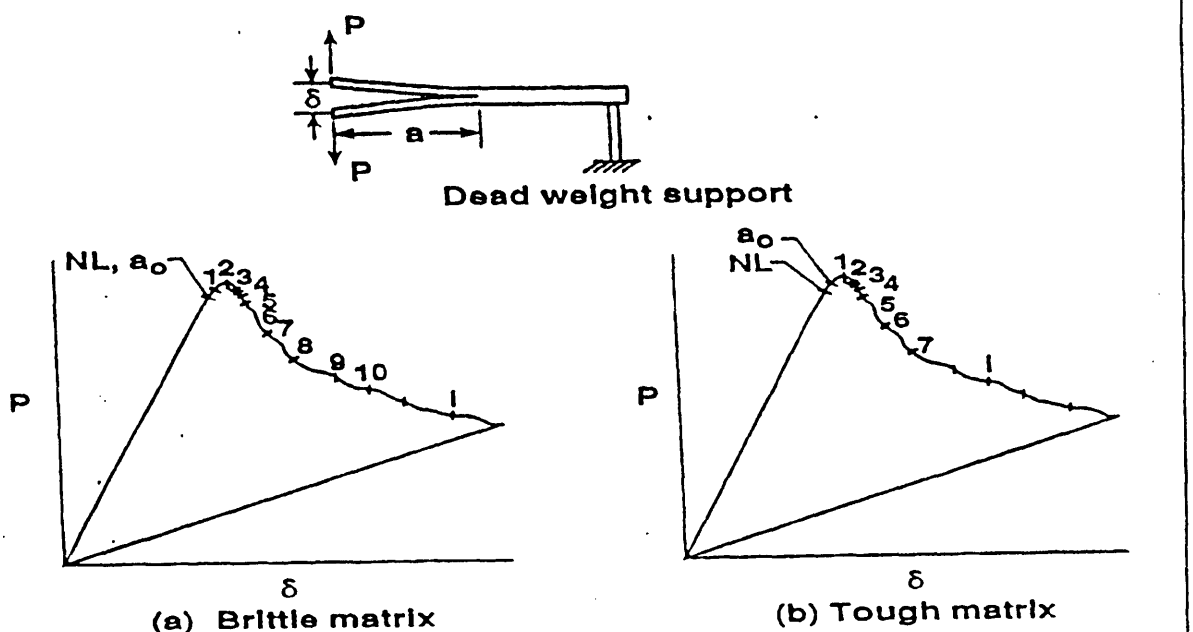


FIGURE 6.6.8.3.1(b) *Load displacement trace from DCB test.*

compliances to Equation 6.6.8.3.1(b). Therefore, if a straight line is made to fit $C^{1/3}$ versus a plot (a being the abscissa then the line when extended will cut the abscissa, x at $x = -\alpha h$) and the slope of the line gives the value of $(64/E_{11}\ b\ h^3)^{1/3}$. These values can then be substituted in Equation 6.6.8.3.1(a) to calculate G_I. It should be pointed out that in many cases αh can be neglected in comparison to length a and the results of Equations 6.6.8.3.1(a) and (b) are the same as those obtained if the two arms are treated as cantilever beams clamped at the end.

Another method, which is suggested in the ASTM standard (Reference 6.6.8.3.1(a)) for calculation of G, when both P and δ can be measured at the point of delamination propagation, is to use the formula

$$G_I = \frac{3P\delta}{2b(a+\alpha h)} \approx \frac{3P\delta}{2ba} \qquad 6.6.8.3.1(d)$$

The second expression is valid when αh can be neglected in comparison to a. This approach is called the modified beam theory method. Various other data reduction methods have been suggested in the literature (Reference 6.6.8.3.1(b)). It may be noted that if P and δ, measured at the point of onset of nonlinearity, are inserted in Equation 6.6.8.3.1(d), the value of G_I yields a conservative estimate of its critical value.

6.6.8.3.2 *Other Mode I tests*. The double cantilever beam test is the most widely used method for determining the Mode I toughness. In cases where joining loading attachments is a problem, wedge

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insertion between the two beams has been suggested. Another method which has been used is liquid pressure loading over a circular delamination between a thin film bonded to a substrate, which may be useful for some special applications.

6.6.8.4 *Mode II tests.*

6.6.8.4.1 End notched flexure (ENF) test. The ENF specimen is schematically shown in Figure 6.6.8.4.1. Typically, the specimens are 6 inches (150 mm) long, 1 inch (25 mm) wide and 0.12 - 0.20 inch (3 to 5 mm) thick. The insert is about 1 inch (25 mm) long. The span 2L of the three-point loaded beam is of the order of 4 inches (100 mm) leaving about a 1 inch (25 mm) overhang beyond the end supports. It is designed such that a delamination will propagate through the midplane of a laminate specimen loaded in three point bending (Reference 6.6.8.4.1(a)). The laminate should be symmetric about the midplane. Implanted and precracked delaminations grow in an unstable manner and, therefore, only the toughness for onset of Mode II fracture can be measured. Recently a stabilized version has been proposed (Reference 6.6.8.4.1(b)), where the test is controlled to a constant shear displacement at the delamination front. However, subcritical shear damage ahead of the front may influence the measurements after the onset.

An asymptotic expression for the energy release rate for large a/h (a being the delamination length and h is the laminate thickness) is given by (Reference 6.6.8.4.1(c)) as

$$G_{II} = \frac{9}{2}\frac{P^2(a+\alpha h)^2}{E_{11}b^2h^3} \qquad 6.6.8.4.1(a)$$

where P is the load at midspan, b and E_{11} are the beam width and axial Young's modulus of the unidirectional composite. α is constant which can be chosen as

$$\alpha = 0.065\sqrt{\frac{E_{11}}{G_{13}}} \qquad 6.6.8.4.1(b)$$

G_{13} being the through the thickness shear modulus. Use of similar expressions for laminates whose top and bottom halves have no bending-twisting and shear-extension coupling, but may have bending-extension coupling have also been suggested (Reference 6.6.8.4.1(c)). Using the procedure described in Section 6.6.8.3.1 for the DCB test, the compliance C for large a/h can be shown to be given by

$$C = C_o + \frac{3(a+\alpha h)^3}{E_{11}bh^3} \qquad 6.6.8.4.1(c)$$

where C_o is an integration constant.

Use of a curve-fitting procedure to fit Equation 6.6.8.4.1(c) with estimated values of α close to that given by Equation 6.6.8.4.1(b) has been suggested in Reference 6.6.8.4.1(c). According to other investigators, C_o can be chosen as the compliance of the beam without any delaminations and fitting a straight line to $C^{1/3}$ vs. length a can, therefore, be performed to determine α and $(3/E_{11}\, b\, h^3)^{1/3}$. Compliance measurements are easily performed by taking a long beam with an implanted delamination and shifting the beam on the supports to obtain various values of a.

An ASTM "round robin" test program is planned for the ENF test and ASTM subcommittee D30.06 plans to draft a proposed standard test method based on this test program. Effects of material nonlinearity and tough adhesive interlayers on Mode II fracture have also been studied (Reference 6.6.8.4.1(c)).

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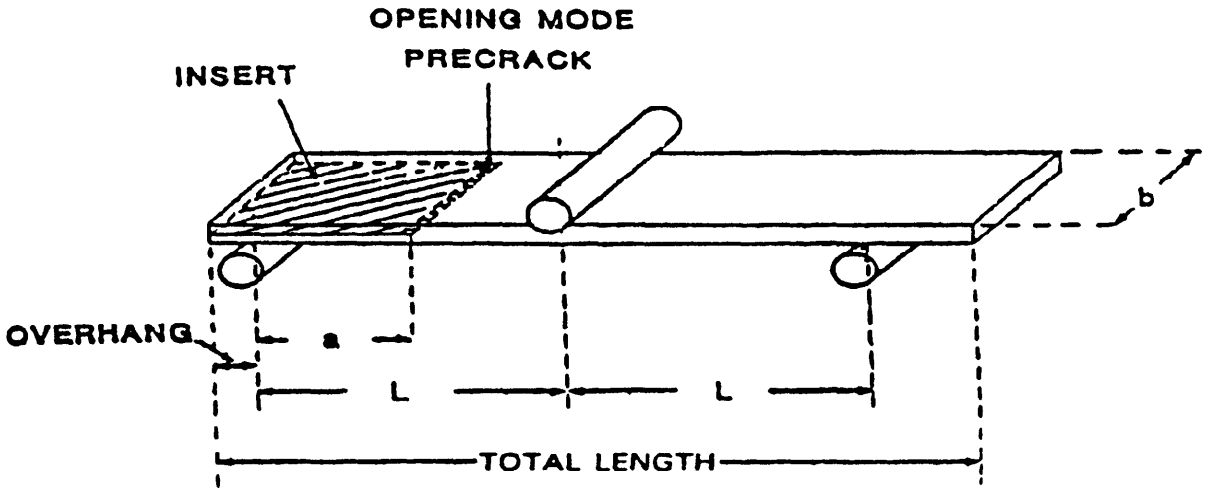


FIGURE 6.6.8.4.1 *Specimen geometry and loading.*

6.6.8.4.2 Other Mode II tests. Flexural loading of a thick laminated beam of the same form used in the ENF test, but with delaminations implanted between the support and the central load, has also been suggested (see Reference 6.6.8.4.1(c)), but compliance measurements and precracking are difficult for this specimen. However, a wider plate type specimen with implanted delaminations of circular and elliptic shapes has been found to be useful to characterize growth of such delaminations under a combined mode, the contribution from each mode varying along the delamination boundary (Reference 6.6.8.4.2).

6.6.8.5 *Mode III tests.* Presently there are no commonly accepted methods for measuring Mode III toughness. A split cantilever beam has been proposed (Reference 6.6.8.5(a)) but there appears to be some Mode II contribution in this specimen (Reference 6.6.8.5(b)). A cone torsion test has been used to characterize adhesive bonds (Reference 6.6.8.5(c)), which showed that Mode III toughness can be higher than that in Mode II. It is a common practice in composites industry to use Mode II value to estimate the criticality of Mode III fracture.

6.6.8.6 Mixed mode tests. As discussed in Section 6.6.8.2, in natural delaminations with curved fronts, the energy release rates not only vary along the front, but mixed mode conditions are also present. Mixed mode effects are also present in delaminations with straight fronts (edge delaminations and adhesive joints are examples). In such cases, Mode I and Mode II contributions are dominant, although small Mode III effects may also be present. For this reason, various attempts have been made to characterize delamination fracture under combined action of Mode I and Mode II effects and some of them are discussed next.

6.6.8.6.1 *Mixed mode coupon or crack lap shear (CLS).* The CLS specimen (Reference 6.6.8.6.1) was patterned after a similar specimen used for bonded joints. The specimen is a tension coupon,

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where some of the plies are terminated in the middle of the coupon (Figure 6.6.8.6.1). This specimen has been used widely in the industry. Mode I and II components are computed based on stress analyses (finite element or other methods). Effects of geometric nonlinearities have also been studied.

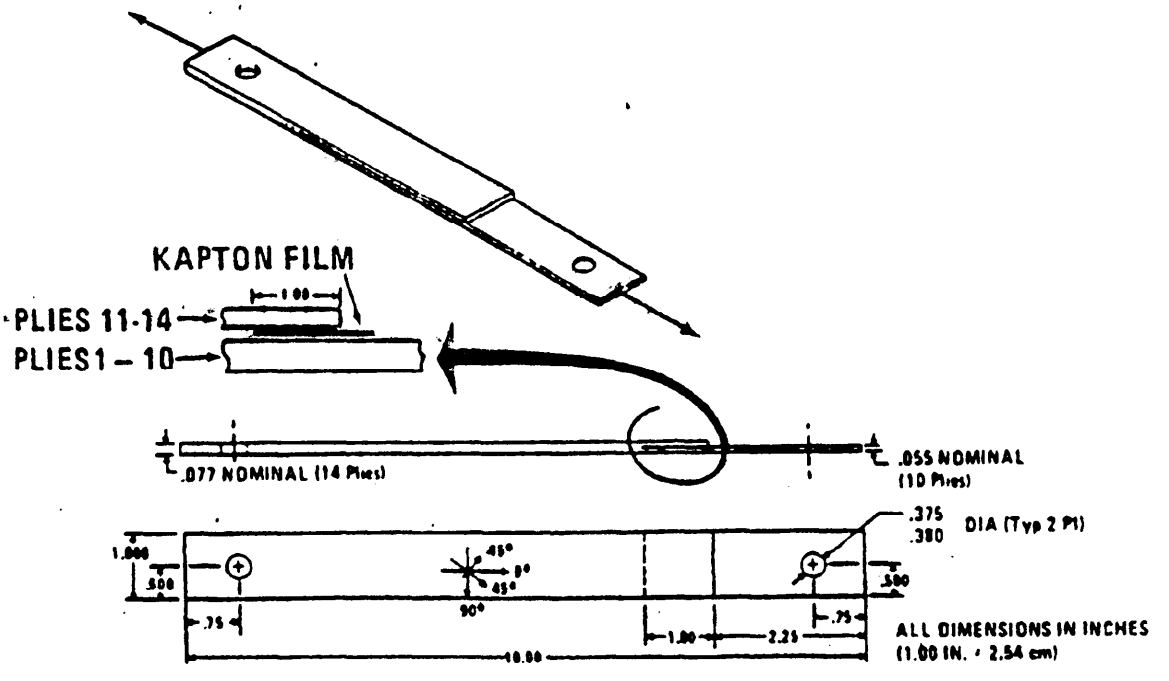


FIGURE 6.6.8.6.1 *Mixed-mode crack lap shear coupon.*

6.6.8.6.2 Mixed mode bending (MMB) test. This test has recently been proposed (References 6.6.8.6.2(a) and 6.6.8.6.2(b)) by combining the schemes used for DCB and ENF tests and the specimen is of the same form as that used in those tests. A special loading device is utilized, which can produce a wide range of the ratio of Mode I and Mode II components by changing the lever arm of the device shown in Figure 6.6.8.6.2. Use of the following equations has been suggested

$$G_I = \frac{3P^2}{4b^2h^3L^2E_{11}} A_1 B_1$$

$$G_{II} = \frac{9P^2}{16b^2h^3L^2E_{11}} A_2 B_2 \qquad 6.6.8.6.2(a)$$

A_1, B_1, A_2, and B_2 are expressed by the following equations when a modified linear beam theory is used for analysis

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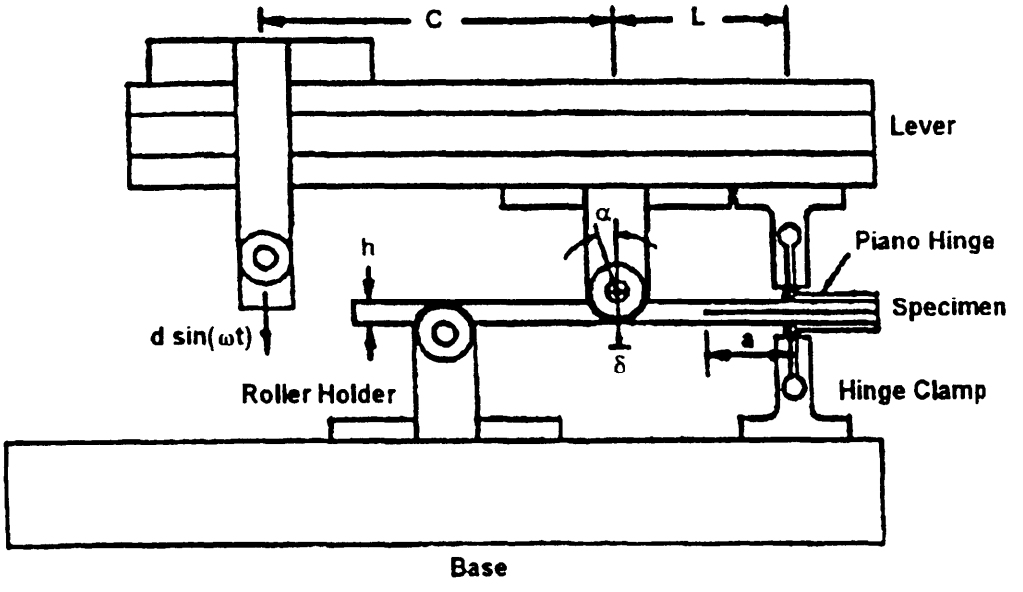


FIGURE 6.6.8.6.2 *Mixed-mode-bearing test setup.*

$$A_1 = a^2 + \frac{2a}{\lambda} + \frac{1}{\lambda^2} + \frac{h^2 E_{11}}{10 G_{13}}$$

$$A_2 = a^2 + \frac{h^2 E_{11}}{5 G_{13}} \qquad 6.6.8.2.6.2(b)$$

$$B_1 = (3C - L)^2$$

$$B_2 = (C + L)^2$$

An ASTM "round robin" test program is planned using the MMB specimen. ASTM subcommittee plans to draft a proposed standard test method based on the results of this test program.

6.6.8.6.3 *Edge delamination test.* This test specimen makes use of a $(\pm \theta_2/90_2)_s$ tension coupon, in which the free edge effect causes growth of delamination from the edges (Reference 6.6.8.6.3(a)). However, the delamination growth is neither uniform nor symmetric, since it does not remain in the midplane, but oscillates vertically between the 90/-θ interfaces. A modified version of the specimen with a starter delamination has been proposed (Reference 6.6.8.6.3(b)). Data reduction procedures are discussed in References 6.6.8.6.3(a) and 6.6.8.6.3(b). The test has not gained wide acceptance for toughness property determination.

6.6.8.7 Fracture toughness tests for MIL-HDBK-17 data submittal. Data produced by the following test methods are currently being accepted by MIL-HDBK-17 for consideration for inclusion in Volume 2.

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TABLE 6.6.8.7 *Fracture toughness test methods for MIL-HDBK-17 data.*

Property	Symbols	Fully Approved, Interim and Screening Data	Screening Data Only
Mode I Toughness	G_{IC}	ASTM D 5528	--
Mode II Toughness	G_{IIC}	ENF	--
Mode III Toughness	G_{IIIC}	--	--
Mixed Mode I, II Fracture	f_c (G_I, G_{II})	--	MMB

6.6.9 Multiaxial testing. Multiaxial tests, including biaxial and triaxial loadings, can be performed to experimentally evaluate the effect of combined stress states on composite material response. No standard test methods exist to guide multiaxial testing, and little data is available. A discussion of multiaxial testing can be found in Volume 3, section 7.2.3.2.

6.7 FILAMENT WINDING MECHANICAL PROPERTY TESTS.

6.7.1 Overview. The mechanical behavior of filament wound structures is typically different from the behavior of flat laminated structures. Some noted differences result from the type of cure, resin void content, microcracking, and free edge construction. However, filament wound structures require the same mechanical property data for design and analysis as used for general laminated structures. The majority of filament wound structures are used in the rocket motorcase community, and consequently, most of the test specimens are in the form of cylinders or bottles that more closely simulate the geometry of the structures to be designed and analyzed.

6.7.2 History. In November 1983, the Joint Army, Navy, NASA, and Air Force (JANNAF) Interagency Propulsion Committee chartered by the Department of Defense formed the Composite Motorcase Subcommittee (CMCS) (Reference 6.7.2.(a)). The CMCS was concerned with the application of composite materials in the construction of rocket motorcases for strategic and tactical missiles, space propulsion systems, and cartridge cases for gun propulsion. The CMCS consisted of four working panels two of which were the Testing and Inspection (T&I) panel and the Design and Analysis (D&A) panel.

The T&I panel surveyed industry on test methods which resulted in seventeen different tension tests, seventeen different compression tests and sixteen different shear tests that were being used to obtain mechanical property data. The T&I and D&A panels joined to evaluate the test methods via a JANNAF Workshop (Reference 6.7.2(b)). A panel of experts in filament wound composites was selected and tasked to make recommendations for test methods. A joint T&I and D&A Workshop was held in April 1986 to discuss the panel of experts' recommendations and to have an industry selection of JANNAF interim test standards to be used for the determination of uniaxial material properties.

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The CMCS conducted a Design Round Robin (DRR) and a Testing Round Robin (TRR) for three of the interim tests: 1) Transverse Tension, 2) Transverse Compression and 3) In-plane Shear of ninety degree filament wound cylindrical specimens. The participants in the DRR and the TRR were paid and were determined through competitive procurement. The manufacturing of the ninety degree filament wound cylinders and strain gaging were also determined through competitive procurement. Each of the test specimens were ultrasonically C-scanned for any anomalies. The TRR was conducted in accordance with ASTM E 691. The DRR and TRR were successful and resulted in three Military Standards in the fall of 1992. The three Military Specifications were put into ASTM format, run through the balloting phases and approved as ASTM test methods in the fall of 1993. The JANNAF efforts were coordinated with MIL-HDBK-17, ASTM Committee D-30, SACMA, and the DOD Standardization Program for Composites Technology.

6.7.3 Tension Tests for Uniaxial Material Properties.

6.7.3.1 *Zero degree tension.* The test method selected for the zero degree tension test is ASTM D 3039 entitled "Standard Test Method for Tensile Properties of Fiber-Resin Composites". It is recommended that the test specimens be obtained from a filament wound laminate. The JANNAF CMCS initially voted on either the pressurized NOL ring or a pressurized ninety degree filament wound tube. There were several attempts to obtain valid data from each technique but with little repeatable success.

6.7.3.2 *Transverse tension.* The test method selected to determine the uniaxial material properties for transverse tension is ASTM D 5450 entitled "Test Method for Transverse Tensile Properties of Hoop Wound Polymer Matrix Composite Cylinders". This test method was approved as MIL-STD-373 entitled "Transverse Tensile Properties of Unidirectional Fiber/Resin Composite Cylinders" in the fall of 1992, and was subsequently approved as an ASTM test method in the fall of 1993.

6.7.4 Compression Tests for Uniaxial Material Properties.

6.7.4.1 *Zero degree compression.*The test method selected to determine zero degree uniaxial material properties is ASTM D 3410 entitled "Test Method for Compressive Properties of Unidirectional or Crossply Fiber-Resin Composites". Method B, also known as the IITRI method, is recommended. It is further recommended that the test specimens be obtained from a filament wound laminate.

6.7.4.2 *Transverse compression.* The test method selected to determine the uniaxial material properties for transverse compression is ASTM D 5449 entitled "Test Method for Transverse Compressive Properties of Hoop Wound Polymer Matrix Composite Cylinders". This test method was approved as MIL-STD-374 entitled "Transverse Compressive Properties of Unidirectional Fiber/Resin Composite Cylinders" in the fall of 1992, and was subsequently approved as an ASTM test method in the fall of 1993.

6.7.5 Shear Tests for Uniaxial Material Properties.

6.7.5.1 *In-plane shear.* The test method selected for the determination of in-plane shear properties is a ninety degree, four inch diameter filament wound torsion tube described in ASTM D 5448 entitled "Test Method for Inplane Shear Properties of Hoop Wound Polymer Matrix Composite Cylinders". This test method was approved as MIL-STD-375 entitled "In-plane Shear Properties of Unidirectional Fiber/Resin Composite Cylinders" in the fall of 1992, and was subsequently approved as an ASTM test method in the fall of 1993.

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6.7.5.2 *Transverse shear.* The test method selected to determine transverse shear material properties is ASTM D-5379 entitled "Test Method for Shear Properties of Composite Materials by the V-notched Beam Method".

6.7.6 Test Methods for MIL-HDBK-17 Data Submittal. Data produced by the following test methods are currently being accepted by MIL-HDBK-17 for consideration for inclusion in Volume 2 (an element of orientation is shown for clarity).

6.8 BRAIDING MECHANICAL PROPERTY TESTS. The goal of testing braided specimens is to produce mechanical property data which mimics the performance of that section of the braided structure in a representative manner. Often this is difficult to achieve due to the local contour or details which are not accurately formed into the test specimens. Therefore certain assumptions as to the applicability of the specimens to the final part must be made prior to the start of the allowables testing.

Most specimens assume pristine manufacturing was used to make the coupon and that this is representative of the final part. This initial coupon may or may not mimic the final product depending upon care and reproducibility in the design of the part. Test specimens for tension, compression, shear, pin bearing, interlaminar shear and interlaminar tension are described in other sections of this handbook. In some cases, specimen geometry may be more accurately represented by the actual shape of the zone -- for example a tapered coupon (preferably symmetric) which contains the essential geometry of the actual tapered braid. In general, the specific mechanical properties of interest are torsional stiffness and strength, shear stiffness and strength, tension, compression and flexural moduli and strengths, bearing strength for bolted assemblies, and adhesive bond strength where required. In the event that the braided parts could be damaged or degraded by the environment, a series of tests to provide a measure of the damage tolerance and environmental effects on braided materials must be included. Damage criteria must include manufacturing defects which are not detectable non-destructive evaluations.

Several candidate configurations for these types of tests are listed in NASA CR 1092 - Revised, 1982 and should be considered as part of a design allowables program based upon intended use and environment.

Fatigue and creep tests should be considered whenever the part is a dynamic component. The highly complex courses followed by the braided yarns through the matrix material cause fiber load paths which are difficult to predict. The chance that the matrix is loaded beyond the level predicted by laminated design and experience are near certain. Therefore, a lower confidence in understanding how a braided part behaves will require a rigorous fatigue and creep test plan owing to the complex interweave of the braided yarns in the part. Tests designed to determine fatigue or creep effects are probably part specific and should be conducted on a braided specimen which is identical to the finished product. Scaling effects and the use of similitude in test specimen design should be avoided whenever practical. Otherwise, the results may not mimic the critical defect which results in a failure of the part prior to the predicted value.

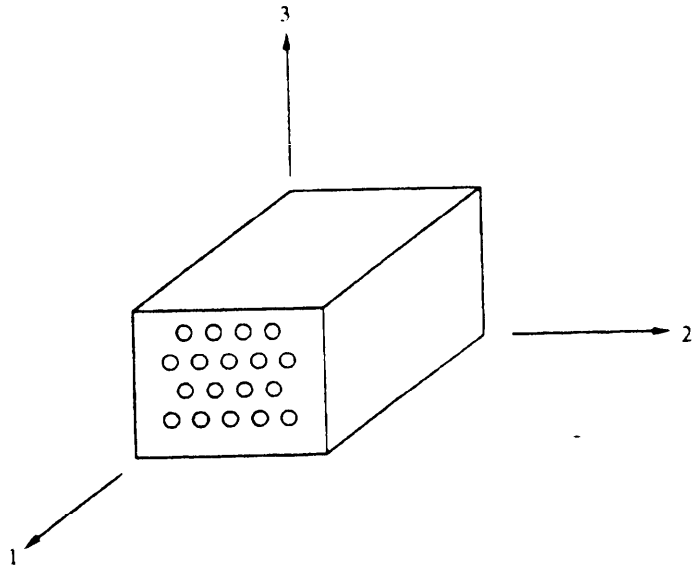
In general, it is accurate to state that for both two- and three-dimensional braided composites, the criticality of defects is the single most important factor in the design of the part and the corresponding application of the allowables which are used. This is due to the nature of complex, net-shape braided preforms as well as physical limitations in the ability of the yarns to fully cover an area of rapid change in contour. The good news is that net-shape braided parts do provide a means to build these parts without intensive hand labor suitable to production environments and fully capable of doing the intended job. This is true provided that the designer or analyst has made the correct choices in the type of specimens used for determining braided allowables.

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TABLE 6.7.6 *Filament wound test methods for MIL-HDBK-17 data.*

Property	Symbols	Fully Approved, Interim, and Screening Data	Screening Data Only
0° Tension*	F_1^{tu}, E_1^t, ν_{12}^t, ϵ_1^{tu}	ASTM D 3039	---
90° Tension*	F_2^{tu}, E_2^t, ν_{21}^t, ϵ_2^{tu}	ASTM D 5450	---
0° Compression*	F_1^{cu}, E_1^c, ν_{12}^c, ϵ_1^{cu}	ASTM D 3410B	---
90° Compression*	F_2^{cu}, E_2^c, ν_{21}^c, ϵ_2^{cu}	ASTM D 5449	---
In-Plane Shear**	F_{12}^{su}, G_{12}, γ_{12}	ASTM D 5448	---
Transverse Shear***	F_{23}^{isu}, G_{23}, γ_{23} F_{31}^{isu}, G_{31}, γ_{31}	ASTM D 5379	---

* Strength, modulus, Poisson's Ratio, and strain
** Strength, modulus, and strain



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4.4.1.6 Examination for dimensional characteristics.

6.9 THICK SECTION COMPOSITES MECHANICAL PROPERTY TESTS. No standard test methods exist to guide thick section testing, and little data is available. Mechanical tests, including uniaxial, biaxial, and triaxial loadings, can be performed to experimentally evaluate the effect of combined stress states on composite material response. A discussion of thick section testing can be found in Volume 3, section 7.2.3.

6.10 THERMAL AND MOISTURE ABSORPTION PROPERTIES.

6.11 ELECTRICAL PROPERTIES. In certain applications, the electrical properties of a composite are important. The properties that are of interest include dielectric constant, dielectric strength, volume resistivity, surface resistivity, are resistance, dissipation and loss factors. The values can be affected by temperature and environment, as well as the type of curing agent, filler, and fiber used in the composite. The following ASTM test methods can be used for determining the electrical properties of polymer matrix composite laminae and laminates:

D 149-87 "Standard Test Method for Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies". Method for determining the dielectric strength of solid insulating materials.

D 150-87 "Standard Test Method for A-C Loss Characteristics and Permittivity (Dielectric Constant) of Solid Electrical Insulating Materials". Method used for determining the relative permittivity, dissipation factor, loss index, power factor, phase angle, and loss angle of solid insulating materials when the standards are lumped impedances.

D 495-84 "Standard Test Method for High-Voltage, Low-Current, Dry Arc Resistance of Solid Electric Insulation". This test method is intended for preliminary screening of material and should not be used in material specifications.

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Chapter 7. STRUCTURAL ELEMENT CHARACTERIZATION

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7.1 INTRODUCTION. Testing and analysis of composite joints and other elements are essential for maintaining the structural integrity of composite structures and ensuring their reliability. Any joint in a composite structure is a potential failure site. Without proper design, a joint can act as a damage initiation point, which can lead to a loss in structural stability and eventual failure of a component. Two types of joints are in common use, namely (1) mechanically-fastened joints and (2) adhesively-bonded joints. These guidelines define the recommended approach for testing bolted joints to determine design properties against various failure modes. A detailed analysis of the stress distribution around a fastener hole is not presented here but is available in Volume 3. Discussion on both theoretical and empirical approaches to the stress analysis of bolted joints in composite materials can be found in Reference 7.1.

In addition to the recommended approaches to obtain bolted joints design properties for the different failure modes, a straightforward test method to determine material bearing strength is provided in Section 7.2.4. The bearing property measured by this test is the upper bound value that can be achieved in a realistic structural joint. The test is useful for qualification purposes or for material-to-material comparisons.

7.2 MECHANICALLY-FASTENED JOINTS.

7.2.1 Definitions. The following definitions are relevant to this section.

Bearing Area -- The diameter of the hole multiplied by the thickness of the specimen.

Bearing Load -- A compressive load on an interface.

Bearing Strain -- The ratio of the deformation of the bearing hole in the direction of the applied force, to the pin diameter.

Bearing Strength -- The maximum bearing stress which can be sustained.

Bearing Stress -- The applied load divided by the bearing area.

Bypass Strength -- The load that transfers around a hole divided by the laminate gross section area.

Edge Distance Ratio -- The distance from the center of the bearing hole to the edge of the specimen in the direction of the applied load, divided by the diameter of the hole.

Offset Bearing Strength -- The bearing stress at the intersection of the bearing load-deformation curve with the tangent modulus drawn from a pre-selected offset value. Offset may be 1, 2 or 4% of the original hole diameter.

Proportional Limit Bearing Stress -- The bearing stress value corresponding to the deviation from linearity of the bearing stress versus hole elongation curve.

Ultimate Bearing Strength -- The bearing stress corresponding to total failure of the test specimen.

Yield Bearing Strength -- The bearing stress at the intersection of the bearing load-deformation curve with the tangent modulus drawn from the 4% elongation value of the original hole diameter.

7.2.2 Failure modes. An important consideration in joint testing and analysis is the selection of the type of test method with due attention to the failure mode which is likely to result with a specific

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joint design in a particular composite system. A brief discussion on various failure modes is provided in this section.

The occurrence of a particular failure mode is dependent primarily on joint geometry and laminate lay-up. Composite bolted joints may fail in various modes as shown in Figure 7.2.2. The likelihood of a particular failure mode is influenced by bolt diameter (D), laminate width (w), edge distance (e),

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The shear-out mode of failure is usually avoided in design by providing sufficient edge distance between the holes or the free edge and balanced laminate configuration. However, in certain rework situations shear-out critical joints cannot be avoided. In those situations, a test program must be undertaken to establish allowables (see Section 7.2.8).

Net tension/compression and shear-out strengths are a strong function of laminate configuration, joint geometry, and hole size, but only marginally dependent on fastener type, joint configuration, or environment. On the other hand, bearing and fastener pull-through strengths are greatly influenced by the type of fastener used and its characteristics such as clamp effects, bolt stiffness, head and tail areas, countersinking, etc. The laminate lay-up and stacking sequence do not affect bearing or pull-through strengths significantly unless extreme lay-ups are used. The extreme being defined as a laminate having highly concentrated plies or plies in only two directions.

7.2.4 Material bearing strength.

7.2.4.1 *Significance*. The proposed test introduces the bearing load in a double shear configuration. In actual applications, load transfer in a single shear configuration is more commonplace, resulting in larger stress concentrations in the thickness direction, and lowering the realizable bearing strength; these single-shear tests are also discussed in Section 7.2.5. In other words, the bearing strength values measured in this test cannot be applied to single shear joints.

Only a tensile loading condition is proposed for evaluating bearing failures; under compression, the larger edge distance ($e >> 3D$) should only influence the bearing stress at failure minimally unless a shear-out mode of failure is possible (e.g., a laminate with a large percent of 0° plies).

Both final bearing failure and bearing deformation are needed for material characterization purposes. Therefore, it is recommended that the bearing stress variation as a function of hole deformation be documented, and that bearing stress values corresponding to the proportional limit, yield bearing (see Section 7.2.1), and ultimate failure be recorded.

In summary, bearing strength, as measured by the test in this section, is considered a material property for relative evaluation and design. In realistic structural joints, factors like geometry, fastener type, and load eccentricity will significantly influence the realizable fraction of the bearing strength measured in the proposed test. Bearing strength tests more appropriate in design of joints are discussed in Section 7.2.5.

7.2.4.2 *Specimen testing and design*. Using standard test equipment, the specimen shall be tested in the double shear arrangement shown in Figure 7.2.4.2. Test specimens shall conform to the dimensions and tolerances shown. Recommended b/D and e/D values are 6 and 3, respectively, and D = 1/4 inch (6.4 mm). A $[+45/0/-45/90]_{ns}$ lay-up, where n is selected so that the total thickness is approximately 0.125 inch (3.2 mm) is also recommended. The bearing hole shall be located as shown in Figure 7.2.4.3. The hole shall be drilled undersized and reamed to size. The bolt shall be made of hardened steel (Rockwell Hardness of C60 - C64) and shall be 0.249 (+0.0000/-0.0010). Thus, a close tolerance fit between the specimen and pin is required since a loose fit will tend to give lower results. Prior to assembly, the specimen, bolt, and adjacent areas should be cleaned of all foreign matter and contamination, especially lubricants.

7.2.4.3 *Replication requirements*. Five replicates shall be tested as a minimum. If three tests are conducted on one specimen, as shown in Figure 7.2.4.3, a total of six tests on two specimens will suffice.

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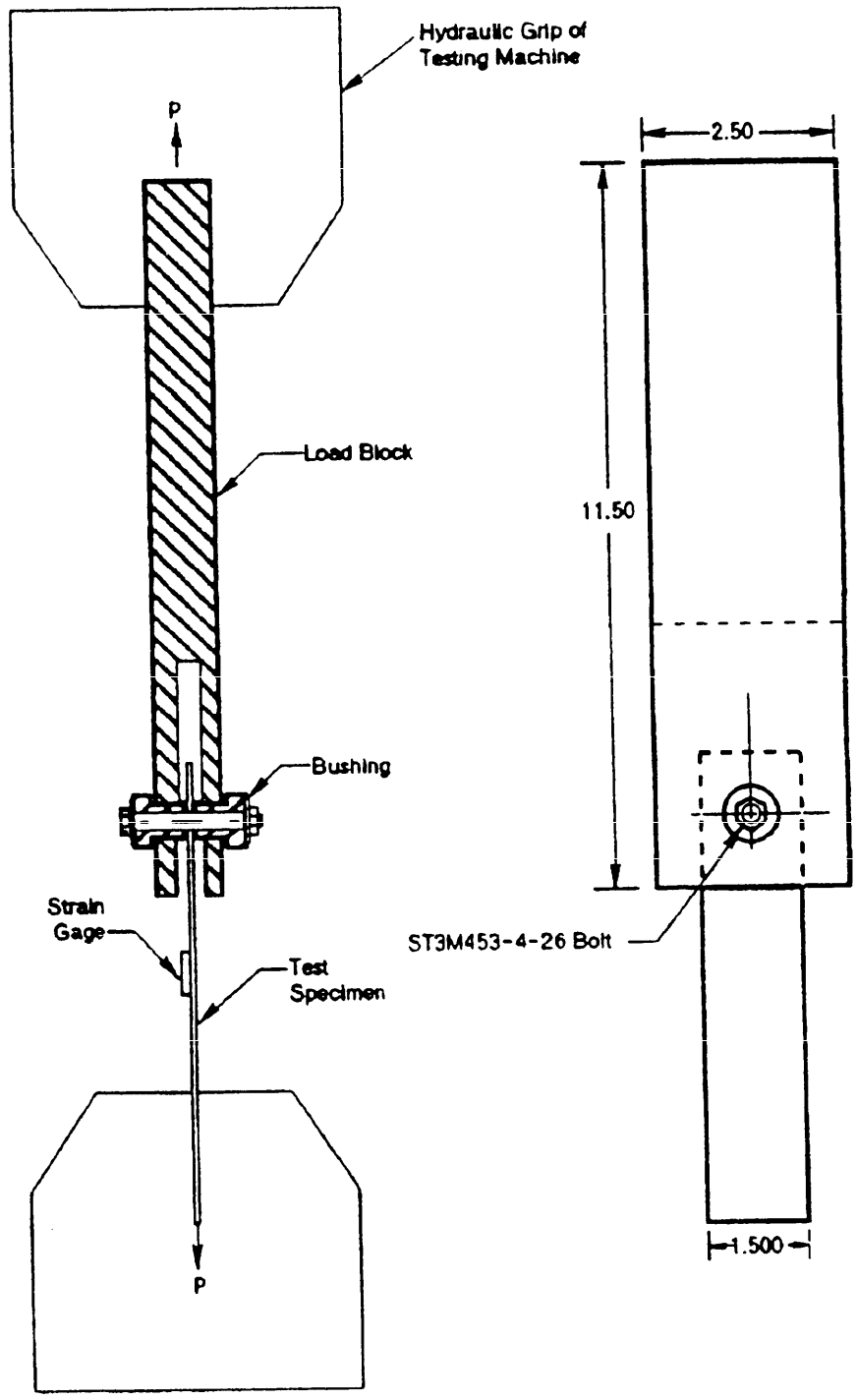
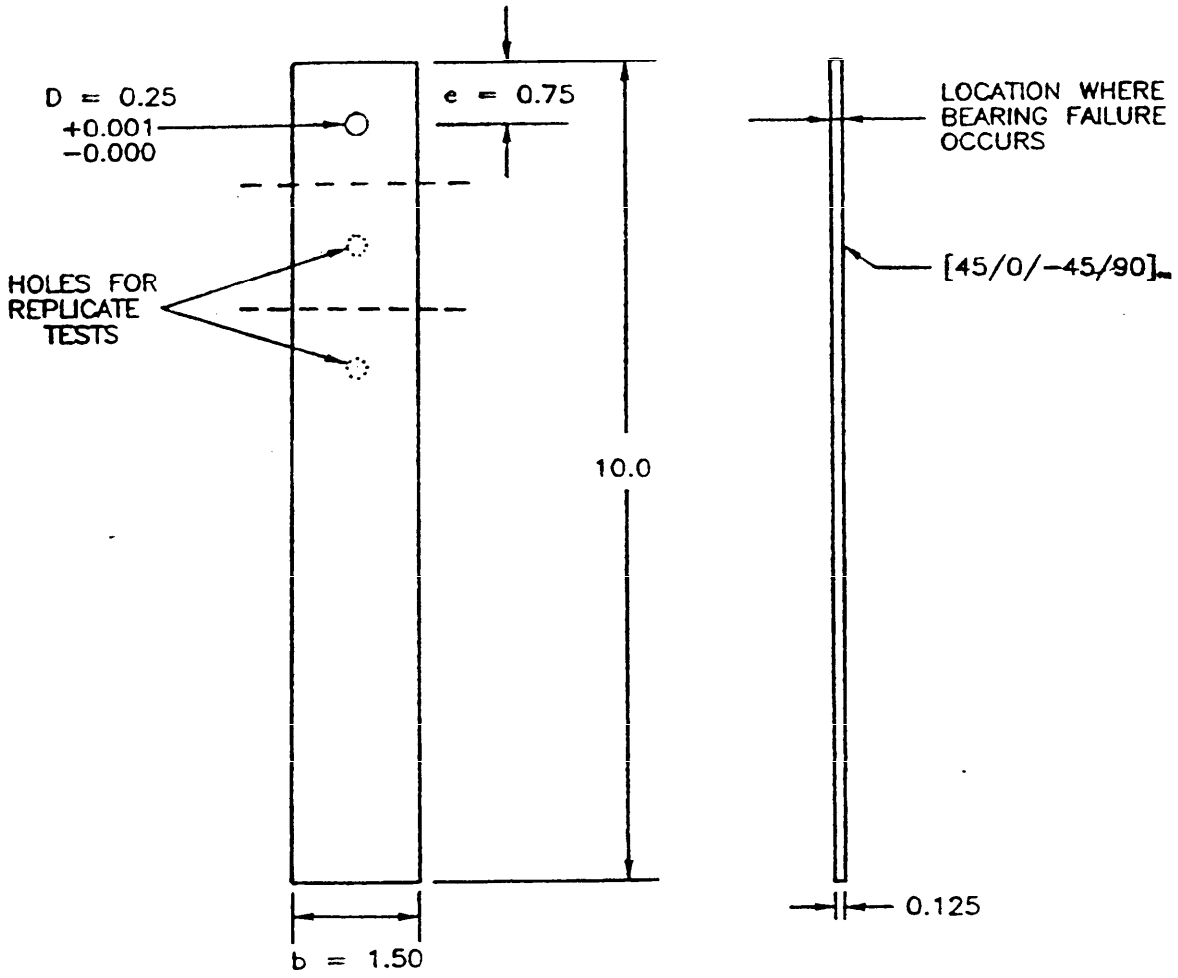


FIGURE 7.2.4.2 *Test arrangement for material bearing strength measurement.*

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ALL DIMENSIONS IN INCHES

FIGURE 7.2.4.3 *Test specimen geometry.*

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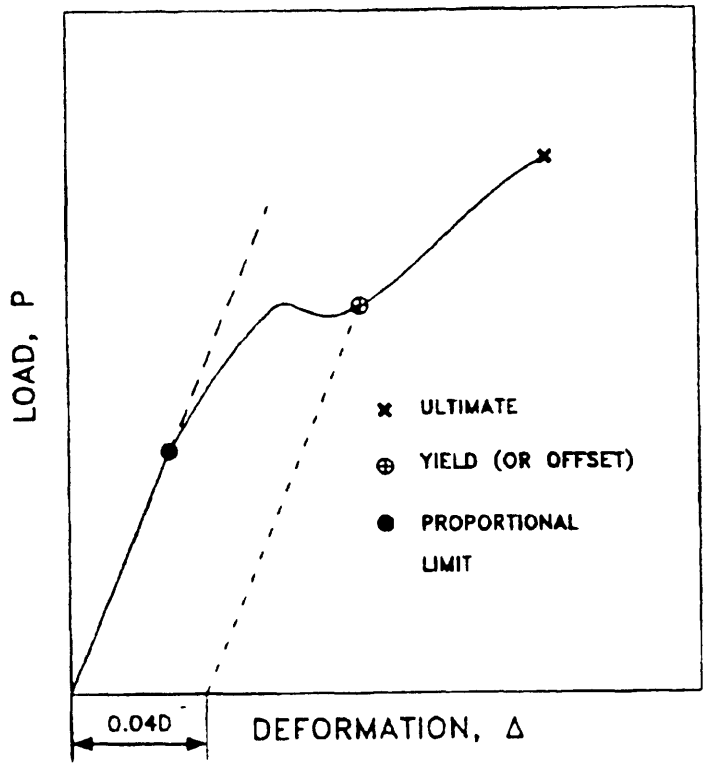


FIGURE 7.2.4.4 *Illustration of a typical load-deformation curve.*

7.2.4.4 *Test conditions*. Tests shall be conducted at room temperature dry (RTD) conditions, and one hot, wet condition. The hot, wet test shall be conducted on specimens after they are preconditioned to equilibrium level moisture content (see Section 2.2.2). The recommended temperature for the hot, wet test is T_g - 50F° (T_g - 28C°), based on the wet glass transition temperature.

Measurement of specimen dimensions, test procedures, etc., shall follow the recommendations of MIL-STD-1312-B/Test Methods, "Fasteners Test Methods, X. Mechanically Attached Composite Shear Joint", (Reference 7.2.4.4). The determination of yield bearing strength is shown for a typical curve in Figure 7.2.4.4 by using an appropriate offset value for consistency of data reduction. Ultimate strength may be limited by excessive displacement rather than catastrophic failure.

7.2.4.5 *Bearing strength calculations*. Bearing strength should be calculated using the following equation:

$$F^{br} = P/tD \qquad 7.2.4.5$$

where

F^{br} = bearing strength, psi (Pa)
P = bearing load, lb_f (N)

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D = bearing hole diameter, in. (m)
t = specimen thickness, in. (m)

Superscripts bry and bru are commonly used to differentiate between yield and ultimate bearing strengths. An offset bearing strength may be determined to represent the yield value. The subscript offset should be used.

Bearing and joint strength values are reported in MIL-HDBK-17 as typical or average values. Therefore, bearing and joint strength values which are available for each specific condition should be analyzed to produce typical property values as described in Chapter 8. Test data must include the data documentation required by Table 8.1.2 and will be published in property tables per Section 8.4.3. Bearing data developed at a specific fiber volume may not be applicable for fiber volumes that are much different because of failure mode changes.

7.2.5 Bearing strength of joints.

7.2.5.1 *Rationale*. This section describes single shear test specimens and the associated test matrices required to obtain bearing strength of single-lap joints. The resulting test data can be applicable either for the selection and screening of fasteners and the design of bolted joints or both. If the actual joint configuration is double shear, the test specimen and procedure of Section 7.2.4 would be more appropriate.

Bearing strength is a function of joint geometry and stiffnesses of the members and the fastener. It should be noted that for a $0/\pm 45/90$ family of laminates with 20-40% of 0° plies and 40-60% of $\pm 45^\circ$ plies the bearing strength is essentially constant. In addition, fastener characteristics such as clamp-up force, and head and tail configuration have a significant effect. However, for a specific laminate family, a specific fastener, and equal thickness lamina joining members, the parameter with the greatest influence is t/D. This was recognized by the aircraft designers and all the bearing data for metals is presented in MIL-HDBK-5 (Reference 7.2.5.1) in terms of the t/D parameter, Figure 7.2.5. The slope of this nondimensional curve is the bearing strength which decreases with increased t/D until for sufficiently thick laminates shear failure occurs in the bolt. The data generated using the recommended test specimens, procedures, and test matrices will produce equivalent data for composite joints.

In the design process there may be instances where the joint configuration may not correspond to the test configurations recommended here, i.e., unequal joining members, gaps, solid shims, fuel sealing provisions. These effects on bearing strengths should be evaluated by modifying the specimen geometry as needed. The test procedures, presented here are still applicable.

7.2.5.2 *Specimen design and testing*. Recommended composite-to-composite and composite-to-metal bearing specimen geometries are shown in Figures 7.2.5.2(a) and (b). Both are single lap geometries. Although it is generally more difficult to test than the double lap configuration, the single lap test configuration is more representative of most critical aircraft bolted joint applications. The single lap induces both bending and shear loads on the fastener, while the double lap induces mostly shear loads.

The joint configurations shown in Figures 7.2.5.2(a) and (b) may be used to generate both design and fastener screening data. In some cases, the fastener supplier and/or the requester may want to evaluate fastener behavior in a single fastener configuration. The recommended single fastener joint configuration is shown in Figure 7.2.5.2(c). This is the same specimen specified in MIL-STD-1312-X (Reference 7.2.5.2). It should be recognized that this joint configuration may be subject to high bending due to the load eccentricity transmitted through the bolt. The bending can be reduced by

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increasing the stiffness of the two laps, through increased thickness, or material stiffness. It should also be noted that the single fastener joint is generally not representative of multi-fastener joint applications (e.g., joint rotation, deflection). Therefore, it should be used mostly for fastener screening purposes.

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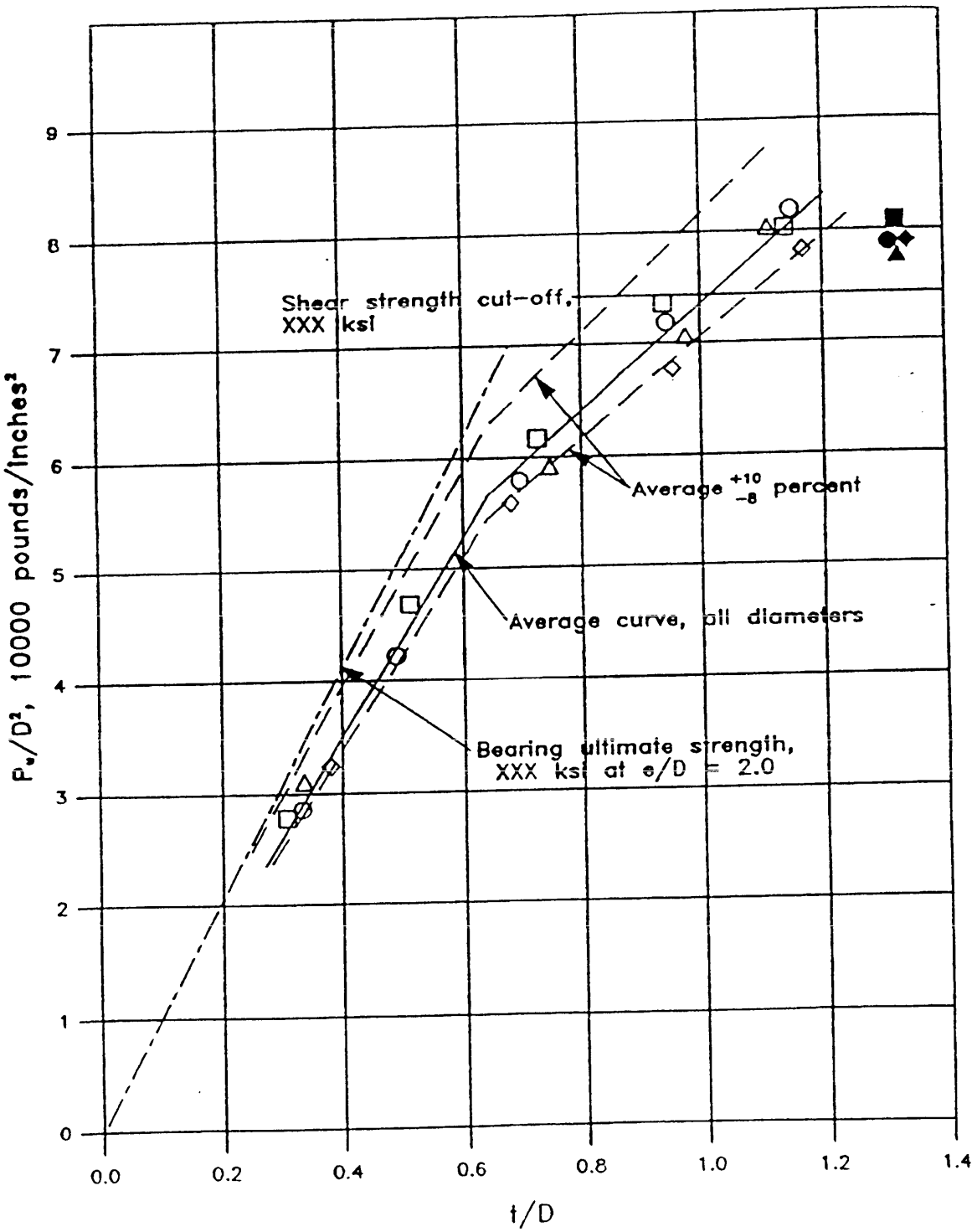


FIGURE 7.2.5 *Average ultimate-load analysis.*

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in Figure 7.2.5.2(a). The test data generated from the full test matrix of Table 7.2.5.3(a) will be sufficient to design composite-to-composite mechanical joints against bearing failure for one material and one fastener type. For other fasteners, the tests with note (1) should be sufficient to provide correction factors which would be applicable to all other not-tested conditions. These are labeled as fastener supplier tests or screening tests. For screening tests, in addition to t2 thickness, a third thickness specimen (t3) is shown so that sufficient test data would be generated to construct Figure 7.2.5.

In the design test matrix two different composite lay-ups are shown at each thickness. The lay-up varies from quasi-isotropic $(45/0\text{-}45/90)_{ns}$ to an orthotropic lay-up of 50% 0° plies in the load direction $(45/0/\text{-}45/90/0_2/45/0/\text{-}45/0)_{ns}$. For a fabric material, the lay-up percentages for the latter laminate have been modified to (40/20/40). One other thickness (t2) and bolt diameter (D2) are left

Table 7.2.5.3(a) *Composite-to-composite mechanically fastened joint test matrix for bearing strength.*

GEOMETRY	SKIN MEMBER THICKNESS (IN)	LAY-UP	BOLT DIAMETER (IN)	ENVIRONMENT (TEMP/% MOIST)	NUMBER OF TESTS
COMPOSITE TO COMPOSITE	0.2	25/50/25	0.25 D2	RT/ambient RT/ambient	10 (1,2) 5 (1)
	0.2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
	t2	25/50/25	0.25 D2	RT/ambient RT/ambient	5 (1) 5 (1)
	t2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
	t3	25/50/25	0.25 D2	RT/ambient RT/ambient	5 (1) only 5 (1) only
COMPOSITE TO COMPOSITE	0.2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	0.2	50/40/10	0.25 D2	hot/wet hot/wet	5 5
	t2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	t2	50/40/10	0.25 D2	hot/wet hot/wet	5 5

Notes:

(1) Supplier fastener screening tests
(2) Add 5 replications with 0.03 ± 0.003 liquid shim gap between members (optional)
(3) Single shear configuration per Figure 7.2.5.1

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unspecified; their choice should be dependent on the application. Two environments should be tested, room temperature as received and hot/wet. The selection of hot/wet temperature and moisture content should be guided by Section 2.2.2 of Volume 1.

The baseline 0.2 inch thick quasi-isotropic lay-up with the 0.25 inch bolt diameter could be used to evaluate the effect of a 0.03 inch thick or thicker liquid shim gap between the two members (option Note (2); also see Section 7.2.5.1). The test specimen configuration in Figure 7.2.5.2(a) shows the location of the liquid shim. A metal spacer can be used instead of the liquid shim if the spacer is unbonded to the composite.

For composite-to-metal bolted joints, the recommended test matrix for single shear bearing strength testing is given in Table 7.2.5.3(b) and the associated test specimen configuration is given in Figure 7.2.5.2(b). The general comments from the composite-to-composite bolted joints section also apply to the composite-to-metal bolted joints since the test matrices are the same. The composite-to-composite configuration is more critical than the composite-to-metal joint with respect to the design of the fastener tail; therefore, the composite-to-composite test specimen is more useful for the evaluation of fasteners by the fastener supplier. Because of the above reason, note (1) in Table 7.2.5.3(b) has been designated as tests required for a different fastener. The metal fixture shown in Figure 7.2.5.2(b) is designed so that the effective load line passes through the joint interface.

The recommended fatigue matrix is given in Table 7.2.5.3(c) and is based on using the test specimen configuration of Figure 7.2.5.2(a). Constant amplitude fatigue is suggested with a stress ratio of R = -0.2 (compressive load is 20 percent of tensile load). Frequency of loading should be selected so as to avoid excessive heating at the joint area of the specimen. For current material systems this translates to 5 Hz. This test matrix should be repeated for each fastener under consideration. The fifteen specimen per test will allow three replicates at five stress levels. A load level of half the static strength is a good starting point. All tests should be conducted at room temperature/ambient environment.

The specimens with specified thickness and bolt diameter (t = 0.2 in. and D = 0.25 in.) have been sized to fail in bearing. Specimens should be selected based on assuring bearing failure and avoiding bolt shearing, or net tension failures either in composite or metal members.

7.2.5.4 *Data reduction.* The procedures of data reduction described for the pure bearing test of Section 7.2.4 apply to the bearing tests in this section. In addition, the failure mode must be adequately described. Standard descriptions for typical failure modes are shown in Figure 7.2.5.6.

7.2.5.5 *Effects of thickness/gaps/shimming*. Although in most composite applications the use of bonded joints appears more weight-efficient, bolted joints still predominate due to their higher joint reliability and the need to disassemble some joints. In the assembly of composite structure, gaps between mating surfaces will occur and the disposition of these gaps is required prior to clamp-up of the fastener. Closing excessive unshimmed gaps when installing fasteners can cause delaminations in the composite structure, however, residual gaps of any size may reduce joint performance.

Test data show that the strength of bolted composite joints depends partially on bolt diameter, composite thickness, shimmed gap thickness, and the type of shimming material used. Examples of strength reduction curves are shown in Figures 7.2.5.4 and 7.2.5.5 for the diameter to thickness ratio and shimmed gap effects for a single shear composite joint in a multiple bolt splice. These are not generic curves and generation of similar data would be required for specific user application. The reduction factors are then used to reduce the nominal allowable bearing stress. The nominal bearing allowable for a particular material system would be obtained using tests with configurations minimizing bolt bending to obtain uniformity of stress through the thickness (a large diameter to thickness ratio

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Table 7.2.5.3(b) *Composite-to-metal mechanically fastened joint test matrix for bearing strength.*

GEOMETRY	SKIN MEMBER THICKNESS (IN)	LAY-UP	BOLT DIAMETER (IN)	ENVIRONMENT (TEMP/% MOIST)	NUMBER OF TESTS
COMPOSITE TO METAL	0.2	25/50/25	0.25 D2	RT/ambient RT/ambient	10 (1,2) 5 (1)
	0.2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
	t2	25/50/25	0.25 D2	RT/ambient RT/ambient	5 (1) 5 (1)
	t2	50/40/10	0.25 D2	RT/ambient RT/ambient	5 5
COMPOSITE TO METAL	0.2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	0.2	50/40/10	0.25 D2	hot/wet hot/wet	5 5
	t2	25/50/25	0.25 D2	hot/wet hot/wet	5 5
	t2	50/40/10	0.25 D2	hot/wet hot/wet	5 5

Notes:
(1) Alternate fastener tests
(2) Add 5 replications with 0.03 ± 0.003 liquid shim gap required between members (optional)
(3) Single shear configuration per Figure 7.2.5.2

Table 7.2.5.3(c) *Mechanically fastened joint fatigue test matrix for bearing fatigue.*

GEOMETRY	THICKNESS (IN)	LAY-UP	TOTAL NUMBER OF TESTS
COMPOSITE TO COMPOSITE	t1	25/50/25	15 (1)
	t1	50/40/10	15
COMPOSITE TO COMPOSITE	t1	25/50/25	15 (1)
	t1	50/40/10	15

Notes:
(1) Supplier fastener screening tests
(2) Constant amplitude fatigue (R = -0.2) to 4% hole elongation measured across a single hole
(3) Single shear configuration; same as static tests

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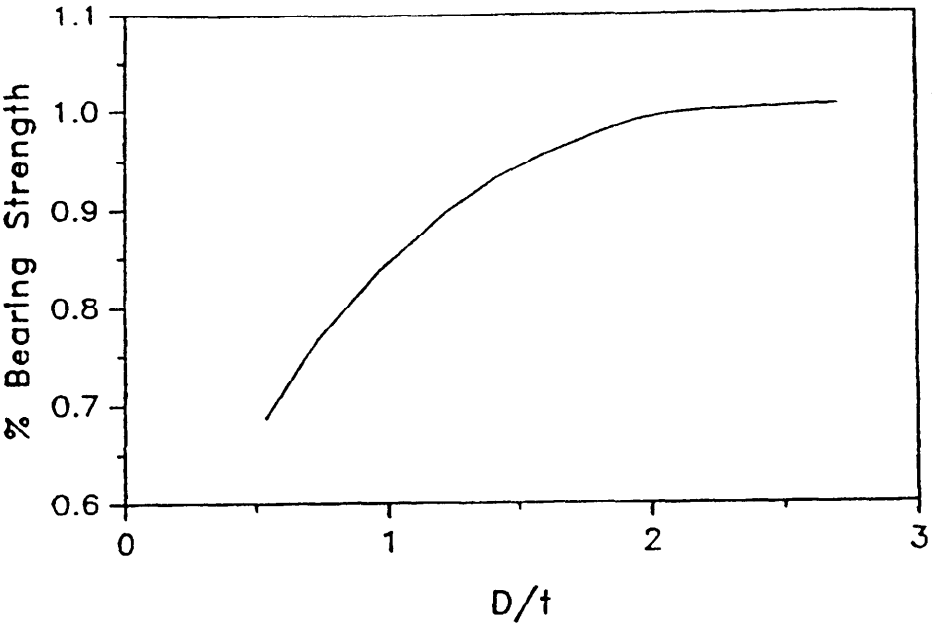


FIGURE 7.2.5.4 *Bearing strength reduction vs D/t.*

clevis or multi-fastener test) and using all pertinent statistical and environmental knockdowns. Joint

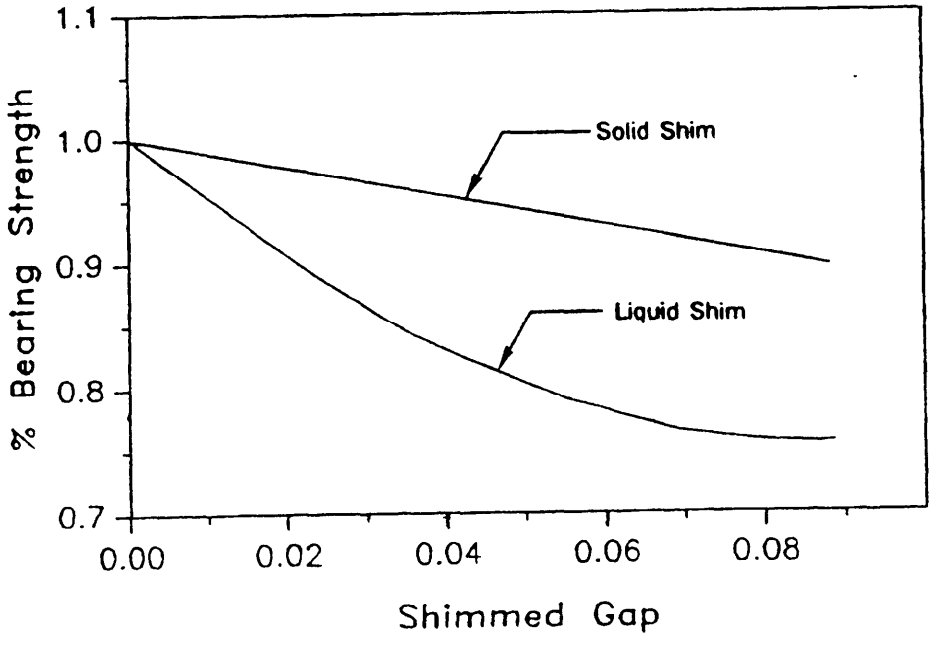


FIGURE 7.2.5.5 *Bearing strength reduction vs shimmed gap.*

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strength reduction factors are greater for joints using liquid shims for filling the gaps than joints using metal or composite solid shims.

7.2.5.6 *Failure modes*. Descriptions for failure modes are provided in Figures 7.2.5.6(a) - (d).

7.2.6 Notch tension/compression strength.

7.2.6.1 *General considerations*. The most common method of assembling composite structure is by the use of mechanical fasteners, even though bolted joints are relatively inefficient. The stress concentration due to the hole will cause substantial reduction in both the notch tension and compression strength of a composite laminate. The magnitude of this reduction varies considerably with a multitude of factors. All composite materials that exhibit a linear elastic stress-strain relationship to failure will be very sensitive to notches. Unlike metallic materials, the effects of the notch on strength will vary with the size of the notch but are relatively independent of notch geometry. Under uniaxial load, large holes will produce a stress concentration factor approaching the theoretical factor for wide plates given by the relationship:

$$K_t = 1 + \left\{2\left[\left(\frac{E_x}{E_y}\right)^{\frac{1}{2}} - \nu_{xy}\right] + \frac{E_x}{G_{xy}}\right\}^{\frac{1}{2}} \qquad 7.2.6.1(a)$$

For a quasi-isotropic laminate, the above relationship reduces to the well-known value $k_t = 3.0$ for a circular hole. This relationship also indicates that holes in high modulus laminates have a much greater effect on strength than holes in low modulus laminates. The stress concentration factor described by the above equation is reasonably proportional to the parameter E/G, the laminate axial modulus divided by the laminate shear modulus.

Considerable research literature exists regarding the influence of holes on the strength of composite laminates. An excellent summary of this literature is given in Reference 7.2.6.1 which includes over 300 citations. While the influence of holes in composites has been researched and reported extensively, there are additional effects to be considered. Two of these effects relate to the influence a fastener has in "filling" a hole in a laminate. The fastener, particularly in tight or interference holes, can induce a biaxial stress field by preventing ovalization of the hole under load. The factor tends to decrease the notch tension strength of 0°-ply dominated laminates and increase the strength of laminates with predominantly 45° plies. The second effect is when clamp-up of the fastener prevents delaminations from occurring around the hole. These delaminations are the result of "free edge" stresses and are very sensitive to stacking sequence. When delaminations are suppressed by the fastener, no stress concentration relief occurs and the notch sensitivity increases.

Filled hole compression strengths are significantly higher than open hole strengths and, in some cases, approach the unnotched strength. This is particularly true with close-fitting holes where load can be transferred through the hole by direct bearing through the fastener. Fabric laminates, because of the balanced nature of fabric materials, tend to have lower stress concentration factors and are less prone to free edge delaminations. The influence of free edge stresses and stacking sequence on delaminations are discussed in Volume 3, Sections 4.6.3 and 4.6.5.

When holes are placed together as in a bolted joint, the stress concentrations at the holes start to interact and the notch strength of the composite laminate decreases. A finite width correction factor is used to account for this interaction effect. For isotropic materials the "finite width correction" factor (FWC) is given by:

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Laminate Failures

L-NT: Laminate Net Section Tension Failure
L-NC: Laminate Net Section Compression Failure
L-OC: Laminate Off-Set Compression Failure
L-BR: Laminate Bearing Failure
L-SO: Laminate Shear-Out Failure
L-MM: Mixed Mode Failure
L-PT: (Laminate allowing) Fastener Pull-Through Failure

Fastener Head/Collar Failures

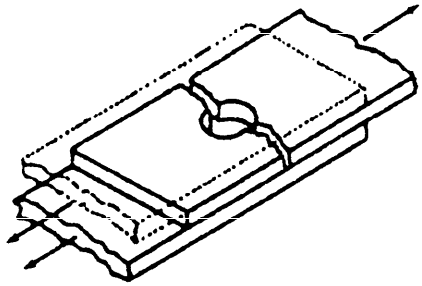
F-HD: Fastener Head Dished
F-FS: Fastener Flange Shear Failure
F-HS: Fastener Head, Blind or Formed Head Shear Failure
F-BH: Fastener Blind Head Deformed
F-NF: Fastener Collar Fracture Failure
F-NS: Fastener Collar Stripped

Fastener Shank Failures

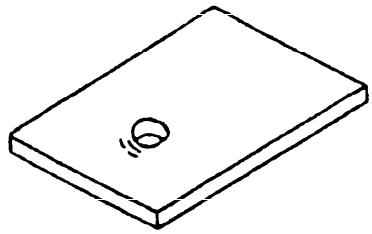
F-STH: Fastener Shank Tension Failure at Shank/Head or Formed Head Junction
F-STT: Fastener Shank Tension Failure in Threads
F-ST: Fastener Shank Tension Failure
F-SST: Fastener Sleeve or Stem Tension Failure
F-SSH: Fastener Shank Shear Failure at Shank/Head Junction
F-SS: Fastener Shank Shear Failure

FIGURE 7.2.5.6(a) *Failure mode descriptions for mechanical fastened joints.*

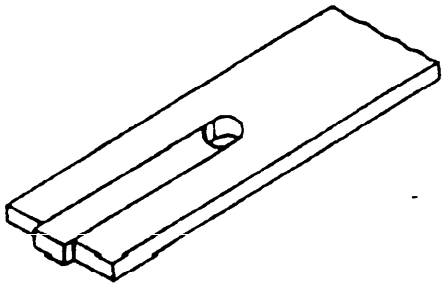
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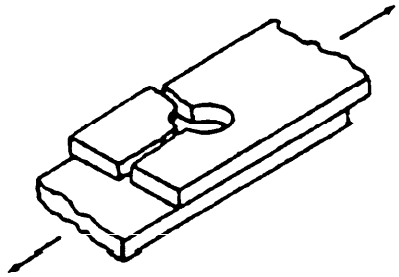
L-NT, L-NC: Laminate Net Section Tension (or Compression) Failure



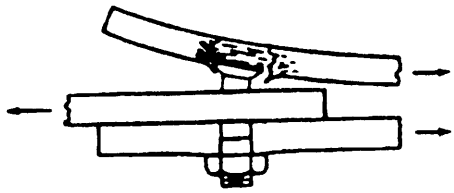
L-BR: Laminate Bearing Failure



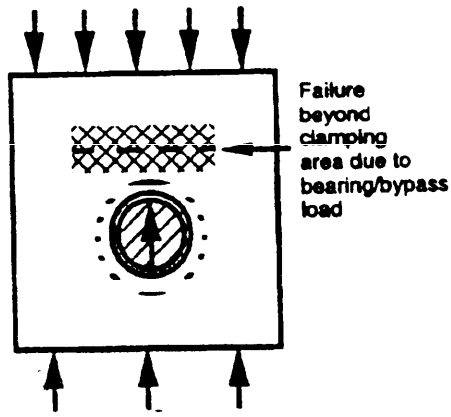
L-SO: Laminate Shear-out Failure



L-MM: Mixed Mode Failure
(unsuitable for data generation)



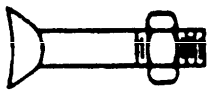
L-PT: Fastener Pull-Through (or Partial Pull-Through) Failure



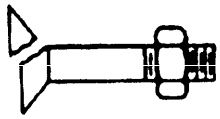
L-OC: Laminate Off-Set Compression Failure

FIGURE 7.2.5.6(b) *Failure mode descriptions for mechanically fastened joints.*

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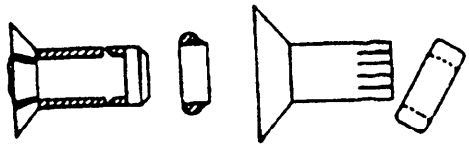
F-HD: Fastener Head Dished



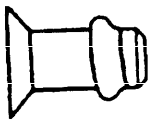
F-FS: Fastener Flange Shear Failure



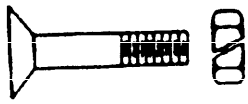
F-HS: Fastener Head Shear Failure



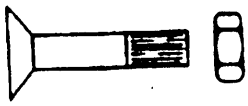
F-HS: Blind or Form Head Shear Failure



F-BH: Fastener Blind Head Deformed



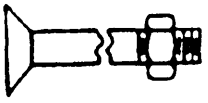
F-NF: Fastener Collar Fracture Failure



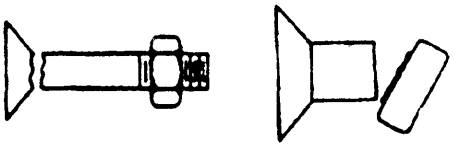
F-NS: Fastener Collar Stripped

FIGURE 7.2.5.6(c) *Failure mode descriptions for mechanically fastened joints.*

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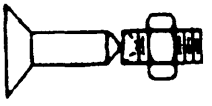
F-ST: Fastener Shank Tension Failure



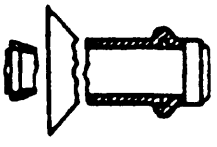
F-STH: Fastener Shank Tension Failure at Shank/Formed Head Junction



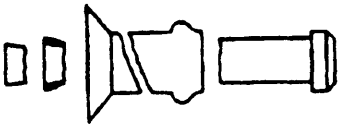
F-SS: Fastener Shank Shear Tension



F-STT: Fastener Shank Tension Failure in Threads



F-SSH Fastener Shank Shear Failure at Shank/Head Junction



F-SST: Fastener Sleeve or Stem Tension Failure

FIGURE 7.2.5.6(d) *Failure mode descriptions for mechanically fastened joints.*

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$$FWC = \frac{2 + \left(1 - \frac{D}{W}\right)^3}{3\left(1 - \frac{D}{W}\right)} \qquad 7.1.6.1(b)$$

where D = fastener diameter
W = fastener spacing

The correction factor for orthotropic materials cannot be expressed in a closed form. In most cases, the isotropic correction has been found to be reasonably accurate.

When the hole diameter is significantly greater than the laminate thickness, the stress concentration is two-dimensional in nature. Most of the research on holes in laminates is for this case. The notch strength of composites is much more difficult to predict when the thickness of the laminate significantly exceeds the hole diameter. The stress concentration at the hole becomes three-dimensional in nature and stacking sequence effects become more dominant.

There have been many failure models proposed for describing the notch strength of composite laminates. All of the models require some form of empirical "calibration" factor such as a "characteristic dimension". Characteristic dimensions have been used as a measure of notch sensitivity. Once calibrated, all of the models are reasonably accurate in describing the notch strength of composites. The drawback to these models is that many parameters such as laminate composition, temperature, and even hole size require re-calibration of the failure model. Some of the calibration factors are reasonably consistent, over a wide range of application laminates, among various material systems of similar characteristics. Low strength or stiffness fibers, or highly nonlinear toughened resins are examples of material constituents which can produce widely different "calibration" factors. Progressive damage failure models have shown some promise in not being overly dependent on empirical factors.

7.2.6.2 *Test matrix*. The minimum recommended test matrix for initial empirical assessment of "calibration" of the various theoretical models and determination of notch strength data for a range of laminates is given in Table 7.2.6.2. This matrix is just part of the overall development test plan. The matrix requires selective tests to be performed under tensile and compressive loadings in various environments applicable to the design of structural components. The test matrix is for open holes but bolted joint design criteria will also require filled hole test data to be generated. It is recommended that portions of the matrix in Table 7.2.6.2 be used to spot test for filled hole strengths, particularly in tension. For filled hole strengths, a reduction factor is applied to the open hole strength and the predictive model is not re-calibrated. The matrix represents the range of laminates commonly used in bolted joint designs. This assures that important interactions between laminate stiffness, failure modes, and joint parameters are assessed. If the laminate of interest is significantly outside the range of behavior of the test laminates, open hole tests for that laminate should be added to that matrix.

The procedure, often used to calibrate single-fastener-hole laminate strength methodology, such as for the "characteristic dimension" approaches, starts by evaluating the effect of hole-size on strength data for the isotropic (25/50/25) laminate, using a baseline specimen width/diameter ratio of six. Three fastener diameter sizes are selected for testing which will span the usual application range of fastener hardware. The trend of the effect of hole size on tensile and compressive strength data is established. The characteristic dimension that produces the trend line which best fits the test data is then selected. All other test case predictions now use that selected characteristic dimension.

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TABLE 7.2.6.2 *Notch tension/compression strength test matrix.*

Lay-up	Diameter (inches)	Width (inches)	W/D Ratio	CTD Tension	RTD Tension	RTD Compression	ETW Compression	Total Number of Tests
(10/80/10)	0.250	1.5	6.0	5	5	5	5	20
(25/50/25)	0.125	1.0	6.0		5	5		10
		1.5	8.0		5	5		10
(25/50/25)	0.250	1.5	6.0	5	5	5	5	20
(25/50/25)	0.500	2.0	4.0		5	5		10
		2.5	6.0		5	5		10
(50/40/10) Tape or (40/20/40) Fabric	0.250	1.5	6.0	5	5	5	5	20
Total				15	35	35	15	100

Lay-up	Ply Stacking Sequence
(10/80/10)	$[45/-45/90/45/-45/45/-45/0/45/-45]_{ns}$
(25/50/25)	$[45/0/-45/90]_{ns}$
(50/40/10)	$[45/0/-45/90/0/0/45/0/-45/0]_{ns}$
(40/20/40)	$[0_f/90_f/0_f/90_f/45_f/-45_f/90_f/0_f/90_f/0_f]_{ns}$

n selected so that total laminate thickness is between 0.1 to 0.2 inches

Conditions

CTD	Cold Temperature Dry
RTD	Room Temperature Dry
ETW	Elevated Temperature Wet

See Volume 1, Section 2.2.2

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Further correlations between the model and the data are then performed to assess the generality of this single characteristic dimension. Additional tests provide data for correlation with predicted effects of laminate composition, temperature variation, and finite width variations. The hole-size effect data, used initially to select a characteristic dimension, "builds in" a correlation for finite width of W/D = 6. If subsequent theory/test correlations are inconsistent or errors too large, further fitting of the "characteristic" dimension may be required. If still unacceptable, for the application range of variables, the test data will be the bases for other analytical or purely empirical approaches, but significantly more testing may be required to offset the loss of predictive methodology which provided an analytical bridge among the limited test conditions defined in Table 7.2.6.2.

7.2.6.3 *Specimen design and testing.* The recommended test specimen in shown in Figure 7.2.6.3(a). There is no need for tabbing or special gripping treatments unless extremely coarse serrated grips or excessive pressure are used. Normally the large stress concentration at the hole will eliminate problems with grip failures. In compression, specimens must be stabilized from general column buckling failures. The fixture shown in Figure 7.2.6.3(b) is in common use throughout industry. This fixture is documented in SACMA recommended method SRM 3-88 (Reference 7.2.6.3) and a proposed ASTM standard to be released in 1992. The test is normally run without instrumentation, recording only ultimate load, specimen dimensions, and failure mode and location. Instrumentation can be added if strains or deformations are of interest.

7.2.6.4 *Calculations.* Results are reported in terms of gross area stress and the effective stress concentration factor (SCF_{eff}) due to the hole is calculated by dividing the material strength (unnotched) by the gross area notch strength.

$$\sigma_{notched} = \frac{P_{ult}}{W t} \qquad 7.2.6.4(a)$$

where P_{ult} = ultimate load
W = specimen width
t = specimen thickness

$$SCF_{eff} = \frac{\sigma_{unnotched}}{\sigma_{notched}} \qquad 7.2.6.4(b)$$

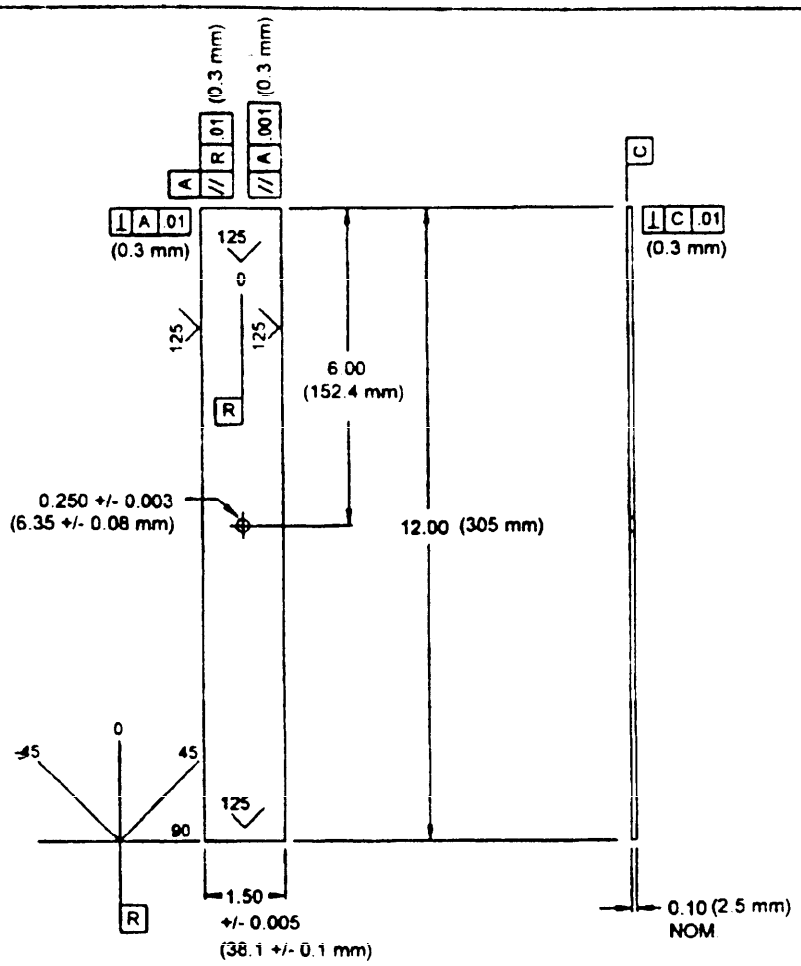
7.2.7 Bearing/by-pass strength.

7.2.7.1 ***Rationale.*** Designs of composite structure containing bolted joints in which the load transfer is greater than 20% of the total load at an individual bolt may require test substantiation. The purpose of this section is to provide guidance on how to obtain these data. Specifically, this section describes specimen geometries, test procedures, and test matrices to sufficiently define experimental lines AC and EC' in Figure 7.2.3 to B-basis significance for the variability and environmental dependence of the material is known *a priori*.

Analytical procedures, *e.g.*, see Reference 7.2.7.1, are being developed to reduce testing requirements. Progress has been made in the net tension/by-pass quadrant (line AC) for the failure mode characterized as net tension. For this failure mode, a good correlation was obtained using linear interaction for combined bearing/by-pass loading.

7.2.7.2 *Specimen design and testing.* Various coupons and text fixtures have been utilized by the aerospace industry to obtain bearing/by-pass strengths. All can be classified into three general categories: (1) passive, (2) independent bolt load, and (3) coupled bolt load/by-pass load. In the

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NOTES:

1. UNLESS NOTED ALL TOLERANCES ARE ±0.100
   EDGE ROUGHNESS IN ACCORDANCE WITH ANSI B46.1
2. HOLE MUST NOT HAVE DELAMINATION OR OTHER DAMAGE.
3. ALL DIMENSIONS IN INCHES. (MILLIMETERS IN PARENTHESES.)
4. CONFIGURATION SHOWN IS FOR 0.25 in. DIAMETER HOLE. FOR ALL OTHER HOLE SIZES, THE WIDTH WOULD CHANGE TO MAINTAIN W/D = 6.

FIGURE 7.2.6.3(a) *Notched tension/compression strength specimen (based on Reference 7.2.6.3).*

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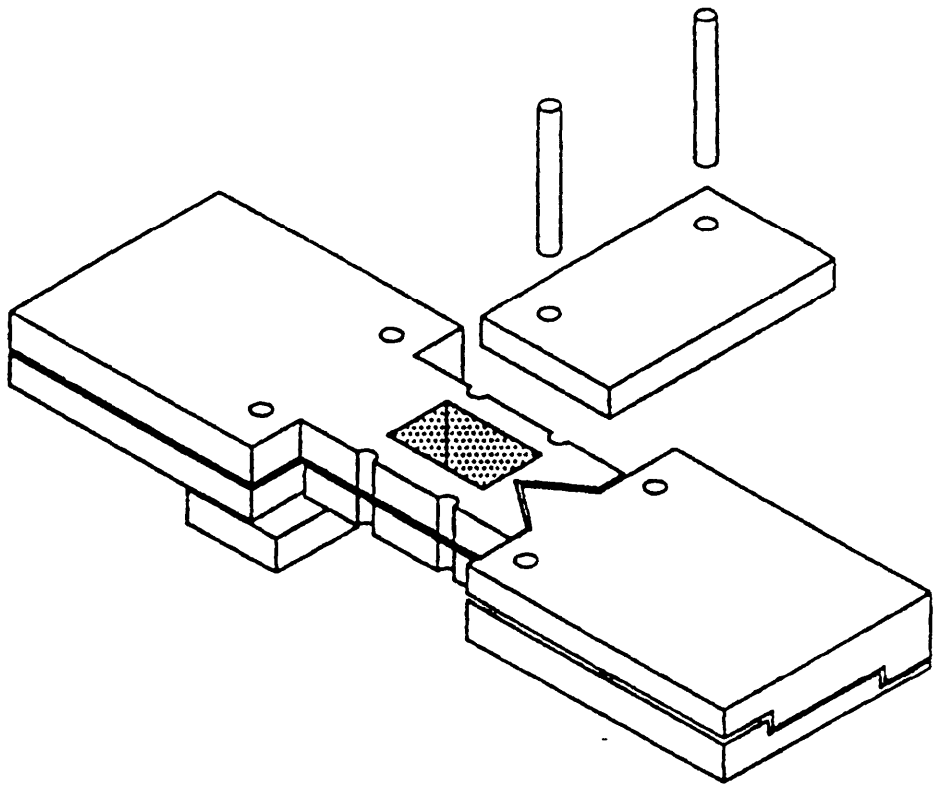


FIGURE 7.2.6.3(b) *Open hole compression fixture assembly (Reference 7.2.6.3).*

passive method, load is transferred through the bolt into an additional strap, as shown in Figure 7.2.7.2(a). The magnitude of the transferred load, and hence the bearing/by-pass ratio, is thus a function of metal strip stiffness and the details of bolt installation. Without a significant amount of strain gaging, it is difficult to establish how much bearing load will be transferred. This method/specimen is not recommended without experimental verification of load transfer parameters. Because of geometrical limitations, this method is most applicable with low load transfer usually not greater than 40%, which may not be where significant interaction effects occur. The major advantage of the passive method is that it does not need special fixturing. The testing itself is equivalent to a standard tension or in-plane stabilized compression test.

In the coupled bolt load/by-pass load method, the bolt is loaded by mechanical linkages attached to the test machine (Figure 7.2.7.2(b)). By locating the vertical link at different locations, different bearing/by-pass ratios can be tested. This ratio will remain constant until failure during each particular test. Because of this constraint and the complexities of test fixturing, this method is also not recommended as the primary method of obtaining bearing/by-pass data.

The recommended test method for bearing/by-pass should load the bolt independently, with the bolt load measured directly, so that the bearing stress can be calculated without resorting to backing out a value from strain gage readings on joining members. Test fixtures to accomplish this require a loading cell(s) separate from the testing machine which complicates the test procedures. Specialized test fixturing has been developed by the industry to synchronize the loading between the bolt and the

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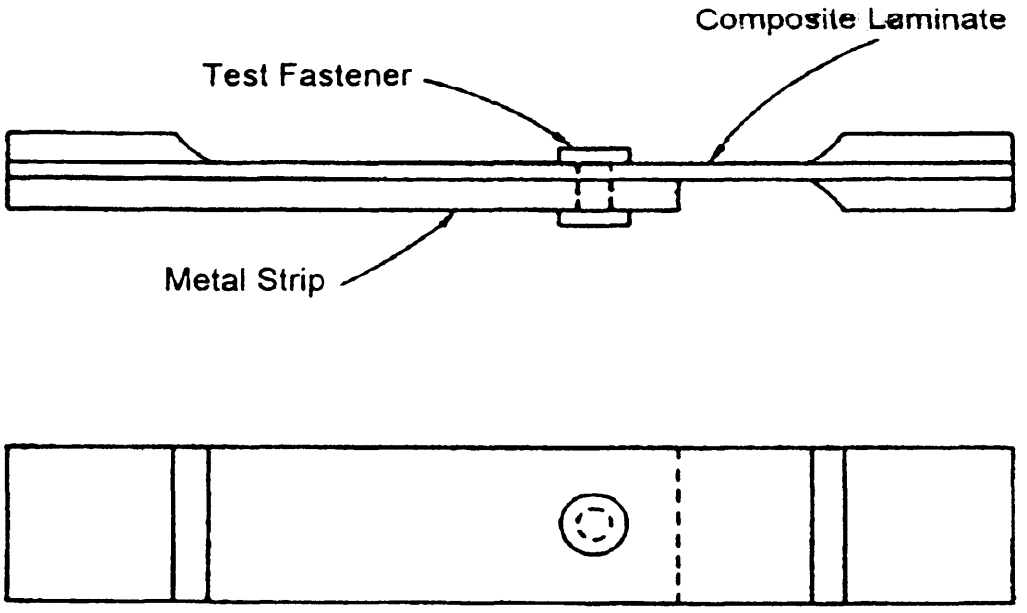


Figure 7.2.7.2(a) Passive by-pass/bearing coupon (single shear).

specimen. One well-documented test system has been developed by the NASA-Langley Research Center (Reference 7.2.7.1). Figure 7.2.7.2(c), taken from this reference, illustrates the complexities of the fixturing. The coupon from Reference 7.2.7.1 is shown in Figure 7.2.7.2(d), modified with an additional hole to alert the tester if any shear-out failures occurred. It is typical of all independently loaded test systems in the industry. It should be noted that for compression loading, the specimen is stabilized to prevent buckling.

7.2.7.3 *Test matrix*. The minimum testing requirements necessary to construct the bearing/by-pass interaction plot of Figure 7.2.3 for a particular polymeric composite material are outlined in Table 7.2.7.3. The test matrix assumes that the end points (A and E) have been or will be obtained from no bolt load notch tension/compression tests recommended in Section 7.2.6. It also assumes that the points C and C' are obtained from the bearing strength tests enumerated in Section 7.2.5 for single shear and Section 7.2.4 for double shear. As no environmental tests other that at room temperature have been specified in Table 7.2.7.3, the environmental effects on the bearing/by-pass strength are to be deduced from the interaction curves' endpoints. The laminate called out are the same as in Section 7.2.5 and 7.2.6. For completeness, the laminate lay-ups should be as follows: $[+45/0/-45/(\pm 45)_3/90]_{ns}$ for 10/80/10, $[+45/0/-45/90]_{ns}$ for 25/50/25, and $[+45/0/-45/90/0_2/+45/0]_{ns}$ for 50/40/10.

The test specimen and procedures to fulfill the test requirements of Table 7.2.7.3 should use an individually loaded bolt method such as described in Reference 7.2.7.1 and shown in Figures 7.2.7.2(c) and 7.2.7.2(d), or similar. The test matrix can be applied to either a single shear or double shear joint. In the event that both types of joints exist in the structure the test matrix should be repeated.

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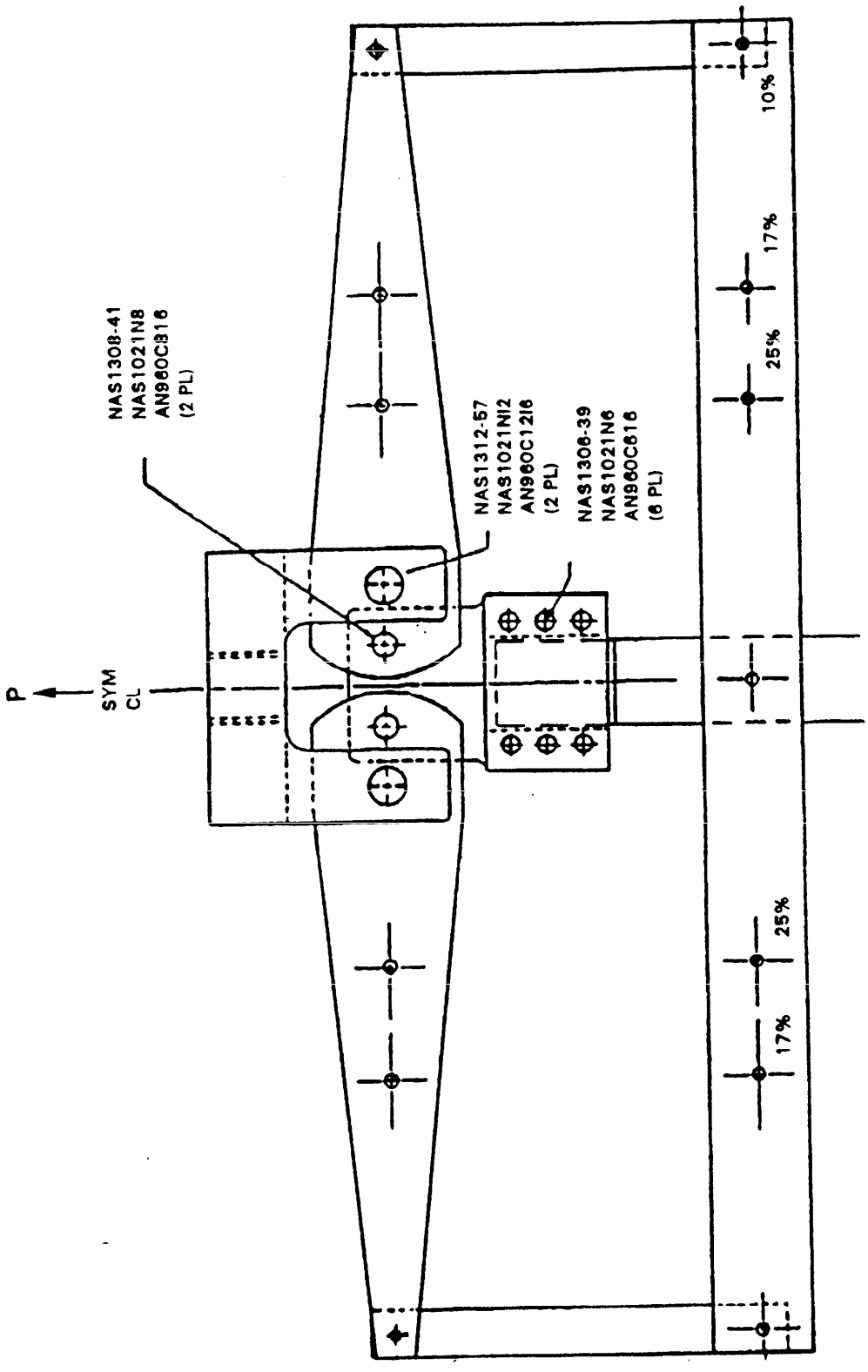


Figure 7.2.7.2(b) *Bolt bearing by-pass test fixture.*

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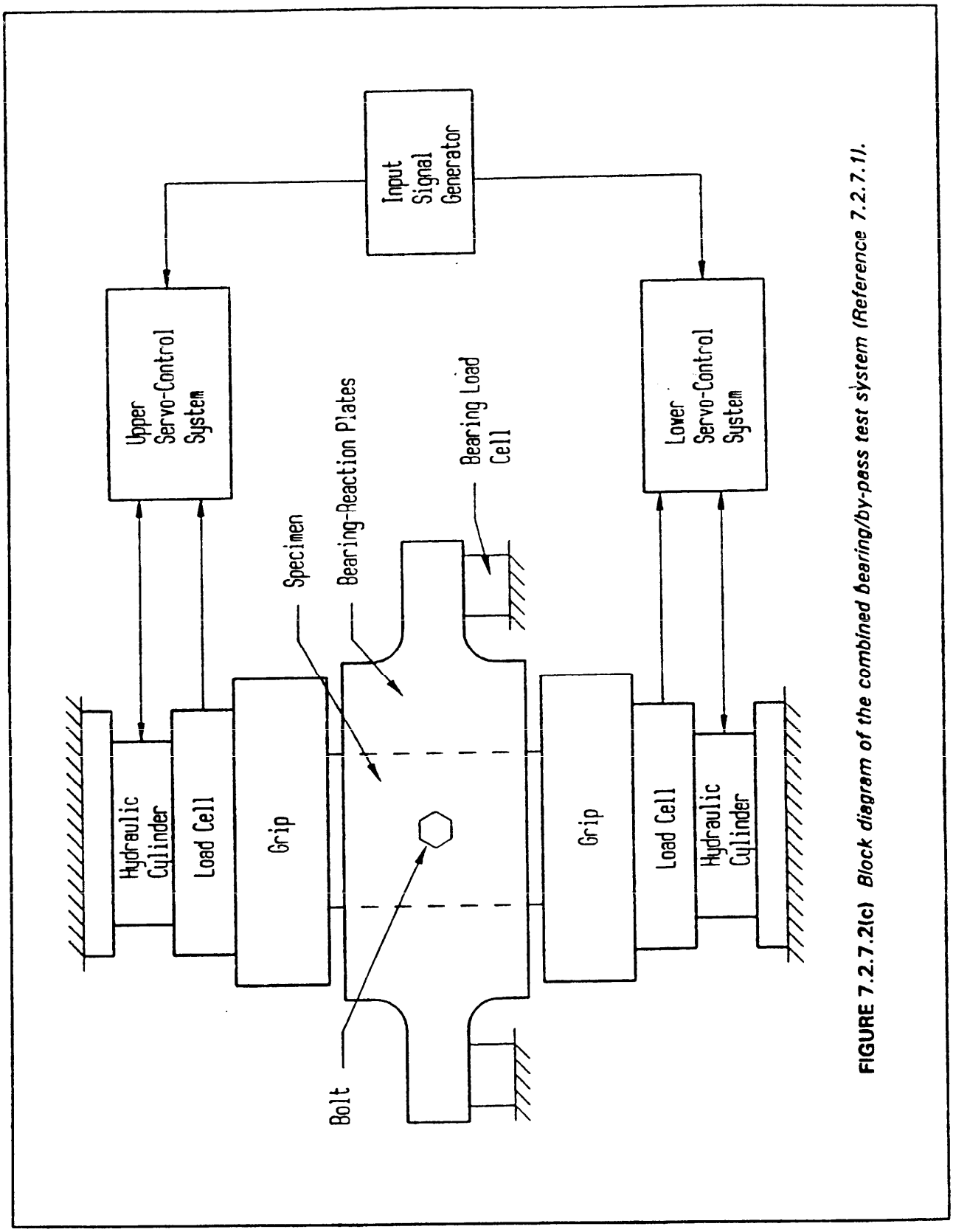
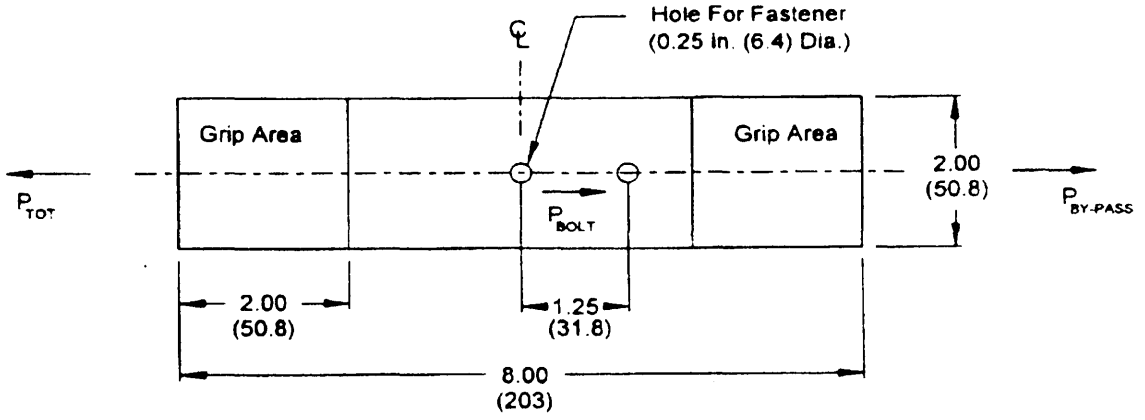


FIGURE 7.2.7.2(c) *Block diagram of the combined bearing/by-pass test system (Reference 7.2.7.1).*

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All Dimensions in Inches (Millimeters in Parentheses).

Figure 7.2.7.2(d) *Specimen for bearing/by-pass test.*

7.2.7.4 *Data reduction.* The data reduction procedures of Sections 7.2.5.4 and 7.2.6 are applicable to bearing/by-pass tests. The bolt load versus displacement plot should be obtained as for the bearing test. In addition, total of by-pass load must be recorded. Failure mode must also be described.

7.2.8 Shear-out strength. The shear-out strength of a material is its ability to withstand shear-out failure of the type shown in Figure 7.2.2. Composite joints are usually designed to avoid this mode of failure. However, by reducing the edge distance from the typical value of three times the fastener diameter (3D), the bearing coupons of Sections 7.2.4 and 7.2.5 can be induced to fail by shear-out. Thus these specimens and procedures are used to determine the joint shear-out strength. The shear-out strength is calculated as P/2et based on the gross section. Definitions of e, D, and t are described in Figure 7.2.8(a).

TABLE 7.2.7.3 *Bearing/by-pass test matrix.*

Lay-up	Environment (Temp/% Moist)	Tension		Compression		Total No. of Tests
		Bearing/By-pass ratio		Bearing/By-pass ratio		
		0.75	0.50	0.75	0.50	
10/80/10	RT/ambient	5	5	5	5	20
25/50/25	RT/ambient	5	5	5	5	20
50/40/10	RT/ambient	5	5	5	5	20
TOTAL		15	15	15	15	60

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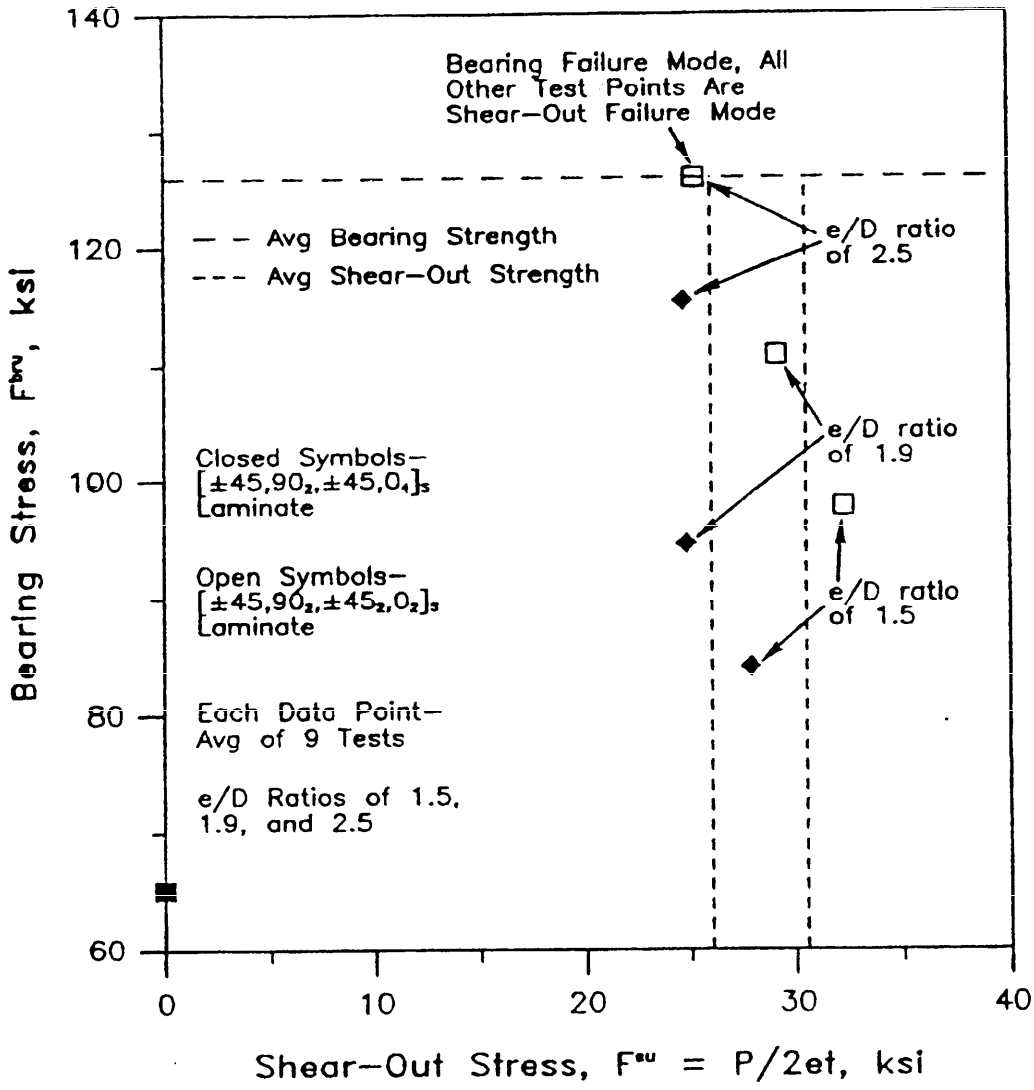


FIGURE 7.2.8(b) *Bearing/shear-out test failure data as a function of bearing and shear-out stresses - AS1/3501-5A, RT, ambient.*

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Deflection Measurement - The measuring device used shall be an averaging, differential transformer extensometer or equivalent and used in conjunction with an autographic recorder and shall have an accuracy of 0.5% of indicated joint deflection at loads equivalent to 70% of the anticipated joint's strength and be calibrated per ASTM E 83. Load and deflection ranges shall be used that give the initial part of the load-extension curve a slope between 45° and 60°. Load and deflection ranges and scales shall be held constant for each test group (test group is defined as specimens of the same configuration, fastener type and size and their baseline counterparts). See Figure 7.2.9.2(a) for a typical load deflection curve.

Test Fixture - The test fixture for the screening test shall be a tension fixture type as described in Figure 7.2.9.2(b) capable of transmitting compression loads to the test specimen. The fixtures shall be parallel within 15 minutes of arc and capable of loading the specimen to fastener failure without experiencing local compressive deformation. The test schematic for a more structure-representative test is shown in Figure 7.2.9.2(c).

7.2.9.3 *Test specimen*. Configuration - Test specimen configuration shall be in accordance with Figure 7.2.9.3(a) or 7.2.9.3(b) and 7.2.9.3(c) for testing or comparing fastener performance characteristics.

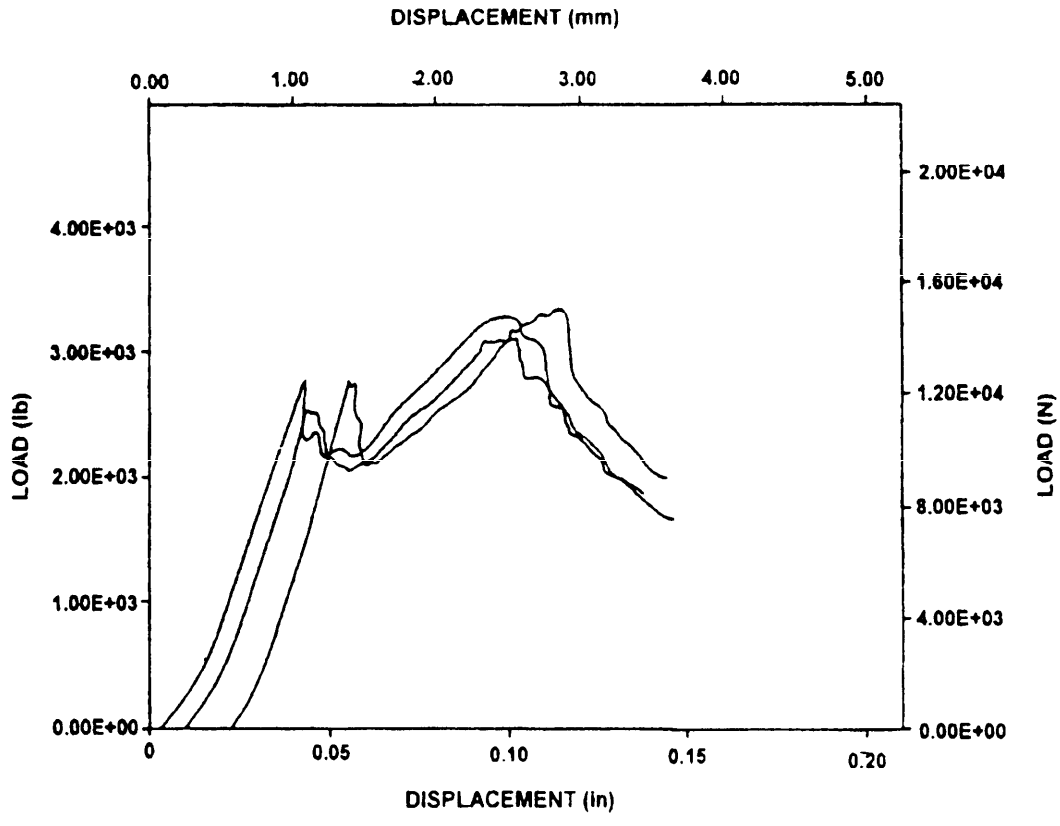
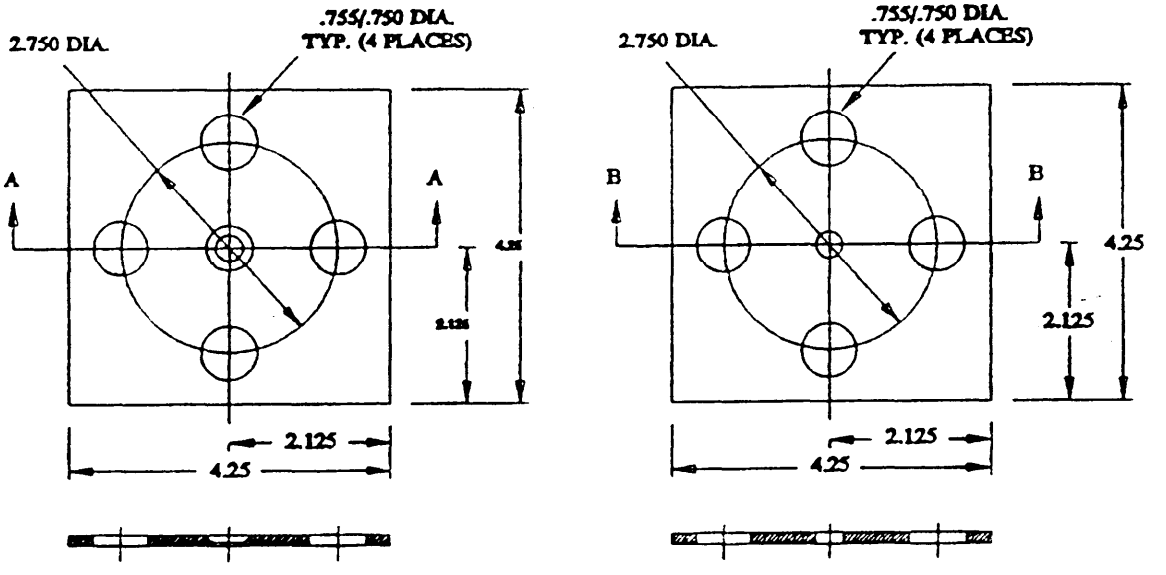


Figure 7.2.9.2(a) *Typical load deflection curve.*

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A-A SECTION (TOP COUPON)

B-B SECTION (BOTTOM COUPON)

100° TENSION HEAD

SHANK DIAMETER	T (1)
5/32	.096
3/16	.111
1/4	.150
5/16	.190
3/8	.229

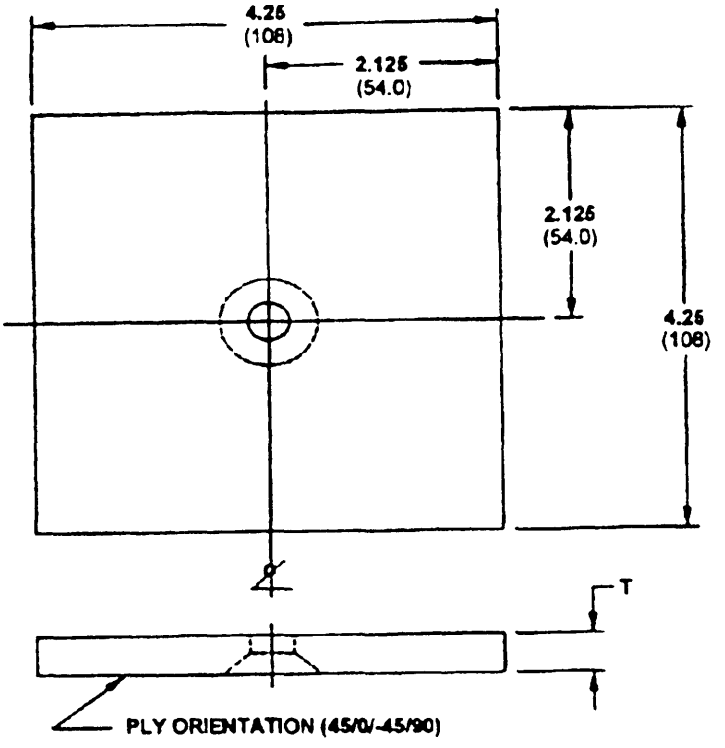
130° SHEAR HEAD

SHANK DIAMETER	T (1)
5/32	.055
3/16	.061
1/4	.080
5/16	.106
3/8	.127

(1) NOTES: "T" IS A SUGGESTED MINIMUM SPECIMEN THICKNESS FOR TENSILE TESTING 100° & 130° FLUSH HEAD FASTENERS. THICKNESS DIMENSIONS REPRESENT STANDARD DESIGN CRITERIA THAT ALLOW THE COUNTERSINK TO PENETRATE A MAXIMUM DEPTH EQUAL TO 70% OF THE SHEET THICKNESS.

Figure 7.2.9.3(a) *Fastener pull-thru test plate.*

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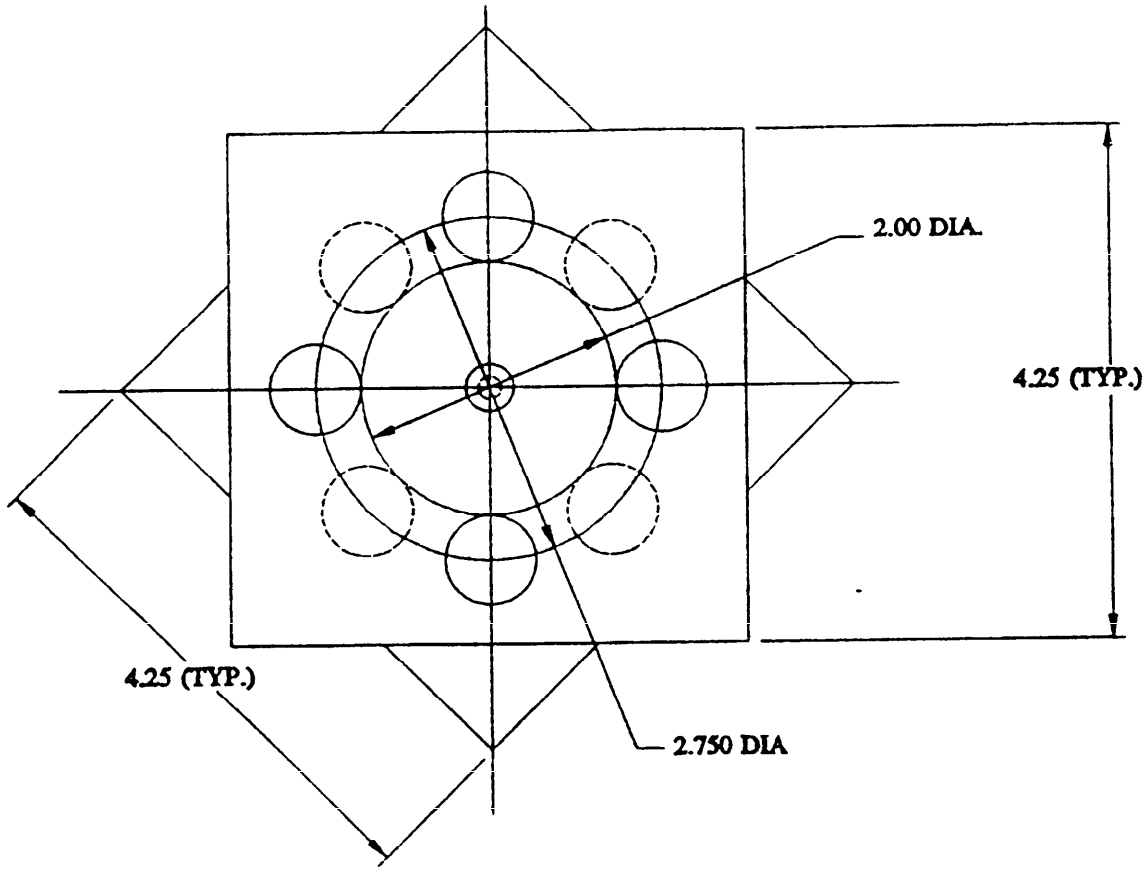
NOMINAL FASTENER SHANK DIA.	100° (1) TENSION HEAD "T"	130° (1) SHEAR HEAD "T"	CLEARANCE HOLE
5/32	0.096	0.055	1.312
3/16	0.111	0.061	1.500
1/4	0.150	0.080	2.000
5/16	0.190	0.106	2.500
3/8	0.229	0.127	3.000

(1) NOTES: "T" IS A SUGGESTED MINIMUM SPECIMEN THICKNESS FOR TENSILE TESTING 100° & 130° FLUSH HEAD FASTENERS. THICKNESS DIMENSIONS REPRESENT STANDARD DESIGN CRITERIA THAT ALLOW THE COUNTERSINK TO PENETRATE A MAXIMUM DEPTH EQUAL TO 70° OF THE SHEET THICKNESS.

ALL DIMENSIONS IN INCHES (MILLIMETERS)

Figure 7.2.9.3(b) *Fastener pull-thru test plate.*

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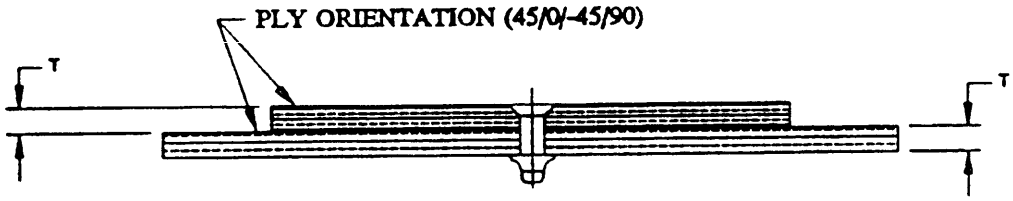


Figure 7.2.9.3(c) *Pull-thru test specimen configuration.*

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7.2.9.5 *Test matrix*. A test matrix to be used with the Figure 7.2.9.2(b) test for one fastener type and head configuration is shown in Table 7.2.9.5. The testing should be performed at room temperature, ambient and hot, wet conditions. The latter is defined as the highest temperature and moisture content for the composite material (see Section 2.2). The test matrix is to be repeated for a different fastener, head or tail configuration, and installation hole clearance. As used in Table 7.2.9.5, Class 1 is reserved for interference fit, Class 2 for aircraft quality, usually +0.003 in., and Class 3 for clearance fit.

A test matrix similar to Table 7.2.9.5 should be constructed for the Figure 7.2.9.2(c) test. However, as the test is more design-oriented, fewer variables need to be tested. The replication of 5 should be maintained.

7.2.10 Fastener-in-composite qualification tests. A first step in design of composite bolted joints is the identification of fasteners that are suitable for use with composites. The data generated by tests outlined in this section will provide a realistic basis for selection as the tests will give a good estimate of joint strength. Composites require fasteners with larger tail footprints (than metals), especially for blind fasteners; the tests described here will interrogate this feature. After fastener selection additional test data, enumerated in Sections 7.2.5 and 7.2.9, will be needed to design bolted joints for other laminates and failure modes which are not a function of specific fastener characteristics.

The test requirements and methods have been extracted from Sections 7.2.5 and 7.2.9, thus details of testing procedures can be obtained from those sections. Testing for fastener-in-composite qualification uses only one laminate lay-up (quasi-isotropic), but more than one thickness. Also, the testing is limited to room temperature as the environment is not a driver for fasteners as it is for composites The test program is based on the assumption that the plates to be joined are both composites. If the particular fastener is also intended to be used in metal/composite combinations, testing should be performed for that configuration. The test matrices reflect two properties most affected by fastener properties: joint bearing and pull-thru strengths. It is suggested that pull-thru tests be conducted first to determine the suitability of the fastener for composites. Once that property is satisfactory, the more expensive bearing tests can be undertaken. It is also a requirement that 25% of bearing and pull-thru tests be tested by someone other than the manufacturer of the fastener. For inclusion of data in the handbook, the fastener must be in-use by a least one airframer. For completeness, test requirements for fastener shear and tension strengths are included here, although these properties are independent of joining members.

Fasteners for use with carbon fiber composites should be titanium, A286 CRES or Monel to reduce the potential for galvanic corrosion. This limitation does not apply to aramid or fiberglass composites.

The data generated by the test program presented here will not be sufficient by itself to qualify a fastener for use in aircraft structures. Fatigue testing, manufacturing tolerances studies (grip lengths, seating angles, hole diameters) are the other criteria which have to be satisfied to complete fastener qualification requirements.

7.2.10.1 *Fastener shear tests*. These test are conducted using steel plates per MIL-STD1312, Test 13 for double shear and Test 20 for single shear. Evidence of previous valid qualification tests could be accepted here.

7.2.10.2 *Fastener tension tests*. These tests are conducted in steel plates per MIL-STD1312, Test 8. Evidence of previous valid qualification tests could be accepted here.

7.2.10.3 *Pull-thru*. Test specimen configuration to determine pull-thru strength shall be in accordance with Figures 7.2.9.3(a), (b), and (c). The test procedures are given in Section 7.2.9. The

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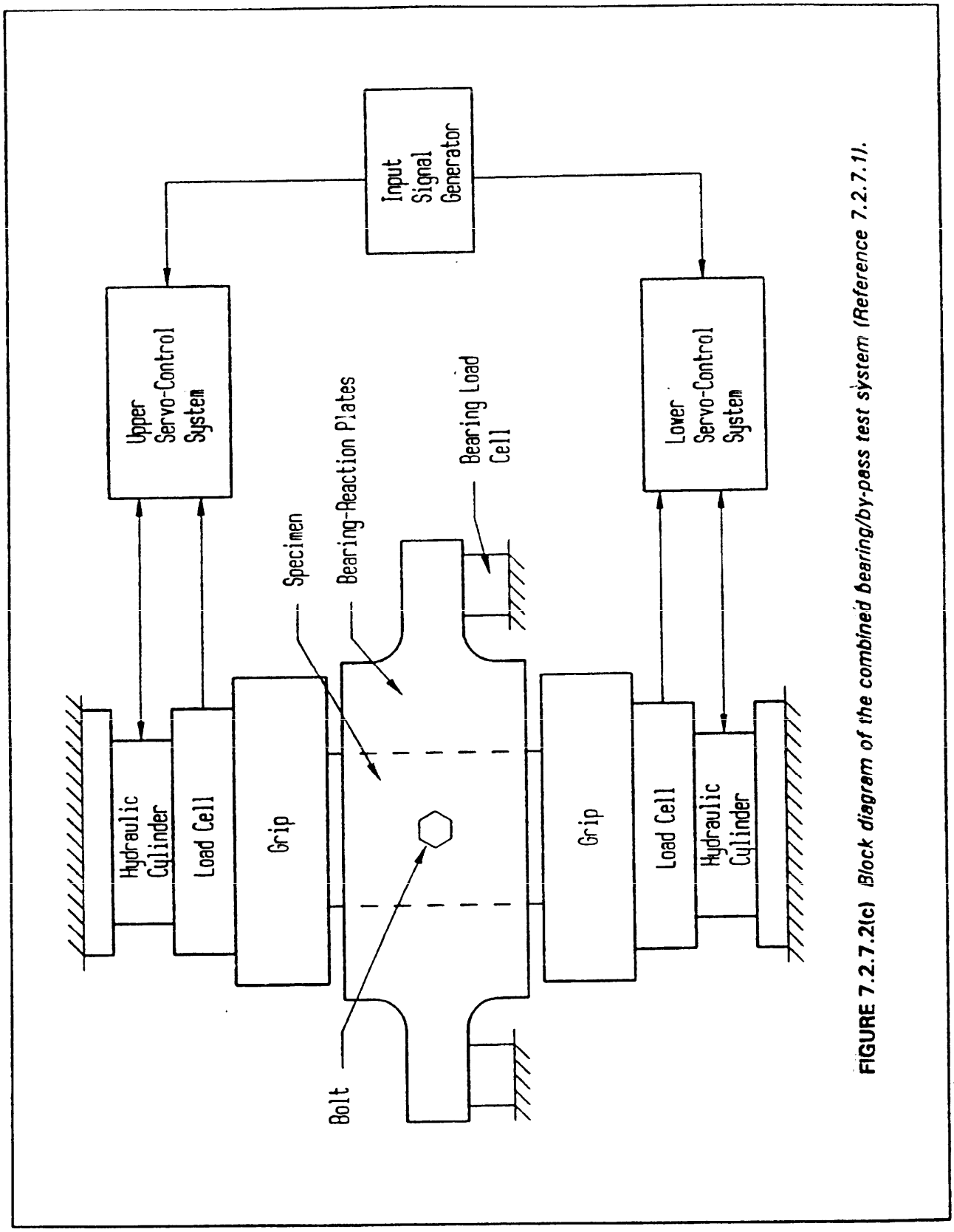
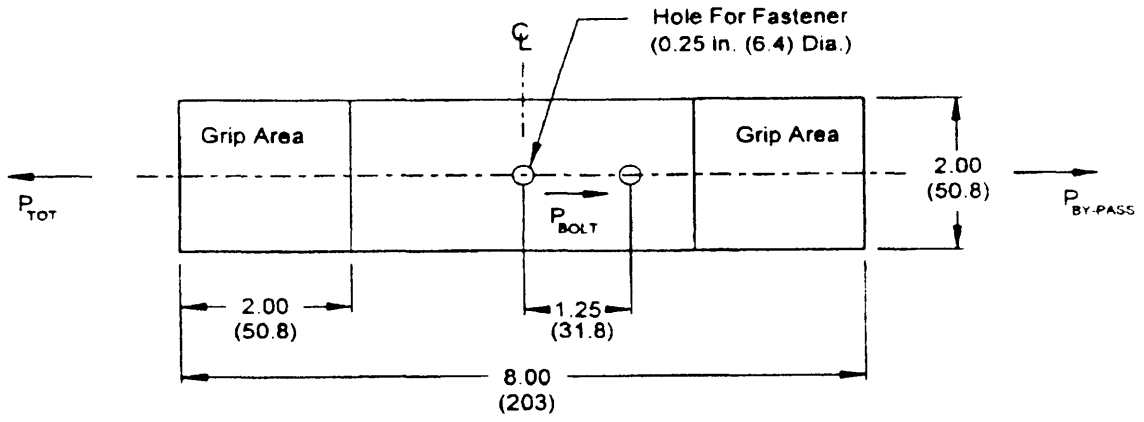


FIGURE 7.2.7.2(c) *Block diagram of the combined bearing/by-pass test system (Reference 7.2.7.1).*

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All Dimensions in Inches (Millimeters in Parentheses).

Figure 7.2.7.2(d) *Specimen for bearing/by-pass test.*

7.2.7.4 *Data reduction.* The data reduction procedures of Sections 7.2.5.4 and 7.2.6 are applicable to bearing/by-pass tests. The bolt load versus displacement plot should be obtained as for the bearing test. In addition, total of by-pass load must be recorded. Failure mode must also be described.

7.2.8 Shear-out strength. The shear-out strength of a material is its ability to withstand shear-out failure of the type shown in Figure 7.2.2. Composite joints are usually designed to avoid this mode of failure. However, by reducing the edge distance from the typical value of three times the fastener diameter (3D), the bearing coupons of Sections 7.2.4 and 7.2.5 can be induced to fail by shear-out. Thus these specimens and procedures are used to determine the joint shear-out strength. The shear-out strength is calculated as P/2et based on the gross section. Definitions of e, D, and t are described in Figure 7.2.8(a).

TABLE 7.2.7.3 *Bearing/by-pass test matrix.*

Lay-up	Environment (Temp/% Moist)	Tension		Compression		Total No. of Tests
		Bearing/By-pass ratio		Bearing/By-pass ratio		
		0.75	0.50	0.75	0.50	
10/80/10	RT/ambient	5	5	5	5	20
25/50/25	RT/ambient	5	5	5	5	20
50/40/10	RT/ambient	5	5	5	5	20
TOTAL		15	15	15	15	60

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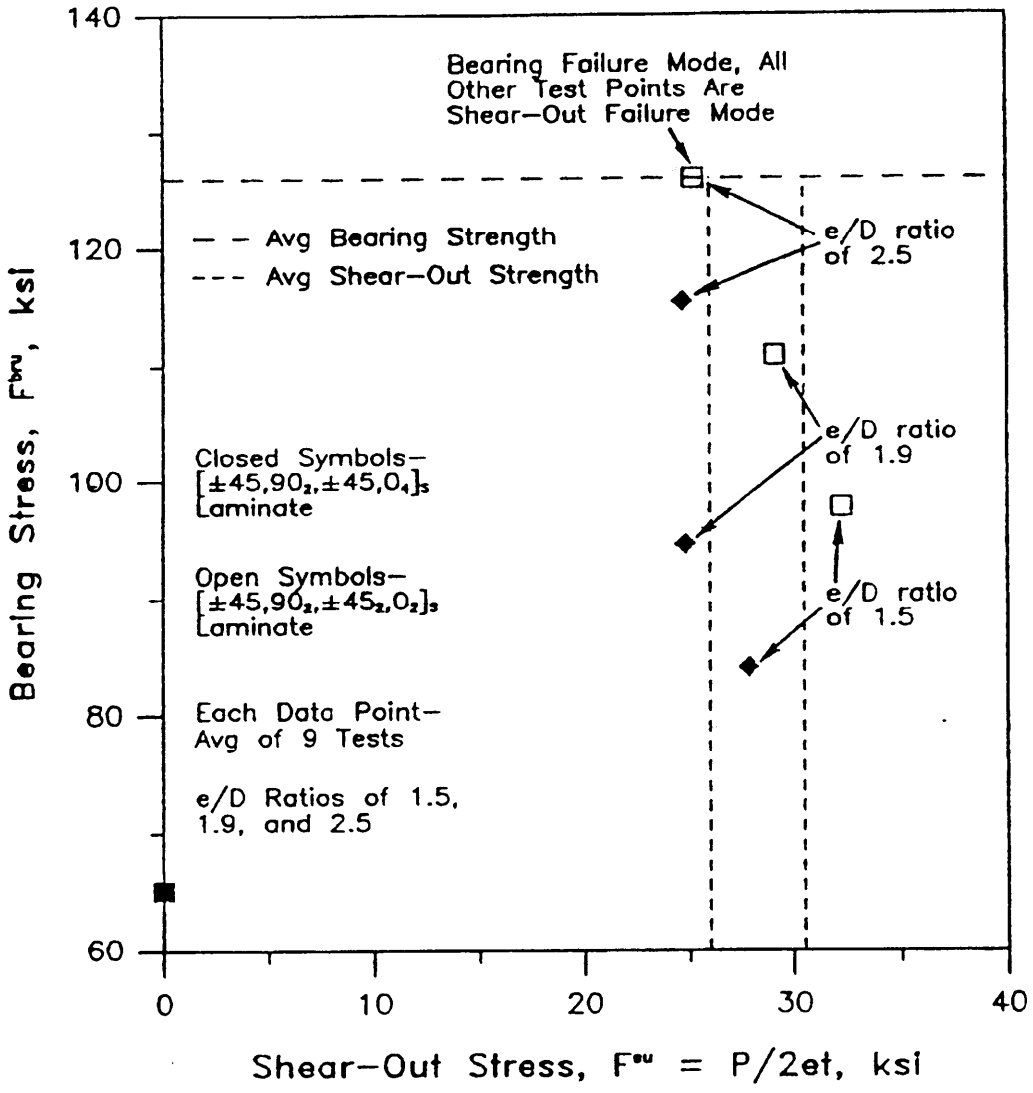


FIGURE 7.2.8(b) *Bearing/shear-out test failure data as a function of bearing and shear-out stresses - AS1/3501-5A, RT, ambient.*

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Deflection Measurement - The measuring device used shall be an averaging, differential transformer extensometer or equivalent and used in conjunction with an autographic recorder and shall have an accuracy of 0.5% of indicated joint deflection at loads equivalent to 70% of the anticipated joint's strength and be calibrated per ASTM E 83. Load and deflection ranges shall be used that give the initial part of the load-extension curve a slope between 45° and 60°. Load and deflection ranges and scales shall be held constant for each test group (test group is defined as specimens of the same configuration, fastener type and size and their baseline counterparts). See Figure 7.2.9.2(a) for a typical load deflection curve.

Test Fixture - The test fixture for the screening test shall be a tension fixture type as described in Figure 7.2.9.2(b) capable of transmitting compression loads to the test specimen. The fixtures shall be parallel within 15 minutes of arc and capable of loading the specimen to fastener failure without experiencing local compressive deformation. The test schematic for a more structure-representative test is shown in Figure 7.2.9.2(c).

7.2.9.3 *Test specimen*. Configuration - Test specimen configuration shall be in accordance with Figure 7.2.9.3(a) or 7.2.9.3(b) and 7.2.9.3(c) for testing or comparing fastener performance characteristics.

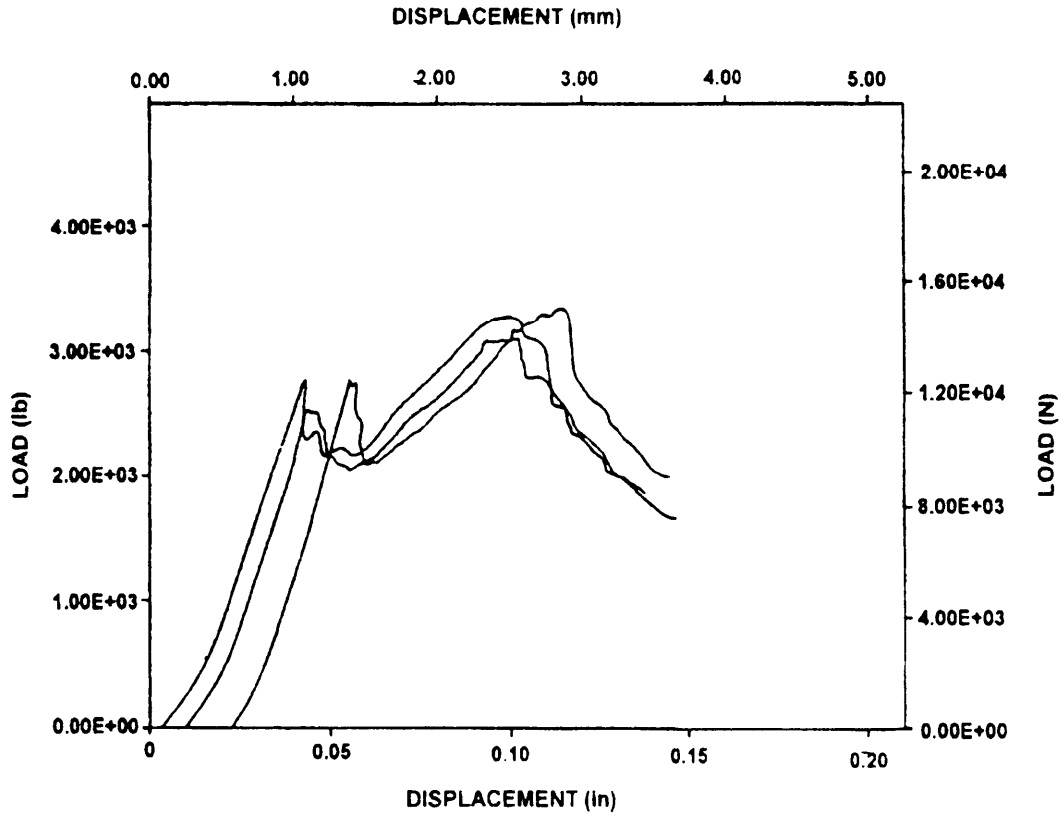
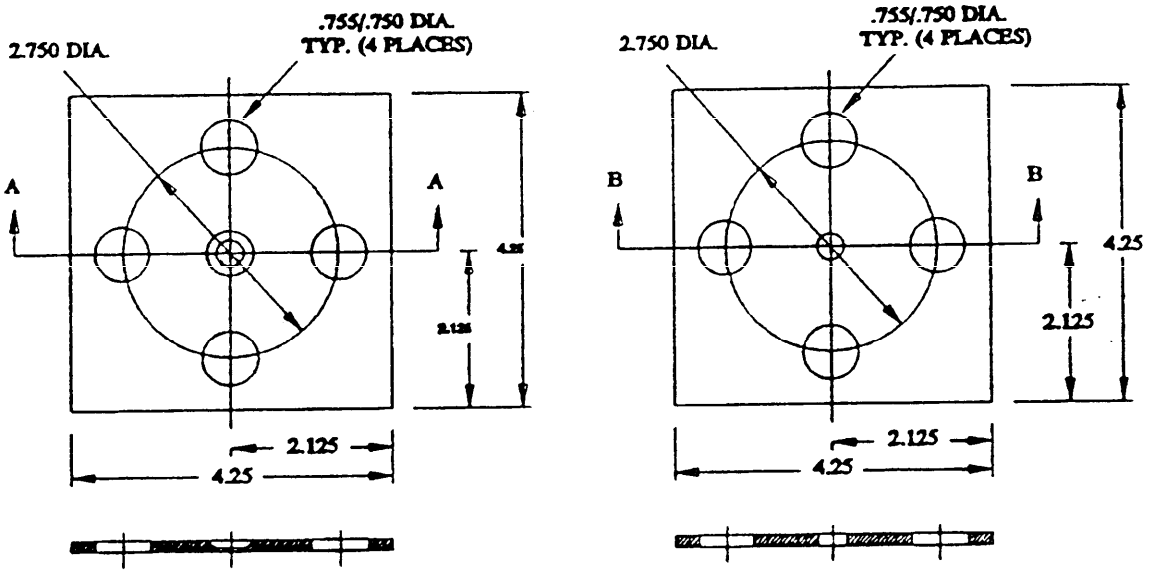


Figure 7.2.9.2(a) *Typical load deflection curve.*

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100° TENSION HEAD

SHANK DIAMETER	T (1)
5/32	.096
3/16	.111
1/4	.150
5/16	.190
3/8	.229

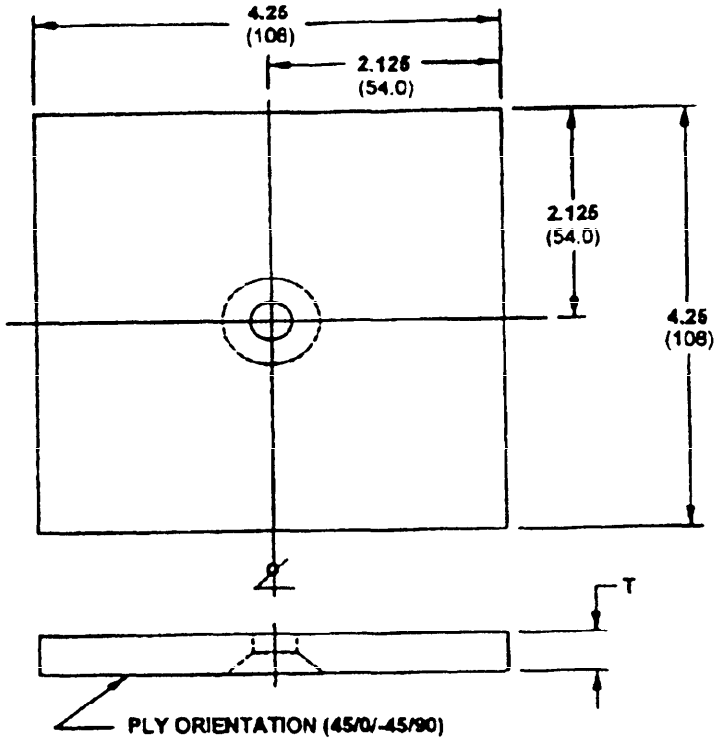
130° SHEAR HEAD

SHANK DIAMETER	T (1)
5/32	.055
3/16	.061
1/4	.080
5/16	.106
3/8	.127

(1) NOTES: "T" IS A SUGGESTED MINIMUM SPECIMEN THICKNESS FOR TENSILE TESTING 100° & 130° FLUSH HEAD FASTENERS. THICKNESS DIMENSIONS REPRESENT STANDARD DESIGN CRITERIA THAT ALLOW THE COUNTERSINK TO PENETRATE A MAXIMUM DEPTH EQUAL TO 70% OF THE SHEET THICKNESS.

Figure 7.2.9.3(a) *Fastener pull-thru test plate.*

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NOMINAL FASTENER SHANK DIA.	100° (1) TENSION HEAD "T"	130° (1) SHEAR HEAD "T"	CLEARANCE HOLE
5/32	0.096	0.055	1.312
3/16	0.111	0.061	1.500
1/4	0.150	0.080	2.000
5/16	0.190	0.106	2.500
3/8	0.229	0.127	3.000

(1) NOTES: "T" IS A SUGGESTED MINIMUM SPECIMEN THICKNESS FOR TENSILE TESTING 100° & 130° FLUSH HEAD FASTENERS. THICKNESS DIMENSIONS REPRESENT STANDARD DESIGN CRITERIA THAT ALLOW THE COUNTERSINK TO PENETRATE A MAXIMUM DEPTH EQUAL TO 70° OF THE SHEET THICKNESS.

ALL DIMENSIONS IN INCHES (MILLIMETERS)

Figure 7.2.9.3(b) *Fastener pull-thru test plate.*

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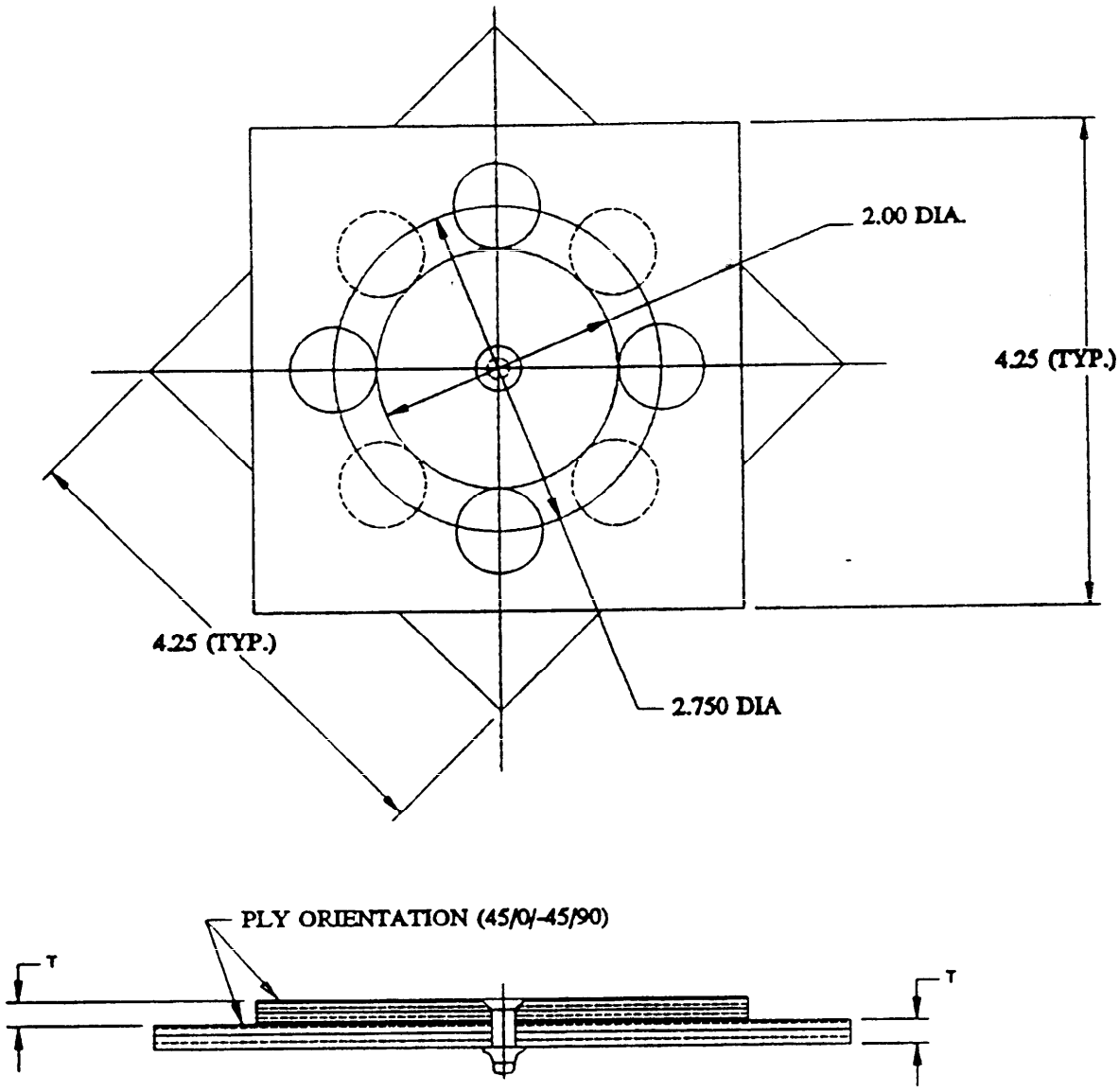


Figure 7.2.9.3(c) *Pull-thru test specimen configuration.*

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7.2.9.5 *Test matrix*. A test matrix to be used with the Figure 7.2.9.2(b) test for one fastener type and head configuration is shown in Table 7.2.9.5. The testing should be performed at room temperature, ambient and hot, wet conditions. The latter is defined as the highest temperature and moisture content for the composite material (see Section 2.2). The test matrix is to be repeated for a different fastener, head or tail configuration, and installation hole clearance. As used in Table 7.2.9.5, Class 1 is reserved for interference fit, Class 2 for aircraft quality, usually +0.003 in., and Class 3 for clearance fit.

A test matrix similar to Table 7.2.9.5 should be constructed for the Figure 7.2.9.2(c) test. However, as the test is more design-oriented, fewer variables need to be tested. The replication of 5 should be maintained.

7.2.10 Fastener-in-composite qualification tests. A first step in design of composite bolted joints is the identification of fasteners that are suitable for use with composites. The data generated by tests outlined in this section will provide a realistic basis for selection as the tests will give a good estimate of joint strength. Composites require fasteners with larger tail footprints (than metals), especially for blind fasteners; the tests described here will interrogate this feature. After fastener selection additional test data, enumerated in Sections 7.2.5 and 7.2.9, will be needed to design bolted joints for other laminates and failure modes which are not a function of specific fastener characteristics.

The test requirements and methods have been extracted from Sections 7.2.5 and 7.2.9, thus details of testing procedures can be obtained from those sections. Testing for fastener-in-composite qualification uses only one laminate lay-up (quasi-isotropic), but more than one thickness. Also, the testing is limited to room temperature as the environment is not a driver for fasteners as it is for composites The test program is based on the assumption that the plates to be joined are both composites. If the particular fastener is also intended to be used in metal/composite combinations, testing should be performed for that configuration. The test matrices reflect two properties most affected by fastener properties: joint bearing and pull-thru strengths. It is suggested that pull-thru tests be conducted first to determine the suitability of the fastener for composites. Once that property is satisfactory, the more expensive bearing tests can be undertaken. It is also a requirement that 25% of bearing and pull-thru tests be tested by someone other than the manufacturer of the fastener. For inclusion of data in the handbook, the fastener must be in-use by a least one airframer. For completeness, test requirements for fastener shear and tension strengths are included here, although these properties are independent of joining members.

Fasteners for use with carbon fiber composites should be titanium, A286 CRES or Monel to reduce the potential for galvanic corrosion. This limitation does not apply to aramid or fiberglass composites.

The data generated by the test program presented here will not be sufficient by itself to qualify a fastener for use in aircraft structures. Fatigue testing, manufacturing tolerances studies (grip lengths, seating angles, hole diameters) are the other criteria which have to be satisfied to complete fastener qualification requirements.

7.2.10.1 *Fastener shear tests*. These test are conducted using steel plates per MIL-STD1312, Test 13 for double shear and Test 20 for single shear. Evidence of previous valid qualification tests could be accepted here.

7.2.10.2 *Fastener tension tests*. These tests are conducted in steel plates per MIL-STD1312, Test 8. Evidence of previous valid qualification tests could be accepted here.

7.2.10.3 *Pull-thru*. Test specimen configuration to determine pull-thru strength shall be in accordance with Figures 7.2.9.3(a), (b), and (c). The test procedures are given in Section 7.2.9. The

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test matrix, Table 7.2.10.3, requires testing for three different diameters representative of the applicability of the fastener. One diameter should be 0.25 in. This may require adjustments in laminate thickness; however, the laminate lay-up must be maintained as $(45/0/-45/90)_{ns}$. The test matrix is to be repeated for each fastener under consideration.

TABLE 7.2.10.3 *Fastener qualification pull-thru test matrix (1).*

Geometry	Composite Sheet Thickness (in)	Lay-Up	Fastener Nominal Shank Diameter	Number of Tests (2)
Composite to Composite	0.190(3) Head Side	25/50/25	0.25	5
	0.120(3) Tail Side	25/50/25	0.25	5
	t2 Head Side	25/50/25	D2	5
	t2 Tail Side	25/50/25	D2	5
	t3 Head Side	25/50/25	D3	5
	t3 Tail Side	25/50/25	D3	5

Notes: (1) All tests to be performed at RT/ambient and with installation hole Class 2.
(2) Each grip condition where applicable (see Section 7.2.9.4).
(3) May be different for other diameters.

7.2.10.4 *Bearing tests.* The composite-to-composite two bolt bearing specimen geometry shown in Figure 7.2.5.2(a) is suggested. This single shear configuration is more representative of multi-fastener joints found in the industry. With an acceptable fastener composite bearing failure should be achieved, although secondary fastener rotation about its longitudinal axis may be evident. The test matrix for fastener qualification is shown in Table 7.2.10.4. Three different thicknesses of one lay-up $(45/0/-45/90)_{ns}$ and three fastener diameters are suggested. One diameter should be 0.25 in. and the other two reflecting the range of available fastener sizes. Selection of additional thicknesses of the composite members should stay within these guidelines to assure maximum usefulness of data: (1) $0.8 < D/t < 2$ and (2) countersink depth shall not exceed 0.67 of total laminate thickness. The goal of the tests is to obtain a family of three curves of bearing stress versus D/t ratio for each diameter tested. There should be 15 data tests for each diameter. The test matrix is to be repeated for each fastener under consideration.

7.2.10.5 *Data presentation.* Data presentation should follow the guidelines of Chapter 8. Additionally, bearing data should be presented as plots of bearing strength vs. D/t for each diameter tested.

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TABLE 7.2.10.4 *Fastener qualification bearing test matrix (1).*

Geometry	Thickness (in)	Lay-Up	Bolt Diameter (in)	Number of Tests
Composite	0.2	25/50/25	0.25	5
to	0.2	25/50/25	D2	5
Composite	0.2	25/50/25	D3	5
Composite	t2	25/50/25	0.25	5
to	t2	25/50/25	D2	5
Composite	t2	25/50/25	D3	5
Composite	t3	25/50/25	0.25	5
to	t3	25/50/25	D2	5
Composite	t3	25/50/25	D3	5

Note: (1) All tests are to be performed at RT, ambient.

7.3 BONDED JOINTS.

7.4 OTHER TOPICS.

7.4.1 Compression after impact tests.

7.4.1.1 *Overview.* The compression after impact (CAI) test is an empirical evaluation of the degradation of laminate compressive strength due to out-of-plane impact. Investigators use many different impact and damage tolerance tests depending on material form, application and expected damage. Although the CAI tests proposed here were developed by the airframe industry for comparing the damage tolerance of candidate composite materials, they may be generally applicable to other industries. The possible damage scenarios the test was designed to simulate include dropped tools, runway debris kickup, etc. Because the impact is relatively low velocity, the test is not commonly used to assess ballistic damage tolerance.

Several methods are commonly used in the composite industries to determine CAI. Though none are currently ASTM standards all of the methods involve impacting a flat laminate plate. The plate is constrained by a support system with a cutout opposite to the impact site. The impactor is normally a hemispherical tup (falling dart, rod or ball). The most common methods are SACMA SRM 2R-94 (Reference 7.4.1.1(a)) and NASA 1092 and 1142, B.11 (References 7.4.1.1(b) and (c)).

Sandwich panels are also commonly evaluated for damage tolerance. There are currently no industry wide standards for CAI on sandwich panels. However, for qualification or screening tests, many firms impact a flat sandwich panel under controlled conditions and then perform an evaluation. This evaluation may consist of NDI, water intrusion, residual compressive strength or shear strength testing. A more detailed discussion may be found in "Impact Damage Tolerance Testing of Bonded Sandwich Panels" by Caldwell et al (Reference 7.4.1.1(d)).

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The impact level is generally selected to cause visual damage to the laminate, but such that the damage is localized at the center of the plate. Other levels of damage such as "barely visible impact damage" (BVID) have been used. If the damage extends to over one half the width of the specimen or if the impactor penetrates through the laminate, the damage level is too large to meaningfully evaluate with a subsequent compression test. Impact levels are specified in the methods but may be varied for experimental purposes.

After impact, the level of damage may be characterized by the apparent damage area (front and back), indentation depth, and nondestructive evaluation by ultrasonic C-Scan or similar techniques. Prior to compression testing, NASA methods require an additional machining step to reduce the specimen size and insure the ends are flat and parallel. Compression testing is then performed in a fixture that stabilizes the specimen near the edges, but does not constrain transverse deformation due to Poisson's effect.

Limitations of CAI testing (all methods) are as follows:

1. Materials with differing thicknesses or lay-ups should not be directly compared.

2. Users should be cautioned that damage mechanisms in these test specimens may not scale up to larger parts. This is particularly true with composites made from toughened resin systems.

3. The level of impact damage is dependent on the rigidity of the specimen support system during impact. Lab to lab variation may be encountered due to differing support systems. Generally less rigid support will result in less impact damage and higher CAI strength.

4. There may be variation among testers regarding the impact mass used to obtain a given energy level. Data and theoretical models are not sufficient to state the significance of varying mass/velocity at a given energy.

5. Reliable results are not obtained if the failure is not in the impact area. Soft laminates may fail by buckling above or below the side supports. End brooming is also possible. Both are unacceptable failure modes.

6. Like most composite tests, proper specimen preparation is critical. End flatness and parallelism are particularly important.

7.4.1.2 *SACMA SRM 2R-94 "Compression after Impact Properties of Oriented Fiber-Resin Composites"*. SACMA SRM 2-88 method was developed from Boeing BSS 7260. The test specimen is a 4"x 6" (100 mm x 150 mm) quasi-isotropic specimen nominally 0.25" (6 mm) thick. If C-Scan will be used after impact, an initial C-Scan should be performed as a baseline. The specimen is clamped to an aluminum support base with a 3"x 5" (76.2 mm x 127 mm) cut out. The specimen is then impacted with an impactor having 0.625" (15.75 mm) diameter hemispherical tup at a height to provide a target impact energy of specimen thickness. The mass of the impactor is not specified but is between 10 and 12 pounds (4.5 and 5.5. kilograms) in normal practice. The impact energy is determined by one of the following methods:

Method 1: Energy = drop weight x drop height/specimen thickness
Method 2: Energy = ½ mass (velocity)^2 /specimen thickness

The specified impact energy level is 1500 inch-pounds/inch thickness (6.7 Joules/mm thickness). The velocity is measured just prior to impact. The velocity measurement is corrected for any travel

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between the flag and the specimen. Since Method 2 takes into account friction losses, it is the preferred method.

Rebound impacts of the specimen must be avoided. If instrumentation is used during impacting, the actual impact energy can be calculated, and impact force versus time can be recorded. The impacted specimen is inspected via an ultrasonic scan. The area and the general configuration of the delamination can be recorded.

Specimen testing -- A compressive loading fixture is used to ensure axial loading in the desired plane. The method requires four axial strain gages to be used to measure the strain although the strain gages are not always used in industry practice. The testing speed is 0.05 inches/min (1 mm/min). The output of each gage is plotted individually to check for unusual loading conditions. CAI is calculated as follows:

$$F^{CAI} = \frac{P}{tw} \qquad 7.4.1.2$$

where

P = load
t = thickness
w = width

Advantages: Requires much less material than the NASA methods and the elimination of a secondary machining step saves cost.

Disadvantage: There is no machining step after impact to remove possible damage in the clamp areas or ends.

7.4.1.3 *NASA 1142, B.11 "Compression After Impact Test".* The NASA CAI methods are described in NASA 1092, ST-1 and NASA 1142, B.11. The NASA 1142, B.11 method is a later version of essentially the same method.

The test specimen is 7"x 12" (180 mm x 300 mm) quasi-isotropic composite plate prior to impacting. Thickness is 0.25" (6 mm) in normal practice. Ultrasonic C-Scan should be performed prior to impact for a baseline. The specimen is clamped to a steel support plate with a 5"x 5" (130 mm x 130 mm) cutout opposite to the impact site. The specimen is impacted with an impactor equipped with a 0.5" (13 mm) diameter hemispherical tup. The mass of the impactor is 10 to 12 lbs (4.5-5.5 kg). The required impact energy is 20 foot-pounds (27 Joules).

Following the impact, the specimen is visually examined, ultrasonically inspected and then machined to its final compression test dimensions of 5"x 10" (130 mm x 250 mm). This final machining step eliminates any damage sustained by the specimen in the clamped area during impact and allows for ends to be machined flat after impact.

The specimen is then instrumented with back to back axial gages. The gages are used to monitor for unusual loading conditions during the test. The strain gages are not always used in industry practice. CAI is calculated as follows:

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$$F^{CAI} = \frac{P}{tw} \qquad 7.4.1.3$$

where

P = load
t = thickness
w = width

7.4.1.4 *Test methods for MIL-HDBK-17 data submittal.* Data produced by the following test methods are currently being accepted by MIL-HDBK-17 for consideration for inclusion in Volume 2:

TABLE 7.4.1.4 *Compression after impact test method for MIL-HDBK-17 data.*

PROPERTY	SYMBOL	ALL DATA TYPES	SCREENING
Compression after impact strength	F^{CAI}		SACMA SRM 2R-94 NASA 1192, B.11

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TABLE 7.2.9.5 *Fastener pull-thru test matrix.*

GEOMETRY	COMPOSITE SHEET THICKNESS (IN)	LAY-UP	FASTENER NOMINAL SHANK DIAMETER	INSTALLATION HOLE CLASS	ENVIRONMENT (TEMP/% MOIST)	NUMBER OF TESTS (1)
COMPOSITE TO COMPOSITE	0.190 Head side	25/50/25	0.250	Class 2	RT/ambient hot/wet	5 5
COMPOSITE TO COMPOSITE	0.120 Tail side	25/50/25	0.250	Class 2	RT/ambient hot/wet	5 5
COMPOSITE TO METAL (2) (Metal on head side)	0.190	25/50/25	0.250	Class 3	RT/ambient hot/wet	5 5
COMPOSITE TO METAL(2) (Metal on tail or nut side)	0.160	25/50/25	0.250	Class 2	RT/ambient hot/wet	5 5

Notes: (1) Each grip condition where applicable (see Section 7.2.9.4).
(2) Metal thickness can be varied to accommodate fastener grip length.

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Chapter 8. ANALYSIS AND PRESENTATION OF DATA

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1. For 0° tension and 0° compression strengths and stiffnesses, normalized values shall be reported. Normalization may be by fiber volume fraction or thickness as appropriate for the material. The method of normalization shall be stated, and the normalizing factor shall be within the material process specification range. Raw data before normalization shall also be reported.

2. For properties other than those in (1.) above, raw data without normalization shall be reported. The fiber volume fraction and thickness for panels used shall be reported for information.

8.1.4 Symbols. The symbols which are used in Chapter 8, particularly those that are used in Sections 8.5 through 8.8, and not commonly used throughout this handbook are listed below, each with its definition and the section in which it is first used. These are primarily used to denote statistical variables.

SYMBOL	DEFINITION	SECTION
A_i	population	8.6.3.1
ADK	k-sample Anderson-Darling statistic	8.6.3.1
B	B-basis value	8.5.1.1
C.V.	coefficient of variation	8.4.3
e	error, residual	8.6.4.4
EV	equality of variances test statistic	8.6.3.2
F	F statistic	8.5.8
IQ	informative quantile function	8.6.5.2
k	number of batches	8.5.4
k	Hanson-Koopmans coefficient	8.5.7.2
k^*	see Equation 8.5.4(i)	8.5.4
k_B	one-sided tolerance limit factor	8.5.5.1
MNR	maximum normed residual test statistic	8.6.2.1
N	total number of observations in several batches	8.5.4
n	number of observations in a sample	8.5.1.1
n'	effective sample size	8.5.4
n^*	see Equation 8.5.4.1(b)	8.5.4
n_i	number of observations in sample i	8.6.6.2
OSL	observed significance level	8.5.3.4
Q	quantile function	8.6.5.1
$\hat{Q}$	quantile function estimate	8.6.5.1
$q_{0.10}$	tenth percentile of the underlying population distribution	8.5.7.1
r	rank of observation	8.5.7.1
r_i	normed residual	8.6.2.1
RME	relative magnitude of error	8.8
S	sample standard deviation	8.5.1.1
S^2	sample variance	8.5.1.1
S_b^2	sample between-batch mean square	8.5.9
S_e^2	sample within-batch variance	8.5.9
S_y	estimated variance of errors from the regression line	8.5.8
T	tolerance limit factor	8.5.4
$t_{\gamma,0.95}$	0.95 quantile of the central t-distribution with degrees of freedom γ	8.5.4

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SYMBOL	DEFINITION	SECTION
$t_{\gamma,0.95}(\delta)$	0.95 quantile of the non-central t-distribution with non-centrality parameter δ and degrees of freedom γ	8.5.8
TIQ	truncated informative quantile function	8.6.5.2
V	one-sided tolerance limit factor for the Weibull distribution	8.5.6.2
V_c	critical value	8.6.2.1
$\bar{X}$	sample mean, overall mean	8.5.1.1
x_i	observation i in a sample	8.5.1.1
x_{ij}	jth observation in sample i	8.6.7
$x_{(r)}$	rth observation, sorted in ascending order, observation of rank r	8.5.7.1
α	(1) significance level	8.8.9
	(2) true intercept of regression equation	8.6.6.1
	(3) scale parameter of Weibull distribution	8.5.6.1
$\hat{\alpha}$	estimate of scale parameter of Weibull distribution	8.5.6.1
β	(1) true slope of regression equation	8.6.6.1
	(2) shape parameter of Weibull distribution	8.5.6.1
$\hat{\beta}$	estimate of shape parameter of Weibull distribution	8.5.6.1
γ	degrees of freedom	8.5.8
δ	noncentrality parameter	8.5.8
ϵ	error	8.6.6.1
μ	population mean	8.5.1.1
σ	population standard deviation	8.6.6.1
σ^2	population variance	8.6.6.1
σ_b^2	population between-batch variance	8.6.7
σ_e^2	population within-batch variance	8.6.7

8.2 MATERIAL AND SPECIMEN REQUIREMENTS.

8.2.1 Statistically-based material properties.. The statistically-based properties included in MIL-HDBK-17 are B-basis values. A B-value, as defined in Section 1.7, is the value above which at least 90 percent of the population of values is expected to fall, with a confidence of 95 percent. In order to obtain a B-value by using a nonparametric method as described in Section 8.5.7.1, twenty-nine measurements are required. For the purposes of obtaining a reasonable evaluation of material variation, properties as presented in this handbook are based on a minimum of six measurements from at least five batches of a material. This criterion provides for a minimum of thirty measurements. Fewer measurements can be evaluated according to the statistical methods described in Section 8.5.7.2; these data will be included in the handbook but B-basis values will not be reported for them. B-values are included for all ultimate strength and strain-to-failure properties.

8.2.2 Typical properties. Typical, sample mean, values are presented for all properties in MIL-HDBK-17. Statistical assurance is associated only with the ultimate strength and strain-to-failure.

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8.3 DETERMINATION OF PROPERTIES.

8.3.1 Introduction. This section outlines the procedures for calculating material property values from the experimental records of the individual specimens. The statistical methods described in Sections 8.5 through 8.8 and the statistical definitions presented in Section 1.7 are used to determine the necessary values.

8.3.2 Mechanical properties.

8.3.2.1 *Ultimate stress and strain.* The ultimate stress and strain-to-failure will receive the full statistical treatment described in Section 8.5.2 including outlier detection, data pooling testing, and determination of distribution.

8.3.2.2 *Transverse tensile properties.* A B-basis value will not be reported for transverse strength in tension for unidirectional tape materials but the remaining statistical parameters will be tabulated.

8.3.2.3 *Elastic constants and Poisson's ratio.* The elastic constants, Young's modulus or shear modulus, shall be collected from each of the experiments, with a minimum of six specimens from each of five batches, for each property. For each experiment, the modulus will be calculated from the slope of the load-deformation curve within the linear portion of the curve or, if the load-deformation curve is nonlinear, the modulus will be calculated from the slope of a secant line between load-deformation points at 0.001 in/in (m/m) and 0.003 in/in (m/m) strain. The results will be examined for batch-to-batch variation and pooled if possible. Minimum and maximum values will be reported in Volume 2. If stress-strain data are provided for each experiment, the data can be fit by algebraic functions as described in Section 8.3.7. Poisson's ratio values will be determined for the same range of data as the modulus values. Typical values of Poisson's ratios determined from the thirty experiments for tensile and compressive loading conditions will be included in the data summary for a given material.

8.3.2.4 *Use of crossply laminate testing to derive lamina strengths in the fiber direction.* The mechanical properties of composites have increased markedly as materials have evolved. Carbon fiber composite tensile strengths and strains at failure, for example, nearly doubled during the 1980's (Reference 8.3.2.4(a)). As properties have improved, however, some test methods that were adequate for previous generations of composites are no longer suitable for characterizing the full capabilities of high strength advanced material systems.

The most serious problems relate to accurate determination of basic lamina (ply) tension and compression strengths which, traditionally, have been characterized using unidirectional test coupons for tape and similar form composites. As material capabilities have advanced, the deficiencies associated with these coupons have been greatly amplified. While it is possible in some cases to generate acceptable strength data with unidirectional coupons, extreme care is required in their design and fabrication, thus adding significant cost. As an alternative, data from the testing of crossply laminates has been used by an increasing number of workers to indirectly calculate lamina properties by classical lamination theory.

NOTE: The term "crossply" is used as defined in Section 1.7, and is synonymous with "angleply" and "multidirectional".

There are numerous arguments that support this approach. The most frequently claimed advantages are higher (more realistic) strength values with lower data scatter, both of which have been demonstrated by a number of investigators (e.g., References 8.3.2.4(a) and 8.3.2.4(b)). Higher values are attributed to reduction or elimination of premature failures stemming from various causes which are discussed later. Lower variability is associated with less sensitivity to coupon quality fluctuation

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and small manufacturing defects. This reduction in sensitivity reflects more closely the response of structural configurations.

Perhaps the most compelling reason for using crossply testing is that it is more closely representative of application laminates used in actual structural components. Since, in general, a ply may respond to loads differently when adjacent to plies of different orientation than when in isolation (or adjacent to plies of the same directionality), it makes sense to characterize ply properties in their end-use setting. In this way, ply values used in laminate analysis will be more representative of properties expected of the ply in the laminate being analyzed, not those of a ply in isolation.

Although this approach is not a panacea for all testing difficulties, it is becoming quite common in the advanced composites industry, and standardization activities are in progress. The method does offer advantages and should be considered when planning test programs. For additional information the reader is referred to References 8.3.2.4(c) and 8.3.2.4(d).

8.3.2.4.1 *Methodology*. The general approach for determining longitudinal lamina strength is to select, fabricate, and test a suitable multidirectional laminate, and then calculate 0° ply tension or compression strength using classical lamination theory. This methodology makes some assumptions:

1. The laminate fails by the same mechanism and at the same strain as the plies in a unidirectional coupon that does not fail prematurely.

2. The stress-strain curves for both the laminate and lamina are essentially linear elastic to failure (a methodology for use when this does not hold is briefly discussed).

3. The values of E_1, E_2, and ν_{12} for the ply used in the equations are valid at incipient failure.

4. Effects of ply residual stresses and damages such as ply cracks are negligible.

Given these assumptions, it is clear that not all laminates are suitable. A family of laminates that has been found useful, and for which the bulk of test data exists, is the $[0_x/90_y]_{ns}$. In this family the $[0/90]_{ns}$ is most widely used. While this laminate is not commonly used for actual structure, it does provide an environment where adjacent plies are of different orientation. In addition, the calculated factor (discussed below) is reasonably low. Quasi-isotropic laminates have also been used successfully, but the factor is almost twice as high as for the 0°/90° laminates, giving somewhat less confidence. Laminates with so many ±45° plies as to cause ν_{12} for the laminate to exceed ν_{12} for the lamina are not preferred because the strain to failure may not be as great as for the unidirectional coupons. Some composites with very brittle resin matrices do not permit the fabrication of quality 0°/90° laminates due to splitting during cool-down after cure. In such cases some ±45° plies must be included. Within a family of laminates, stacking sequence will have an effect. Laminates with several plies of the same orientation stacked together (thick layers) will generally yield lower compression strength values than more homogeneous lay-ups (Reference 8.3.2.4.1). Obviously, symmetric laminates must be used in all cases to preclude bending.

The third assumption presumes that E_1, E_2, and ν_{12} have been obtained from other tests (most likely, unidirectional coupons). This does not present a serious problem, since the shortcomings of unidirectional coupons do not affect modulus measurements to the same degree as strength measurements. It can be argued that E_2 (and to some degree E_1) is not linear to failure, and is usually calculated significantly below the failure load. However, as discussed in detail later, this is not a significant issue due to the rather low sensitivity of this methodology to variation in E_2.

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To calculate lamina strength, the measured test laminate strength is multiplied by a factor (F) generated from classical lamination theory:

$$\sigma_1 = F\sigma_x \qquad 8.3.2.4.1(a)$$

For the $[0_x/90_y]_{ns}$ family of laminates this factor, based on the assumption of uniform strains in each ply, is calculated according to the following formula:

$$F = \frac{E_1[mE_2 + (1-m)E_1] - (\nu_{12}E_2)^2}{[mE_1 + (1-m)E_2][mE_2 + (1-m)E_1] - (\nu_{12}E_2)^2} \qquad 8.3.2.4.1(b)$$

where m is the fraction of 0° plies in the laminate (E_1, E_2, and ν_{12} are for tension or compression as appropriate).

For $[0/90]_{ns}$ laminates (equal numbers of 0° and 90° plies), the formula reduces to:

$$F = \frac{E_1\left(\frac{E_1+E_2}{2}\right) - (\nu_{12}E_2)^2}{\frac{(E_1+E_2)^2}{4} - (\nu_{12}E_2)^2} \qquad 8.3.2.4.1(c)$$

As stated above, these equations are not very sensitive to variability in E_2, and show very little response to changes in ν_{12}. For $[0/90]_{ns}$ laminates with E_1 = 20 Msi, a 20% change in E_2 results in less than 2% change in the factor, and a 20% change in ν_{12} has negligible effect. Therefore, E_2 and ν_{12} do not have to be quantified with great accuracy (the precise effect on the factor will, of course, depend on the actual ratio of E_1 to E_2). The shear moduli of a ply are a function of stress. Since these moduli affect stability, and hence compression strength, there may be some difficulties with soft matrix materials.

Many times a value for E_2 may not be available at all. If this is the case, there is an alternate approach, which may be preferable even if E_2 has been determined. This method involves measuring only E_1 from a unidirectional coupon, and E_x of the crossply laminate being tested. Under assumption 1 that the test laminate fails at the same strain as a unidirectional coupon, the lamina strength may be calculated as follows:

$$\sigma_1 = \frac{E_1}{E_x}\sigma_x \qquad 8.3.2.4.1(d)$$

Very good agreement has been reported between the E_1/E_x ratio and the factor, F, obtained as described above (Reference 8.3.2.4.(a)).

All of the methodology described thus far assumes linear stress-strain behavior to failure. If this is not the case (as in some crossply glass/epoxy laminates, for example) the fiber direction lamina strength can be calculated as follows:

$$\sigma_1 = E_1(e_x + \nu_{21}e_y)/(1 - \nu_{12}\nu_{21}) \qquad 8.3.2.4.1(e)$$

where the strains (e) in the x and y directions are those measured at failure.

If Poisson effects can be neglected, the above equation reduces to:

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$$\sigma_1 = E_1 e_x = \sigma_x \frac{E_1}{E_x^*} \qquad 8.3.2.4.1(f)$$

where E_x^* is the secant modulus of the laminate at failure.

This equation is useful when ν_{12} and E_2 are not known.

8.3.2.4.2 *Tensile strength tests*. Well designed and fabricated unidirectional tensile specimens can give good results for advanced composites, but this is generally the exception rather than the rule. One major problem is premature failure at the tips of adhesively bonded tabs, particularly when they are tapered gently rather than square cut and gripped over their entire length. The higher loads required to test advanced composites often result in high peel forces at the tab ends and subsequent interlaminar tension failure of the first ply of the composite. Once this has occurred, most of the load is taken by this outer ply, which then fails in tension and results in tab loss. Since lower loads are required to test crossply coupons, this is much less likely to happen.

Rawlinson (Reference 8.3.2.4(a)) and others have investigated various laminate stacking sequences. Both 0°/90° and 0°/±45° balanced laminates yielded mean strength values comparable to those measured from the best quality unidirectional coupons, and had significantly less data scatter. In addition, it has been demonstrated that some of these laminates can be tested successfully without bonded tabs using hydraulic grips, thus offering additional testing economy. What have sometimes been referred to as "tabless" specimens actually require an interlayer between the grips and the coupon: for example, a sheet of emery paper with the abrasive side in contact with the specimen, or an abrasive-coated wire mesh with a sheet of plastic to protect the jaws of the test machine. If bonded tabs are not used, 0° plies should not be on the outside surfaces of the laminate since damage may be inflicted by the grips. Thus, for crossply testing without bonded tabs, a $[90/0]_{ns}$ laminate would be preferred over $[0/90]_{ns}$. It should be noted that surface strain measurements are more sensitive to matrix cracking of the outer 90° plies in the $[90/0]_{ns}$ configuration. If bonded tabs are used, the stacking sequence of the tab material is important to consider (Reference 8.3.2.4(b)) (see section 6.6.2). For coupons with tabs there appears to be little difference between results from $[90/0]_{ns}$ and $[0/90]_{ns}$ laminates.

8.3.2.4.3 *Compression strength tests*. As in tensile testing, the high loads needed to test advanced composites cause problems in compression testing of unidirectional coupons. In compression, end "brooming" and longitudinal splitting are common modes of premature failure. Occurrence of these modes is greatly reduced or eliminated by crossply coupons, which tend to fail in microbuckling or ply buckling (Reference 8.3.2.4.3). Furthermore, low sensitivity to methods of loading and end constraint has been reported for quasi-isotropic laminates (Reference 8.3.2.4.3). The same result has been reported by others for $[0/90]_{ns}$ laminates. This suggests that the capability of the material is being evaluated, not the capability of the test method.

There is currently no consensus regarding the "best" laminate stacking sequence to be used, although $[0/90]_{ns}$ has been commonly employed. These specimens are reported to give high strength values and low data scatter. Data from several sources (yet unpublished) indicates that $[90/0]_{ns}$ laminates yield higher mean values than $[0/90]_{ns}$. The reason for this increase has not been conclusively established, but has been attributed to several factors. First, there is speculation that the 90° outer plies act to protect the load bearing 0° plies from damage which might be inflicted during coupon fabrication or testing. Such damage could provide sites for initiation of premature failure if inflicted on 0° plies. Second, it is thought that the presence of the outer 90° plies enhances the stability of the otherwise outer 0° plies. If this is true, structural analysts will have to determine if

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design properties derived from $[90/0]_{ns}$ laminates are appropriate for specific applications where outer 0° plies are aligned with the primary compressive load direction. Third, it is known that 0° outer plies increase stress concentrations at the ends of the gage area for tabbed test specimens, and this is suspected to contribute to premature failure. Fourth, outer 0° plies might split as a result of transverse tensile stresses induced by Poisson effects.

8.3.2.4.4 *Other properties.* Transverse strengths of unidirectional composites have always been difficult to characterize because of premature failures due to extreme notch sensitivity. In an effort to improve this situation, a few studies using crossply laminates have been undertaken, but these are not well documented. There is relatively low interest in pursuing this since, for the analysis of most structure, the accuracy of these strength values does not significantly affect the result unless the transverse strength used in the analysis is so low as to cause a false prediction of a "first ply" failure.

$[+45/-45]_{ns}$ laminates tested in tension have commonly been used to derive [0/90] (matrix dominated) in-plane shear strength and modulus properties. This method generally produces a strength result that is a lower bound of the true material shear capability. See Section 6.6.5 of this volume for more detail.

8.3.2.5 *Bearing and joint strengths.* Bearing and joint strength values will be reported as typical or average values. The failure mode can be recorded using the descriptions in Figures 7.2.5.6(a) - (d). Test data involving different failure modes must be reported separately in different property tables.

8.3.3 Chemical properties.

8.3.4 Physical properties. The following requirements have been established for determining physical properties from experimental results and reporting those properties in MIL-HDBK-17.

8.3.4.1 *Density.* The density will be reported for each batch used in the experiments to determine any of the mechanical properties. Typical density values will be part of the material description presented with each set of data.

8.3.4.2 *Composition.*

8.3.4.3 *Water absorption.* The equilibrium moisture content for the relative humidity values from Tables 2.4(a) and (b) will be determined from a minimum of three specimens. This value will be presented in the data summary for each material. If additional information is available for moisture content as a function of temperature and relative humidity, those values will be presented graphically.

Values for the moisture expansion coefficient will be treated in the same way as those for the thermal expansion coefficient.

8.3.4.4 *Glass transition temperature.* The glass transition temperature will be determined from a minimum of three measurements per dry and wet test condition. This information will be a basis for determining the maximum use temperature.

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8.3.5 **Thermal properties.** The coefficient of linear thermal expansion will be determined from a minimum of five measurements at each condition. Room temperature values will be listed in the data summary and additional values of the thermal expansion coefficient will be presented graphically as a function of temperature.

The specific heat and the thermal conductivity will be presented in a similar fashion. When data are adequate to present curves showing specific heat, thermal conductivity, and mean coefficient of thermal expansion over a range of temperatures, graphical presentation is used in addition to tabular presentation. A smooth curve is drawn through the plotted points to depict the overall trend of the data. The smooth curves for specific heat, thermal conductivity, and thermal expansion are then shown in a single figure as shown in Figure 8.3.5. The reference temperature for thermal expansion should be shown on the figure. In Figure 8.3.5, the reference temperature of 70°F (21°C) indicates that the mean coefficient of expansion between 70°F (21°C) and the indicated temperature is plotted.

8.3.6 **Electrical properties.**

8.3.7 **Typical stress-strain curves.** Typical stress-strain curves will be determined from a minimum of two stress-strain curves from each batch. It is highly preferable to have stress-strain records from all of the experiments completed to evaluate a given mechanical property. The best fit of the stress-strain data for each experiment to one of seven algebraic functions will be obtained as described in References 8.3.7(a) and (b). The seven functions evaluated for best fit are:

1) linear
2) parabolic
3) inverse parabolic
4) Ramberg-Osgood exponential

Figure 8.3.5 *Thermal properties as a function of temperature.*

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8.4 **PRESENTATION OF DATA.** All data are presented in Volume 2 of the handbook (MIL-HDBK-17-2). This section describes how the data are presented and organized in that volume.

8.4.1 **Properties and definitions.** The properties and their definitions are found in the appropriate chapters of Volume 1. Fiber properties and methods for obtaining them are discussed in Chapter 3. Resin properties are presented in Chapter 4. Methods for characterizing prepreg materials are discussed in Chapter 5 and properties and definitions for laminae and laminates are presented in Chapter 6. The statistical methods used in determining these properties are discussed in this chapter. Material system codes and laminate orientation codes are defined in Chapter 1 of Volume 2.

8.4.1.1 *Sign convention.* All compressive values, represented by a superscript c, are reported as positive numbers. Thus, a positive compression strength indicates failure due to a load applied in the opposite direction of a positive tensile failure.

8.4.2 **Organization of data in handbook.** The data in Volume 2 is divided into chapters of fiber properties, resin properties, and composite properties organized by fiber and then resin.

8.4.2.1 *Fiber properties.* Chapter 2 in Volume 2 provides data for fiber properties. Sections are included for different types of fiber, e.g., glass fibers and carbon fibers. In each section, the general characteristics of the type of fiber are given, as well as an index of suppliers, designations, and abbreviations. For each specific fiber, data are organized in the following manner. The X's in the subsection number are determined by the type of fiber and the specific fiber described.

2.X.X.1	Supplier and product data
2.X.X.2	Chemical and physical properties
2.X.X.2.1	Typical range of chemical constituents
2.X.X.2.2	Expected bound in physical properties
2.X.X.3	Thermal-mechanical properties
2.X.X.3.1	Stress-strain curves
2.X.X.3.2	Environmental effects

8.4.2.2 *Matrix properties.* Matrix or resin properties are included in Chapter 3 which is divided into sections according to the type of resin. For example, Section 3.2 gives data for epoxies and Section 3.3 provides data for polyester resins. The subsections for each specific resin are the same as those in Chapter 2 given above.

8.4.2.3 *Composite properties.* The remaining chapters of Volume 2 provide data for prepreg, lamina, laminate, and joint properties. There are individual chapters for each family of composites based on fiber type. For example, Chapter 4 describes carbon fiber composites. Within each chapter, there is an index of suppliers, designations, and abbreviations. Sections are included based on the resin type used with the fiber described in the chapter, e.g., Section 4.3 provides properties for epoxy-carbon composites.

Properties are organized in the following manner for each specific composite:

X.X.X.1	Supplier and product data
X.X.X.2	Prepreg chemical and physical properties
X.X.X.2.1	Physical description
X.X.X.2.2	Resin content
X.X.X.2.3	Fiber content
X.X.X.2.4	Volatiles content
X.X.X.2.5	Moisture content

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X.X.X.2.5 Moisture content
X.X.X.2.6 Inorganic fillers and additives content
X.X.X.2.7 Areal weight
X.X.X.2.8 Tack and drape
X.X.X.2.9 Resin flow
X.X.X.2.10 Gel time
X.X.X.3 Lamina chemical properties
X.X.X.4 Lamina physical properties
X.X.X.5 Lamina mechanical properties
X.X.X.5.1 Data Summaries
X.X.X.5.2 Typical stress-strain curves
X.X.X.6 Thermal properties
X.X.X.7 Electrical properties
X.X.X.8 Laminate thermal-mechanical properties
X.X.X.8.1 Index of properties by lay-up
X.X.X.8.2 Strength properties
a. Lay-up No. 1
b. Lay-up No.2
X.X.X.8.3 Thermal properties
X.X.X.8.4 Electrical properties
X.X.X.9 Joint thermal-mechanical properties
X.X.X.9.1 Index of properties by joint and composite system
X.X.X.9.2 Bearing strength
a. System No. 1
b. System No.2

8.4.3 Sample tables. Sample tables for mechanical property data presentation in Volume 2 are given in Tables 8.4.3(a) through (f). Table 8.4.3(a) is a summary page giving information about the material system and the mean strength and modulus for each test and environmental condition. Table 8.4.3(b) shows the format for data which are normalized (see Section 8.1.3) and Table 8.4.3(c) is the format for data which are not normalized. Table 8.4.3(d) is the format for bearing data, Table 8.4.3(e) is for notched laminate data and Table 8.4.3(f) is for bearing/bypass data. In these tables, a number of values are given for several properties. For ultimate stress and strain, the following values are reported:

Typical
Maximum
Minimum
Standard deviation
B-basis
Type of distribution
Distribution constants, C_1 and C_2
Number of specimens
Number of batches
Test method
Source code

The B-basis value is determined for the type of distribution which has been found to fit the data as described in Sections 8.5.2 and 8.5.5 - 7. The type of distribution which was used is listed as:

Weibull = Two-parameter Weibull distribution
Normal = Normal distribution

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Lognormal = Lognormal distribution
Nonpara = Nonparametric method
ANOVA = Analysis of variance method

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TABLE 8.4.3(d) *Format for bearing strength property table.*

{Table Number}
{Material Class}/{Form}
{Material Designation}
{Property and Direction}
{Laminate Family}
{Test Temp/Moisture Conditions}
{Data Approval Status}

MATERIAL: {Fiber} {Fil. Count} / {Matrix} {tape/weave type}

RESIN CONTENT: XX-XX wt% COMP. DENSITY: 0.0XX lb/in^3
FIBER VOLUME: XX-XX % VOID CONTENT: X.X - X.X %
PLY THICKNESS: 0.00XX - 0.00XX in.

TEST METHOD: {Org. Method - Date}

TYPE OF BEARING TEST: {material/joint}

JOINT CONFIGURATION
Member 1 (t,w,layup): {thickness, width, layup}
Member 2 (t,w,layup): {thickness, width, layup}
FASTENER TYPE: { }
TORQUE: { }
NORMALIZED BY: Not normalized
FAILURE DESCRIPTION (3):

Temperature (°F) / Moisture Content (%) / Equilibrium at T, RH (°F, %) / Source Code		75 (1)		ET (2)	
Diameter		0.25	D2	0.25	D2
F^{bru} (ksi)	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
	No. Specimens				
	No. Batches				
F^{br}_{offset} (ksi)	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
	No. Specimens				
	No. Batches				
F^{br}_{pl} (ksi)	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
	No. Specimens				
	No. Batches				

(1) As fabricated.
(2) Conditioned at XXX°F, XX% RH for XX days.
(3) See Table 8.3.2.1 for failure mode description.

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TABLE 8.4.3(e) *Format for notched laminate strength property table.*

MATERIAL:	{Fiber} {Fil. Count} / {Matrix} {tape/weave type}		{Table Number} {Material Class}/{Form} {Material Designation} {Property and Direction} {Laminate Family} {Test Temp/Moisture Conditions} {Data Approval Status}
RESIN CONTENT:	XX-XX wt%	COMP. DENSITY:	0.0XX lb/in^3
FIBER VOLUME:	XX-XX %	VOID CONTENT:	X.X - X.X %
PLY THICKNESS:	0.00XX - 0.00XX in.		
TEST METHOD:		{Org. Method - Date}	
SPECIMEN GEOMETRY:		{thickness}	
FASTENER TYPE:		{ }	
TORQUE:		{ }	
NORMALIZED BY:		Not normalized	

Temperature (°F)		75			ET		
Moisture Content (%)		(1)			(2)		
Equilibrium at T, RH (°F, %)							
Source Code							
Width to Diameter Ratio		W1/D	W2/D	W3/D	W1/D	W2/D	W3/D
Diameter = 0.25	Mean						
	Minimum						
F_x^{ou}	Maximum						
	C.V. (%)						
(ksi)	No. Specimens						
	No. Batches						
Diameter = D2	Mean						
	Minimum						
F_x^{ou}	Maximum						
	C.V. (%)						
(ksi)	No. Specimens						
	No. Batches						
Diameter = D3	Mean						
	Minimum						
	Maximum						
F_x^{ou}	C.V. (%)						
(ksi)	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) Conditioned at XXX°F, XX% RH for XX days.

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TABLE 8.4.3(f) *Format for bearing/bypass property table.*

MATERIAL:	{Fiber} {Fil. Count} / {Matrix} {tape/weave type}		{Table Number} {Material Class}/{Form} {Material Designation} {Property and Direction} {Laminate Family} {Test Temp/Moisture Conditions} {Data Approval Status}
RESIN CONTENT:	XX-XX wt%	COMP. DENSITY:	0.0XX lb/in^3
FIBER VOLUME:	XX-XX %	VOID CONTENT:	X.X - X.X %
PLY THICKNESS:	0.00XX - 0.00XX in.		
TEST METHOD:		{Org. Method - Date}	
SPECIMEN GEOMETRY:		{thickness}: { }	
FASTENER TYPE:		{ }	
TORQUE:		{ }	
NORMALIZED BY:		Not normalized	

Temperature (°F)		75	
Moisture Content (%)		(1)	
Equilibrium at T, RH (°F, %)			
Source Code			
Bearing/Bypass Ratio		0.50	0.75
F^{su}_{byp}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
F^{br}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
	No. Specimens		
	No. Batches		
F^{su}_{byp}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
F^{br}	Mean		
	Minimum		
(ksi)	Maximum		
	C.V. (%)		
	No. Specimens		
	No. Batches		

(1) As fabricated.

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The constants C_1 and C_2 are the mean and standard deviation for the normal distribution, the mean and standard deviation of the log-data for the lognormal distribution, and σ and β for the two-parameter Weibull distribution. For the ANOVA method, C_1 is the estimate of the population standard deviation (S), and C_2 is the B-base tolerance limit factor (T).

Typical, minimum, and maximum values are give for the elastic constants. Typical values only are provided for Poisson's ratio and any other properties shown. For each set of properties, the test method, the number of specimens, the number of batches, and a source code are given.

8.4.4 Sample graphs. Sample graphs for the stress-strain curves are given in Section 8.3.7. Figure 8.4.4 shows a graph of the tangent modulus and the secant modulus as a function of strain for the data of Figure 8.3.7(a).

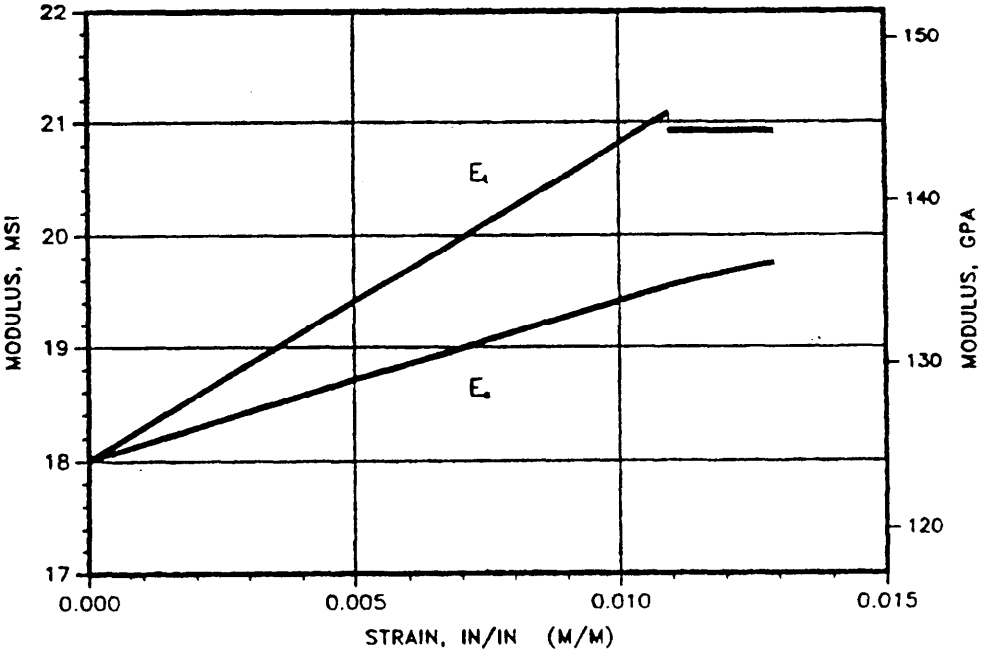


Figure 8.4.4 *Modulus Curves for AS4/3501-6, see Figure 8.3.7(a).*

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8.5 CALCULATION OF STATISTICALLY-BASED MATERIAL PROPERTIES.

8.5.1 Introduction. Section 8.5 contains computational methods for obtaining B-basis values from composite material failure data. In Section 8.5.1.1, definitions are provided for a number of statistical terms which are used frequently in these guidelines. A discussion of the data requirements for publishing a B-basis value in MIL-HDBK-17 is presented in Section 8.5.1.2.

A step-by-step procedure for selecting the appropriate computational method is outlined in Section 8.5.2 with a flowchart showing the steps which must be taken. Section 8.5.3 contains procedures which evaluate several different statistical models and determine which, if any, of the models adequately describes the data. These procedures include methods for detecting outliers, for testing the compatibility of several batches of data, and for investigating the form of the underlying population from which a sample is drawn.

There are various methods for computing B-basis values. The assumption of a statistical model for the data determines which of these methods should be used. Sections 8.5.4 through 8.5.8 contain methods for computing B-basis values under the various models. Statistical tools used in these methods, as well as supplemental methods and tables, are presented in Section 8.6.

Finally, Section 8.7 contains examples illustrating the computational methods. Illustrative sets of data are used to demonstrate the steps that must be followed and the calculations which must be performed in calculating B-basis values.

8.5.1.1 *Statistical terms*. Proper use of the following statistical terms and equations will do much to alleviate misunderstanding in the presentation of data analyses. Definitions for other statistical terms are included in Section 1.7.

Population - The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for Composite Material A, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population.

Sample - The collection of measurements or observations taken from a specified population.

Sample size - The number of measurements or observations in a sample.

A-basis value - A statistically-based material property; a 95% lower confidence bound on the first percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 99% of a specified population.

B-basis value - A statistically-based material property; a 95% lower confidence bound on the tenth percentile of a specified population of measurements. Also a 95% lower tolerance bound for the upper 90% of a specified population.

Compatible - Descriptive term referring to different batches which may be treated as coming from the same population.

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Location parameters and statistics:

Population mean - The average of all potential measurements in a given population weighted by their relative frequencies in the population. The population mean is the limit of the sample mean as the sample size increases.

Sample mean - The average of all observations in a sample and an estimate of the population mean. If the notation $X_1, X_2, \ldots, X_n$ is used to denote the n observations in a sample, then the sample mean is defined by:

$$\bar{X} = \frac{X_1 + X_2 + \ldots + X_n}{n} \qquad 8.5.1.1(a)$$

or

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \qquad 8.5.1.1(b)$$

Sample median - After ordering the observations in a sample from least to greatest, the sample median is the value of the middle-most observation if the sample size is odd and the average of the two middle-most observations if the sample size is even. If the population is symmetric about its mean, the sample median is also a satisfactory estimator of the population mean.

Dispersion statistics:

Sample variance - The sum of the squared deviations from the sample mean, divided by n - 1, where n denotes the sample size. The sample variance is defined by:

$$S^2 = \frac{\sum_{i=1}^{n} (X_i - \bar{X})^2}{n - 1} \qquad 8.5.1.1(c)$$

or

$$S^2 = \frac{n \sum_{i=1}^{n} (X_i)^2 - \left(\sum_{i=1}^{n} X_i \right)^2}{n(n - 1)} \qquad 8.5.1.1(d)$$

Sample standard deviation - The square root of the sample variance. The sample standard deviation is denoted by S.

Probability distribution terms:

Probability distribution - A formula which gives the probability that a value will fall within prescribed limits. When the word *distribution* is used in this handbook, it should be interpreted to mean *probability distribution*.

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Normal Distribution - A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-(x-\mu)^2/2\sigma^2\right] \qquad 8.5.1.1(e)$$

between a and b. A normal distribution with parameters (μ,σ) has population mean μ and variance σ^2.

Lognormal Distribution - A probability distribution for which the probability that an observation selected at random from this population falls between a and b $(0 < a < b < \infty)$ is given by the area under the normal distribution between ℓog a and ℓog b.

Two-Parameter Weibull Distribution - A probability distribution for which the probability that a randomly selected observation from this population lies between a and b $(0 < a < b < \infty)$ is given by

$$\exp\left[-(a/\alpha)^\beta\right] - \exp\left[-(b/\alpha)^\beta\right] \qquad 8.5.1.1(f)$$

where α is called the scale parameter and β is called the shape parameter.

Probability function terms:

Cumulative Distribution Function - A function, usually denoted by F(x), which gives the probability that a random variable lies between any prescribed pair of numbers, that is

$$P\{a < x \leq b\} = F(b) - F(a) \qquad 8.5.1.1(g)$$

Such functions are non-decreasing and satisfy

$$\lim_{x\to-\infty} F(x) = 0 \quad \text{and} \quad \lim_{x\to\infty} F(x) = 1 \qquad 8.5.1.1(h)$$

The cumulative distribution function, F, is related to the probability density function, f, by

$$f(x) = \frac{d}{dx} F(x) \qquad 8.5.1.1(i)$$

provided that F(x) is differentiable.

F-distribution - A probability distribution which is employed in the analysis of variance, regression analysis, and tests for equality of variance. Tables of this distribution are readily available.

Probability Density Function - A function $f(x) \geq 0$ for all x with

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \qquad 8.5.1.1(k)$$

The probability density function determines the cumulative distribution function F(x) by

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$$F(x) = \int_{-\infty}^{x} f(t)\,dt \qquad 8.5.1.1(l)$$

Note that the limits $(-\infty, \infty)$ may be conventional; for example, the exponential distribution satisfies the definition by defining its probability density function as

$$f(x) = \begin{cases} 0 & x \le 0 \\ e^{-x} & x > 0 \end{cases} \qquad 8.5.1.1(m)$$

The probability density function is used to calculate probabilities as follows:

$$P\{a \le x \le b\} = \int_{a}^{b} f(x)\,dx \qquad 8.5.1.1(n)$$

Error and Variability:

Fixed Effect - A systematic shift in a measured quantity due to a particular level change of a treatment or condition. The change in level for a treatment or condition is often under the control of the experimenter. A measured quantity could be compressive strength or tensile modulus. A treatment or condition could be test temperature, fabricator, etc. For a fixed effect, the shift in the measured quantities is to be interpreted as a consistent change not only in the context of the observed data but also with respect to future data under the same treatment or condition.

Random Effect - A shift in a measured quantity due to a particular level change of an external, usually uncontrollable, factor. The level of this factor is regarded as a random draw from an infinite population. The specific level of a random effect is never under the control of the experimenter however it may remain fixed within a limited subgroup of observed data. A measured quantity could be compressive strength or tensile modulus. An external factor could be batch production leading to batch-to-batch differences. Fabricator-to-fabricator differences may be considered a random effect if the number of fabricators involved are considered to be a small random sample of all present and future fabricators. For a random effect, the shift in the measured quantities is viewed as a random variable having mean zero and a non-zero variance. Within a subgroup experiencing a fixed level of an external factor, the measured quantities are correlated (shifting as a cluster around a population average with the magnitude of the shift depending on the level of the factor). Therefore, to obtain the most independent information concerning the population of response values, it is better to have more subgroups than to have more measurements per subgroup.

Random Error - That part of the data variation which is due to unknown or uncontrolled external factors and which affects each observation independently and unpredictably. It is the residual error in a model under analysis (i.e., the variability remaining after the variability due to fixed and random effects has been removed). Random error is a special case of a random effect. In both cases, the level of the random effect or error is uncontrollable but random errors vary independently from measurement to measurement (i.e., there are no random error shifts shared in common by several measurements). An important example of random error is the specimen-to-specimen variability occurring within a subgroup experiencing constant levels of treatment, condition, batch, and other external factors (fixed and random effects).

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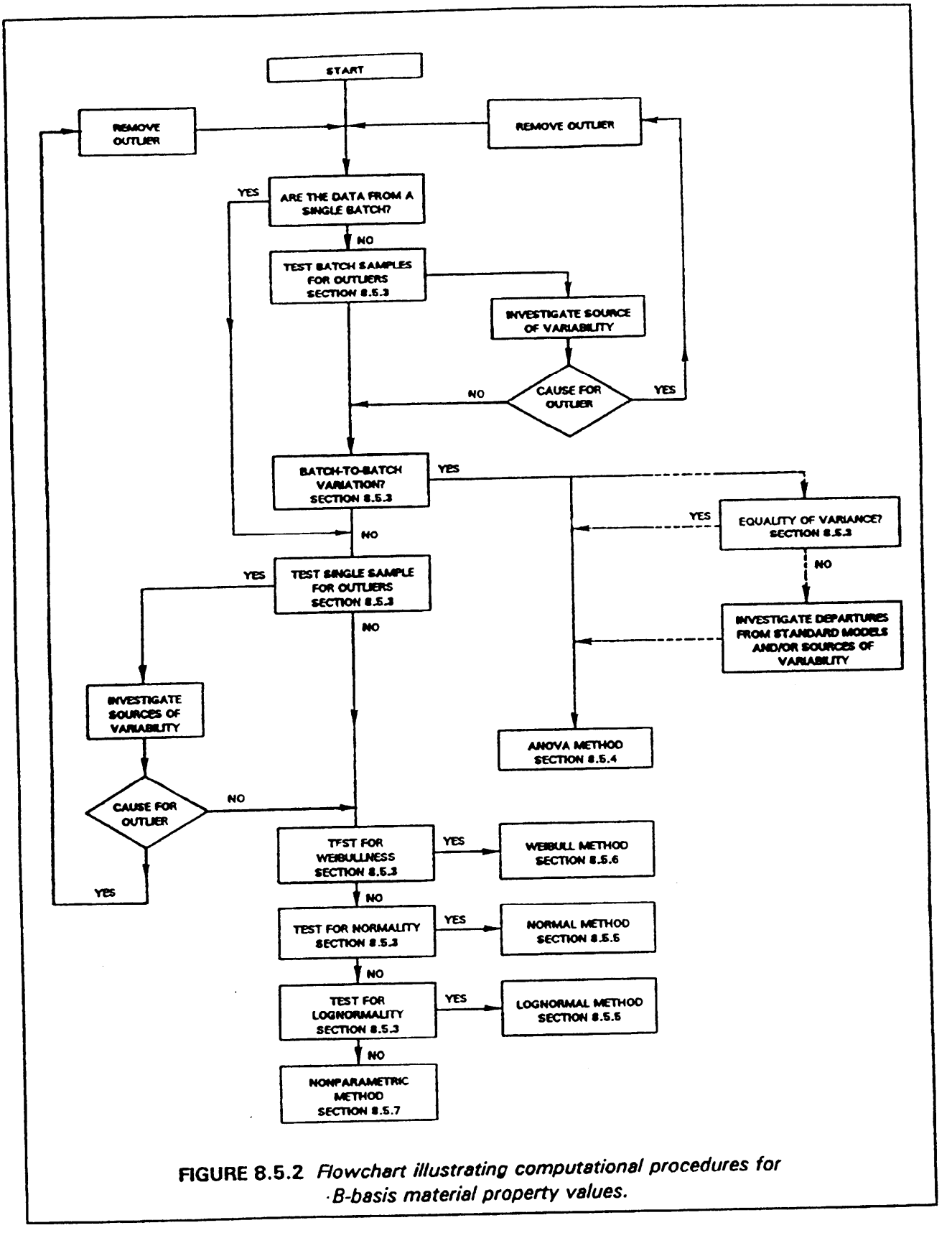


FIGURE 8.5.2 *Flowchart illustrating computational procedures for B-basis material property values.*

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should also be corrected or discarded. If any observations are corrected or discarded, both the statistical outlier procedure and the visual inspection should be repeated.

8.5.3.2 *The k-sample Anderson-Darling test.* The k-sample Anderson-Darling test is used to test the hypothesis that the mechanical property data from different batches are independent random samples from the same population. The procedure for performing this test is presented in Section 8.6.3.1. If the test statistic is less than the critical value from Equation 8.6.3.1(j), then the batches should be treated as a single sample and one should proceed to the goodness-of-fit tests for investigating the form of the distribution discussed in Section 8.5.3.4.

If the test statistic is greater than or equal to the critical value, it is concluded that the batches are not identically distributed and the test for equality of variance should be performed. This test is discussed in the next section.

8.5.3.3 *Equality of variance test.* The ANOVA method (Section 8.5.4) is derived under the assumption that the variances within each batch are equal. The recommended procedure for testing this assumption is presented in Section 8.6.3.2. If the test does reject the hypothesis that the variances are equal, it is recommended that an investigation of the reason for the non-equal variances be carried out. This may reveal problems in the generation of the data or in the fabrication of the material. Basis values caculated using the ANOVA method are likely to be conservative if the variances differ substantially.

8.5.3.4 *Investigating the form of the distribution.* The method employed in calculating the B-basis value for a single sample depends on the distributional form which is assumed for the data. The most frequently used parametric procedures involve the two-parameter Weibull, normal, or lognormal distributions. Section 8.6.4 contains procedures for performing a goodness-of-fit test for each of these distributions. These goodness-of-fit tests yield an observed significance level (OSL) for each of the distributions listed above. The OSL from each of the tests measures the probability of observing an Anderson-Darling statistic as extreme as the value calculated assuming that the given distribution is the correct one.

As shown in Figure 8.5.2, the test for the Weibull model (Section 8.6.4.3) is to be performed first. If the Weibull model cannot be demonstrated to adequately fit the data, then the normal and lognormal (Section 8.6.4.2) tests are performed in succession. This preference for the Weibull distribution is based on several factors:

1) Theoretical considerations suggest that the Weibull distribution is appropriate for the strength distribution of brittle materials such as composite fibers. For an elementary account of this result, see Reference 8.5.3.4(a).

2) The "Chain-of-Bundles" model for the strength of two- and three-dimensional unidirectional composites suggests that the Weibull model is appropriate for the strength distribution of such composites. This result is stated in References 8.5.3.4(b) and (c).

3) Empirical support for the Weibull model is provided by the model's good fit to data.

To determine which method should be used in calculating the B-basis value, the goodness-of-fit tests should be employed in the following manner. The two-parameter Weibull distribution is tested first. If the OSL is larger than 0.05, the Weibull distribution method of Section 8.5.6 should be used to compute the B-basis value. If the OSL is less than or equal to 0.05, the hypothesis that the data is a sample from a normal distribution is tested next. If this test for normality gives an OSL larger than 0.05, the normal distribution method of Section 8.5.5 should be used to compute the B-basis value.

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If the normal OSL is less than or equal to 0.05, the lognormal distribution is tested next. If this results in an OSL greater than 0.05, the lognormal distribution method of Section 8.5.5 should be used to compute the B-basis value. If this OSL is less than or equal to 0.05, it is concluded that the data is not a sample from any of these families of parametric distributions, and the nonparametric procedure of Section 8.5.7.1 should be used to compute the B-basis value, provided that the sample size is greater than or equal to 29. If the sample size is less than 29, then the Hanson-Koopmans method, described in Section 8.5.7.2, may be employed.

The exploratory data analysis (EDA) techniques of Section 8.6.5 may be used to graphically display the distribution of measurements in the sample and to provide graphical evidence for the conclusions obtained through the goodness-of-fit tests.

8.5.4 Normal analysis procedure in the presence of batch-to-batch variation. When data from different batches are to be analyzed, it is possible that the variation from batch to batch will be significant. This will prevent pooling the data and using the methods of Section 8.5.5, 8.5.6 or 8.5.7 to compute basis values from a single sample. This section contains a method, referred to as the ANOVA method, for computing B-basis values when there is significant batch-to-batch variation. This method is documented in Reference 8.5.4(a). It is related to a method suggested in Reference 8.5.4(b). The original paper on this topic (Reference 8.5.4(c)) is of some historical interest, but should not be used for computation.

The method is based on the one-way analysis of variance (ANOVA) random effects model discussed in Section 8.6.7. The assumptions are that

1. the data from each batch are normally distributed,

2. the within-batch variance is the same from batch to batch, and

3. the batch means are normally distributed.

There is no test available for the first assumption. Simulation studies, however, suggest that moderate violation of this assumption does not have an adverse effect on the properties of the ANOVA method. The second assumption should be validated by performing the test described in Section 8.6.3.2. This test is currently recommended as a diagnostic, since extensive simulation suggests that violation of this assumption will likely result in conservatism, although non-conservatism can arise in some situations. There is no useful test for the third assumption unless data from many (20 or more) batches are available.

In the analysis, all batches are treated the same, and no distinction is made between batches from different fabricators. If the batches are not from a single fabricator, then the method of Section 8.5.9 is recommended.

The organization of this subsection is as follows. First a method for three or more batches, which should cover most cases of practical importance, is presented. The case of two batches is discussed separately.

In what follows, let x_{ij} denote measurements on a material property, where the first subscript, $i = 1, \ldots, k$, indicates the batch from which the specimen came, and the second subscript, $j = 1, \ldots, n_i$, denotes the values within a batch.

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A numerical approximation to the k_B values is given in Equation 8.8.1.

8.5.5.2 *Lognormal procedure.* If the sample is assumed to follow a lognormal distribution, then the computational procedure presented in Section 8.5.5.1 is used to calculate the B-basis value. However, the calculations are performed using the logarithms of the data rather than the original observations. Either the natural or the common logarithm may be used. The computed B-basis value must then be transformed back to the original units by applying the inverse of the log transformation which was used. See Section 8.5.1.1 for definition of the lognormal distribution.

If, for example, the natural logarithm is used to transform the data, the B-basis value would be calculated as follows:

$$B = \exp\ (\bar{X}_L - k_B S_L) \qquad 8.5.5.2(a)$$

where

$$\bar{X}_L = \frac{\sum_{i=1}^{n} \ln(X_i)}{n} \qquad 8.5.5.2(b)$$

$$S_L = \sqrt{\frac{\sum_{i=1}^{n} [\ln(X_i) - \bar{X}_L]^2}{n-1}} \qquad 8.5.5.2(c)$$

and k_B is the appropriate one-sided tolerance-limit factor from Table 8.8.1.

8.5.6 Two-parameter Weibull analysis procedure for a single population. This procedure should be used when a B-basis value is to be computed from a single sample of data which is assumed to follow a two-parameter Weibull distribution. See Section 8.5.1.1 for a definition of the two-parameter Weibull distribution.

In order to compute a B-basis value for a two-parameter Weibull population, it is first necessary to obtain estimates of the population shape and scale parameters. Section 8.5.6.1 contains a step-by-step procedure for calculating maximum likelihood estimates of these parameters. The computational procedure for calculating B-basis values using these estimates and the tolerance limit factors in Table 8.8.2 is outlined in Section 8.5.6.2. For further information on these procedures, see Reference 8.5.6.

8.5.6.1 *Estimating the shape and scale parameters of a Weibull distribution.* In order to calculate the B-basis value for a random sample from the two-parameter Weibull distribution, estimates of the shape and scale parameters are needed. The procedure described below is the maximum likelihood method, and the estimates obtained are known as the maximum likelihood estimates. The shape parameter estimate is denoted by $\hat{\beta}$, and the scale parameter by $\hat{\alpha}$. The estimates are obtained by solving the pair of likelihood equations given below for $\hat{\beta}$ and $\hat{\alpha}$.

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$$\hat{\alpha}\hat{\beta}n - \frac{\hat{\beta}}{\hat{\alpha}^{\hat{\beta}-1}}\sum_{i=1}^{n} X_i^{\hat{\beta}} = 0 \qquad 8.5.6.1(a)$$

$$\frac{n}{\hat{\beta}} - n \ln \hat{\alpha} + \sum_{i=1}^{n} \ln X_i - \sum_{i=1}^{n} \left[\frac{X_i}{\hat{\alpha}}\right]^{\hat{\beta}} (\ln X_i - \ln \hat{\alpha}) = 0 \qquad 8.5.6.1(b)$$

Equation 8.5.6.1(a) can be rewritten as

$$\hat{\alpha} = \left(\frac{\sum_{i=1}^{n} X_i^{\hat{\beta}}}{n}\right)^{1/\hat{\beta}} \qquad 8.5.6.1(c)$$

By substituting Equation 8.5.6.1(c) into Equation 8.5.6.1(b), the following equation is obtained.

$$\frac{n}{\hat{\beta}} + \sum_{i=1}^{n} \ln X_i - \frac{n}{\sum_{i=1}^{n} X_i^{\hat{\beta}}} \sum_{i=1}^{n} X_i^{\hat{\beta}} \ln X_i = 0 \qquad 8.5.6.1(d)$$

Since everything in Equation 8.5.6.1(d) is known except $\hat{\beta}$, this equation can be solved numerically, and the solution of $\hat{\beta}$ that is obtained is substituted into Equation 8.5.6.1(c) to obtain $\hat{\alpha}$, the estimate of α.

Figure 8.5.6 shows FORTRAN source code for three routines which compute the estimates of $\hat{\alpha}$ and $\hat{\beta}$ by the method described above. WBLEST is a subroutine which returns the estimates of the parameters, $\hat{\beta}$ and $\hat{\alpha}$. FNALPH is a function which calculates the estimate of the scale parameter, $\hat{\alpha}$. GFUNCT is a function which evaluates Equation 8.5.6.1(d). Arguments to WBLEST are

X	=	a vector of length NOBS containing the data (input),
NOBS	=	the number of data values, N (input),
BETA	=	estimate of the shape parameter (output),
ALPHA	=	estimate of the scale parameter (output).

The algorithm by which the FORTRAN code computes the estimates is described in the following paragraph.

Equation 8.5.6.1(d) is a monotonically increasing continuous function of $\hat{\beta}$. Designate the left-hand side of Equation 8.5.6.1(d) divided by n as $G(\hat{\beta})$ and obtain a solution for $\hat{\beta}$ by the following iterative procedure. Let S_y denote the standard deviation of $Y_1,\ldots,Y_n$ where $Y_1 = \ell n(X_i)$ for $i = 1,\ldots,n$. Calculate $I = 1.28/S_y$ as an initial guess at the solution and calculate G(I). If G(I) > 0, then find the smallest positive integer k such that $G(I/2^k) < 0$ and let $L = I/2^k$ and $H = I/2^{k-1}$. If G(I) < 0, then find the smallest positive integer k such that $G(2^kI) > 0$ and let $L = 2^{k-1}I$ and $H = 2^kI$. In either case, the interval (L,H) contains the solution to $G(\hat{\beta}) = 0$. Now calculate G(M) where M = (L + H)/2. If G(M) = 0, then the solution is $\hat{\beta} = M$. If G(M) > 0, then let H = M. If G(M) < 0 then let L = M. The

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$$SST = \sum_{i=1}^{n} (Y_i - \bar{Y})^2 \qquad 8.5.8(n)$$

and

$$SSR = SST - SSE \qquad 8.5.8(o)$$

Then define

$$F = \frac{SSR}{S_y^2} \qquad 8.5.8(p)$$

which has the F-distribution with 1 and n-2 degrees of freedom. The regression is considered significant if the value in Equation 8.5.8(p) exceeds the $1-\alpha$ quantile of the F-distribution with $\gamma_1 = 1$ and $\gamma_2 = n - 2$ degrees of freedom. Table 8.8.5 provides these values for $\alpha = 0.05$.

For given X_0, the B-basis value should satisfy the following condition: $B(X_0)$ is a B-basis value for the normal population with mean $\mu(X_0) = \beta_0 + \beta_1 X_0$ and variance σ^2.

A B-basis value, in the case of simple linear regression, can be determined as follows. For $X = X_0$, compute B as

$$B = (\hat{\beta}_0 + \hat{\beta}_1 X_0) - k_B S_y \qquad 8.5.8(q)$$

where S_y is the square root of S_y^2 in 8.5.8(l),

$$k_B = \left[\frac{1+\Delta}{n}\right]^{\frac{1}{2}} t_{\gamma,0.95}(\delta) \qquad 8.5.8(r)$$

and $t_{\gamma,0.95}(\delta)$ is the 95th percentile of the non-central t-distribution with $\gamma = n-2$ degrees of freedom and noncentrality parameter

$$\delta = \frac{1.282}{\left[\frac{1+\Delta}{n}\right]^{\frac{1}{2}}} \qquad 8.5.8(s)$$

with

$$\Delta = \frac{n(X_0 - \bar{X})^2}{\sum_{i=1}^{n} (X_i - \bar{X})^2} \qquad 8.5.8(t)$$

The following approximation to k_B may be used when n is greater than or equal to 10 and $0 \le \Delta \le 10$:

$$k_B = 1.282 + \exp\left[0.595 - 0.508 \ln(n) + \frac{4.62}{n} + \left(0.486 - \frac{0.986}{n}\right) \ln(1.82 + \Delta)\right] \qquad 8.5.8(u)$$

Under the stated conditions, this approximation has a relative error not exceeding 1%. A numerical illustration is given in Problem 8, Section 8.7.8.

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To adapt Equation 8.5.8(q) to A-basis values, replace 1.282 in Equation 8.5.8(s) with 2.326. For A-basis values, k_A can be approximated by

$$k_B = 2.326 + \exp\left[0.659 - 0.514\ln(n) + \frac{6.58}{n} + \left(0.481 - \frac{1.42}{n}\right)\ln(3.71 + \Delta)\right] \qquad 8.5.8(v)$$

8.5.9 Basis values for multiple batches with a fixed effect.

8.5.9.1 *Description of the problem.* This section is concerned with the following situation. Several batches have been obtained for each level of a fixed effect. A single basis value is desired for all of the data, but concern over the impact that the fixed effect might have on the basis value precludes regarding all of the data as just a collection of batches and applying the ANOVA method of Sections 8.5.4 and 8.6.7.

For example, consider the strength measurements in Table 8.5.9.1. The specimens tested to give these data were taken from a group of three consecutive batches and a group of five consecutive batches. The second group of batches was produced more than a year after the first group. Because of this time difference, one should not consider these data to be eight random batches from a single population without justification. A more prudent approach is to regard these test results as a random sample of three batches from one population and a random sample of five batches from a possible different population.

The two groups would then constitute two levels of a fixed effect. Another example of a fixed effect would be where the data came from material manufactured at two sites. Note that the level of a fixed effect at which data is collected is, at least in principle, controllable by the experimenter; he can test batches sooner rather than later, at site A rather than site B, etc. In contrast, the level of a random effect, for example, which particular batch of infinitely many possible batches will be produced next, is never under experimental control.

8.5.9.2 *Details of the method.* In the formulas of this section, the number of levels of the fixed effect and the number of batches at the ith level of the fixed effect are denoted ℓ and m_i respectively for $i = 1, \ldots, \ell$. The number of values for the jth batch at the ith level of the fixed effect is denoted n_{ij}, for $i = 1, \ldots, \ell$ and $j = 1, \ldots, m_i$. It is desirable that the n_{ij} be as nearly equal as possible.

TABLE 8.5.9.1 *Strength measurements from two groups of consecutive batches.*

Set 1			Set 2		
Mean	Variance	n	Mean	Variance	n
402.2	138.7	5	408.4	40.8	5
387.8	1002.2	5	395.8	113.2	5
389.4	321.8	5	357.2	451.7	5
			376.2	119.7	5
			377.0	189.5	5

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8.5.9.3 *An example.* For the example introduced at the beginning of this section, we have $\ell = 2$, $m_1 = 5$, $m_2 = 3$, and $n_{ij} = 5$ for all i and j. Hence $n_{..} = 40$, $m_{.} = 8$, and

$$\bar{X}_{1..} = 393.13 \qquad 8.5.9.3(a)$$

$$\bar{X}_{2.} = 382.92 \qquad 8.5.9.3(b)$$

$$S_b^2 = 8407.17/(3+5-2) = 1401.20 \qquad 8.5.9.3(c)$$

$$S_e^2 = 9510.4/(40-8) = 297.2 \qquad 8.5.9.3(d)$$

We also have

$$\theta_1 = 1/3 \qquad 8.5.9.3(e)$$

$$\theta_2 = 1/5 \qquad 8.5.9.3(f)$$

$$e_1 = 1/9 \qquad 8.5.9.3(g)$$

$$e_2 = 1/25 \qquad 8.5.9.3(h)$$

$$\phi = 1/5 \qquad 8.5.9.3(i)$$

$$\eta_0 = 38 \qquad 8.5.9.3(j)$$

$$\eta_1 = 30^2/150 = 6 \qquad 8.5.9.3(k)$$

$$K_{01} = T_{38}^{-1}(0.95, 1.282\sqrt{15})/\sqrt{15} = 1.8384 \qquad 8.5.9.3(l)$$

$$K_{02} = T_{38}^{-1}(0.95, 1.282\sqrt{25})/\sqrt{25} = 1.7558 \qquad 8.5.9.3(m)$$

$$K_{11} = T_6^{-1}(0.95, 1.282\sqrt{3})/\sqrt{3} = 3.0562 \qquad 8.5.9.3(n)$$

$$K_{12} = T_6^{-1}(0.95, 1.282\sqrt{5})/\sqrt{5} = 2.8540 \qquad 8.5.9.3(o)$$

$$S^2 = 517.999 \qquad 8.5.9.3(p)$$

$$S_*^2 = 471.515 \qquad 8.5.9.3(q)$$

$$S_{\bar{X}_{1.}}^2 = S_b^2/15 = 93.413 \qquad 8.5.9.3(r)$$

$$9S_{\bar{X}_{2.}}^2 = S_b^2/25 = 56.048 \qquad 8.5.9.3(s)$$

$$c_{11} = 3.8157 \qquad 8.5.9.3(t)$$

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$$c_{12} = 4.4422 \qquad 8.5.9.3(u)$$

$$c_{21} = .8532 \qquad 8.5.9.3(v)$$

$$c_{22} = .8674 \qquad 8.5.9.3(w)$$

$$T_1 = 3.8157\sqrt{\frac{93.413}{471.515}} + 0.8532\sqrt{\frac{517.999}{471.515}} = 2.6544 \qquad 8.5.9.3(x)$$

$$T_2 = 4.4422\sqrt{\frac{56.048}{471.515}} + 0.8674\sqrt{\frac{517.999}{471.515}} = 2.5027 \qquad 8.5.9.3(y)$$

Since $T_1 > K_{01}$ and $T_2 > K_{02}$, we have the basis values

$$L_1 = 393.13 - 2.5926\sqrt{471.515} = 336.84 \qquad 8.5.9.3(z)$$

and

$$L_2 = 382.92 - 2.4406\sqrt{471.515} = 329.92 \qquad 8.5.9.3(aa)$$

Using the approximate tolerance limit factors of the following subsection, the values corresponding to 8.5.9.3(z) and 8.5.9.3(aa) are 336.57 and 329.61, respectively. For comparison, the ANOVA method applied to the first set of data gives a B-basis value of 348.66, applied to the second set of data it leads to 313.77, and for the two sets of data combined and treated as eight batches of five, the ANOVA method gives 338.47.

8.5.9.4 *Approximate formulas for normal tolerance limit factors.* Let

$$K_A = T_{\nu 2}^{-1}(0.95, 2.327\sqrt{\nu_1})/\sqrt{\nu_1} \qquad 8.5.9.4(a)$$

$$K_B = T_{\nu 2}^{-1}(0.95, 1.282\sqrt{\nu_1})/\sqrt{\nu_1} \qquad 8.5.9.4(b)$$

denote A- and B-basis tolerance limit factors, respectively. These factors are proportional to the 95th percentiles of noncentral t distributions with ν_2 degrees of freedom and noncentrality parameters $2.327\sqrt{\nu_1}$ (A-basis), or $1.282\sqrt{\nu_1}$ (B-basis). If ν_1 and ν_2 are positive integers with $\nu_1 = \nu_2 + 1$, then the factors K_A and K_B are tabulated in Table 8.8.12 and Table 8.8.1, respectively. For the method discussed in the previous subsection, however, we need approximations suitable for any pair of positive numbers $\nu_1 \geq 1$ and $\nu_2 \geq 1$.

In order to achieve acceptable accuracy over the entire range of (ν_1, ν_2), it is helpful to develop separate approximations for four cases:

1. $1 \leq \nu_1 \leq 5$ and $1 \leq \nu_2 \leq 5$,

2. $1 \leq \nu_1 \leq 5$ and $\nu_2 > 5$,

3. $\nu_1 > 5$ and $1 \leq \nu_2 \leq 5$, and

4. $\nu_1 > 5$ and $\nu_2 > 5$.

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TABLE 8.5.9.4(c). *Constants C_{ijk} for determining tolerance limit factors when $1 \leq \nu_1 \leq 5$ and $\nu_2 > 5$*

i	k	j	
		1	2
1	1	-2.5701	2.6053
2	1	-2.5341	2.7948
3	1	-2.4956	2.9812
4	1	-2.4562	3.1191
5	1	-2.4176	3.2191
6	1	0	0
1	2	-1.4329	1.6882
2	2	-1.4240	1.4591
3	2	-1.4183	1.4330
4	2	-1.4127	1.4526
5	2	-1.4069	1.4855
6	2	0	0

TABLE 8.5.9.4(d). *Constants C_{ijk} for determining tolerance limit factors when $1 \leq \nu_2 \leq 5$ and $\nu_1 > 5$*

i	k	j	
		1	2
1	1	-1.0972	2.3539
2	1	-0.0282	0.9105
3	1	-0.0154	1.0462
4	1	-0.0080	1.1440
5	1	-0.0020	1.2244
6	1	0	0
1	2	-0.6158	1.3109
2	2	-0.0153	1.5568
3	2	-0.0095	1.7374
4	2	-0.0296	1.8383
5	2	-0.0477	1.9103
6	2	0	0

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8.6 STATISTICAL PROPERTIES.

8.6.1 Introduction. This section includes a number of statistical tools for use in the analysis of data for MIL-HDBK-17. These tools are intended to supplement the basic procedures described in Section 8.5.

Various statistical techniques are employed in the analysis of material-property data. This section presents brief descriptions of procedures that are used most frequently in this application. More detailed descriptions of these and other statistical techniques and tables can be found in a number of references and texts, some of which are given as references.

When procedures other than those described below are employed in the preparation of data proposals, they should be described adequately in the proposal to allow a proper evaluation of their validity and relevance.

8.6.2 Outlier detection procedure. An observation is said to be an outlier if it is an observation that has been recorded in error. Since outliers may have a substantial influence on the statistical analysis of a set of data, it is often desirable that a data set be screened for outliers prior to data analysis. The maximum normed residual method for detecting potential outliers is described below. For more information on this test procedure, see References 8.6.2(a) and 8.6.2(b).

8.6.2.1 *The maximum normed residual method.* The maximum normed residual (MNR) test is a screening procedure for identifying outliers in a single set of data. It involves an examination of the residuals (signed distances from the mean divided by the standard deviation) to determine whether or not they are unusually large. The test assumes that the non-erroneous data values are independently and identically normally distributed. Also, the outlier procedure searches for one outlier at a time and, therefore, the significance level pertains to a single decision.

The first step is to compute the normed residuals and the MNR statistic. If $X_1, X_2, \ldots X_n$ denote the data values in the sample of size n, the normed residuals are defined as:

$$r_i = \frac{X_i - \bar{X}}{S}, \quad i = 1,2,\ldots,n \qquad 8.6.2.1(a)$$

where $\bar{X}$ and S are the sample mean and sample standard deviation, respectively, defined in Section 8.5.1.1. The MNR statistic is the maximum of the absolute values of the normed residuals:

$$MNR = \max_i \{|r_i|\} \qquad 8.6.2.1(b)$$

Next, the MNR statistic is compared to the critical value for the sample size n from Table 8.8.9. The critical values for this test are computed from the following formula:

$$V_c = \frac{n-1}{\sqrt{n}} \sqrt{\frac{t^2}{n-2+t^2}} \qquad 8.6.2.1(c)$$

where t is the $[1 - \alpha/(2n)]$ quantile of the t distribution with n - 2 degrees of freedom and α is the significance level of the test. Critical values for sample sizes 3 through 200 and a significance level of $\alpha = 0.05$ are tabulated in Table 8.8.9. Values for other sample sizes or other significance levels may be computed using the above formula.

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8.6.4.3 *Anderson-Darling test for Weibullness.* The Anderson-Darling test for two-parameter Weibullness is used to test the hypothesis that the population from which a given sample of data was drawn has a two-parameter Weibull distribution. See Section 8.5.1.1 for a definition of the two-parameter Weibull distribution.

The test compares the cumulative Weibull distribution function that fits the data best with the cumulative distribution function of the data. The first step is to compute the parameter estimates $\hat{\alpha}$ and $\hat{\beta}$. A procedure for computing these estimates is given in Section 8.5.6.1. Then let

$$Z_{(i)} = [X_{(i)} / \hat{\alpha}]^{\hat{\beta}} \qquad i = 1, \ldots, n \qquad 8.6.4.3(a)$$

The Anderson-Darling test statistic is then

$$AD = \sum_{i=1}^{n} \frac{1-2i}{n} \{ \ln [1 - \exp(-Z_{(i)})] - Z_{(n+1-i)} \} - n \qquad 8.6.4.3(b)$$

and the observed significance level is calculated as

$$OSL = 1 / \{1 + \exp [-0.10 + 1.24 \ln (AD^*) + 4.48 (AD^*)]\} \qquad 8.6.4.3(c)$$

where

$$AD^* = (1 + 0.2/\sqrt{n})\, AD \qquad 8.6.4.3(d)$$

The OSL is a measure of the goodness-of-fit of the two-parameter Weibull distribution to the data. Specifically, the OSL measures the probability of observing an Anderson-Darling statistic at least as extreme as the value calculated if the two-parameter Weibull distribution is in fact the underlying distribution.

8.6.4.4 *Testing for normality in a regression setting.* This section contains a test for normality of the dependent variable in a regression setting. The test is an Anderson-Darling test for normality performed on the residuals from the regression (see Section 8.6.6) assuming equality of variance of the residuals over the range of the independent variable. Letting

$$Z_{(i)} = e_{(i)} / s_y \qquad i = 1, \ldots, n \qquad 8.6.4.4$$

where $e_{(1)}, \ldots, e_{(n)}$ are the ordered residuals from smallest to largest and S_y is defined in Section 8.6.6. The Anderson-Darling test statistic is computed using Equation 8.6.4.1(b), and the observed significance level (OSL) is computed as in Equation 8.6.4.1(c).

If OSL $\leq$ 0.05, conclude that the dependent variable (for an arbitrary set of fixed values of the independent variables within the range of the data) is not normally distributed. Otherwise, this normality assumption is not rejected. The justification for this procedure may be found in Reference 8.6.4.5.

8.6.5 Exploratory Data Analysis. Exploratory Data Analysis (EDA) techniques can be used to help familiarize the analyst with a set of data. The advantage of such techniques is that they are simple and visual and often point out problems in the data which can be corrected, or at least noted before more involved techniques are undertaken. The disadvantage of these techniques is that they are only

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exploratory in nature and yield no quantitative measure upon which conclusions can be based. Due to the relative unfamiliarity with the materials being analyzed, the use of EDA as both an educational tool and a data screening technique can be invaluable; however, due to the subjectivity involved with the use of these techniques, it is essential to confirm any and all conclusions based on EDA with other statistical procedures. Two EDA techniques are described below; the Quantile Box Plot and the Informative Quantile Functions. Those readers desiring a more complete treatment of the subject are referred to Reference 8.6.5.

8.6.5.1 *The quantile box plot.* The quantile box plot provides a graphical summary of the sample values. This procedure depicts the symmetry, tail sizes, and median value of the sample as well as indicating the possible existence of outliers and inhomogeneous data.

Let F(x) be the underlying distribution function. The uth quantile of F(x), q_u, is the solution to the equation $F(q_u) = u$. The quantile function, Q(u), is defined by

$$Q(u) = F^{-1}(u) \qquad 0 < u < 1 \qquad 8.6.5.1(a)$$

(see Figure 8.6.5.1(a)). Letting $X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)}$ denote the ordered measurements for a sample of size n, Q(u) is estimated by the piecewise linear function

$$\hat{Q}(u) = (nu - j + \frac{1}{2}) X_{(j+1)} + (j + \frac{1}{2} - nu) X_{(j)} \qquad 8.6.5.1(b)$$

for

$$\frac{2j - 1}{2n} \leq u < \frac{2j + 1}{2n} \qquad 8.6.5.1(c)$$

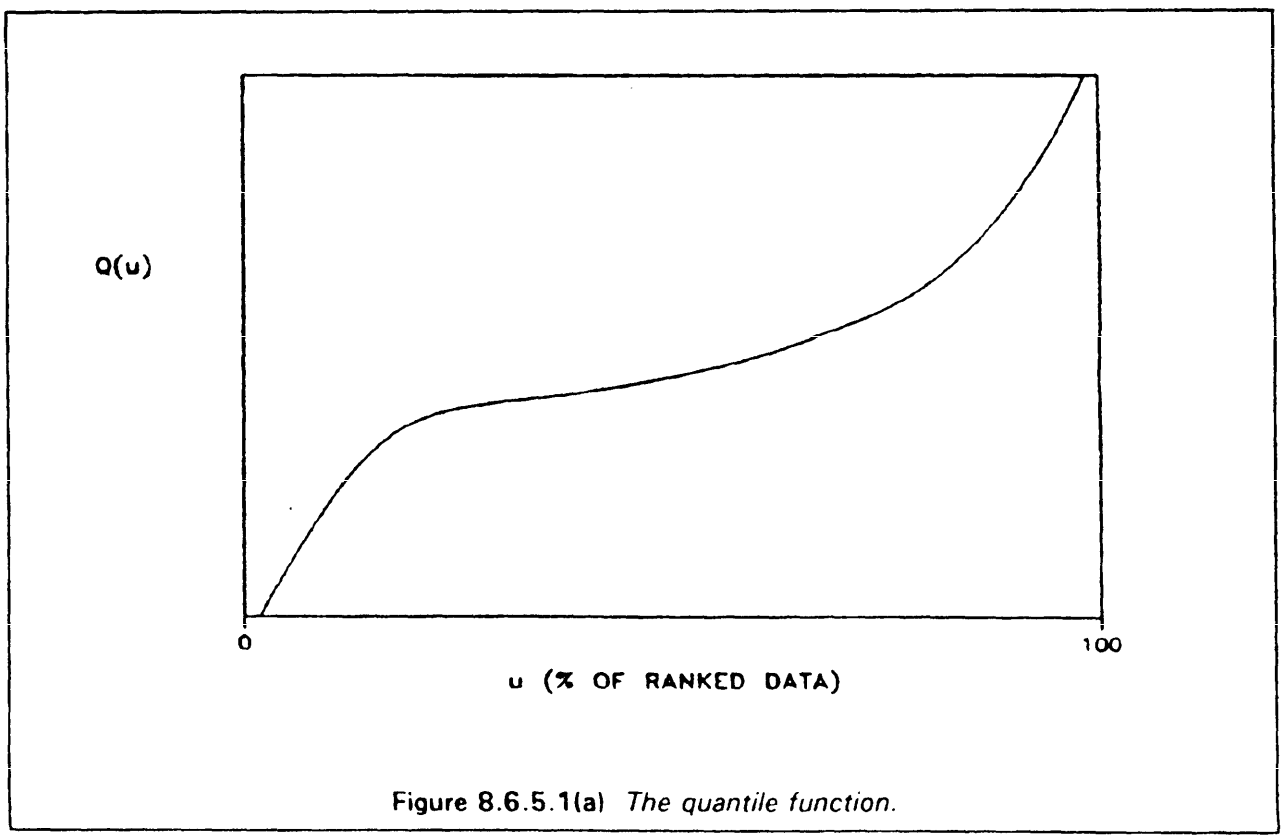


Figure 8.6.5.1(a) *The quantile function.*

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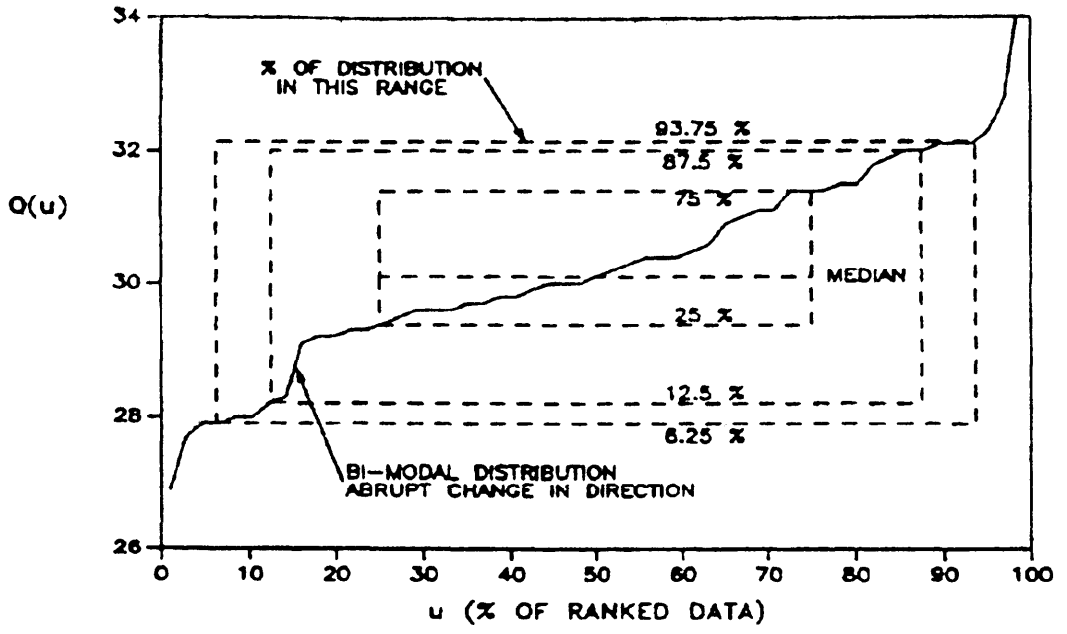


Figure 8.6.5.1(b) *Example of a quantile box plot.*

The quantile box plot is a graph of the above function, with three rectangles (boxes) drawn to represent various quantiles and a line segment indicating the sample median. The innermost box has boundaries of 0.25 and 0.75 on the u axis, and boundaries of Q(0.25) and Q(0.75) on the Q(u) axis. The other two boxes are drawn in a similar manner with boundaries of 0.125 to 0.875 and 0.0625 to 0.9375, respectively, on the u axis. The line segment indicating the sample median is drawn from coordinates (0.25, Q(0.5)) to (0.75, Q(0.5)).

Figure 8.6.5.1(b) is an example of a quantile box plot. The boxes are used to examine the symmetry and tail sizes of the underlying distribution. Flat spots in Q(u) indicate modal values. Sharp rises in Q(u) for u in the vicinity of 0 or 1 indicate the possible presence of outliers in the data. Sharp rises in Q(u) within the boxes indicate the possible existence of two (or more) populations or gaps in the data. A thorough treatment of the use of the Quantile Box plot can be found in Reference 8.6.5.1.

8.6.5.2 *The Informative Quantile function.* Techniques for obtaining B-basis values for unimodal data can be divided into two main categories: techniques for specific parametric families, and nonparametric techniques. The Informative Quantile (IQ) function may be used as an aid in identifying a parametric model which provides a satisfactory fit to the data. Parametric techniques have been most thoroughly discussed for the normal, lognormal, and two-parameter Weibull parametric families; thus only these techniques will be considered here. Henceforth in this section, any reference to the Weibull parametric family should be interpreted as a reference to the two-parameter Weibull parametric family.

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The IQ function was developed to identify which univariate location-scale parametric distribution best describes an ordered group of data. A univariate location-scale parametric distribution is one whose distribution function F(x) can be expressed as

$$F(x) = F_o[(x - a)/b] \qquad 8.6.5.2(a)$$

where a and b are the location and scale parameters respectively, and $F_o(x)$ is the "standard" distribution with a = 0 and b = 1. The IQ function identifies the standard distributional form and is thus independent of the values of the location and scale parameters.

The Weibull and lognormal parametric families are not location-scale parametric families. However, these distributions are simply related to two location-scale families: the normal and the extreme value families.

The IQ function is defined as

$$IQ(u) = \frac{Q(u) - Q(0.5)}{2[Q(0.75) - Q(0.25)]} \qquad 8.6.5.2(a)$$

where Q(u) is the quantile function defined in Equation 8.6.5.1(a). In order to determine whether the data can be adequately modeled by either the normal or extreme value distribution, a plot of the truncated IQ function, defined by

$$TIQ(u) \begin{cases} -1 & \text{if } IQ(u) \le -1 \\ IQ(u) & \text{if } -1 < IQ(u) \le 1 \\ 1 & \text{if } IQ(u) > 1 \end{cases} \qquad 8.6.5.2(b)$$

is compared to the graph of the exact TIQ plots for these distributions (see Figures 8.6.5.2(a) and (b)). Though the TIQ plots for the data will be considerably less smooth than the exact TIQ plots, they may be compared for general shape and tail behavior.

In order to determine the adequacy of either the lognormal or the Weibull distribution, use the natural logarithms of the data to define the quantile function. Thus, Equation 8.6.5.1(b) becomes

$$\hat{Q}(u) = (nu - j + \frac{1}{2}) \ln(X_{(j+1)}) + (j + \frac{1}{2} - nu) \ln(X_{(j)}) \qquad 8.6.5.2(c)$$

for

$$\frac{2j - 1}{2n} \le u < \frac{2j + 1}{2n} \qquad 8.6.5.2(d)$$

The IQ and TIQ functions in Equations 8.6.5.2(a) and 8.6.5.2(b) are defined using this quantile function.

Thus, to determine whether the data can be adequately modeled by the normal distribution, compare the TIQ plot for the original data to the TIQ plot for the normal distribution in Figure 8.6.5.2(a). To determine whether the data can be adequately modeled by the lognormal distribution, compare the TIQ plot for the log data to the TIQ plot for the normal distribution in Figure 8.6.5.2(a). The adequacy of the two-parameter Weibull distribution is determined by comparing the TIQ plot for the log data to the exact TIQ plot for the extreme value distribution in Figure 8.6.5.2(b). For further

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$$SST = (Y - \bar{Y})^T (Y - \bar{Y}) \qquad 8.6.6(f)$$

where

$$(Y - \bar{Y}) = \begin{bmatrix} Y_1 - \bar{Y} \\ Y_2 - \bar{Y} \\ \vdots \\ Y_n - \bar{Y} \end{bmatrix}$$

and let

$$SSE = (Y - \bar{X}\hat{\beta})^T (Y - \bar{X}\hat{\beta}) \qquad 8.6.6(g)$$

Then

$$SSR = SST - SSE \qquad 8.6.6(h)$$

and

$$S_y^2 = \frac{SSE}{n-k} \qquad 8.6.6(i)$$

$$F = \frac{SSR/(k-1)}{S_y^2} \qquad 8.6.6(j)$$

has the F-distribution with k-1 and n-k degrees of freedom.

To test the significance of the regression one compares the value in Equation 8.6.6(j) with the $1-\alpha$ quantile of the F-distribution with $\gamma_1 = k-1$ and $\gamma_2 = n-k$ degrees of freedom. If the value in Equation 8.6.6(j) exceeds this quantile, the regression is judged significant at significance level α, i.e., the risk is 100α % that the regression is judged significant when in fact it is not. Table 8.8.5 produces these quantile values for the customary significance level of $\alpha = 0.05$.

The regression is significant if, assuming $\beta_2 = \beta_3 = \ldots = \beta_k = 0$, then the probability of obtaining an F value (Equation 8.6.6(h)) as large or larger than the $1 - \alpha$ quantile of the F distribution does not exceed α. Then, significance of the regression tends to contradict the assumption that $\beta_2 = \beta_3 = \ldots = \beta_k = 0$, with a probability no greater than α of falsely reaching this conclusion.

In the regression context, the A- and B-basis values are A- and B-basis values for normal populations with mean $X\beta$ and variance σ^2. Thus, for any vector $X_0^T = (1, X_{02}, \ldots, X_{0k})$, which need not be any of the experimental X-values, a B-basis value is a B-basis for the normal population with mean $X_0\beta$ and variance σ^2. Such a value is determined as follows.

For any vector $X_0^T = (1, X_{02}, \ldots, X_{0k})$, which need not be any of the experimental values, compute B as follows:

$$B = X_0\hat{\beta} - k_B S_y \qquad 8.6.6(k)$$

where

$$k_B = \sqrt{X_0^T (X^T X)^{-1} X_0}\, t_{\gamma,0.95}(\delta) \qquad 8.6.6(l)$$

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$$\delta = \frac{1.282}{\sqrt{X_0^T (X^T X)^{-1} X_0}} \qquad 8.6.6(m)$$

and $\gamma = n - k$ where $t_{\gamma,0.95}(\delta)$ is the 95th percentile of the non-central t-distribution with n - k degrees of freedom and non-centrality parameter δ.

To adapt Equation 8.6.6(i) to A-basis values, replace 1.282 by 2.326 in Equation 8.6.6(k). The approximation in Equations 8.5.8(u) and (v) are not generally applicable here and the reader is directed to either the approximation in Section 8.5.9.4 or the algorithm in Reference 8.6.6(a) for evaluating the non-central t-distribution.

8.6.6.1 *Example: Simple linear regression problems.* In order to present the simple linear regression model of Equation 8.5.8(d) in matrix notation Equation 8.6.6(a), set k = 2 and let β_1 be the intercept and β_2 the slope of the regression line. The matrices are as follows.

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \quad e = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}$$

Thus, $X\beta + e$ is the n-element column vector whose rows are

$$Y_i = \beta_1 + \beta_2 X_i + \epsilon_i, \quad i = 1, 2, \ldots, n \qquad 8.6.6.1(a)$$

which is the model for the simple linear regression described in Section 8.5.8. (It is customary when using the matrices to label the parameters $\beta_1, \beta_2, \ldots, \beta_k$, which differs slightly from the notation used in Section 8.5.8. There the intercept, denoted here β_1, was denoted β_0 and the slope, denoted here β_2, was denoted β_1. The present notation describes the parameters as elements of the β vector, the subscripts denoting their position in the vector.)

Thus

$$X^T X = \begin{bmatrix} n & \sum_{i=1}^{n} X_i \\ \sum_{i=1}^{n} X_i & \sum_{i=1}^{n} X_i^2 \end{bmatrix} \qquad 8.6.6.1(b)$$

so that,

$$(X^T X)^{-1} = \frac{1}{n \sum_{i=1}^{n} X_i^2 - \left(\sum_{i=1}^{n} X_i\right)^2} \begin{bmatrix} \sum_{i=1}^{n} X_i^2 & -\sum_{i=1}^{n} X_i \\ -\sum_{i=1}^{n} X_i & n \end{bmatrix} \qquad 8.6.6.1(c)$$

Equations 8.6.6.1(a), 8.6.6.1(b) and 8.6.6.1(c) can easily be obtained by performing the indicated matrix multiplications. Consequently, it can be verified that $\hat{\beta}^T = (\hat{\beta}_1, \hat{\beta}_2)$, where

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$$\hat{\beta}_1 = \bar{Y} - \hat{\beta}_2 \bar{X} \qquad 8.6.6.1(d)$$

and

$$\hat{\beta}_2 = \frac{\sum_{j=1}^{n} (X_j - \bar{X})(Y_j - \bar{Y})}{\sum_{j=1}^{n} (X_j - \bar{X})^2} \qquad 8.6.6.1(e)$$

as was previously obtained in Equations 8.5.8(g) and 8.5.8(h).

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8.6.6.2 *Example: One-way fixed-effects analysis of variance model.* This section is concluded with a discussion of the one-way fixed effects analysis of variance model using matrix notation. The model is given by

$$Y_{ij} = \mu + \tau_i + e_{ij}, \qquad \begin{matrix} i = 1, 2, \ldots, r \\ j = 1, 2, \ldots, m \end{matrix} \qquad 8.6.6.2(a)$$

where each of the parameters τ_i, i = 1, 2, . . ., r, can be chosen to denote the difference between the mean of the ith group of data elements and the grand mean, averaged over all r groups. (See Equation 8.6.6.2(d).) This model can be written in the form of Equation 8.6.6(b) as follows: write the matrix X by

$$X = \begin{bmatrix} 1 & 1 & 0 & \ldots & 0 \\ 1 & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 0 & \ldots & 0 \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 0 & \ldots & 1 \end{bmatrix} \qquad 8.6.6.2(b)$$

where X has rm rows and r + 1 columns. Every entry in the first column is a "1". The first m elements of the second column are "1" and the rest are zero. The third column has all elements zero except for the second group of m elements, and so on. Also, the matrices Y, β, and e can be written

$$Y = \begin{bmatrix} Y_{11} \\ \vdots \\ Y_{1m} \\ Y_{21} \\ \vdots \\ Y_{2m} \\ \vdots \\ \vdots \\ Y_{rm} \end{bmatrix} \quad \beta = \begin{bmatrix} \mu \\ \tau_1 \\ \tau_2 \\ \vdots \\ \tau_r \end{bmatrix} \quad e = \begin{bmatrix} e_{11} \\ \vdots \\ e_{1m} \\ e_{21} \\ \vdots \\ e_{2m} \\ \vdots \\ \vdots \\ e_{rm} \end{bmatrix} \qquad 8.6.6.2(c)$$

This exhibits the principal distinction between an analysis of variance model and a regression model. For analysis of variance models, the X matrix consists of "zeros" and "ones," which, except for the first column, serve as indicators for the various treatments, blocks, and so on. Note further that the matrix X in Equation 8.6.6.2(b) is not of full rank. This can be resolved in this specific example by introducing the restriction

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$$\sum_{i=1}^{n} \tau_i = 0 \qquad 8.6.6.2(d)$$

so that the parameter τ_{1i} describes the mean departure of the ith group from the grand mean of all groups. The linear restriction of Equation 8.6.6.2(d) permits the reduction in the number of parameters by one; since, for example, $\tau_r = \tau_1 - \tau_2 - \ldots - \tau_{r-1}$. Thus X can be replaced by X^*, where

$$X^* = \begin{bmatrix} 1 & 1 & 0 & \ldots & 0 \\ 1 & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 0 & \ldots & 0 \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 0 & 1 & \ldots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & -1 & -1 & \ldots & -1 \end{bmatrix} \qquad 8.6.6.2(e)$$

where X^* has rm rows and r columns. Also X^* has (full) rank r. The least square estimates are given $(X^{*T}X^*)^{-1}X^{*T}Y$ (see Equation 8.6.6(d)). By direct calculation, one obtains

$$(X^{*T}X^*) = \begin{bmatrix} rm & 0 & 0 & \ldots & 0 \\ 0 & 2m & m & \ldots & m \\ 0 & m & 2m & \ldots & m \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & m & m & \ldots & 2m \end{bmatrix} \qquad 8.6.6.2(f)$$

and therefore,

$$(X^{*T}X^*)^{-1} = \begin{bmatrix} \frac{1}{rm} & 0 & 0 & \ldots & 0 \\ 0 & \frac{r-1}{rm} & -\frac{1}{rm} & \ldots & -\frac{1}{rm} \\ 0 & -\frac{1}{rm} & \frac{r-1}{rm} & \ldots & -\frac{1}{rm} \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & -\frac{1}{rm} & -\frac{1}{rm} & \ldots & \frac{r-1}{rm} \end{bmatrix} \qquad 8.6.6.2(g)$$

Let $T_i = \sum_{j=1}^{m} Y_{ij}$. Then

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$$X^{*}Y = \begin{pmatrix} \sum_{i=1}^{r} T_i \\ T_1 - T_r \\ T_2 - T_r \\ \vdots \\ T_2 - T_r \end{pmatrix} \qquad 8.6.6.2(h)$$

Thus

$$\hat{\mu} = \bar{Y} = \sum_{j=1}^{m} \sum_{i=1}^{r} Y_{ij}/mr \qquad 8.6.6.2(i)$$

and

$$\hat{\tau}_i = \sum_{j=1}^{n} Y_{ij}/n - \hat{\mu}, \qquad i = 1, 2, \ldots, k \qquad 8.6.6.2(j)$$

As before (Section 8.5.8), define

$$SSE = \sum_{ij} (Y_{ij} - \hat{\mu} - \hat{\tau}_i)^2 \qquad 8.6.6.2(k)$$

$$S_y^2 = \frac{SSE}{mr - r} \qquad 8.6.6.2(l)$$

and

$$SST = \sum_{ij} (Y_{ij} - \bar{Y})^2 \qquad 8.6.6.2(m)$$

Then, defining

$$SSR = SST - SSE \qquad 8.6.6.2(n)$$

a test of the hypothesis, $\tau_i = 0$, $i = 1, 2, \ldots, r$ may be carried out by defining

$$F = \frac{SSR/(r-1)}{S_y^2} \qquad 8.6.6.2(o)$$

which has the F-distribution with r - 1 and mr - r degrees of freedom. Thus, the treatments $\{\tau_i\}$ would be considered significant at level α if Equation 8.6.6.2(o) exceeds the $1-\alpha$ quantile for $\gamma_1 = r - 1$ and $\gamma_2 = mr - r$ degrees of freedom. Table 8.8.5 provides these values for $\alpha = 0.05$. A B-basis value is given by

$$B = (\hat{\mu} + \hat{\tau}_i) - k_B S_y \qquad i = 1, 2, \ldots, r \qquad 8.6.6.2(p)$$

when the ith treatment applies, where

$$k_B = [X_0^T (X^T X)^{-1} X_0]^{\frac{1}{2}} t_{\gamma, 0.95}(\delta) \qquad 8.6.6.2(q)$$

By direct calculation, using Equation 8.6.6.2(g), it follows that

$$k_B = \frac{1}{\sqrt{m}} t_{\gamma, 0.95}(\delta) \qquad 8.6.6.2(r)$$

where $\gamma = rm - r$ (the number of degrees of freedom) and δ, the non-centrality parameter is given by

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(2) Single shear configuration per figure 4.4.2.1

$$\delta = 1.282\sqrt{m} \qquad 8.6.6.2(s)$$

8.6.7 Analysis of variance procedures. This section contains a discussion of the standard one-way analysis of variance (ANOVA) procedures for analyzing multiple samples of data. Two different models are referred to: the fixed effects model and the random effects model. The fixed effects model is appropriate when inferences are to be made only to the populations from which the actual samples were drawn. The random effects model is appropriate when the samples are viewed as representing a random selection from a population. In the random effects model, inferences are made to the whole collection of populations, not just those from which samples were actually drawn. Although the two models involve different assumptions and lead to different interpretations of the data, the computational formulas for computing sums of squares are the same.

The basic model can be written as

$$X_{ij} = \mu + b_i + e_{ij}, \qquad \begin{matrix} i = 1, \ldots, k \\ j = 1, \ldots, n_i \end{matrix} \qquad 8.6.7(a)$$

where k is the number of different samples, n_i is the number of observations in the ith sample, and X_{ij} represents the jth observation in the ith sample.

The ANOVA models represent each observation as the sum of three components; μ is the overall average of the population, b_i is the effect attributed to the ith sample, and e_{ij} is a random error term which represents unexplainable sources of variation. The error terms, e_{ij}, are assumed to be independently distributed normal random variables with a mean of zero and a variance of σ_e^2 (the within-sample variance). In the fixed effects model, the b_i values are assumed to be population parameters, taking on a specific value for each sample. In the random effects model, the b_i are assumed to be independent random variables following a normal distribution with a mean of zero and a variance of σ_b^2 (the between-sample variance). See Reference 8.6.7 for more information on analysis of variance procedures.

8.6.7.1 *Computational procedures.* The analysis of data based on either of the above models involves computing sums of squares which measure variability between samples and within samples. Two procedures for computing these quantities are presented. The first assumes that all of the original observations are available. The second assumes that only the batch means, standard deviations and sample sizes are available. Both of these procedures yield the same results for the computed sums of squares.

8.6.7.1.1 *Computations based on individual measurements.* When all of the observations in a sample are available, the first step is to compute averages. The overall average is computed as

$$\bar{X} = \sum_{i=1}^{k} \sum_{j=1}^{n_1} X_{ij} / N \qquad 8.6.7.1.1(a)$$

$$N = \sum_{i=1}^{k} n_i \qquad 8.6.7.1.1(b)$$

where N is the total sample size. The sample averages are computed as

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nditions minimizing allowable for a particular material system would be obtained using tests with con

or more specimens) or, to some extent, by the Hanson-Koopmans method (which can be quite conservative for small samples).

Consequently, the procedures outline herein for determining A-basis values should be regarded as providing a safety level for the B-basis values, but not as 95% confidence limits on the first percentile.

8.6.8.1 *Weibull model.* Proceed as in Section 8.5.6, except replace the value V with the corresponding value V_A from Table 8.8.11. These factors are appropriate for both two-parameter and three-parameter Weibull models.

8.6.8.2 *Normal model.* Proceed as in Section 8.5.5, except replace the value k_B with the corresponding value from Table 8.8.12.

8.6.8.3 *Lognormal model.* Proceed as in Section 8.5.5.2, except replace the value k_B with the corresponding value from Table 8.8.12.

8.6.8.4 *Nonparametric method.* Proceed as in Section 8.5.7.1, except replace MIL-HDBK-17 Table 8.8.3 with Table 8.8.13. If $n < 299$ (which will nearly always be the case), then use the Hanson-Koopmans method below.

8.6.8.5 *Hanson-Koopmans method.* Find the value k_A corresponding to the sample size n in Table 8.8.14, then proceed as follows. If x_n denotes the largest sample value, and x_1 denotes the smallest sample value, then the A-basis value is:

$$A = x_{(n)} \left[\frac{x_{(1)}}{x_{(n)}} \right]^{k_A} \qquad 8.6.8.5$$

8.6.8.6 *ANOVA method.* Section 8.5.4 describes the ANOVA method for both A- and B- basis values.

8.6.8.7 *Linear regression.* The method for calculating A-basis values for linear regression are noted at the end of Section 8.6.6 and Equation 8.5.8(v).

8.6.9 Sample size guidelines for determining basis values. Material basis values are often regarded as material properties, that is, these values are interpreted as constants which can be used to help characterize the material and processing. Since basis values will *always* vary from one set of data to the next, even if the material, conditioning, and test remain unchanged, treating them as material constants is always an approximation.

However, if the calculations are based on 'enough' data, the basis values should be reproducible, to within engineering accuracy, across comparable data sets. The objective of this section is to illustrate the small-sample reproducibility problem and to provide guidance on how much data is necessary in basis value calculations in order for these values to be approximately reproducible.

How much data is 'enough' depends on many factors, including

- The statistical model which is used to approximate the population from which the data is sampled,
- The degree of reproducibility which is desired,

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- The variability in the property being measured, and
- Variability in measurements of the property due to the test method

Because of this, it is impossible to give firm recommendations. The discussion in this section has another purpose. It is intended to provide background information and guidelines to assist the user of this handbook in making a sample size decision. We emphasize that this section deals *only* with the stability of basis values with respect to sample size. Another important issue relevant to the choice of a sample size, which deserves separate consideration, is the effect on basis values of statistical model assumptions - since there is considerable uncertainty in model selection from small samples.

8.6.9.1 *Example.* Table 8.6.9.1 presents actual tensile strength data (in ksi) for a unidirectional composite material, tested under room temperature dry conditions.

TABLE 8.6.9.1 *Room temperature dry tensile strength for a unidirectional composite material*

226	227	226	232	252

The mean and standard deviation for these data are

$$\bar{X} = 232.6$$

and

$$s = 11.13$$

respectively. Using the normal model, a B-basis value for these data is

$$B_1 = \bar{X} - k_B s = 232.6 - 3.407(11.13) = 194.68$$

The first point to be made is that a B-basis value determined from as few as five specimens is not likely to be sufficiently reproducible for it to be regarded as a material constant for most applications. For the present discussion, we make the plausible assumption that the above data are a sample from a normal distribution with a mean of 230 and a standard deviation of 10.

The theoretical population of B-basis values which corresponds to this assumed normal population of strength measurements can be calculated, and is displayed in Figure 8.6.9.1. Note that the observed basis value is near the mean of this population of basis values. This is to be expected since we chose the parameters of our hypothetical normal distribution from the same set of data from which we determined the basis value. However, note also that values within +20 ksi of the basis value are also likely to be observed. Based on this analysis, we can not rule out the B-basis value of our next sample of five to be as low as 180 ksi or as high as 220 ksi.

8.6.9.2 *Mean and standard deviations of normal basis values.* Basis values calculated from small samples exhibit high variability. One way of quantifying this is to calculate the theoretical mean, standard deviation, and coefficient of variation of basis values from hypothetical populations as functions of the number of specimens. Of course, these calculations are going to depend on the statistical model chosen and the parameters selected for this model. However, the objective of these

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calculations are not to provide rigid criteria, but rather to inform the user of the qualitative behavior of basis values.

A normal population with a mean of 100 and a standard deviation of 10 will be considered for the discussion in this subsection. The 10% coefficient of variation is typical of what is observed for many material properties, and the mean of 100 is within an order of magnitude of most strength measurements (in ksi) for unidirectional composite materials. The choice of the normal population is made because the normal basis values procedures have broad appeal, and because the required calculations can be done in closed form. Sample sizes for basis values from Weibull populations should as a rule be larger than those for normal populations in order to achieve the same degree of reproducibility. Only B-basis values for a simple random sample are considered here; ANOVA basis values and A-Basis values are discussed in two following subsections.

The mean and one standard deviation limits for B-basis values from a normal population with a mean of 100 and a standard deviation of 10 is displayed in Figure 8.6.9.2(a) as a function of the number of specimens. Note the extremely high variability for sample sizes of ten or less.

The coefficient of variation (CV) is the ratio of the standard deviation to the mean. It is, therefore, easy to obtain the CV as a function of sample size from the information in Figure 8.6.9.2(a). Figure 8.6.9.2(b) displays these CV values, with a horizontal line at 10% provided for reference.

8.6.9.3 *Basis values using the ANOVA method.* When the data comes from several batches, and the between batch variability is substantial, the flowchart (Figure 8.5.2) might indicate that the ANOVA method of Section 8.5.4 should be used. To decide how many specimens are required when the data is to come from several batches begin by acting as if the data were from a single batch, and

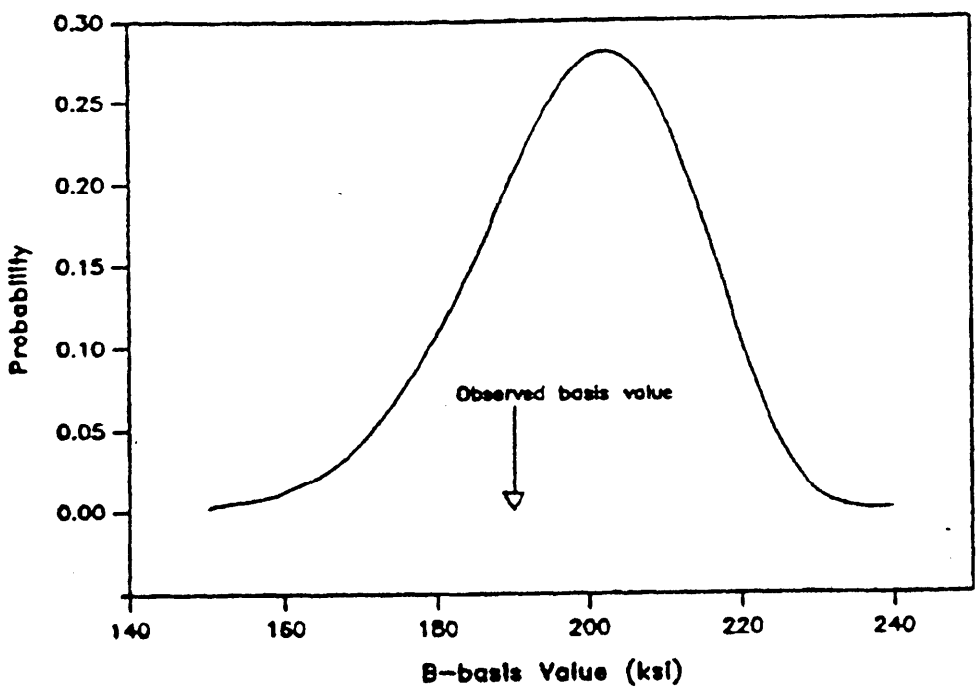


FIGURE 8.6.9.1 *B-basis value population for a sample of size five.*

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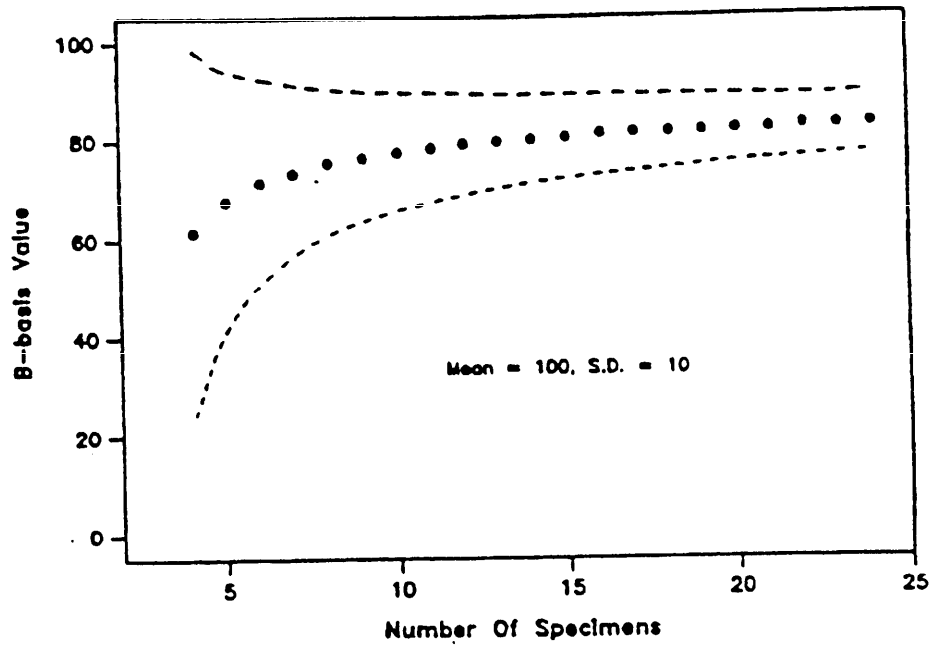


FIGURE 8.6.9.2(a) *Normal B-basis values with one-sigma limits.*

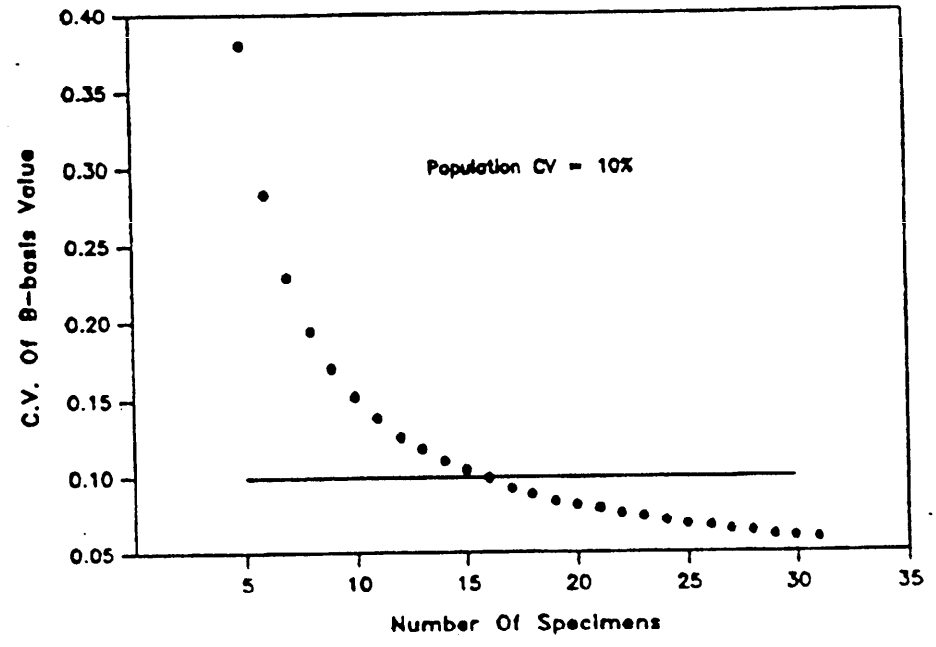


FIGURE 8.6.9.2(b) *C.V. of B-basis values: normal model.*

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selecting a sample size, say n, based on the discussion of the previous subsection. If J is the number of specimens per batch (assumed equal for all batches) and ρ is the correlation between any two measurements taken on specimens from the same batch, then the number of specimens required for comparable reproducibility in the multi-batch case is approximately

$$\tilde{n} = [J\rho + 1 - \rho]\, n \qquad 8.6.9.3$$

If $\rho = 0$, there is no between-batch variability; hence $\tilde{n} = n$. At the other extreme, if $\rho = 1$, there is perfect correlation within each batch (that is, each batch consists of J copies of a single value), and $\tilde{n} = Jn$, one needs n batches to have the same degree of reproducibility as n specimens in the uncorrelated ($\rho = 0$) case. In practice, ρ is unknown. For sample size guidelines, letting $\rho = 1/2$ in Equation (8.6.9.3) is adequate for most applications. This suggests that $[(J + 1)/(2J)] + n$ batches of size J are necessary for the same degree of reproducibility as a single sample of size n.

It is usually preferable to divide a fixed number of specimens among as many batches as is possible. However, testing a new batch is much more expensive than testing several more specimens within a single batch. It is sometimes the case that the variability between two panels from the same batch, processed and tested separately, is comparable to the variability between two panels from different batches. When this is the case, it is reasonable to substitute multiple panels within a batch for multiple batches.

8.6.9.4 *Sample size guidelines for A-basis values.* Since an A-basis value is a 95% lower confidence limit on the *first* population percentile, while a B-basis value is a 95% lower confidence limit on the *tenth* percentile, it is obvious that, for a given amount of reproducibility in the basis values, substantially more data is required for A-basis than for B-basis. If one assumes that the measurements are a sample from a normal distribution, then it is reasonable to decide on the number of specimens as for B-basis and then multiply the resulting n (or, for multiple batches, n) by three to get an A-basis sample size. This is based on the assumption that the population coefficient of variation is less than 15%.

The following example illustrates how these sample size guidelines can be used in practice. Suppose that an A-basis ANOVA value is desired which has the same degree of reproducibility as a B-basis value would have for a single sample of size $n_B = 5$. First, make the adjustment to an A-basis sample size: $n_A = 3 \cdot 5 = 15$. Next, assuming moderate between-batch variability and a batch size of (say) $J = 3$, calculate that $n_A[(J + 1)/(2J)] = 10$ batches are required for the desired degree of reproducibility, for a total of 30 specimens.

8.6.10 Confidence intervals for the coefficient of variation. The coefficient of variation is defined to be the ratio of the population standard deviation to the population mean. This subsection provides a method for calculating confidence intervals for a coefficient of variation, assuming that the underlying distribution is normal. The coefficient of variation of the population is estimated by the *sample* coefficient of variation

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$$\frac{1}{n_1}\sum_{j=1}^{L} h_j \frac{(NF_{1j} - n_1H_j)^2}{H_j(N-H_j) - Nh_j/4} = \frac{1}{7}\sum_{j=1}^{47} h_j \frac{(53F_{1j} - 7H_j)^2}{H_j(53-H_j) - 53h_j/4} = 363.33 \qquad 8.6.3.6(d)$$

When these calculations are repeated for the remaining eight batches, the k-sample Anderson-Darling statistic is computed as

$$\begin{aligned} ADK &= \frac{N-1}{N^2(k-1)}\sum_{i=1}^{k}\left[\frac{1}{n_i}\sum_{j=1}^{L} h_j \frac{(NF_{ij} - n_iH_j)^2}{H_j(N-H_j) - Nh_j/4}\right] \\ &= \frac{52}{53^2(9-1)}(363.33 + \ldots + 324.11) \\ &= 2.67 \end{aligned} \qquad 8.6.3.1(a)$$

The computed value of the statistic is compared to the critical value from Equation 8.6.3.1(j), which is 1.44. Since the computed value of 2.67 is greater than the critical value of 1.44, the hypothesis that the populations from which these groups were drawn are identical is rejected.

Problem 1 - Step 3. The equality of variance test described in Section 8.6.3.2 is used to determine if the within-batch variances are significantly different. Relevant sample statistics for the nine batches are given in the table below.

Batch	n_i	S_i^2	Y_i	$z_i = \ell n\ (S_i^2)$
1	7	3.46	6	1.24
2	5	3.61	4	1.28
3	5	5.34	4	1.68
4	7	6.25	6	1.83
5	6	22.37	5	3.11
6	5	19.01	4	2.95
7	5	65.29	4	4.18
8	6	33.76	5	3.52
9	7	44.09	6	3.79

The test statistic is then calculated as follows.

$$\begin{aligned} EV &= \frac{1}{2}\sum_{i=1}^{k}\gamma_i Z_i^2 - \frac{1}{2}(n-k)\bar{Z}^2 \\ &= \frac{1}{2}\sum_{i=1}^{9}\gamma_i Z_i - \frac{1}{2}(53-9)(2.61) \\ &= 24.63 \end{aligned}$$

The critical chi-square value from Table 8.8.8 corresponding to k - 1 = 8 degrees of freedom is 15.51. Since the test statistic is greater than the critical value, the hypothesis that the variances are equal is rejected. Since the equality of variance test is a diagnostic test, a B-basis value can be

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calculated using the ANOVA method. However, a nonconservative B-basis value can result is some instances. Potential problems with consistency in fabrication or processing of the different batches should be examined.

Problem 1 - Step 4. The ANOVA method of Section 8.5.4 is used to calculate a B-basis value. (See Problem 2 for the details of the computations using the ANOVA method.) The B-basis value is calculated as

$$B = \bar{X} - TS = 69.78 - 2.123(6.641) = 55.68$$

For presentation in MIL-HDBK-17, this B-basis value would be rounded to 55.7.

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8.8 CALCULATIONS FOR STATISTICAL TABLES. This section contains a number of tables which are required for the analyses described in Sections 8.5 and 8.6. Tables 8.8.4 through 8.8.7, 8.8.10, 8.8.15, and 8.8.16 were generated specifically for MIL-HDBK-17. The remaining tables were adapted from MIL-HDBK-5 (see Reference 8.5.5).

For some of the tabulated values, theoretical derivations and numerical approximations are provided below. The approximations are useful in computer applications when the software required to generate the tabulated values is not available. The accuracy of the approximations is measured by the relative magnitude of error (RME). The RME is defined as

$$\text{RME} = \frac{|\text{approximate value} - \text{actual value}|}{\text{actual value}} \qquad 8.8$$

and measures the percentage error in the approximate value with respect to the actual value.

8.8.1 One-sided tolerance factors, k_B, for the normal distribution, 0.95 confidence. The k_B values in Table 8.8.1 are calculated as $1/\sqrt{n}$ times the 0.95th quantile of the noncentral t-distribution with noncentrality parameter $1.282\sqrt{n}$ and n - 1 degrees of freedom. An approximation to the k_B values in Table 8.8.1 is:

$$k_B \approx 1.282 + \exp\{0.958 - 0.520\,\ell n(n) + 3.19/n\} \qquad 8.8.1$$

This approximation is accurate to within 0.2% of the tabulated values for n greater than or equal to 16.

8.8.2 One-sided tolerance factors, V, for the Weibull distribution, 0.95 confidence. The V values in Table 8.8.2 are calculated using the following statistical results. First, define the random variables

$$A_i = \frac{\ell n(X_i) - \ell n(\hat{\alpha})}{1/\hat{\beta}} \qquad i = 1, \ldots, n \qquad 8.8.2(a)$$

where X_i is a Weibull random variable with unknown shape and scale parameters β and α and $\hat{\alpha}$ and $\hat{\beta}$ are the maximum likelihood estimators (MLE's) of β and α given by Equations 8.5.6.1(a) and 8.5.6.1(c). For a particular n, the V value is the 0.95th quantile of the conditional distribution of the random variable

$$V = \frac{\sqrt{n}[\ell n(\hat{Q}) - \ell n(Q)]}{1/\hat{\beta}} \qquad 8.8.2(b)$$

given that

$$A_i = \frac{\ell n(x_i') - \ell n(\hat{\alpha}')}{1/\hat{\beta}'} \qquad 8.8.2(c)$$

where

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$$x_i' = -\ell n\left(1 - \frac{i - 0.5}{n + 0.25}\right) \qquad i = 1,\ldots,n \qquad 8.8.2(d)$$

$$\hat{Q} = \hat{\alpha}(0.10536)^{1/\beta} \qquad 8.8.2(e)$$

$$Q = \alpha(0.10536)^{1/\beta} \qquad 8.8.2(f)$$

and $\hat{\alpha}'$ and $\hat{\beta}'$ are the MLE's of the two-parameter Weibull scale and shape parameters for the sample x_1', ..., x_n'. The conditional distribution of V is determined by the relationship

$$V = \sqrt{n}\,[Z + \ell n(0.10536)] \qquad 8.8.2(g)$$

where the distribution of Z is given in Theorem 4.1.3 on page 150 of reference 8.6.4.1. Numerical integration was used to determine the V values in Table 8.8.2 based on these results.

An approximation to the V values in Table 8.8.2 is:

$$V \approx 3.803 + \exp\left\{1.79 - 0.516\,\ell n(n) + \frac{5.1}{n-1}\right\} \qquad 8.8.2(h)$$

This approximation is accurate to within 0.5% of the tabulated values for n greater than or equal to 16.

8.8.3 Ranks, r, of observations for determining B-values for an unknown distribution. For $n > 29$, an approximation to the ranks for B-basis values in Table 8.8.3 is

$$r = \frac{n}{10} - 1.645\sqrt{\frac{9n}{100}} + 0.23 \qquad 8.8.3$$

rounded to the nearest integer. This approximation is exact for all but 12 values of n in the range of the table ($29 \le n \le 10499$). For this small percentage of n values (0.1%), the approximation errs by one rank on the conservative side.

8.8.4 0.20 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. This table and its associated calculations are no longer required.

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8.8.5 0.95 Quantiles of the F-distribution associated with γ_1 and γ_2 degrees of freedom. An approximation to the $F_{0.95}$ values in Table 8.8.5 is

$$F_{0.95} = \exp\left(2\delta\left[1 + \frac{z^2-1}{3} - \frac{4\sigma^2}{3}\right] + 2\sigma z\sqrt{1 + \frac{\sigma^2(z^2-3)}{6}}\right) \qquad 8.8.5(a)$$

where

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$$\delta = 0.5\{1/(\gamma_2 - 1) - 1/(\gamma_1 - 1)\} \qquad 8.8.5(b)$$

$$\sigma^2 = 0.5\{1/(\gamma_2 - 1) + 1/(\gamma_1 - 1)\} \qquad 8.8.5(c)$$

$z = 1.645$

γ_1 = numerator degrees of freedom

γ_2 = denominator degrees of freedom.

(See Reference 8.8.4.)

Equations 8.8.5(a-d) are not valid when either γ_1 or γ_2 equals one. The following equations are to be used for these special cases:

For $\gamma_1 = 1$

$$F_{0.95} = \left[1.95996400 + \frac{2.37227200}{\gamma_2} + \frac{2.82250000}{\gamma_2^2} + \frac{2.555585200}{\gamma_2^3} + \frac{1.58953600}{\gamma_2^4}\right]^2 \qquad 8.8.5(e)$$

For $\gamma_2 = 1$

$$F_{0.95} = \left[0.06270671 + \frac{0.01573832}{\gamma_1} + \frac{0.00200073}{\gamma_1^2} - \frac{0.00243852}{\gamma_1^3} - \frac{0.00064811}{\gamma_1^4}\right]^{-2} \qquad 8.8.5(f)$$

8.8.6 One-sided tolerance factors, T, for the one-way random effects ANOVA, 0.95 confidence. This information is no longer used in the handbook.

8.8.7 Critical values for the k-sample Anderson-Darling test at the 0.05 significance level. The k-sample Anderson-Darling test critical values in Table 8.8.7 were calculated using Equation 8.6.3.1(j) for the case of samples of equal size n.

8.8.8 0.95 Quantiles of the chi-squared distribution associated with γ degrees of freedom. An approximation to the chi-squared quantiles ($\chi^2_{0.95}$) in Table 8.8.8 is:

$$\chi^2_{0.95} = \gamma\left[1 - \frac{2}{9\gamma} + 1.645\left(\frac{2}{9\gamma}\right)^{\frac{1}{2}}\right]^3 + \frac{9}{100\gamma} \qquad 8.8.8$$

where γ is the degrees of freedom. This approximation is accurate to within 0.2% of the tabulated values. (See Reference 8.8.8.)

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8.8.9 Critical values for the MNR outlier detection procedure ($\gamma = 0.05$). The critical values in Table 8.3.9 are computed by the following formula:

$$V_c = \frac{n-1}{\sqrt{n}} \sqrt{\frac{t^2}{n-2+t^2}} \qquad 8.8.9$$

where t is the $[1 - \gamma/(2n)]$ quantile of the t-distribution with n - 2 degrees of freedom, γ is the significance level of the test, and n is the sample size. Numbers in Table 8.8.9 are computed with a significance level of $\gamma = 0.05$. (See Reference 8.6.2(b).)

8.8.10 Nonparametric B-basis material properties for small samples. The values in Table 8.8.10 are based on Reference 8.5.7.2.

8.8.11 One-sided tolerance limit factors, V_A, for the Weibull distribution, 0.95 confidence. The V_A values in Table 8.8.11 are calculated as described in Section 8.8.2 (Reference 8.8.11). An approximation to the V_A values is:

$$V = 6.649 + \exp\left[2.55 - 0.526 \ln(n) + \frac{4.76}{n}\right] \qquad 8.8.12$$

This approximation is accurate within 0.5% of the tabulated values for n greater than or equal to 16.

8.8.12 One-sided tolerance limit factors, k_A, for the normal distribution, 0.95 confidence. The k_A values in Table 8.8.12 are calculated as $1/\sqrt{n}$ times the 0.95th quantile of the noncentral t-distribution with noncentrality parameter $2.326\sqrt{n}$ and n - 1 degrees of freedom (Reference 8.8.12). An approximation to the k_B values in Table 8.8.12 is:

$$k_B = 2.326 + \exp\left(1.34 - 0.522 \ln(n) + \frac{3.87}{n}\right) \qquad 8.8.13$$

This approximation is accurate to within 0.2% of the tabulated values for n greater than or equal to 16.

8.8.13 Ranks, r, of observations for determining A-basis values for an unknown distribution. For $n \geq 299$, an approximation to the ranks for A-basis values in Table 8.8.13 is:

$$r = \frac{n}{100} - 1.645\sqrt{\frac{99n}{10{,}000}} + 0.29 + \frac{19.1}{n} \qquad 8.8.14$$

For n less than 299, an A-allowable cannot be computed. This approximation is exact for all but 23 values of n in the range of the table ($299 \leq n \leq 11691$). For this small percentage of n values (0.2%), the approximation errs by one rank on the conservative side (Reference 8.8.13).

8.8.14 The Hanson-Koopmans method exponents for A-basis values. The values in Table 8.8.14 are based on Reference 8.8.14.

8.8.15 Two-tail probabilities for t-distribution. Table 8.8.15 was generated specifically for MIL-HDBK-17.

8.8.16 Upper-tail probabilities for the standard normal distribution. Table 8.8.16 was generated specifically for MIL-HDBK-17.

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8.8.17 **Upper tail quantiles for the t-distribution with γ degrees of freedom.** Table 8.8.17 was generated specifically for MIL-HDBK-17.

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TABLE 8.8.17 *Upper tail quantiles for the t-distribution with γ degrees of freedom.*

γ	0.75	0.90	0.95	0.975	0.99	0.995
1	1.0000	3.0777	6.3137	12.7062	31.8205	63.6568
2	0.8165	1.8856	2.9200	4.3027	6.9646	9.9248
3	0.7649	1.6377	2.3534	3.1825	4.5407	5.8409
4	0.7407	1.5332	2.1318	2.7764	3.7470	4.6041
5	0.7267	1.4759	2.0150	2.5706	3.3649	4.0322
6	0.7176	1.4398	1.9432	2.4469	3.1427	3.7074
7	0.7111	1.4149	1.8946	2.3646	2.9980	3.4995
8	0.7064	1.3968	1.8595	2.3060	2.8965	3.3554
9	0.7027	1.3830	1.8331	2.2622	2.8214	3.2498
10	0.6998	1.3722	1.8125	2.2281	2.7638	3.1693
11	0.6974	1.3634	1.7959	2.2010	2.7181	3.1058
12	0.6955	1.3562	1.7823	2.1788	2.6810	3.0545
13	0.6938	1.3502	1.7709	2.1604	2.6503	3.0123
14	0.6924	1.3450	1.7613	2.1448	2.6245	2.9768
15	0.6912	1.3406	1.7530	2.1314	2.6025	2.9467
16	0.6901	1.3368	1.7459	2.1199	2.5835	2.9208
17	0.6892	1.3334	1.7396	2.1098	2.5669	2.8982
18	0.6884	1.3304	1.7341	2.1009	2.5524	2.8784
19	0.6876	1.3277	1.7291	2.0930	2.5395	2.8609
20	0.6870	1.3253	1.7247	2.0860	2.5280	2.8453
21	0.6864	1.3232	1.7207	2.0796	2.5176	2.8314
22	0.6858	1.3212	1.7171	2.0739	2.5083	2.8188
23	0.6853	1.3195	1.7139	2.0687	2.4999	2.8073
24	0.6848	1.3178	1.7109	2.0639	2.4922	2.7969
25	0.6844	1.3163	1.7081	2.0595	2.4851	2.7874
30	0.6828	1.3104	1.6973	2.0423	2.4573	2.7500
40	0.6807	1.3031	1.6839	2.0211	2.4233	2.7045
60	0.6786	1.2958	1.6706	2.0003	2.3901	2.6603
120	0.6765	1.2886	1.6577	1.9799	2.3578	2.6174
∞	0.6745	1.2816	1.6449	1.9600	2.3263	2.5758

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Navy - AS
Air Force - 1
DOT - FAA

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Air Force - 19

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